EnergyPlus™ Version 25.1.0 Documentation

Input Output Reference

U.S. Department of Energy

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Build: 68a4a7c774

1.7.32.3 Zone Outdoor Air Wind Speed [m/s]

The outdoor wind speed calculated at the height above ground of the zone centroid.

1.7.32.4 Surface Ext Outdoor Dry Bulb [C]

The outdoor air dry-bulb temperature calculated at the height above ground of the surface centroid.

1.7.32.5 Surface Ext Outdoor Wet Bulb [C]

The outdoor air wet-bulb temperature calculated at the height above ground of the surface centroid.

1.7.32.6 Surface Ext Wind Speed [m/s]

The outdoor wind speed calculated at the height above ground of the surface centroid.

1.7.32.7 System Node Temperature [C]

When reporting for the **OutdoorAir:Node** object, this is the outdoor air dry-bulb temperature calculated at the height above ground of the node, if height is specified in the input.

1.8 Group – Schedules

This group of objects allows the user to influence scheduling of many items (such as occupancy density, lighting, thermostatic controls, occupancy activity). In addition, schedules are used to control shading element density on the building.

EnergyPlus schedules consist of three pieces: a day description, a week description, and an annual description. An optional element is the schedule type. Each description level builds off the previous sub-level. The day description is simply a name and the values that span the 24 hours in a day to be associated with that name. The week description also has an identifier (name) and twelve additional names corresponding to previously defined day descriptions. There are names for each individual day of the week plus holiday, summer design day, winter design day and two more custom day designations. Finally, the annual schedule contains an identifier and the names and FROM-THROUGH dates of the week schedules associate with this annual schedule. The annual schedule can have several FROM-THROUGH date pairs. One type of schedule reads the values from an external file to facilitate the incorporation of monitored data or factors that change throughout the year.

Schedules are processed by the EnergyPlus Schedule Manager, stored within the Schedule Manager and are accessed through module routines to get the basic values (timestep, hourly, etc). Values are resolved at the Zone Timestep frequency and carry through any HVAC timesteps.

1.8.1 Day Type

A brief description of "Day Type" which is used in the SizingPeriod objects, RunPeriodControl:SpecialDays object, the Sizing objects and also used by reference in the Schedule:Week:Daily object (discussed later in this section).

Schedules work on days of the week as well as certain specially designated days. Days of the week are the normal – Sunday, Monday, Tuesday, Wednesday, Thursday, Friday and Saturday. Special day types that can be designated are: Holiday, SummerDesignDay, WinterDesignDay, CustomDay1, CustomDay2. These day types can be used at the user's convenience to designate special scheduling (e.g. lights, electric equipment, set point temperatures) using these days as reference.

For example, a normal office building may have normal "occupancy" rules during the weekdays but significantly different use on weekend. For this, you would set up rules/schedules based on the weekdays (Monday through Friday, in the US) and different rules/schedules for the weekend (Saturday and Sunday, in the US). However, you could also specially designate SummerDesignDay and WinterDesignDay schedules for sizing calculations. These schedules can be activated by setting the Day Type field in the Design Day object to the appropriate season (**SummerDesignDay** for cooling design calculations; **WinterDesignDay** for heating design calculations).

In a different building, such as a theater/playhouse, the building may only have occupancy during certain weeks of the year and/or certain hours of certain days. If it was every week, you could designate the appropriate values during the "regular" days (Sunday through Saturday). But this would also be an ideal application for the "CustomDay1" and/or "CustomDay2". Here you would set the significant occupancy, lighting, and other schedules for the custom days and use unoccupied values for the normal weekdays. Then, using a weather file and setting special day periods as appropriate, you will get the "picture" of the building usage during the appropriate periods.

1.8.2 ScheduleTypeLimits

Schedule types can be used to validate portions of the other schedules. Hourly day schedules, for example, are validated by range—minimum/maximum (if entered)—as well as numeric type (continuous or discrete). Annual schedules, on the other hand, are only validated for range—as the numeric type validation has already been done.

1.8.2.1 Inputs

1.8.2.1.1 Field: Name

This alpha field should contain a unique (within the schedule types) designator. It is referenced wherever Schedule Type Limits Names can be referenced.

1.8.2.1.2 Field: Lower Limit Value

In this field, the lower (minimum) limit value for the schedule type should be entered. If this field is left blank, then the schedule type is not limited to a minimum/maximum value range.

1.8.2.1.3 Field: Upper Limit Value

In this field, the upper (maximum) limit value for the schedule type should be entered. If this field is left blank, then the schedule type is not limited to a minimum/maximum value range.

1.8.2.1.4 Field: Numeric Type

This field designates how the range values are validated. Using **Continuous** in this field allows for all numbers, including fractional amounts, within the range to be valid. Using **Discrete** in this field allows only integer values between the minimum and maximum range values to be valid.

1.8.2.1.5 Field: Unit Type

This field is used to indicate the kind of units that may be associated with the schedule that references the ScheduleTypeLimits object. It is used by IDF Editor to display the appropriate SI and IP units. This field is not used by EnergyPlus. The available options are shown below. If none of these options are appropriate, select **Dimensionless.**

- Dimensionless
- Temperature
- DeltaTemperature
- PrecipitationRate
- Angle
- Convection Coefficient
- Activity Level
- Velocity
- Capacity
- Power
- Availability
- Percent
- Control
- Mode

Several IDF Examples will illustrate the use:

```
ScheduleTypeLimits,Any Number;  ! Not limited
ScheduleTypeLimits,Fraction, 0.0 , 1.0 ,CONTINUOUS;
ScheduleTypeLimits,Temperature,-60,200,CONTINUOUS;
ScheduleTypeLimits,Control Type,0,4,DISCRETE;
ScheduleTypeLimits,On/Off,0,1,DISCRETE;
```

1.8.3 Day Schedules

The day schedules perform the assignment of pieces of information across a 24 hour day. This can occur in various fashions including a 1-per hour assignment, a user specified interval scheme or a list of values that represent an hour or portion of an hour.

1.8.4 Schedule:Day:Hourly

The Schedule:Day:Hourly contains an hour-by-hour profile for a single simulation day.

1.8.4.1 Inputs

1.8.4.1.1 Field: Name

This field should contain a unique (within all DaySchedules) designation for this schedule. It is referenced by WeekSchedules to define the appropriate schedule values.

1.8.4.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.4.1.3 Field: Hour Values (1-24)

These fields contain the hourly values for each of the 24 hours in a day. (Hour field 1 represents clock time 00:00:01 AM to 1:00:00 AM, hour field 2 is 1:00:01 AM to 2:00:00 AM, etc.) The values in these fields will be passed to the simulation as indicated for "scheduled" items.

An IDF example:

```
Schedule:Day:Hourly, Day On Peak, Fraction,
  0.,0.,0.,0.,0.,0.,0.,0.,0.,1.,1.,1.,1.,1.,1.,1.,1.,1.,0.,0.,0.,0.,0.,0.;
```

1.8.5 Schedule:Day:Interval

The Schedule:Day:Interval introduces a slightly different way of entering the schedule values for a day. Using the intervals, you can shorten the "hourly" input of the "Schedule:Day:Hourly" object to 2 fields. And, more importantly, you can enter an interval that represents only a portion of an hour. Schedule values are "given" to the simulation at the zone timestep, so there is also a possibility of "interpolation" from the entries used in this object to the value used in the simulation.

1.8.5.1 Inputs

1.8.5.1.1 Field: Name

This field should contain a unique (within all DaySchedules) designation for this schedule. It is referenced by WeekSchedules to define the appropriate schedule values.

1.8.5.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.5.1.3 Field: Interpolate to Timestep

Three possible inputs are available for this field: Average, Linear, and No. The default value is No.

If "Average" is entered, it is used to apply values that aren't coincident with the given timestep (ref: Timestep) intervals. If "Average" is entered, then any intervals entered here will be interpolated/averaged and that value will be used at the appropriate minute in the hour. For example, if "Average" is entered and the interval is every 15 minutes (say a value of 0 for the first 15 minutes, then 0.5 for the second 15 minutes) AND there is a 10 minute timestep for the simulation: the value at 10 minutes will be 0 and the value at 20 minutes will be 0.25. In earlier versions of EnergyPlus, this was the Yes option for the field.

If "No" is entered, then the value that occurs on the appropriate minute in the hour will be used as the schedule value. For the same input entries but "no" for this field, the value at 10 minutes will be 0 and the value at 20 minutes will be 0.5. No is the default for this field.

If "Linear" is entered, then the value that is used is based on the linear interpolation between successive values. With linear, if the value at 1:00 is 0.0 and the value at 2:00 is 10.0, with fifteen minute timesteps, the value at 1:15 would be 2.5, the value at 1:30 would be 5.0 and the value at 1:45 would be 7.5.

1.8.5.1.4 Field-Set: Time and Value (extensible object)

To specify each interval, both an "until" time (which includes the designated time) and the value must be given. This object is extensible, so additional pairs of the following two fields can be added to the end of this object.

1.8.5.1.5 Field: Time

The value of each field should represent clock time (standard) in the format "Until: HH:MM". 24 hour clock format (i.e. 1PM is 13:00) is used. Note that Until: 7:00 includes all times up through 07:00 or 7am.

1.8.5.1.6 Field: Value

This represents the actual value to be passed to the simulation at the appropriate timestep. (Using interpolation value as shown above). Limits on the values are indicated by the Schedule Type Limits Name field of this object.

And an example of use:

```
Schedule:Day:Interval,
  dd winter rel humidity, !- Name
  Percent,                !- Schedule Type Limits Name
  No,                     !- Interpolate to Timestep
  until: 24:00,           !- Time 1
  74;                     !- Value Until Time 1
```

1.8.6 Schedule:Day:List

To facilitate possible matches to externally generated data intervals, this object has been included. In similar fashion to the Schedule:Day:Interval object, this object can also include "sub-hourly" values but must represent a complete day in its list of values.

1.8.6.1 Inputs

1.8.6.1.1 Field: Name

This field should contain a unique (within all day schedules) designation for this schedule. It is referenced by week schedules to define the appropriate schedule values.

1.8.6.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.6.1.3 Field: Interpolate to Timestep

Three possible inputs are available for this field: Average, Linear, and No. The default value is No.

If "Average" is entered, it is used to apply values that aren't coincident with the given timestep (ref: Timestep) intervals. If "Average" is entered, then any intervals entered here will be interpolated/averaged and that value will be used at the appropriate minute in the hour. For example, if "Average" is entered and the interval is every 15 minutes (say a value of 0 for the first 15 minutes, then 0.5 for the second 15 minutes) AND there is a 10 minute timestep for the simulation: the value at 10 minutes will be 0 and the value at 20 minutes will be 0.25. In earlier versions of EnergyPlus, this was the Yes option for the field.

If "No" is entered, then the value that occurs on the appropriate minute in the hour will be used as the schedule value. For the same input entries but "no" for this field, the value at 10 minutes will be 0 and the value at 20 minutes will be 0.5. No is the default for this field.

If "Linear" is entered, then the value that is used is based on the linear interpolation between successive values. With linear, if the value at 1:00 is 0.0 and the value at 2:00 is 10.0, with fifteen minute timesteps, the value at 1:15 would be 2.5, the value at 1:30 would be 5.0 and the value at 1:45 would be 7.5.

1.8.6.1.4 Field: Minutes Per Item

This field allows the "list" interval to be specified in the number of minutes for each item. The value here must be <= 60 and evenly divisible into 60 (same as the timestep limits).

1.8.6.1.5 Field Value 1 (same definition for each value – up to 1440 (24*60) allowed)

This is the value to be used for the specified number of minutes.

For example:

```
Schedule:Day:List,
  Myschedule, ! name
  Fraction,   ! Schedule type
  No,         ! Interpolate value
  30,         ! Minutes per item
  0.0,        ! from 00:01 to 00:30
  0.5,        ! from 00:31 to 01:00
  <snipped>
```

1.8.7 Week Schedule(s)

The week schedule object(s) perform the task of assigning the day schedule to day types in the simulation. The basic week schedule is shown next.

1.8.8 Schedule:Week:Daily

1.8.8.1 Inputs

1.8.8.1.1 Field: Name

This field should contain a unique (within all WeekSchedules) designation for this schedule. It is referenced by Schedules to define the appropriate schedule values.

1.8.8.1.2 Field: Schedule Day Name Fields (12 day types – Sunday, Monday, ...)

These fields contain day schedule names for the appropriate day types. Days of the week (or special days as described earlier) will then use the indicated hourly profile as the actual schedule value.

An IDF example:

```
Schedule:Week:Daily, Week on Peak,
  Day On Peak,Day On Peak,Day On Peak,
  Day On Peak,Day On Peak,Day On Peak,
  Day On Peak,Day On Peak,Day On Peak,
  Day On Peak,Day On Peak,Day On Peak;
```

1.8.9 Schedule:Week:Compact

Further flexibility can be realized by using the Schedule:Week:Compact object. In this the fields, after the name is given, a "for" field is given for the days to be assigned and then a dayschedule name is used.

1.8.9.1 Inputs

1.8.9.1.1 Field:Name

This field should contain a unique (within all WeekSchedules) designation for this schedule. It is referenced by Schedules to define the appropriate schedule values.

1.8.9.1.2 Field-Set – DayType List#, Schedule:Day Name

Each assignment is made in a pair-wise fashion. First the "days" assignment and then the dayschedule name to be assigned. The entire set of day types must be assigned or an error will result.

1.8.9.1.3 Field: DayType List

This field can optionally contain the prefix "For" for clarity. Multiple choices may then be combined on the line. Choices are: Weekdays, Weekends, Holidays, Alldays, SummerDesignDay, WinterDesignDay, Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, CustomDay1, CustomDay2. In fields after the first "for", AllOtherDays may also be used. Note that the colon (:) after the For is optional but is suggested for readability.

1.8.9.1.4 Field: Schedule:Day Name

This field contains the name of the day schedule (any of the Schedule:Day object names) that is to be applied for the days referenced in the prior field.

Some IDF examples:

```
Schedule:Week:Compact, Week on Peak,
  For: AllDays,
  Day On Peak;

Schedule:Week:Compact, WeekDays on Peak,
  WeekDays,
  Day On Peak,
  AllOtherDays
  Day Off Peak;
```

1.8.10 Schedule:Year

The yearly schedule is used to cover the entire year using references to week schedules (which in turn reference day schedules). If the entered schedule does not cover the entire year, a fatal error will result.

1.8.10.1 Inputs

1.8.10.1.1 Field: Name

This field should contain a unique (between Schedule:Year, Schedule:Compact, and Schedule:File) designation for the schedule. It is referenced by various "scheduled" items (e.g. Lights, People, Infiltration) to define the appropriate schedule values.

1.8.10.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.10.1.3 Field Set (WeekSchedule, Start Month and Day, End Month and Day)

Each of the designated fields is used to fully define the schedule values for the indicated time period). Up to 53 sets can be used. An error will be noted and EnergyPlus will be terminated if an incomplete set is entered. Missing time periods will also be noted as warning errors; for these time periods a zero (0.0) value will be returned when a schedule value is requested. Each of the sets has the following 5 fields:

1.8.10.1.4 Field: Schedule Week Name

This field contains the appropriate WeekSchedule name for the designated time period.

1.8.10.1.5 Field: Start Month

This numeric field is the starting month for the schedule time period.

1.8.10.1.6 Field: Start Day

This numeric field is the starting day for the schedule time period.

1.8.10.1.7 Field: End Month

This numeric field is the ending month for the schedule time period.

1.8.10.1.8 Field: End Day

This numeric field is the ending day for the schedule time period.

Note that there are many possible periods to be described. An IDF example with a single period:

```
Schedule:Year, On Peak, Fraction,
  Week On Peak, 1,1, 12,31;
```

And a multiple period illustration:

```
Schedule:Year,CoolingCoilAvailSched,Fraction,
  FanAndCoilAllOffWeekSched, 1,1, 3,31,
  FanAndCoilSummerWeekSched, 4,1, 9,30,
  FanAndCoilAllOffWeekSched, 10,1, 12,31;
```

The following definition will generate an error (if any scheduled items are used in the simulation):

```
Schedule:Year,MySchedule,Fraction,4,1,9,30;
```

1.8.11 Schedule:Compact

For flexibility, a schedule can be entered in “one fell swoop”. Using the Schedule:Compact object, all the features of the schedule components are accessed in a single command. Like the “regular” schedule object, each schedule:compact entry must cover all the days for a year. Additionally, the validations for DaySchedule (i.e. must have values for all 24 hours) and WeekSchedule (i.e. must have values for all day types) will apply. Schedule values are “given” to the simulation at the zone timestep, so there is also a possibility of “interpolation” from the entries used in this object to the value used in the simulation.

This object is an unusual object for description. For the data the number of fields and position are not set, they cannot really be described in the usual Field # manner. Thus, the following description will list the fields and order in which they must be used in the object. The name and schedule type are the exceptions:

1.8.11.1 Inputs

1.8.11.1.1 Field: Name

This field should contain a unique (between Schedule:Year, Schedule:Compact, and Schedule:File) designation for the schedule. It is referenced by various “scheduled” items (e.g. Lights, People, Infiltration) to define the appropriate schedule values.

1.8.11.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.11.1.3 Field-Set (Through, For, Interpolate, Until, Value)

Each compact schedule must contain the elements Through (date), For (days), Interpolate (optional), Until (time of day) and Value. In general, each of the “titled” fields must include the “title”. Note that the colon (:) after these elements (Through, For, Until) is optional but is suggested for readability.

1.8.11.1.4 Field: Through

This field starts with "Through:" and contains the ending date for the schedule period (may be more than one). Refer to Table 1.9. Date Field Interpretation for information on date entry – note that only Month-Day combinations are allowed for this field. Each "through" field generates a new WeekSchedule named "Schedule Name"_wk_# where # is the sequential number for this compact schedule.

1.8.11.1.5 Field: For

This field starts with "For:" and contains the applicable days (reference the compact week schedule object above for complete description) for the 24 hour period that must be described. Each "for" field generates a new DaySchedule named "Schedule Name"_dy_# where # is the sequential number for this compact schedule.

1.8.11.1.6 Field: Interpolate (optional)

This field, if used, starts with "Interpolate:" and contains the word "Average", "Linear" or "No". If this field is not used, it should not be blank – rather just have the following field appear in this slot.

If "Average" is entered, it is used to apply values that aren't coincident with the given timestep (ref: Timestep) intervals. If "Average" is entered, then any intervals entered here will be interpolated/averaged and that value will be used at the appropriate minute in the hour. For example, if "Average" is entered and the interval is every 15 minutes (say a value of 0 for the first 15 minutes, then .5 for the second 15 minutes) AND there is a 10 minute timestep for the simulation: the value at 10 minutes will be 0 and the value at 20 minutes will be .25. In earlier versions of EnergyPlus, this was the Yes option for the field.

If "No" is entered, then the value that occurs on the appropriate minute in the hour will be used as the schedule value. For the same input entries but "no" for this field, the value at 10 minutes will be 0 and the value at 20 minutes will be .5. No is the default for this field.

If "Linear" is entered, then the value that is used is based on the linear interpolation between successive values. With linear, if the value at 1:00 is 0.0 and the value at 2:00 is 10.0, with fifteen minute timesteps, the value at 1:15 would be 2.5, the value at 1:30 would be 5.0 and the value at 1:45 would be 7.5.

1.8.11.1.7 Field: Until

This field contains the ending time (again, reference the interval day schedule discussion above) for the current days and day schedule being defined.

1.8.11.1.8 Field: Value

Finally, the value field is the schedule value for the specified time interval.

And, some IDF examples:

```
Schedule:Compact,
  POFF,       !- Name
  Fraction,   !- Schedule Type Limits Name
  Through: 4/30,
  For: AllDays,
  Until: 24:00, 1.0,
  Through: 12/31,
  For: Weekdays,
    Until: 7:00,   .1,
```

```
    Until: 17:00, 1.0,
    Until: 24:00,  .1,
  For: Weekends Holidays,
    Until: 24:00,  .1,
  For: AllOtherDays,
    Until: 24:00,  .1;

! Schedule Continuous
Schedule:Compact,
  Continuous,
  on/off,
  Through: 12/31,
  For: AllDays,
    Until: 24:00, 1.0;

! Schedule Daytime Ventilation
Schedule:Compact,
  Daytime Ventilation,
  Fraction,
  Through: 12/31,
  For: Weekdays SummerDesignDay,
    Until: 08:00, 0.0,
    Until: 18:00, 1.0,
    Until: 24:00, 0.0,
  For: Weekends WinterDesignDay,
    Until: 10:00, 0.0,
    Until: 16:00, 1.0,
    Until: 24:00, 0.0,
  For: Holidays AllOtherDays,
    Until: 24:00, 0.0;
```

1.8.12 Schedule:Constant

The constant schedule is used to assign a constant hourly value. This schedule is created when a fixed hourly value is desired to represent a period of interest (e.g., always on operation mode for supply air fan).

1.8.12.1 Inputs

1.8.12.1.1 Field: Name

This field should contain a unique name designation for this schedule. It is referenced by Schedules to define the appropriate schedule value.

1.8.12.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.12.1.3 Field: Hourly Value

This field contains a constant real value. A fixed value is assigned as an hourly value.

An IDF example:

```
Schedule:Constant,
  AlwaysOn,      !- Name
  On/Off,        !- Schedule Type Limits Name
  1.0;           !- Hourly Value

ScheduleTypeLimits,
```

```
On/Off,       !- Name
0,            !- Lower Limit Value
1,            !- Upper Limit Value
DISCRETE,     !- Numeric Type
Availability; !- Unit Type
```

1.8.13 Schedule:File

At times, data is available from a building being monitored or for factors that change throughout the year. The Schedule:File object allows this type of data to be used in EnergyPlus as a schedule. Schedule:File can also be used to read in hourly or sub-hourly schedules computed by other software or developed in a spreadsheet or other utility.

The format for the data file referenced is a text file with values separated by commas (or other optional delimiters) with one line per hour. The file may contain header lines that are skipped. The file should contain values for an entire year (8760 or 8784 hours of data) or a warning message will be issued. Multiple schedules may be created using a single external data file or multiple external data files may be used. The first row of data must be for January 1, hour 1 (or timestep 1 for subhourly files).

Schedule:File may be used along with the FuelFactors object and TDV files in the DataSets directory to compute Time Dependent Valuation based source energy as used by the California Energy Commission's Title 24 Energy Code. See Fuel Factor for more discussion on Time Dependent Valuation.

Two optional fields: **Interpolate to Timestep** and **Minutes per Item** allow for the input of sub-hourly schedules (similar to the Schedule:Day:List object).

1.8.13.1 Inputs

1.8.13.1.1 Field: Name

This field should contain a unique (between Schedule:Year, Schedule:Compact, and Schedule:File) designation for the schedule. It is referenced by various "scheduled" items (e.g. Lights, People, Infiltration, FuelFactors) to define the appropriate schedule values.

1.8.13.1.2 Field: Schedule Type Limits Name

This field contains a reference to the Schedule Type Limits object. If found in a list of Schedule Type Limits (see ScheduleTypeLimits object above), then the restrictions from the referenced object will be used to validate the hourly field values (below).

1.8.13.1.3 Field: File Name

This field contains the name of the file that contains the data for the schedule. The field should include a full path with file name, for best results. The field must be $\leq$ 100 characters. The file name must not include commas or an exclamation point. A relative path or a simple file name should work with version 7.0 or later when using EP-Launch even though EP-Launch uses temporary directories as part of the execution of EnergyPlus. If using RunEPlus.bat to run EnergyPlus from the command line, a relative path or a simple file name may work if RunEPlus.bat is run from the folder that contains EnergyPlus.exe.

1.8.13.1.4 Field: Column Number

The column that contains the value to be used in the schedule. The first column is column one. If no data for a row appears for a referenced column the value of zero is used for the schedule value for that hour.

1.8.13.1.5 Field: Rows to Skip at Top

Many times the data in a file contains rows (lines) that are describing the files or contain the names of each column. These rows need to be skipped and the number of skipped rows should be entered for this field. The next row after the skipped rows must contain data for January 1, hour 1.

1.8.13.1.6 Field: Number of Hours of Data

The value entered in this field should be either 8760 or 8784 as the number of hours of data. 8760 does not include the extra 24 hours for a leap year (if needed). 8784 will include the possibility of leap year data which can be processed according to leap year indicators in the weather file or specified elsewhere. Note if the simulation does not have a leap year specified but the schedule file contains 8784 hours of data, the first 8760 hours of data will be used. The schedule manager will not know to skip the 24 hours representing February 29.

1.8.13.1.7 Field: Column Separator

This field specifies the character used to separate columns of data if the file has more than one column. The choices are: Comma, Tab, Fixed (spaces fill each column to a fixed width), or Semicolon. The default is Comma.

1.8.13.1.8 Field: Interpolate to Timestep

The value contained in this field directs how to apply values that aren't coincident with the given timestep (ref: TimeStep object) intervals. If "Yes" is entered, then any intervals entered here will be interpolated/averaged and that value will be used at the appropriate minute in the hour. If "No" is entered, then the value that occurs on the appropriate minute in the hour will be used as the schedule value.

For example, if "yes" is entered and the minutes per item is 15 minutes (say a value of 0 for the first 15 minutes, then 0.5 for the second 15 minutes) AND there is a 10 minute timestep for the simulation: the value at 10 minutes will be 0 and the value at 20 minutes will be 0.25. For the same input entries but "no" for this field, the value at 10 minutes will be 0 and the value at 20 minutes will be 0.5.

1.8.13.1.9 Field: Minutes Per Item

This field represents the number of minutes for each item – in this case each line of the file. The value here must be ≤ 60 and evenly divisible into 60 (same as the timestep limits).

Here is an IDF example:

```
Schedule:File,
  elecTDVfromCZ01res, !- Name
  Any Number,         !- ScheduleType
  TDV_kBtu_CTZ01.csv, !- Name of File
  2,                  !- Column Number
  4,                  !- Rows to Skip at Top
  8760,               !- Number of Hours of Data
  Comma;              !- Column Separator
```

or with a relational path:

```
Schedule:File,
  elecTDVfromCZ01res, !- Name
  Any Number,         !- ScheduleType
  DataSets\TDV\TDV_kBtu_CTZ01.csv,  !- Name of File
  2,                  !- Column Number
  4;                  !- Rows to Skip at Top
```

A sub-hourly indication. Note that this is identical to an hourly file because there are 60 minutes per item – the number of hours defaults to 8760 and the column separator defaults to a comma. If the number of minutes per item had been, say, 15, then the file would need to contain 8760*4 or 35,040 rows for this item.

```
Schedule:File,
  elecTDVfromCZ06com, !- Name
  Any Number,         !- Schedule Type Limits Name
  DataSets\TDV\TDV_2008_kBtu_CTZ06.csv, !- File Name
  1,                  !- Column Number
  4,                  !- Rows to Skip at Top
  ,                   !- Number of Hours of Data
  ,                   !- Column Separator
  ,                   !- Interpolate to Timestep
  60;                 !- Minutes per Item
```

1.8.13.1.10 Field: Adjust Schedule for Daylight Savings

This field indicates whether or not to shift the schedule forward an hour during daylight savings time. If "Yes" the schedule skips forward an hour in the CSV on the "spring forward" day and repeats an hour on the "fall back" day. If "No", the schedule is read and applied in standard time, without adjusting for daylight savings time (**RunPeriodControl:DaylightSavingTime**). This indicator is defaulted to "Yes" if no input is provided.

Here is an IDF example:

```
        Schedule:File,
        TestName,        !- Name
        Any Number,      !- Schedule Type Limits Name
        schedulefile.csv, !- File Name
        2,               !- Column Number
        1,               !- Rows to Skip at Top
        8760,            !- Number of Hours of Data
        Comma,           !- Column Separator
        No,              !- Interpolate to Timestep
        60,              !- Minutes per Item
        No;              !- Adjust Schedule for Daylight Savings
```

1.8.13.2 Outputs

An optional report can be used to gain the values described in the previous Schedule objects. This is a condensed reporting that illustrates the full range of schedule values—in the style of input: DaySchedule, WeekSchedule, Annual Schedule.

```
! will give them on hourly increments (day schedule resolution)
Output:Schedules, Hourly;
! will give them at the timestep of the simulation
Output:Schedules, Timestep;
```

This report is placed on the eplusout.eio file. Details of this reporting are shown in the Output Details and Examples document.

1.8.13.2.1 Schedule Value Output

```
Zone,Average,Schedule Value {[]}
```

1.8.13.2.2 Schedule Value

This is the schedule value (as given to whatever entity that uses it). It has no units in this context because values may be many different units (i.e. temperatures, fractions, watts). For best results, you may want to apply the schedule name when you use this output variable to avoid output proliferation. For example, the following reporting should yield the values shown above, depending on day of week and day type:

```
Output:Variable,People_Shopping_Sch,Schedule Value,hourly;
Output:Variable,Activity_Shopping_Sch,Schedule Value,hourly;
```

A wild card * would also work in place of the schedule name. For example, the following syntax would report all the schedule values for all the schedules in an IDF file:

```
Output:Variable,*,Schedule Value,hourly;
```

1.8.14 Schedule:File:Shading

The **Schedule:File:Shading** object allows shading schedules to be imported altogether from a file exported using ShadowCalculation Output External Shading Calculation Results. The object can also be used to read in hourly or sub-hourly schedules of the sunlit fraction of all exterior surfaces computed by other software or developed in a spreadsheet or other utility.

The format for the data file is Comma-separated values (CSV). The CSV file referenced is a text file with values separated by commas (or other optional delimiters) with one line per timestep. Each column stores the annual shading sunlit fraction schedule data of an exterior surface. The sunlit fraction is only overwritten and used when the sun is above horizon at a certain time step. To map to the surface, each column should name its column header exactly the same with the surface name defined in the **Surface:Detailed** object. If any surface name is missing, the sunlit fraction of this surface will be set to 1.0 during the run period, which means no shading will be assigned for this surface, and a warning message will be issued. The file should contain values for an entire year with the exact same number of rows as the total number of time steps in a year (number of days in a year $\times$ number of hours per day $\times$ number of time steps per hour). Time step should be consistent with the setting in the **Timestep** object. Otherwise, an error will be issued.

The first row of the CSV file should be the header, and the first column of the CSV file should be the timestamp and is not imported.

With a **Schedule:File:Shading** object defined, the shading schedules for all exterior surfaces (if defined) can be imported altogether without repeatedly defining Schedule:File objects.

1.8.14.1 Inputs

1.8.14.1.1 Field: File Name

This field contains the name of the file that contains the data for the shading schedules. The field should include a full path with file name, for best results. The field must be ≤ 100 characters. The file name must not include commas or an exclamation point. A relative path or a simple file name should work with version 7.0 or later when using EP-Launch even though EP-Launch uses temporary directories as part of the execution of EnergyPlus. If using RunEPlus.bat to run

```
Construction:WindowDataFile,
  DoubleClear,          !- Name of a Window on the Window Data File
  C:\EnergyPlusData\DataSets\MyWindow.dat;
```

1.9.52.2 Outputs

An optional report (contained in **eplusout.eio**) gives calculational elements for the materials and constructions used in the input. These reports are explained fully in the Output Details and Examples document.

1.10 Group – Thermal Zone Description/Geometry

Without spaces, thermal zones, and surfaces, the building can't be simulated. This group of objects (Space, Zone, BuildingSurface) describes the thermal zone characteristics as well as the details of each surface to be modeled. Included here are shading surfaces.

1.10.1 Space

This object defines a space (or room) in the building. All Spaces are part of a Zone. Every Zone contains one or more spaces.

Space is an optional input. If a Zone has no Space(s) specified in the input then a default Space named <Zone Name> will be created. If some surfaces in a Zone are assigned to a Space and some are not, then a default Space named <Zone Name>-Remainder will be created. Input references to Space Names must have a matching Space object. Default space names may not be referenced in the input except for Output:Variable keys).

Space geometry is specified by attaching surfaces to a space using the "Space Name" field. If a Space has only floor surface(s) and internal mass assigned to it, the Space is assumed to share the same enclosure with other such Spaces in the same Zone (and with any surfaces that are not explicitly assigned to a space). If a Space also has other types of surfaces (e.g. wall, ceiling, or roof) then the Space forms its own enclosure unless it is connected to other spaces with air boundaries (see Construction:AirBoundary).

1.10.1.1 Inputs

1.10.1.1.1 Field: Name

The name of the Space object. This name must be unique across Zone, Space, ZoneList, and SpaceList names.

1.10.1.1.2 Field: Zone Name

The name of the Zone this Space is a part of. This field is required.

1.10.1.1.3 Field: Ceiling Height

Space ceiling height is used in several areas within EnergyPlus (such as various room models, some convection coefficient calculations and, primarily, in calculating the space volume in the absence of other parameters). Energyplus automatically calculates the space ceiling height (m) from the average height of the space. If this field is 0.0, negative or **autocalculate**, then the calculated space ceiling height will be used in subsequent calculations. If this field is positive, then the calculated space ceiling height will not be used – the number entered here will be used as the space ceiling height. If this number differs significantly from the calculated ceiling height, then a warning message will be issued. If a space ceiling height and floor area are entered, but no volume is entered, then the volume will be the ceiling height times floor area.

Note that the Ceiling Height is the distance from the Floor to the Ceiling in the Space, not an absolute height from the ground.

NOTE: At the moment, the autocalculated Space ceiling height is set to the Zone ceiling height.

1.10.1.1.4 Field: Volume

Space volume is used in several areas within EnergyPlus (such as calculating air change rates for reporting or flow when air change rates are chosen as input, daylighting calculations, some convection coefficient calculations). EnergyPlus automatically calculates the space volume (m^3) from the space geometry given by the surfaces that belong to the space. If this field is 0.0, negative or **autocalculate**, then the calculated space volume will be used in subsequent calculations. If this field is positive, then it will be used as the space volume. If this number differs significantly from the calculated space volume a warning message will be issued. For autocalculate to work properly, the space must be enclosed by the entered walls. Note that indicating the volume to be calculated but entering a positive ceiling height in the previous field will cause the volume to be calculated as the floor area (if > 0) times the entered ceiling height; else the volume will be calculated from the described surfaces. If this field is positive, any ceiling height positive value will not be used in volume calculations.

NOTE: At the moment, the autocalculated Space volume is the Zone volume apportioned by floor area.

1.10.1.1.5 Field: Floor Area

Space floor area is used in many places within EnergyPlus. EnergyPlus automatically calculates the space floor area (m^2) as the total area of surfaces with Surface Type = Floor. If this field is 0.0, negative or **autocalculate** (the default), then the calculated space floor area will be used in subsequent calculations. If this field is positive, then it will be used as the space floor area. If the calculated floor area differs from the user input floor area by more than 5%, a warning message will be issued.

For each Zone, the calculated floor area is the sum of the space floor areas belonging to the zone. If the input for Zone Floor Area does not equal the calculated floor area, the input zone floor area will take precedence, and the space floor areas will be adjusted proportionately. If the calculated floor area differs from the user input floor area by more than 5%, a warning message will be issued.

1.10.1.1.6 Field: Space Type

Space type is used to categorize spaces by activity type, such as office, classroom, storage, etc. There is no predefined list of space types; any valid input string may be used. Several table outputs include a subtable by Space Type. There are also SpaceType Output:Meters. See the meter details (mtd) output file for a list of all available meters for a given simulation. If left blank, this field defaults to “General”’.

1.10.1.1.7 Field: Tag <#>

Tags are used to group spaces by additional characteristics that may be more or less detailed than Space Type. For example, codes and standards designations for ventilation or occupancy may use specific terms that are different from one standard to the next. There is no predefined list of tags; any valid input string may be used. Several table outputs include a subtable by Space Tag.

1.10.2 SpaceList

The SpaceList object defines a list of Space objects. It is primarily used with Internal Gains objects (People, Lights, etc.) to apply the same gain across a group of Spaces.

Space lists are not exclusive. A Space can be referenced by more than one SpaceList object.

1.10.2.1 Inputs

1.10.2.1.1 Field: Name

The name of the SpaceList object. This name must be unique across Zone, Space, ZoneList, and SpaceList names.

1.10.2.1.2 Field: Space <#> Name

Reference to a Space object.

```
SpaceList,
  Mid Floor List,   !- Name
  Mid West Space,   !- Space 1 Name
  Mid Center Space, !- Space 2 Name
  Mid East Space;   !- Space 3 Name
```

1.10.3 Zone

This element sets up the parameters to simulate each thermal zone of the building. This object defines a thermal zone of the building. Every Zone contains one or more Spaces.

Space is an optional input. If a Zone has no Space(s) specified in the input then a default Space named <Zone Name> will be created. If some surfaces in a Zone are assigned to a Space and some are not, then a default Space named <Zone Name>-Remainder will be created. Input references to Space Names must have a matching Space object. Default space names may not be referenced in the input except for Output:Variable keys).

1.10.3.1 Inputs

1.10.3.1.1 Field: Name

The name of the Zone object. This name must be unique across Zone, Space, ZoneList, and SpaceList names.

1.10.3.1.2 Field: Direction of Relative North

The Zone North Axis is specified **relative to the Building North Axis**. This value is specified in degrees (clockwise is positive). For more information, see the figure below as well as the description under "GlobalGeometryRules".

1.10.3.1.3 Field(s): (X,Y,Z) Origin

The X,Y,Z coordinates of a zone origin can be specified, for convenience in vertices entry. Depending on the values in "GlobalGeometryRules" (see description later in this section), these will be used to completely specify the building coordinates in "world coordinate" or not. Zone Origin coordinates are specified **relative to the Building Origin (which always 0,0,0)**. The following figure illustrates the use of Zone North Axis as well as Zone Origin values.

1.10.3.1.4 Field: Type

Zone type is currently unused.

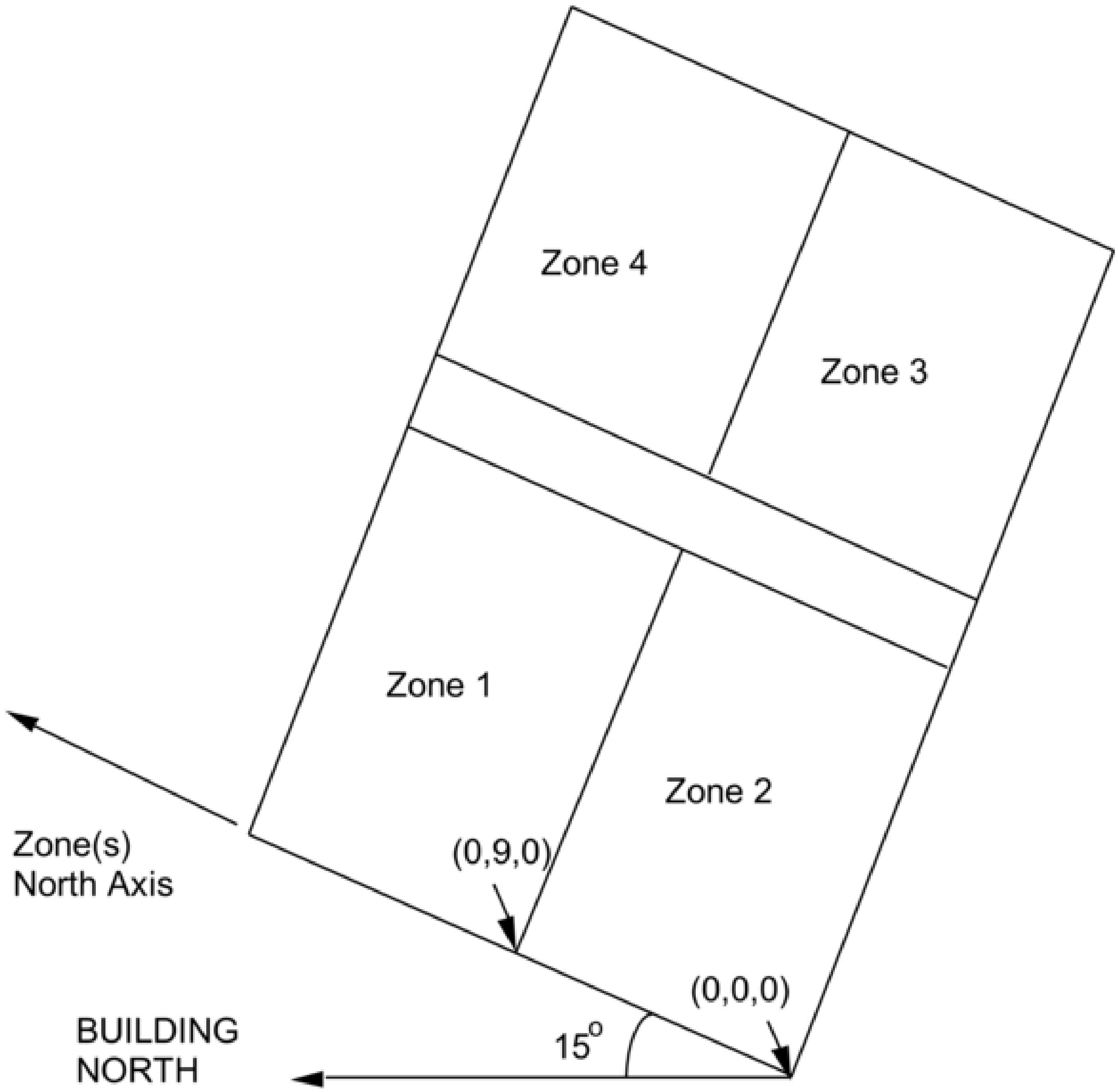

Figure 1.29: Illustration of Zone North Axis and Origins

1.10.3.1.5 Field: Multiplier

Zone Multiplier is designed as a "multiplier" for floor area, zone loads, and energy consumed by internal gains. It takes the calculated load for the zone and multiplies it, sending the multiplied load to the attached HVAC system. The HVAC system size is specified to meet the entire multiplied zone load and will report the amount of the load met in the Zone Air System Sensible Heating or Cooling Energy/Rate output variable. Autosizing automatically accounts for multipliers. Metered energy consumption by internal gains objects such as Lights or Electric Equipment will be multiplied. The default is 1.

1.10.3.1.6 Field: Ceiling Height

Zone ceiling height is used in several areas within EnergyPlus (such as various room models, some convection coefficient calculations and, primarily, in calculating zone volume in the absence of other parameters). Energyplus automatically calculates the zone ceiling height (m) from the average height of the zone. If this field is 0.0, negative or **autocalculate**, then the calculated zone ceiling height will be used in subsequent calculations. If this field is positive, then the calculated zone ceiling height will not be used – the number entered here will be used as the zone ceiling height. If this number differs significantly from the calculated ceiling height, then a warning message will be issued. If a zone ceiling height is entered, but no Volume is entered, then the floor area (if there is one) times the zone ceiling height will be used as the volume.

Note that the Zone Ceiling Height is the distance from the Floor to the Ceiling in the Zone, not an absolute height from the ground.

1.10.3.1.7 Field: Volume

Zone volume is used in several areas within EnergyPlus (such as calculating air change rates for reporting or flow when air change rates are chosen as input, daylighting calculations, some convection coefficient calculations). EnergyPlus automatically calculates the zone volume (m^3) from the zone geometry given by the surfaces that belong to the zone. If this field is 0.0, negative or **autocalculate**, then the calculated zone volume will be used in subsequent calculations. If this field is positive, then it will be used as the zone volume. If this number differs significantly from the calculated zone volume a warning message will be issued. For autocalculate to work properly, the zone must be enclosed by the entered walls. Note that indicating the volume to be calculated but entering a positive ceiling height in the previous field will cause the volume to be calculated as the floor area (if > 0) times the entered ceiling height; else the volume will be calculated from the described surfaces. If this field is positive, any ceiling height positive value will not be used in volume calculations.

1.10.3.1.8 Field: Floor Area

Zone floor area is used in many places within EnergyPlus. EnergyPlus automatically calculates the zone floor area (m^2) as the sum of the floor areas for the Spaces attached to this Zone. If this field is 0.0, negative or **autocalculate**, then the calculated zone floor area will be used in subsequent calculations. If this field is positive, then it will be used as the zone floor area. If the input zone floor area does not equal the calculated floor area, the input zone floor area will take precedence, and the space floor areas will be adjusted proportionately. If the calculated floor area differs from the user input floor area by more than 5%, a warning message will be issued.

1.10.3.1.9 Field: Zone Inside Convection Algorithm

The Zone Inside Convection Algorithm field is optional. This field specifies the convection model to be used for the inside face of heat transfer surfaces associated with this zone. The choices are: **Simple** (constant natural convection - ASHRAE), **TARP** (combines natural and wind-driven convection correlations from laboratory measurements on flat plates), **CeilingDiffuser** (ACH based forced and mixed convection correlations for ceiling diffuser configuration with simple natural convection limit), **AdaptiveConvectionAlgorithm** (complex arrangement of various models that adapt to various zone conditions and can be customized), **TrombeWall** (variable natural convection in an enclosed rectangular cavity), and **ASTMC1340** (mixed convection coefficients specified for attic zone). See the Inside Convection Algorithm object for further descriptions of the available models.

If omitted or blank, the algorithm specified in the SurfaceConvectionAlgorithm:Inside object is the default.

1.10.3.1.10 Field: Zone Outside Convection Algorithm

The Zone Outside Convection Algorithm field is optional. This field specifies the convection model to be used for the outside face of heat transfer surfaces associated with this zone. The choices are: **SimpleCombined**, **TARP**, **DOE-2**, **MoWiTT, and AdaptiveConvectionAlgorithm**. The simple convection model applies heat transfer coefficients depending on the roughness and windspeed. This is a combined heat transfer coefficient that includes radiation to sky, ground, and air. The correlation is based on Figure 1.153, Page 25.1 (Thermal and Water Vapor Transmission Data), 2001 ASHRAE Handbook of Fundamentals.

The other convection models apply heat transfer coefficients depending on the roughness, windspeed, and terrain of the building's location. These are *convection only* heat transfer coefficients; radiation heat transfer coefficients are calculated automatically by the program. The TARP algorithm was developed for the TARP software and combines natural and wind-driven convection correlations from laboratory measurements on flat plates. The DOE-2 and MoWiTT were derived from field measurements. The AdaptiveConvectionAlgorithm model is an dynamic algorithm that organizes a large number of different convection models and automatically selects the one that best applies. The adaptive convection algorithm can also be customized using the SurfaceConvectionAlgorithm:Outside:AdaptiveModelSelections input object. All algorithms are described more fully in the Engineering Reference.

If omitted or blank, the algorithm specified in the SurfaceConvectionAlgorithm:Outside object is the default.

1.10.3.1.11 Field: Part of Total Floor Area

This optional field defaults to Yes if not specified. The field is used to show when a zone is not part of the Total Floor Area as shown in the Annual Building Utility Performance Summary tables. Specifically, when No is specified, the area is excluded from both the conditioned floor area and the total floor area in the Building Area sub table and the Normalized Metrics sub tables.

And, an IDF example:

```
Zone,
    DORM ROOMS AND COMMON AREAS,  !- Name
    0.0000000E+00,           !- Direction of Relative North {deg}
    0.0000000E+00,           !- X Origin {m}
    6.096000,                !- Y Origin {m}
    0.0000000E+00,           !- Z Origin {m}
    1,                       !- Type
    1,                       !- Multiplier
    autocalculate,           !-Ceiling Height {m}
    autocalculate;           !- Volume {m3}
```

1.10.3.2 Zone Outputs

- Zone,Average,Zone Outdoor Air Drybulb Temperature [C]
- Zone,Average,Zone Outdoor Air Wetbulb Temperature [C]

- Zone,Average,Zone Outdoor Air Wind Speed [m/s]

1.10.3.2.1 Zone Outdoor Air Drybulb Temperature [C]

The outdoor air dry-bulb temperature calculated at the height above ground of the zone centroid.

1.10.3.2.2 Zone Outdoor Air Wetbulb Temperature [C]

The outdoor air wet-bulb temperature calculated at the height above ground of the zone centroid.

1.10.3.2.3 Zone Outdoor Air Wind Speed [m/s]

The outdoor wind speed calculated at the height above ground of the zone centroid.

1.10.4 Space and Zone Thermal Output(s)

1.10.4.1 Space Thermal Outputs Overview

- Zone,Sum,Space Total Internal Radiant Heating Energy [J]
- Zone,Average,Space Total Internal Radiant Heating Rate [W]
- Zone,Sum,Space Total Internal Visible Radiation Heating Energy [J]
- Zone,Average,Space Total Internal Visible Radiation Heating Rate [W]
- Zone,Sum,Space Total Internal Convective Heating Energy [J]
- Zone,Average,Space Total Internal Convective Heating Rate [W]
- Zone,Sum,Space Total Internal Latent Gain Energy [J]
- Zone,Average,Space Total Internal Latent Gain Rate [W]
- Zone,Sum,Space Total Internal Total Heating Energy [J]
- Zone,Average,Space Total Internal Total Heating Rate [W]
- HVAC,Average,Space Air Temperature [C]
- Zone,Average,Space Mean Air Dewpoint Temperature [C]
- Zone,Average,Space Mean Radiant Temperature [C]
- Zone,Average,Enclosure Mean Radiant Temperature [C]
- Zone,Average,Space Operative Temperature [C]
- HVAC,Average,Space Air Heat Balance Internal Convective Heat Gain Rate [W]
- HVAC,Average,Space Air Heat Balance Surface Convection Rate [W]
- HVAC,Average,Space Air Heat Balance InterSpace Air Transfer Rate [W]
- HVAC,Average,Space Air Heat Balance Outdoor Air Transfer Rate [W]
- HVAC,Average,Space Air Heat Balance System Air Transfer Rate [W]
- HVAC,Average,Space Air Heat Balance System Convective Heat Gain Rate [W]

- HVAC,Average,Space Air Heat Balance Air Energy Storage Rate [W]
- HVAC,Average,Space Air Heat Balance Deviation Rate [W]
- HVAC,Sum,Space Air System Sensible Heating Energy [J]
- HVAC,Sum,Space Air System Sensible Cooling Energy [J]
- HVAC,Average,Space Air System Sensible Heating Rate [W]
- HVAC,Average,Space Air System Sensible Cooling Rate [W]
- HVAC,Average,Space Air Humidity Ratio[kgWater/kgDryAir]
- HVAC,Average,Space Air Relative Humidity[%]
- HVAC,Sum,Space Air System Latent Heating Energy [J]
- HVAC,Sum,Space Air System Latent Cooling Energy [J]
- HVAC,Average,Space Air System Latent Heating Rate [W]
- HVAC,Average,Space Air System Latent Cooling Rate [W]
- HVAC,Average,Space Air System Sensible Heat Ratio []
- HVAC,Average,Space Air Vapor Pressure Difference [Pa]

1.10.4.2 Zone Thermal Outputs Overview

In addition to the canned Surface reports (view the Reports section later in this document) and surface variables (above), the following variables are available for all zones:

- Zone,Sum,Zone Total Internal Radiant Heating Energy [J]
- Zone,Average,Zone Total Internal Radiant Heating Rate [W]
- Zone,Sum,Zone Total Internal Visible Radiation Heating Energy [J]
- Zone,Average,Zone Total Internal Visible Radiation Heating Rate [W]
- Zone,Sum,Zone Total Internal Convective Heating Energy [J]
- Zone,Average,Zone Total Internal Convective Heating Rate [W]
- Zone,Sum,Zone Total Internal Latent Gain Energy [J]
- Zone,Average,Zone Total Internal Latent Gain Rate [W]
- Zone,Sum,Zone Total Internal Total Heating Energy [J]
- Zone,Average,Zone Total Internal Total Heating Rate [W]
- Zone Phase Change Material Melting Enthalpy[J/kg]
- Zone Phase Change Material Freezing Enthalpy[J/kg]
- HVAC,Average,Zone Exfiltration Heat Transfer Rate [W]

- HVAC,Average,Zone Exfiltration Sensible Heat Transfer Rate [W]
- HVAC,Average,Zone Exfiltration Latent Heat Transfer Rate [W]
- HVAC,Average,Zone Exhaust Air Heat Transfer Rate [W]
- HVAC,Average,Zone Exhaust Air Sensible Heat Transfer Rate [W]
- HVAC,Average,Zone Exhaust Air Latent Heat Transfer Rate [W]
- HVAC,Average,Zone Air Temperature [C]
- Zone,Average,Zone Mean Air Dewpoint Temperature [C]
- Zone,Average,Zone Mean Radiant Temperature [C]
- Zone,Average,Zone Operative Temperature [C]
- HVAC,Average,Zone Air Heat Balance Internal Convective Heat Gain Rate [W]
- HVAC,Average,Zone Air Heat Balance Surface Convection Rate [W]
- HVAC,Average,Zone Air Heat Balance Interzone Air Transfer Rate [W]
- HVAC,Average,Zone Air Heat Balance Outdoor Air Transfer Rate [W]
- HVAC,Average,Zone Air Heat Balance System Air Transfer Rate [W]
- HVAC,Average,Zone Air Heat Balance System Convective Heat Gain Rate [W]
- HVAC,Average,Zone Air Heat Balance Air Energy Storage Rate [W]
- HVAC,Average,Zone Air Heat Balance Deviation Rate [W]
- HVAC,Sum,Zone Air System Sensible Heating Energy [J]
- HVAC,Sum,Zone Air System Sensible Cooling Energy [J]
- HVAC,Average,Zone Air System Sensible Heating Rate [W]
- HVAC,Average,Zone Air System Sensible Cooling Rate [W]
- HVAC,Average,Zone Air Humidity Ratio[kgWater/kgDryAir]
- HVAC,Average,Zone Air Relative Humidity[%]
- HVAC,Sum,Zone Air System Latent Heating Energy [J]
- HVAC,Sum,Zone Air System Latent Cooling Energy [J]
- HVAC,Average,Zone Air System Latent Heating Rate [W]
- HVAC,Average,Zone Air System Latent Cooling Rate [W]
- HVAC,Average,Zone Air System Sensible Heat Ratio []
- HVAC,Average,Zone Air Vapor Pressure Difference [Pa]

Two of these are of particular interest:

```
Zone,Average,Zone Mean Air Temperature [C]
HVAC,Average,Zone Air Temperature [C]
```

These two variable outputs are/should be identical. However, note that they can be reported at different time intervals. "Zone Mean Air Temperature" is only available on the Zone/HB timestep (Number of Timesteps per Hour) whereas "Zone Air Temperature" can be reported at the HVAC timestep (which can vary).

1.10.4.3 Space and Zone Thermal Outputs Details

1.10.4.3.1 Zone Mean Air Temperature [C]

From the code definition, the zone mean air temperature is the average temperature of the air temperatures at the system timestep. Remember that the zone heat balance represents a "well stirred" model for a zone, therefore there is only one mean air temperature to represent the air temperature for the zone unless a non-mixing RoomAirModel is specified.

1.10.4.3.2 Zone Wetbulb Globe Temperature [C]

The indoor wet-bulb globe temperature (WBGT) is a measure of indoor heat stress as it affects humans. It is calculated as a weighted average of the zone wet-bulb temperature (WBT) and the operative temperature (OT) as follows

WBGT = 0.7*WBT + 0.3*OT

1.10.4.3.3 Zone Air Temperature [C]

This is very similar to the mean air temperature in the last field. The "well stirred" model for the zone is the basis, but this temperature is also available at the "detailed" system timestep.

1.10.4.3.4 Zone Mean Air Dewpoint Temperature [C]

This is the dewpoint temperature of the zone calculated from the Zone Mean Air Temperature (above), the Zone Air Humidity Ratio (below) and the outdoor barometric pressure.

1.10.4.3.5 Zone Thermostat Air Temperature [C]

This is the zone air node temperature for the well-mixed room air model, which is the default room air model type (RoomAirModelType = Mixing). But for other types of Room Air Model (the RoomAir:TemperaturePattern:* and RoomAirSettings:* objects) the zone thermostat air temperature may depend on the Thermostat Height and Thermostat Offset.

1.10.4.3.6 Zone Mean Radiant Temperature [C]

1.10.4.3.7 Space Mean Radiant Temperature [C]

1.10.4.3.8 Enclosure Mean Radiant Temperature [C]

The Mean Radiant Temperature (MRT) in degrees Celsius is the surface area $\times$ emissivity weighted average of the Surface Inside Face Temperature, where emissivity is the Thermal Absorptance of the inside material layer of each surface. Zone Mean Radiant Temperature is based on all surfaces in the zone (regardless of what enclosure each surface is in). Enclosure Mean Radiant Temperature is based on all surfaces in an enclosure (one or more Spaces, see Construction:AirBoundary). Space Mean Radiant Temperature is based on all surfaces in the enclosure which contains the space (same as its Enclsoure Mean Radiant Temperature).

1.10.4.3.9 Zone Operative Temperature [C]

Zone Operative Temperature (OT) is the average of the Zone Mean Air Temperature (MAT) and Zone Mean Radiant Temperature (MRT), OT = 0.5·MAT + 0.5·MRT. This output variable is not affected by the type of thermostat controls in the zone, and does not include the direct effect of high temperature radiant systems. See also Zone Thermostat Operative Temperature.

1.10.4.3.10 Zone Air Heat Balance Internal Convective Heat Gain Rate [W]

The Zone Air Heat Balance Internal Convective Heat Gain Rate is the sum, in watts, of heat transferred to the zone air from all types of internal gains from energyplus objects (e.g., people, lights, devices, appliances, equipment) listed in Tables 1.14 and 1.15. In addition to the convective heat gains listed in these tables, if a zone or a space is served by a zone HVAC equipment, i.e., the zone has no return air node, or served by an AirloopHVAC system that has cycling fan operating mode, then the convective heat gain specified to the return air is re-directed to the space or the zone air heat balance as convective internal heat gain. This and the following provide results on the load components of the zone air heat balance. This field is not multiplied by zone or group multipliers.

Table 1.14: Space or Zone Zone Air Heat Balance Internal Convective Heat Gain Constituents

EnergyPlus Object	Convective Heat Gain Type
People	People
Lights	Lights
ElectricEquipment	ElectricEquipment
ElectricEquipment:ITE:AirCooled	ElectricEquipmentITEAirCooled
GasEquipment	GasEquipment
HotWaterEquipment	HotWaterEquipment
SteamEquipment	SteamEquipment
OtherEquipment	OtherEquipment
Refrigeration:Case	RefrigerationCase
Refrigeration:CompressorRack	RefrigerationCompressorRack
Refrigeration:System:Condenser:AirCooled	RefrigerationSystemAirCooledCondenser
Refrigeration:System:SuctionPipe	RefrigerationSystemSuctionPipe
Refrigeration:SecondarySystem:Receiver	RefrigerationSecondaryReceiver
Refrigeration:SecondarySystem:Pipe	RefrigerationSecondaryPipe
Refrigeration:WalkIn	RefrigerationWalkIn
Refrigeration:TranscriticalSystem:GasCooler:AirCooled	RefrigerationTransSysAirCooledGasCooler
Refrigeration:TranscriticalSystem:SuctionPipeMT	RefrigerationTransSysSuctionPipeMT
Refrigeration:TranscriticalSystem:SuctionPipeLT	RefrigerationTransSysSuctionPipeLT
WaterUse:Equipment	WaterUseEquipment
WaterHeater:Mixed	WaterHeaterMixed
WaterHeater:Stratified	WaterHeaterStratified
ZoneBaseboard:OutdoorTemperatureControlled	ZoneBaseboardOutdoorTemperatureControlled
ThermalStorage:ChilledWater:Mixed	ThermalStorageChilledWaterMixed

Table 1.14: Space or Zone Zone Air Heat Balance Internal Convective Heat Gain Constituents

EnergyPlus Object	Convective Heat Gain Type
ThermalStorage:ChilledWater:Stratified	ThermalStorageChilledWaterStratified
Pipe:Indoor	PipeIndoor
Pump:VariableSpeed	Pump_VarSpeed
Pump:ConstantSpeed	Pump_ConSpeed
Pump:VariableSpeed:Condensate	Pump_Cond
HeaderedPumps:VariableSpeed	PumpBank_VarSpeed
HeaderedPumps:ConstantSpeed	PumpBank_ConSpeed
PlantComponent:UserDefined	PlantComponentUserDefined
Coil:UserDefined	CoilUserDefined
ZoneHVAC:ForcedAir:UserDefined	ZoneHVACForcedAirUserDefined
AirTerminal:SingleDuct:UserDefined	AirTerminalUserDefined
Coil:Cooling:DX:SingleSpeed:ThermalStorage	PackagedTESCoilTank
Fan:SystemModel	FanSystemModel
Coil:Cooling:DX:SingleSpeed	SecCoolingDXCoilSingleSpeed
Coil:Heating:DX:SingleSpeed	SecHeatingDXCoilSingleSpeed
Coil:Cooling:DX:TwoSpeed	SecCoolingDXCoilTwoSpeed
Coil:Cooling:DX:MultiSpeed	SecCoolingDXCoilMultiSpeed
Coil:Heating:DX:MultiSpeed	SecHeatingDXCoilMultiSpeed
Generator:FuelCell	GeneratorFuelCell
Generator:MicroCHP	GeneratorMicroCHP
ElectricLoadCenter:Transformer	ElectricLoadCenterTransformer
ElectricLoadCenter:Inverter:Simple	ElectricLoadCenterInverterSimple
ElectricLoadCenter:Inverter:FunctionOfPower	ElectricLoadCenterInverterFunctionOfPower
ElectricLoadCenter:Inverter:LookUpTable	ElectricLoadCenterInverterLookUpTable
ElectricLoadCenter:Storage:LiIonNMCBattery	ElectricLoadCenterStorageLiIonNmcBattery
ElectricLoadCenter:Storage:Battery	ElectricLoadCenterStorageBattery
ElectricLoadCenter:Storage:Simple	ElectricLoadCenterStorageSimple
ElectricLoadCenter:Storage:Converter	ElectricLoadCenterConverter

Table 1.15: Space or Zone Zone Air Heat Balance Internal Convective Heat Gain from fenestration

FenestrationSurface:Detailed Components	Internal Convective Heat Gain Sources
WindowProperty:FrameAndDivider	Convective heat gain from window frame and divider
Construction:WindowEquivalentLayer	Other convective heat gain from equivalent layer window model
Construction	Natural convection heat gain from a gap

Table 1.15: Space or Zone Zone Air Heat Balance Internal Convective Heat Gain Constituents

FenestrationSurface:Detailed Components	Internal Convective Heat Gain Sources
	between glass and interior shade or blind
Construction:ComplexFenestrationState	Convective heat gain from air flow through a gap between glass and interior shade or blind

1.10.4.3.11 Zone Air Heat Balance Surface Convection Rate [W]

The Zone Air Heat Balance Surface Convection Rate is the sum, in watts, of heat transferred to the zone air from all the surfaces. This field is not multiplied by zone or group multipliers.

1.10.4.3.12 Zone Air Heat Balance Interzone Air Transfer Rate [W]

The Zone Air Heat Balance Interzone Air Transfer Rate is the sum, in watts, of heat transferred to the zone air from all the transfers of air from other thermal zones. This field is not multiplied by zone or group multipliers.

1.10.4.3.13 Zone Air Heat Balance Outdoor Air Transfer Rate [W]

The Zone Air Heat Balance Outdoor Air Transfer Rate is the sum, in watts, of heat transferred to the zone air from all the transfers of air from the out side, such as infiltration. This field is not multiplied by zone or group multipliers.

1.10.4.3.14 Zone Air Heat Balance System Air Transfer Rate [W]

The Zone Air Heat Balance System Air Transfer Rate is the sum, in watts, of heat transferred to the zone air by HVAC forced-air systems and air terminal units. Such HVAC systems are connected to the zone by an inlet node (see ZoneHVAC:EquipmentConnections input field called Zone Air Inlet Node or Node List Name) This field is not multiplied by zone or group multipliers. The Zone Air Heat Balance System Air Transfer Rate may not agree exactly with the equipment-level delivered energy transfer rate when the zone temperature is changing significantly over a timestep (e.g. during thermostat setback and setup), but the energy will balance out over time.

1.10.4.3.15 Zone Air Heat Balance System Convective Heat Gain Rate [W]

The Zone Air Heat Balance System Convective Heat Gain Rate is the sum, in watts, of heat transferred directly to the zone air by "non-air" HVAC systems. Such HVAC systems are not connected to the zone by an inlet node but rather add or subtract heat directly to the zone air in a manner similar to internal gains. These include the convective fraction of zone HVAC baseboards and high temperature radiant systems, zone HVAC refrigeration chiller set, and the extra convective cooling provided by the cooled beam air terminal unit. This field is not multiplied by zone or group multipliers.

1.10.4.3.16 Zone Air Heat Balance Air Energy Storage Rate [W]

The Zone Air Heat Balance Air Energy Storage Rate is the heat stored, in watts, in the zone air as result of zone air temperature changing from one timestep to the next. This field is not multiplied by zone or group multipliers.

1.10.4.3.17 Zone Air Heat Balance Deviation Rate [W]

The Zone Air Heat Balance Deviation Rate is the imbalance, in watts, in the energy balance for zone air. The value should be near zero but will become non-zero if zone conditions are changing rapidly or erratically. This field is not multiplied by zone or group multipliers. (This output variable is only generated if the user has set a computer system environment variable DisplayAdvancedReportVariables equal to “yes”.)

1.10.4.3.18 Zone Air System Sensible Heating Energy [J]

This output variable represents the sensible heating energy in Joules that is actually supplied by the system to that zone for the timestep reported. This is the sensible heating rate multiplied by the simulation timestep. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This field is not multiplied by zone or group multipliers.

Zone Air System Sensible Heating (and Cooling) Energy (and Rate) all report the heating or cooling delivered by the HVAC system to a zone. These values are calculated by multiplying the supply air mass flow rate by the difference between the supply air temperature and the zone air temperature. This does not always indicate the operation of heating or cooling coils. For example, cooling will be reported if the supply air is cooled due to the introduction of outside air, even if all coils are off.

In addition, certain “non-air” zone-based systems will also add their heating or cooling contribution to this output variable. For example, the following zone equipment will also add their output to the appropriate heating or cooling energy/rate output variable:

- ZoneHVAC:Baseboard:RadiantConvective:Water
- ZoneHVAC:Baseboard:RadiantConvective:Steam
- ZoneHVAC:Baseboard:RadiantConvective:Electric
- ZoneHVAC:Baseboard:Convective:Water
- ZoneHVAC:Baseboard:Convective:Electric
- ZoneHVAC:CoolingPanel:RadiantConvective:Water
- ZoneHVAC:RefrigerationChillerSet

This means that system output from equipment such as high and low temperature radiant systems are NOT included in this output variable as the control of these radiant systems are handled separately with their own unique controls.

Finally, note that these variables are calculated at the system timestep. When reported at the “detailed” reporting frequency, these variables will never show heating and cooling both in the same system timestep. If reported at a frequency less than “Detailed” (for example, Hourly) values may appear in both the heating and cooling variable for the same hour if the system cooled the zone for part of the reporting period and heated the zone for another part of the reporting period.

1.10.4.3.19 Zone Air System Sensible Cooling Energy [J]

This output variable represents the sensible cooling energy in Joules that is actually supplied by the system to that zone for the timestep reported. This is the sensible cooling rate multiplied by the simulation timestep. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module.

This field is not multiplied by zone or group multipliers. For additional information on this output variable, see the note that accompanies the Zone Air System Sensible Heating Energy output variable above.

1.10.4.3.20 Zone Air System Sensible Heating Rate [W]

This output variable represents the sensible heating rate in Watts that is actually supplied by the system to that zone for the timestep reported. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This field is not multiplied by zone or group multipliers. For additional information on this output variable, see the note that accompanies the Zone Air System Sensible Heating Energy output variable above.

1.10.4.3.21 Zone Air System Sensible Cooling Rate [W]

This output variable represents the sensible cooling rate in Watts that is actually supplied by the system to that zone for the timestep reported. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This field is not multiplied by zone or group multipliers. For additional information on this output variable, see the note that accompanies the Zone Air System Sensible Heating Energy output variable above.

1.10.4.3.22 Zone Air Humidity Ratio [kgWater/kgDryAir]

This output variable represents the air humidity ratio after the correct step for each zone. The humidity ratio is the mass of water vapor to the mass of dry air contained in the zone in (kg water/kg air) and is unitless.

1.10.4.3.23 Zone Air Relative Humidity [%]

This output variable represents the air relative humidity after the correct step for each zone. The relative humidity is in percent and uses the Zone Air Temperature, the Zone Air Humidity Ratio and the Outside Barometric Pressure for calculation.

1.10.4.3.24 Zone Air System Latent Heating Energy [J]

1.10.4.3.25 Zone Air System Latent Cooling Energy [J]

These output variables represent the latent heating/cooling energy in Joules that is actually supplied by the system to that zone for the timestep reported. This is the sensible heating/cooling rate multiplied by the simulation timestep. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This field is not multiplied by zone or group multipliers.

1.10.4.3.26 Zone Air System Latent Heating Rate [W]

This output variable represents the sensible heating rate in Watts that is actually supplied by the system to that zone for the timestep reported. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This field is not multiplied by zone or group multipliers. For additional information on this output variable, see the note that accompanies the Zone Air System Sensible Heating Energy output variable above.

1.10.4.3.27 Zone Air System Latent Cooling Rate [W]

This output variable represents the sensible cooling rate in Watts that is actually supplied by the system to that zone for the timestep reported. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This field is not multiplied by zone or group multipliers. For additional information on this output variable, see the note that accompanies the Zone Air System Sensible Heating Energy output variable above.

1.10.4.3.28 Zone Air System Sensible Heat Ratio []

This output variable represents the zone sensible heat transfer divided by the total heat transfer for the timestep reported. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module.

1.10.4.3.29 Zone Air Vapor Pressure Difference [Pa]

This output variable represents the zone vapor pressure difference, or depression, in pascals for the timestep reported. This is calculated and reported from the Correct step in the Zone Predictor-Corrector module. This value represents the saturated air vapor pressure minus the actual zone air vapor pressure and is indicative of the dryness of the zone air when latent loads are of interest.

1.10.4.3.30 Space or Zone Total Internal Radiant Heating Rate [W]

1.10.4.3.31 Space or Zone Total Internal Radiant Heating Energy [J]

These output variables represent the sum of radiant gains from specific internal sources (e.g. equipment) throughout the space or zone in Watts (for rate) or joules. This includes radiant gain from People, Lights, Electric Equipment, Gas Equipment, Other Equipment, Hot Water Equipment, and Steam Equipment.

1.10.4.3.32 Space or Zone Total Internal Visible Radiation Heating Rate [W]

1.10.4.3.33 Space or Zone Total Internal Visible Radiation Heating Energy [J]

These output variables express the sum of heat gain in Watts (for rate) or joules that is the calculated short wavelength radiation gain from lights in the space or zones. This calculation uses the total energy from lights and the fraction visible to realize this value, summed over the space or zones in the simulation.

1.10.4.3.34 Space or Zone Total Internal Convective Heating Rate [W]

1.10.4.3.35 Space or Zone Total Internal Convective Heating Energy [J]

These output variables represent the sum of convective gains from specific sources (e.g. equipment) throughout the space or zone in Watts (for rate) or joules. This includes convective gain from Energyplus objects that are listed in Table 1.16:

Table 1.16: Space or Zone Total Internal Heat Gain Constituents

EnergyPlus Object	Internal Heat Gain Type
People	People
Lights	Lights
ElectricEquipment	ElectricEquipment
ElectricEquipment:ITE:AirCooled	ElectricEquipmentITEAirCooled
GasEquipment	GasEquipment
HotWaterEquipment	HotWaterEquipment
SteamEquipment	SteamEquipment
OtherEquipment	OtherEquipment

1.10.4.3.36 Space or Zone Total Internal Latent Gain Rate [W]

1.10.4.3.37 Space or Zone Total Internal Latent Gain Energy [J]

These output variables represent the sum of latent gains from specific internal sources (e.g. equipment) throughout the space or zone in Watts (for rate) or joules. This includes latent gain from Energyplus objects that are listed in Table 1.16.

1.10.4.3.38 Space or Zone Total Internal Total Heating Rate [W]

1.10.4.3.39 Space or Zone Total Internal Total Heating Energy [J]

These output variables represent the sum of all heat gains throughout the space or zone in Watts (for rate) or joules. This includes all heat gains from Energyplus objects that are listed in Table 1.16.

1.10.4.3.40 Zone Phase Change Material Melting Enthalpy [J/kg]

This output variable tracks the summation of all energy within a zone that is associated with phase change materials melting during the current time period in J/kg. This report is only produced when the user enters phase change materials (see MaterialProperty:PhaseChange), opts to use the Conduction Finite Difference (aka CondFD) solution algorithm (see HeatBalanceAlgorithm), and requests DisplayAdvancedReportVariables (see Output:Diagnostics).

1.10.4.3.41 Zone Phase Change Material Freezing Enthalpy [J/kg]

This output variable tracks the summation of all energy within a zone that is associated with phase change materials freezing during the current time period in J/kg. This report is only produced when the user enters phase change materials (MaterialProperty:PhaseChange), opts to use the Conduction Finite Difference (aka CondFD) solution algorithm (see HeatBalanceAlgorithm), and requests DisplayAdvancedReportVariables (see Output:Diagnostics).

1.10.4.3.42 Zone Exfiltration Heat Transfer Rate [W]

1.10.4.3.43 Zone Exfiltration Sensible Heat Transfer Rate [W]

1.10.4.3.44 Zone Exfiltration Latent Heat Transfer Rate [W]

These output variables represent the heat emitted to ambient from exfiltration in Watts. The exfiltration rate is calculated by solving a mass flow balance on the zone including infiltration, ventilation, outdoor air mixing, zone-to-zone mixing, and all of the zone inlet, exhaust, and return nodes. The latent parts are determined by taking the difference between the total and the sensible rate. Positive values indicate the zone injects heat to the environment, while negative values indicate the building extracts heat from the environment.

1.10.4.3.45 Zone Exhaust Air Heat Transfer Rate [W]

1.10.4.3.46 Zone Exhaust Air Sensible Heat Transfer Rate [W]

1.10.4.3.47 Zone Exhaust Air Latent Heat Transfer Rate [W]

These output variables represent the heat emitted to ambient from exhaust air in Watts. The zone exhaust air flow rate is aggregated from the zone exhaust nodes. Positive values indicate the building injects heat to the environment, while negative values indicate the building extracts heat from the environment.

1.10.4.3.48 Site Total Zone Exfiltration Heat Loss [W]

These output variables represent the total amount of heat emitted to ambient from all zones by exfiltration in Watts. Positive values indicate the building injects heat to the environment, while negative values indicate the building extracts heat from the environment. This includes both sensible and latent heat loss from zone exfiltration.

1.10.4.3.49 Site Total Zone Exhaust Air Heat Loss [W]

These output variables represent the total amount of heat emitted to ambient from all zones by exhaust air in Watts. Positive values indicate the building injects heat to the environment, while negative values indicate the building extracts heat from the environment. This includes both sensible and latent heat loss from zone exhaust air.

1.10.5 ZoneList

The ZoneList object defines a list of Zone objects. It is primarily used with the ZoneGroup object to provide a generalized way for doing "Floor Multipliers". (See the ZoneGroup description below.) The associated ZoneList output variables also provide a way to aggregate and organize zone loads.

Zone lists are not exclusive. A zone can be referenced be more than one ZoneList object.

1.10.5.1 Inputs

1.10.5.1.1 Field: Name

The name of the ZoneList object. This name must be unique across Zone, Space, ZoneList, and SpaceList names.

1.10.5.1.2 Field: Zone <#> Name

Reference to a Zone object. This object is extensible, so additional fields of this type can be added to the end of this object.

```
ZoneList,
  Mid Floor List,  !- Name
  Mid West Zone,   !- Zone 1 Name
  Mid Center Zone, !- Zone 2 Name
  Mid East Zone;   !- Zone 3 Name
```

1.10.5.2 Outputs

The following output variables are reported by the ZoneList object:

- HVAC,Average,Zone List Sensible Heating Rate [W]
- HVAC,Average,Zone List Sensible Cooling Rate [W]
- HVAC,Sum,Zone List Sensible Heating Energy [J]
- HVAC,Sum,Zone List Sensible Cooling Energy [J]

All ZoneList variables are the sum of the corresponding Zone variables. *Zone Multiplier* fields in the Zone objects are also taken into account.

1.10.6 ZoneGroup

The ZoneGroup object adds a multiplier to a ZoneList. This can be used to reduce the amount of input necessary for simulating repetitive structures, such as the identical floors of a multi-story building. To create a "Floor Multiplier", use the ZoneList object to organize several zones into a typical floor. Then use the *Zone List Multiplier* field in the ZoneGroup object to multiply the system load for the zones in the list will also be multiplied. Zones with a *Multiplier* field greater than one in the Zone object are effectively double-multiplied.

NOTE: Although ZoneLists are not exclusive by themselves, ZoneLists used to form a ZoneGroup are exclusive; the ZoneLists used with a ZoneGroup must not have any zones in common.

1.10.6.1 Inputs

1.10.6.1.1 Tips for Multi-Story Simulations:

- For floors that are multiplied, connect exterior boundary conditions of the floor to the ceiling and vice versa.
- Since exterior convection coefficients vary with elevation, locate the typical middle floor zones mid-height between the lowest and highest middle floors to be modeled.
- Shading must be identical for all multiplied floors or less accurate results may be obtained by using the zone list multiplier.

ZoneGroup and ZoneList can also be used to simulate other repetitive cases, such as clusters of zones on the ground.

1.10.6.1.2 Field: Zone Group Name

The name of the ZoneGroup object. This must be unique across ZoneGroups.

1.10.6.1.3 Field: Zone List Name

Reference to a ZoneList object. The zones in the list constitute the zones in the group.

1.10.6.1.4 Field: Zone List Multiplier

An integer multiplier. Zone List Multiplier is designed as a "multiplier" for floor area, zone loads, and energy consumed by internal gains. It takes the calculated load for the zone and multiplies it, sending the multiplied load to the attached HVAC system. The HVAC system size is specified to meet the entire multiplied zone load and will report the amount of the load met in the Zone Air System Sensible Heating or Cooling Energy/Rate output variable. Autosizing automatically accounts for multipliers. Metered energy consumption by internal gains objects such as Lights or Electric Equipment will be multiplied. The default is 1.

```
ZONE GROUP,
    Mid Floor,  !- Zone Group Name
    Mid Floor List,  !- Zone List Name
    8;  !- Zone List Multiplier
```

1.10.6.2 Outputs

The following output variables are reported by the ZoneGroup object:

- HVAC,Average,Zone Group Sensible Heating Rate [W]
- HVAC,Average,Zone Group Sensible Cooling Rate [W]
- HVAC,Sum,Zone Group Sensible Heating Energy [J]
- HVAC,Sum,Zone Group Sensible Cooling Energy [J]

All ZoneGroup variables report the associated ZoneList value multiplied by the *Zone List Multiplier*.

1.10.7 Surface(s)

What's a building without surfaces?

EnergyPlus allows for several surface types:

- **BuildingSurface:Detailed**
- **FenestrationSurface:Detailed**
- **Shading:Site:Detailed**
- **Shading:Building:Detailed**
- **Shading:Zone:Detailed**

Each of the preceding surfaces has "correct" geometry specifications. BuildingSurface and Fenestration surfaces (heat transfer surfaces) are used to describe the important elements of the building (walls, roofs, floors, windows, doors) that will determine the interactions of the building surfaces with the outside environment parameters and the internal space requirements. These surfaces are also used to represent "interzone" heat transfer. All surfaces are modeled as a thin plane (with no thickness) except that material thicknesses are taken into account for heat transfer calculations.

During specification of surfaces, several "outside" environments may be chosen:

- **Ground** – when the surface is in touch with the ground (e.g. slab floors)
- **Outdoors** – when the surface is an external surface (e.g. walls, roofs, windows directly exposed to the outdoor conditions)
- **Surface** – when the surface is
  - An adiabatic internal zone surface
  - A interzone surface
- **Zone** – when the surface is
  - A interzone surface in which the other surface is not put in the input file.
- **OtherSideCoefficients** – when using a custom profile to describe the external conditions of the surface (advanced concept – covered in subject: SurfaceProperty:OtherSideCoefficients)

- **OtherSideConditionsModel** – when using specially modeled components, such as active solar systems, that cover the outside surface and modify the conditions it experiences.

Surface items must also specify an "outside face object". This is

- Current surface name – for adiabatic internal surfaces.
- A surface name in another zone – for interzone heat transfer.
- An "opposing" surface name (in the current zone) – for representing "middle" zones.

Zone items must also specify an "outside face object". This is

- The zone that contains the other surface that is adjacent to this surface but is not entered in input.

Note that heat transfer surfaces are fully represented with each description. As stated earlier in the Construction description, materials in the construction (outside to inside) are included but film coefficients neither inside nor outside are used in the description – these are automatically calculated during the EnergyPlus run. Interzone surfaces which do not have a symmetrical construction (such as a ceiling/floor) require two Construction objects with the layers in reverse order. For example, CEILING with carpet, concrete, ceiling tile and FLOOR with ceiling tile, concrete, carpet. If interzone surfaces have a symmetrical construction, the specification for the two surfaces can reference the same Construction. When a surface is connected as the outside boundary condition for another surface, the two surfaces may be in the same plane, or they may be separated to imply thickness.

Note also that if interzone constructions include either WindowMaterial:Glazing or WindowMaterial:Glazing:EquivalentLayer then the materials properties of layers consisting of those material types will have front and back side values that may differ from one another. So, if using such materials in the definition of an interzone construction, reversed materials must also have their front and back properties inverted so as to match in reverse order. For example, if there is a single layer of glass of type WindowMaterial:Glazing defined for an interzone construction and the front and back properties differ, the user must define two materials with equivalent properties except that the front and back values must be reversed. This means that one material used for the definition on one side might have a front side solar reflectance at normal incidence of 0.35 and a back side solar reflectance at normal incidence of 0.25. The material used for the other side of the interzone construction must then have a front side solar reflectance at normal incidence of 0.25 and a back side solar reflectance at normal incidence of 0.35 to match properly in reverse. EnergyPlus checks for all of the user input properties for single and multiple layer interzone constructions that use these two material types to confirm that this has been done correctly so that the simulation accurately reflects the properties of the materials on each side.

Shading surfaces are used to describe aspects of the site which do not directly impact the physical interactions of the environmental parameters but may significantly shade the building during specific hours of the day or time so the year (e.g. trees, bushes, mountains, nearby buildings which aren't being simulated as part of this facility, etc.)

Note that surfaces which are part of the simulated building automatically shade other parts of the building as geometry and time of day dictate – there is no need on the user's part to include surfaces that might be in other zones for shading.

Another surface type:

- **InternalMass**

is used to specify the construction/material parameters and area of items within the space that are important to heat transfer calculations but not necessarily important geometrically. (For example, furniture within the space – particularly for large spaces). Internal mass can also be used for internal walls that are not needed (when FullInteriorAndExterior Solar Distribution is in effect) for solar distribution or to represent many, if not all, interior walls when solar is distributed to the floors only.

1.10.8 Interzone Surfaces

EnergyPlus can quite accurately simulate the surface heat exchange between two zones. However, this accuracy is not always required and using interzonal heat transfer does add to the complexity of the calculations – thus requiring more CPU time to simulate. More information about interzonal heat transfer calculations is contained in the Engineering Reference. Some simple guidelines are presented here – for three cases: adiabatic surfaces, surfaces in "middle" zones, and surfaces where heat transfer is "expected" (e.g. between a residence and an unheated, attached garage).

- Adiabatic Surfaces – These surfaces would be represented as common surfaces (between two zones) where both zones are typically the same temperature. Thus, no transfer is expected in the surface from one zone to the next. These surfaces should be described as simply internal surfaces for the zone referencing as their Outside Boundary Condition Object (see later description in individual surface objects) their own surface names.

- Surfaces in Middle Zones – Middle zones in a building can be simulated using a judicious use of surfaces and zone multipliers to effect the correct "loads" for the building. Thus, middle zone behavior can be simulated without modeling the adjacent zones. This is done by specifying a surface within the zone. For example, a middle floor zone can be modeled by making the floor the Outside Boundary Condition Object for the ceiling, and the ceiling the Outside Boundary Condition Object for the floor.

- Surfaces between Zones with differing temperatures – These zones represent the true use of interzone surfaces. In a residence that has an attached garage, the garage may be unheated/uncooled or at least not conditioned to the same degree as the residence interior. In this case, EnergyPlus can be used to accurately calculate the effects of the differently conditioned space to the other spaces.

1.10.9 Surface View Factors

EnergyPlus uses an area weighted approximation to calculate "view factors" between surfaces within a thermal zone. Each surface uses the total area that it can "see" among the other surfaces. The approximate view factor from this surface to each other surface is then the area of the receiving surface over the sum of areas that is visible to the sending surface.

In order to account in some limited way for the fact that certain surfaces will not see each other, several assumptions have been built into this simple view factor approximation. First, a surface cannot see itself. Second, surfaces with approximately the same azimuth (facing direction) and tilt ("same" being within a built in limit) will not see each other. This means that a window will not see the wall that it is placed on, for example. Third, floors cannot see each other. Fourth, if the surface is a floor, ceiling, roof, or internal mass, the rule for the same azimuth and tilt eliminating radiant exchange between surface is waived when the receiving surface is floor, roof, ceiling, or internal mass as long as both surfaces are not floors.

Note that this does not take into account that surfaces may be "around the corner" from each other and in reality not see each other at all. Rooms are assumed to be convex rather than concave in this method.

To summarize, using the Surface "Class", the approximate view factors have:

1. No surface sees itself.
2. No Floor sees another floor.
3. All other surface types see Internal Mass.
4. All other surface types see floors.
5. Floors always see ceilings.
6. Floors always see roofs.
7. All other surfaces whose tilt or facing angle differences are greater than 10 degrees see each other.

If geometry is correct, conditions 1, 3, and 7 should take care of all surfaces, but the other conditions supply common sense when the geometry is incorrect. More information about the EnergyPlus view factor calculation is contained in the Engineering Reference document.

1.10.10 GlobalGeometryRules

Before the surface objects are explained in detail, a description of geometric parameters used in EnergyPlus will be given. Since the input of surface vertices is common to most of the surface types, it will also be given a separate discussion.

Some flexibility is allowed in specifying surface vertices. This flexibility is embodied in the GlobalGeometryRules class/object in the input file. Note that the parameters specified in this statement are used for all surface vertices inputs—there is no further "flexibility" allowed.

In order to perform shadowing calculations, the building surfaces must be specified. EnergyPlus uses a three dimensional (3D) Cartesian coordinate system for surface vertex specification. This Right Hand coordinate system has the X-axis pointing east, the Y-axis pointing north, and the Z-axis pointing up. See figure below.

1.10.10.1 Inputs

1.10.10.1.1 Field: Starting Vertex Position

The shadowing algorithms in EnergyPlus rely on surfaces having vertices in a certain order and positional structure. Thus, the surface translator needs to know the starting point for each surface entry. The choices are: UpperLeftCorner, LowerLeftCorner , UpperRightCorner, or LowerRightCorner. Since most surfaces will be 4 sided, the convention will specify this position as though each surface were 4 sided. Extrapolate 3 sided figures to this convention. For 5 and more sided figures, again, try to extrapolate the best "corner" starting position.

1.10.10.1.2 Field: Vertex Entry Direction

Surfaces are always specified as being viewed from the outside of the zone to which they belong. (Shading surfaces are specified slightly differently and are discussed under the particular types). EnergyPlus needs to know whether the surfaces are being specified in counterclockwise or clockwise order (from the Starting Vertex Position). EnergyPlus uses this to determine the outward facing normal for the surface (which is the *facing angle* of the surface – very important in shading and shadowing calculations.

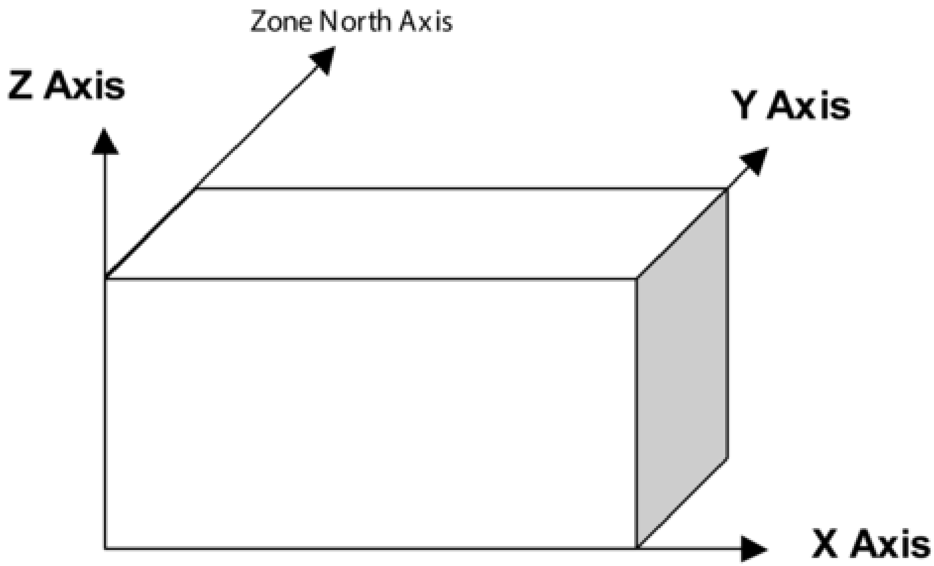

Figure 1.30: EnergyPlus Coordinate System

1.10.10.1.3 Field: Coordinate System

Vertices can be specified in two ways: using **World** coordinates, or a **Relative** coordinate specification. Relative coordinates allow flexibility of rapid change to observe changes in building results due to orientation and position. **World** coordinates will facilitate use within a CADD system structure.

Relative coordinates make use of both Building and Zone North Axis values as well as Zone Origin values to locate the surface in 3D coordinate space. **World** coordinates do not use these values.

Typically, all zone origin values for "World" coordinates will be (0,0,0) but Building and Zone North Axis values may be used in certain instances (namely the Daylighting Coordinate Location entries).

1.10.10.1.4 Field: Daylighting Reference Point Coordinate System

Daylighting reference points need to be specified as well. Again, there can be two flavors; **Relative** and **World**. Daylighting reference points must fit within the zone boundaries.

Relative coordinates make use of both Building and Zone North Axis values as well as Zone Origin values to locate the reference point in 3D coordinate space. **World** coordinates do not use these values.

1.10.10.1.5 Field: Rectangular Surface Coordinate System

Simple, rectangular surfaces (Wall:Exterior, Wall:Adiabatic, Wall:Underground, Wall:Interzone, Roof, Ceiling:Adiabatic, Ceiling:Interzone, Floor:GroundContact, Floor:Adiabatic, Floor:Interzone) can be specified with their Lower Left Corner as **relative** or **world**.

Relative (default) corners are specified relative to the Zone Origin for each surface. **World** corners would specify the absolute/world coordinate for this corner.

1.10.11 Surfaces

Surfaces make up the buildings and the elements that shade buildings. There are several methods to inputting surfaces, ranging from simple rectangular surfaces to detailed descriptions that describe each vertex in the order specified in the GlobalGeometryRules object. The simple, rectangular surface objects are described first with the more detailed descriptions following.

1.10.12 Walls

Walls are usually vertical (tilt = 90 degrees). These objects are used to describe exterior walls, interior walls (adiabatic), underground walls, and walls adjacent to other zones.

1.10.13 Wall:Exterior

The Wall:Exterior object is used to describe walls that are exposed to the external environment. They receive sun, wind – all the characteristics of the external world.

1.10.13.1 Inputs

1.10.13.1.1 Field: Name

This is a unique name associated with the exterior wall. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.13.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.13.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.13.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.13.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction that the wall faces (outward normal). The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.13.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Normally, walls are tilted 90 degrees and that is the default for this field.

Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.13.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.13.1.8 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.13.1.9 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.13.1.10 Field: Length

This field is the length of the wall in meters.

1.10.13.1.11 Field: Height

This field is the height of the wall in meters.

1.10.14 Wall:Adiabatic

The Wall:Adiabatic object is used to describe interior walls and partitions. Adiabatic walls are used to describe walls next to zones that have the same thermal conditions (thus, no heat transfer).

1.10.14.1 Inputs

1.10.14.1.1 Field: Name

This is a unique name associated with the interior wall. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.14.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the “full” construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.14.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.14.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.14.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction that the wall faces (outward normal). The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.14.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Normally, walls are tilted 90 degrees and that is the default for this field.

Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.14.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.14.1.8 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.14.1.9 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.14.1.10 Field: Length

This field is the length of the wall in meters.

1.10.14.1.11 Field: Height

This field is the height of the wall in meters.

1.10.15 Wall:Underground

The Wall:Underground object is used to describe walls with ground contact. The temperature at the outside of the wall is the temperature in the GroundTemperature:BuildingSurface object.

1.10.15.1 Inputs

1.10.15.1.1 Field: Name

This is a unique name associated with the underground wall. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.15.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones. Note that if the construction is **Construction:CfactorUndergroundWall** then the GroundFCfactoreMethod will be used for this wall.

1.10.15.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.15.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.15.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction that the wall faces (outward normal). The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.15.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Normally, walls are tilted 90 degrees and that is the default for this field.

1.10.15.1.7 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.15.1.8 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.15.1.9 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.15.1.10 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.15.1.11 Field: Length

This field is the length of the wall in meters.

1.10.15.1.12 Field: Height

This field is the height of the wall in meters.

1.10.16 Wall:Interzone

The Wall:Interzone object is used to describe walls adjacent to zones that are significantly different conditions than the zone with this wall.

1.10.16.1 Inputs

1.10.16.1.1 Field: Name

This is a unique name associated with the interzone wall. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.16.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.16.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.16.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.16.1.5 Field: Outside Boundary Condition Object

The Outside Boundary Condition Object field is the name of a wall in an adjacent zone or the name of the adjacent zone. If the adjacent zone option is used, the adjacent wall is automatically generated in the adjacent zone. If the surface name is used, it must be in the adjacent zone.

1.10.16.1.6 Field: Azimuth Angle

The Azimuth Angle indicates the direction that the wall faces (outward normal). The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.16.1.7 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Normally, walls are tilted 90 degrees and that is the default for this field.

1.10.16.1.8 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.16.1.9 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.16.1.10 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.16.1.11 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.16.1.12 Field: Length

This field is the length of the wall in meters.

1.10.16.1.13 Field: Height

This field is the height of the wall in meters.

1.10.17 Roofs/Ceilings

Roofs and ceilings are, by default, flat (tilt = 0 degrees). These objects are used to describe roofs, interior ceilings (adiabatic) and ceilings adjacent to other zones.

1.10.18 Roof

The Roof object is used to describe roofs that are exposed to the external environment.

1.10.18.1 Inputs

1.10.18.1.1 Field: Name

This is a unique name associated with the roof. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.18.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.18.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.18.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.18.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction of the outward normal for the roof. The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.18.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Flat roofs are tilted 0 degrees and that is the default for this field.

1.10.18.1.7 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.18.1.8 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.18.1.9 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.18.1.10 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.18.1.11 Field: Length

This field is the length of the roof in meters.

1.10.18.1.12 Field: Width

This field is the width of the roof in meters.

1.10.19 Ceiling:Adiabatic

The Ceiling:Adiabatic object is used to describe interior ceilings that separate zones of like conditions.

1.10.19.1 Inputs

1.10.19.1.1 Field: Name

This is a unique name associated with the ceiling. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.19.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.19.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.19.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.19.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction of the outward normal for the roof. The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.19.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Flat ceilings are tilted 0 degrees and that is the default for this field.

1.10.19.1.7 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.19.1.8 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.19.1.9 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.19.1.10 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.19.1.11 Field: Length

This field is the length of the ceiling in meters.

1.10.19.1.12 Field: Width

This field is the width of the ceiling in meters.

1.10.20 Ceiling:Interzone

The Ceiling:Interzone object is used to describe interior ceilings that separate zones of differing conditions (and expect heat transfer through the ceiling from the adjacent zone).

1.10.20.1 Inputs

1.10.20.1.1 Field: Name

This is a unique name associated with the interzone ceiling. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.20.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.20.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.20.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.20.1.5 Field: Outside Boundary Condition Object

The Outside Boundary Condition Object field is the name of a floor in an adjacent zone or the name of the adjacent zone. If the adjacent zone option is used, the adjacent floor is automatically generated in the adjacent zone. If the surface name is used, it must be in the adjacent zone.

1.10.20.1.6 Field: Azimuth Angle

The Azimuth Angle indicates the direction of the outward normal for the roof. The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.20.1.7 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Flat ceilings are tilted 0 degrees and that is the default for this field.

1.10.20.1.8 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.20.1.9 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.20.1.10 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.20.1.11 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.20.1.12 Field: Length

This field is the length of the ceiling in meters.

1.10.20.1.13 Field: Width

This field is the width of the ceiling in meters.

1.10.21 Floors

Floors are, by default, flat (tilt = 180 degrees). These objects are used to describe floors on the ground, interior floors (adiabatic) and floors adjacent to other zones.

1.10.22 Floor:GroundContact

The Floor:GroundContact object is used to describe floors that have ground contact (usually called slabs). The temperature at the outside of the floor is the temperature in the GroundTemperature:BuildingSurface object.

1.10.22.1 Inputs

1.10.22.1.1 Field: Name

This is a unique name associated with the floor.

1.10.22.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones. Note that if the construction is **Construction:FfactorGroundFloor,** then the GroundFCfactorMethod will be used with this floor.

1.10.22.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.22.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.22.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction of the outward normal for the roof. The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.22.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Flat floors are tilted 180 degrees and that is the default for this field.

1.10.22.1.7 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.22.1.8 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.22.1.9 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.22.1.10 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.22.1.11 Field: Length

This field is the length of the floor in meters.

1.10.22.1.12 Field: Width

This field is the width of the floor in meters.

1.10.23 Floor:Adiabatic

The Floor:Adiabatic object is used to describe interior floors or floors that you wish to model with no heat transfer from the exterior to the floor.

1.10.23.1 Inputs

1.10.23.1.1 Field: Name

This is a unique name associated with the floor.

1.10.23.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.23.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.23.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.23.1.5 Field: Azimuth Angle

The Azimuth Angle indicates the direction of the outward normal for the roof. The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.23.1.6 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Flat floors are tilted 180 degrees and that is the default for this field.

1.10.23.1.7 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.23.1.8 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.23.1.9 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.23.1.10 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.23.1.11 Field: Length

This field is the length of the floor in meters.

1.10.23.1.12 Field: Width

This field is the width of the floor in meters.

1.10.24 Floor:Interzone

The Floor:Interzone object is used to describe floors that are adjacent to other zones that have differing conditions and you wish to model the heat transfer through the floor.

1.10.24.1 Inputs

1.10.24.1.1 Field: Name

This is a unique name associated with the floor.

1.10.24.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.24.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.24.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.24.1.5 Field: Outside Boundary Condition Object

The Outside Boundary Condition Object field is the name of a ceiling in an adjacent zone or the name of the adjacent zone. If the adjacent zone option is used, the adjacent ceiling is automatically generated in the adjacent zone. If the surface name is used, it must be in the adjacent zone.

1.10.24.1.6 Field: Azimuth Angle

The Azimuth Angle indicates the direction of the outward normal for the roof. The angle is specified in degrees where East = 90, South = 180, West = 270, North = 0.

1.10.24.1.7 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the wall is tilted from horizontal (or the ground). Flat floors are tilted 180 degrees and that is the default for this field.

1.10.24.1.8 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. This is specified with (x,y,z) and can be relative to the zone origin or in world coordinates, depending on the value for rectangular surfaces specified in the GlobalGeometryRules object.

1.10.24.1.9 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.24.1.10 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.24.1.11 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.24.1.12 Field: Length

This field is the length of the floor in meters.

1.10.24.1.13 Field: Width

This field is the width of the floor in meters.

1.10.25 Windows/Doors

The following window and door objects can be used to specify simple, rectangular doors and windows. In each case, the lower left corner (locator coordinate) of the window or door is specified **relative** to the surface it is on. Viewing the base surface as a planar surface, base the relative location from the lower left corner of the base surface. Vertex entry description as well as provisions for a few other surface types can be entered with the FenestrationSurface:Detailed object.

1.10.26 Window

The Window object is used to place windows on surfaces that can have windows, including exterior walls, interior walls, interzone walls, roofs, floors that are exposed to outdoor conditions, interzone ceiling/floors. These, of course, can be entered using the simple rectangular objects or the more detailed vertex entry objects. Shades and screens may be applied by referencing this subsurface in a window shading control (ref: WindowShadingControl object). To assign a shade to a window or glass door, see WindowMaterial: Shade. To assign a screen, see WindowMaterial:Screen. To assign a blind, see WindowMaterial:Blind. To assign switchable glazing, such as electrochromic glazing, see WindowShadingControl.

1.10.26.1 Inputs

1.10.26.1.1 Field: Name

This is a unique name associated with the window.

1.10.26.1.2 Field: Construction Name

This is the name of the subsurface's construction (ref: objects: Construction, Construction:WindowDataFile, Construction:ComplexFenestrationState).

For windows, if Construction Name is not found among the constructions on the input (.idf) file, the Window Data File (ref. Construction:WindowDataFile object) will be searched for that Construction Name (see "Importing Windows from WINDOW"). If that file is not present or if the Construction Name does not match the name of an entry on the file, an error will result. If there is a match, a window construction and its corresponding glass and gas materials will be created from the information read from the file.

1.10.26.1.3 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. For example, a surface in contact with the ground (e.g., Outside Boundary Condition = Ground) cannot contain a window. The window assumes the outward facing angle as well as the tilt angle of the base surface.

1.10.26.1.4 Field: Frame and Divider Name

This field, if not blank, can be used to specify window frame, divider and reveal-surface data (ref: WindowProperty:FrameAndDivider object). It is used only for exterior GlassDoors and rectangular exterior Windows, i.e., those with OutsideFaceEnvironment = Outdoors.

This field should be blank for triangular windows.

1.10.26.1.5 Field: Multiplier

This field is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.26.1.6 Starting Corner for the surface

The rectangular subsurfaces specify the lower left corner of the surface for their starting coordinate. This corner is specified relative to the lower left corner of the base surface by specifying the X and Z values from that corner.

1.10.26.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.26.1.8 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.26.1.9 Field: Length

This field is the length of the window in meters.

1.10.26.1.10 Field: Height

This field is the height of the window in meters.

1.10.27 Door

The Door object is used to place opaque doors on surfaces that can have doors, including exterior walls, interior walls, interzone walls, roofs, floors that are exposed to outdoor conditions, interzone ceiling/floors. These, of course, can be entered using the simple rectangular objects or the more detailed vertex entry objects.

1.10.27.1 Inputs

1.10.27.1.1 Field: Name

This is a unique name associated with the door.

1.10.27.1.2 Field: Construction Name

This is the name of the subsurface's construction (ref: Construction object)

1.10.27.1.3 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. The door assumes the outward facing angle as well as the tilt angle of the base surface.

1.10.27.1.4 Field: Multiplier

This field is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.27.1.5 Starting Corner for the surface

The rectangular subsurfaces specify the lower left corner of the surface for their starting coordinate. This corner is specified relative to the lower left corner of the base surface by specifying the X and Z values from that corner.

1.10.27.1.6 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.27.1.7 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.27.1.8 Field: Length

This field is the length of the door in meters.

1.10.27.1.9 Field: Height

This field is the height of the door in meters.

1.10.28 GlazedDoor

The GlazedDoor object is used to place doors on surfaces that can have doors, including exterior walls, interior walls, interzone walls, roofs, floors that are exposed to outdoor conditions, interzone ceiling/floors. These, of course, can be entered using the simple rectangular objects or the more detailed vertex entry objects. Shades and screens may be applied by referencing this subsurface in a window shading control (ref: WindowShadingControl object). To assign a shade to a window or glass door, see WindowMaterial: Shade. To assign a screen, see WindowMaterial:Screen. To assign a blind, see WindowMaterial:Blind. To assign switchable glazing, such as electrochromic glazing, see WindowShadingControl.

1.10.28.1 Inputs

1.10.28.1.1 Field: Name

This is a unique name associated with the glass door.

1.10.28.1.2 Field: Construction Name

This is the name of the subsurface's construction (ref: objects: Construction, Construction:WindowDataFile, Construction:ComplexFenestrationState).

For windows, if Construction Name is not found among the constructions on the input (.idf) file, the Window Data File (ref. Construction:WindowDataFile object) will be searched for that Construction Name (see "Importing Windows from WINDOW"). If that file is not present or if the Construction Name does not match the name of an entry on the file, an error will result. If there is a match, a window construction and its corresponding glass and gas materials will be created from the information read from the file.

1.10.28.1.3 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. For example, a surface in contact with the ground (Outside Boundary Condition = Ground) cannot contain a window. The door assumes the outward facing angle as well as the tilt angle of the base surface.

1.10.28.1.4 Field: Frame and Divider Name

This field, if not blank, can be used to specify window frame, divider and reveal-surface data (ref: WindowProperty:FrameAndDivider object). It is used only for exterior GlassDoors and rectangular exterior Windows, i.e., those with OutsideFaceEnvironment = Outdoors.

This field should be blank for triangular windows.

1.10.28.1.5 Field: Multiplier

This field is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.28.1.6 Starting Corner for the surface

The rectangular subsurfaces specify the lower left corner of the surface for their starting coordinate. This corner is specified relative to the lower left corner of the base surface by specifying the X and Z values from that corner.

1.10.28.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.28.1.8 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.28.1.9 Field: Length

This field is the length of the door in meters.

1.10.28.1.10 Field: Height

This field is the height of the door in meters.

1.10.29 Window:Interzone

The Window:Interzone object is used to place windows on surfaces that can have windows, including interzone walls, interzone ceiling/floors. These, of course, can be entered using the simple rectangular objects or the more detailed vertex entry objects.

1.10.29.1 Inputs

1.10.29.1.1 Field: Name

This is a unique name associated with the window.

1.10.29.1.2 Field: Construction Name

This is the name of the subsurface's construction (ref: objects: Construction, Construction:WindowDataFile, Construction:ComplexFenestrationState).

For windows, if Construction Name is not found among the constructions on the input (.idf) file, the Window Data File (ref. Construction:WindowDataFile object) will be searched for that Construction Name (see "Importing Windows from WINDOW"). If that file is not present or if the Construction Name does not match the name of an entry on the file, an error will result. If there is a match, a window construction and its corresponding glass and gas materials will be created from the information read from the file.

1.10.29.1.3 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. For example, a surface in contact with the ground (Outside Boundary Condition = Ground) cannot contain a window. The window assumes the outward facing angle as well as the tilt angle of the base surface.

1.10.29.1.4 Field: Outside Boundary Condition Object

The Outside Boundary Condition Object field is the name of a window in an adjacent zone or the name of the adjacent zone. If the adjacent zone option is used, the adjacent ceiling is automatically generated in the adjacent zone. If the surface name is used, it must be in the adjacent zone.

1.10.29.1.5 Field: Multiplier

This field is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.29.1.6 Starting Corner for the surface

The rectangular subsurfaces specify the lower left corner of the surface for their starting coordinate. This corner is specified relative to the lower left corner of the base surface by specifying the X and Z values from that corner.

1.10.29.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.29.1.8 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.29.1.9 Field: Length

This field is the length of the window in meters.

1.10.29.1.10 Field: Height

This field is the height of the window in meters.

1.10.30 Door:Interzone

The Door:Interzone object is used to place opaque doors on surfaces that can have doors, including interzone walls, interzone ceiling/floors. These, of course, can be entered using the simple rectangular objects or the more detailed vertex entry objects.

1.10.30.1 Inputs

1.10.30.1.1 Field: Name

This is a unique name associated with the door.

1.10.30.1.2 Field: Construction Name

This is the name of the subsurface's construction (ref: Construction object).

1.10.30.1.3 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. The door assumes the outward facing angle as well as the tilt angle of the base surface.

1.10.30.1.4 Field: Outside Boundary Condition Object

The Outside Boundary Condition Object field is the name of a door in an adjacent zone or the name of the adjacent zone. If the adjacent zone option is used, the adjacent ceiling is automatically generated in the adjacent zone. If the surface name is used, it must be in the adjacent zone.

1.10.30.1.5 Field: Multiplier

This field is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.30.1.6 Starting Corner for the surface

The rectangular subsurfaces specify the lower left corner of the surface for their starting coordinate. This corner is specified relative to the lower left corner of the base surface by specifying the X and Z values from that corner.

1.10.30.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.30.1.8 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.30.1.9 Field: Length

This field is the length of the door in meters.

1.10.30.1.10 Field: Height

This field is the height of the door in meters.

1.10.31 GlazedDoor:Interzone

The GlazedDoor:Interzone object is used to place doors on surfaces that can have doors, including interzone walls, interzone ceiling/floors. These, of course, can be entered using the simple rectangular objects or the more detailed vertex entry objects.

1.10.31.1 Inputs

1.10.31.1.1 Field: Name

This is a unique name associated with the glass door.

1.10.31.1.2 Field: Construction Name

This is the name of the subsurface's construction (ref: objects: Construction, Construction:WindowDataFile, Construction:ComplexFenestrationState).

For windows, if Construction Name is not found among the constructions on the input (.idf) file, the Window Data File (ref. Construction:WindowDataFile object) will be searched for that Construction Name (see "Importing Windows from WINDOW"). If that file is not present or if the Construction Name does not match the name of an entry on the file, an error will result. If there is a match, a window construction and its corresponding glass and gas materials will be created from the information read from the file.

1.10.31.1.3 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. For example, a surface in contact with the ground (Outside Boundary Condition = Ground) cannot contain a window. The door assumes the outward facing angle as well as the tilt angle of the base surface.

1.10.31.1.4 Field: Outside Boundary Condition Object

The Outside Boundary Condition Object field is the name of a glazed (glass) door in an adjacent zone or the name of the adjacent zone. If the adjacent zone option is used, the adjacent ceiling is automatically generated in the adjacent zone. If the surface name is used, it must be in the adjacent zone.

1.10.31.1.5 Field: Multiplier

This field is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.31.1.6 Starting Corner for the surface

The rectangular subsurfaces specify the lower left corner of the surface for their starting coordinate. This corner is specified relative to the lower left corner of the base surface by specifying the X and Z values from that corner.

1.10.31.1.7 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.31.1.8 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.31.1.9 Field: Length

This field is the length of the door in meters.

1.10.31.1.10 Field: Height

This field is the height of the door in meters.

Examples of the rectangular surfaces are found in the example files 4ZoneWithShading_Simple_1.idf and 4ZoneWithShading_Simple_2. Some examples:

```
Wall:Exterior,
    Zn001:Wall001,          !- Name
    EXTERIOR,               !- Construction Name
    ZONE 1,                 !- Zone Name
    180,                    !- Azimuth Angle {deg}
    90,                     !- Tilt Angle {deg}
    0,                      !- Starting X Coordinate {m}
    0,                      !- Starting Y Coordinate {m}
    0,                      !- Starting Z Coordinate {m}
    20,                     !- Length {m}
```

```
    10;                      !- Height {m}

  Window,
    Zn001:Wall001:Win001,    !- Name
    SINGLE PANE HW WINDOW,   !- Construction Name
    Zn001:Wall001,           !- Building Surface Name
    ,                        !- Frame and Divider Name
    1,                       !- Multiplier
    4,                       !- Starting X Coordinate {m}
    3,                       !- Starting Z Coordinate {m}
    3,                       !- Length {m}
    5;                       !- Height {m}

  Door,
    Zn001:Wall001:Door001,   !- Name
    HOLLOW WOOD DOOR,        !- Construction Name
    Zn001:Wall001,           !- Building Surface Name
    1,                       !- Multiplier
    14,                      !- Starting X Coordinate {m}
    0,                       !- Starting Z Coordinate {m}
    3,                       !- Length {m}
    5;                       !- Height {m}

  Wall:Adiabatic,
    Zn001:Wall004,           !- Name
    INTERIOR,                !- Construction Name
    ZONE 1,                  !- Zone Name
    90,                      !- Azimuth Angle {deg}
    90,                      !- Tilt Angle {deg}
    20,                      !- Starting X Coordinate {m}
    0,                       !- Starting Y Coordinate {m}
    0,                       !- Starting Z Coordinate {m}
    20,                      !- Length {m}
    10;                      !- Height {m}

  Floor:Adiabatic,
    Zn001:Flr001,            !- Name
    FLOOR,                   !- Construction Name
    ZONE 1,                  !- Zone Name
    90,                      !- Azimuth Angle {deg}
    180,                     !- Tilt Angle {deg}
    0,                       !- Starting X Coordinate {m}
    0,                       !- Starting Y Coordinate {m}
    0,                       !- Starting Z Coordinate {m}
    20,                      !- Length {m}
    20;                      !- Width {m}

  Ceiling:Interzone,
    Zn001:Roof001,           !- Name
    CEILING34,               !- Construction Name
    ZONE 1,                  !- Zone Name
    Zn003:Flr001,            !- Outside Boundary Condition Object
    180,                     !- Azimuth Angle {deg}
    0,                       !- Tilt Angle {deg}
    0,                       !- Starting X Coordinate {m}
    0,                       !- Starting Y Coordinate {m}
    10,                      !- Starting Z Coordinate {m}
    20,                      !- Length {m}
    20;                      !- Width {m}

  Window,
    Zn002:Wall001:Win001,    !- Name
    SINGLE PANE HW WINDOW,   !- Construction Name
    Zn002:Wall001,           !- Building Surface Name
    ,                        !- Frame and Divider Name
    1,                       !- Multiplier
    4,                       !- Starting X Coordinate {m}
    3,                       !- Starting Z Coordinate {m}
    3,                       !- Length {m}
    5;                       !- Height {m}
```

1.10.32 Surface Vertices

Each of the following surfaces:

- BuildingSurface:Detailed
- Wall:Detailed
- RoofCeiling:Detailed
- Floor:Detailed
- FenestrationSurface:Detailed
- Shading:Site:Detailed
- Shading:Building:Detailed
- Shading:Zone:Detailed

use the same vertex input. The numeric parameters indicated below are taken from the BuildingSurface:Detailed definition; the others may not be exactly the same but are identical in configuration. They are also "extensible" – so, to define more vertices for these surfaces, simply add the required number of vertices (X, Y, and Z coordinates for each vertex) to the input file. Note that FenestrationSurface:Detailed is not extensible and is limited to 4 (max) vertices. If the Number of Surface Vertex groups is left blank or entered as **autocalculate**, EnergyPlus looks at the number of groups entered and figures out how many coordinate groups are entered.

Note that the resolution for surface vertex input is 1 centimeter (0.01 meter). Vertices that are < 1 cm apart will be combined.

The figure above will help illustrate Surface Vertex entry. The convention used in "GlobalGeometryRules" dictates the order of the vertices (ref: GlobalGeometryRules). In this example, the conventions used are Starting Vertex Position = UpperLeftCorner and Vertex Entry Direction = CounterClockwise. The surfaces for this single zone are:

```
4,0,0,H, 0,0,0, A,0,0, A,0,H; ! (4 vertices, South Wall)
4,A,0,H,A,0,0,A,B,0,A,B,H; ! (4 vertices, East Wall)
     ignore other walls that are not shown in this figure
4,C,0,J,A,0,H,A,B,H,C,B,J; ! (4 vertices, roof)
3,C,0,J,0,0,H,A,0,H; ! (3 vertices, gable end)
4,0,0,H, 0,0,0, A,0,0, A,0,H; ! (4 vertices, South Wall)
```

Note that in this example, point 1 of the entry is the Upper Left Corner of the rectangular surfaces and the point of the triangle for the 3 sided surface. The east wall shows the order of vertex entry. For horizontal surfaces, any vertex may be chosen as the starting position, but the Vertex Entry Direction convention must be followed. The surface details report (Output: Surfaces:List, Details;) is very useful for reviewing the accuracy of surface geometry inputs (ref: Surface Output Variables/Reports and Variable Dictionary Reports).

From the detailed vertices, EnergyPlus tries to determine the "height" and "width" of the surface. Obviously, this doesn't work well for >4 sided surfaces; for these, if the calculated height and width are not close to the gross area for the surface, the height and width shown will be the square root of the area (and thus a square).

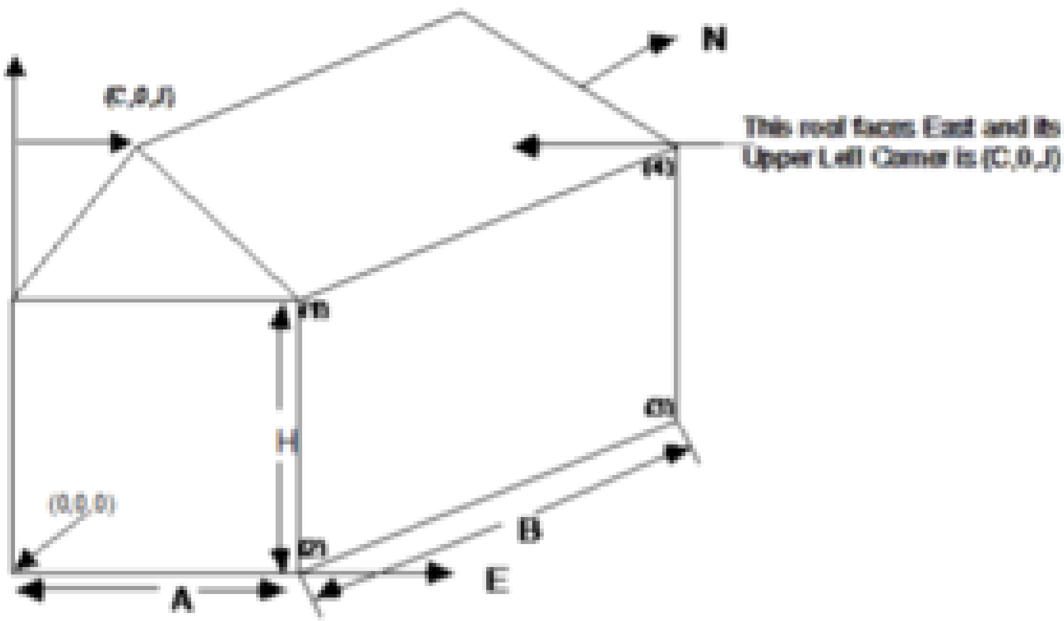

Figure 1.31: Illustration for Surface Vertices

1.10.33 Building Surfaces - Detailed

A building surface is necessary for all calculations. There must be at least one building surface per zone. You can use the detailed descriptions as shown below or the simpler, rectangular surface descriptions shown earlier.

1.10.34 Wall:Detailed

The Wall:Detailed object is used to describe walls.

1.10.34.1 Inputs

1.10.34.1.1 Field: Name

This is a unique name associated with each building surface. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.34.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.34.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.34.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.34.1.5 Field: Outside Boundary Condition

This value can be one of several things depending on the actual kind of surface.

1. **Surface** – if this surface is an internal surface, then this is the choice. The value will either be a surface in the base zone or a surface in another zone. The heat balance between two zones can be accurately simulated by specifying a surface in an adjacent zone. EnergyPlus will simulate a group of zones simultaneously and will include the heat transfer between zones. However, as this increases the complexity of the calculations, it is not necessary to specify the other zone unless the two zones will have a significant temperature difference. If the two zones will not be very different (temperature wise), then the surface should use itself as the outside environment or specify this field as **Adiabatic**. The surface name on the "outside" of this surface (adjacent to) is placed in the next field.

2. **Adiabatic** – an internal surface in the same Zone. This surface will not transfer heat out of the zone, but will still store heat in thermal mass. Only the inside face of the surface will exchange heat with the zone (i.e. two adiabatic surfaces are required to model internal partitions where both sides of the surface are exchanging heat with the zone). The Outside Boundary Condition Object can be left blank.

3. **Zone** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent zone when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

4. **Space** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent space when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

5. **Outdoors** – if this surface is exposed to outside temperature conditions, then this is the choice. See Sun Exposure and Wind Exposure below for further specifications on this kind of surface.

6. **Foundation** - uses an alternative model (currently only the Kiva™ model) to account for the multi-dimensional heat transfer of foundation surfaces. The Outside Boundary Condition Object will refer to the name of a Foundation:Kiva object (or be left blank to use the default foundation without extra insulation).

7. **Ground** - The temperature on the outside of this surface will be the Site:GroundTemperature:Surface value for the month.

8. **GroundFCfactorMethod** – if this surface is exposed to the ground and using the **Construction:CfactorUnd** then this is the choice. The temperature on the outside of this surface will be the Site:GroundTemperature:Fcfacto value for the month.

9. **OtherSideCoefficients** – if this surface has a custom, user specified temperature or other parameters (See SurfaceProperty:OtherSideCoefficients specification), then this is the choice. The outside boundary condition will be the name of the SurfaceProperty:OtherSideCoefficients specification.

10. **OtherSideConditionsModel** – if this surface has a specially-modeled multi-skin component, such as a transpired collector or vented photovoltaic panel, attached to the outside (See SurfaceProperty:OtherSideConditionsModel specification), then this the choice. The outside face environment will be the name of the SurfaceProperty:OtherSideConditionsModel specification.

11. **GroundSlabPreprocessorAverage** – uses the average results from the Slab preprocessor calculations.

12. **GroundSlabPreprocessorCore** – uses the core results from the Slab preprocessor calculations.

13. **GroundSlabPreprocessorPerimeter** – uses the perimeter results from the Slab preprocessor calculations.

14. **GroundBasementPreprocessorAverageWall** – uses the average wall results from the Basement preprocessor calculations.

15. **GroundBasementPreprocessorAverageFloor** – uses the average floor results from the Basement preprocessor calculations.

16. **GroundBasementPreprocessorUpperWall** – uses the upper wall results from the Basement preprocessor calculations.

17. **GroundBasementPreprocessorLowerWall** – uses the lower wall results from the Basement preprocessor calculations.

1.10.34.1.6 Field: Outside Boundary Condition Object

If neither Surface, Zone, Space, Foundation, OtherSideCoefficients, or OtherSideConditionsModel are specified for the Outside Boundary Condition (previous field), then this field should be left blank.

As stated above, if the Outside Boundary Condition is "Surface", then this field's value must be the surface name whose inside face temperature will be forced on the outside face of the base surface. This permits heat exchange between adjacent zones (interzone heat transfer) when multiple zones are simulated, but can also be used to simulate middle zone behavior without modeling the adjacent zones. This is done by specifying a surface within the zone. For example, a middle floor zone can be modeled by making the floor the Outside Boundary Condition Object for the ceiling, and the ceiling the Outside Boundary Condition Object for the floor.

If the Outside Boundary Condition is Zone or Space, then this field should contain the zone or space name for the adjacent surface.

> Note: Zones with interzone heat transfer are not adiabatic and the internal surfaces contribute to gains or losses. Adiabatic surfaces are modeled by specifying the base surface itself in this field. Also, for interzone heat transfer, both surfaces must be represented – for example, if you want interzone heat transfer to an attic space, the ceiling in the lower zone must have a surface object with the outside face environment as the floor in the attic and, likewise, there must be a floor surface object in the attic that references the ceiling surface name in the lower zone.

Equally, if the Outside Boundary Condition is "OtherSideCoefficients", then this field's value must be the SurfaceProperty:OtherSideCoefficients name. Or if the Outside Boundary Condition is "OtherSideConditionsModel" then this field's value must be the SurfaceProperty:OtherSideConditionsModel name.

1.10.34.1.7 Field: Sun Exposure

If the surface is exposed to the sun, then "SunExposed" should be entered in this field. Otherwise, "NoSun" should be entered.

Note, a cantilevered floor could have "Outdoors" but "NoSun" exposure.

1.10.34.1.8 Field: Wind Exposure

If the surface is exposed to the Wind, then "WindExposed" should be entered in this field. Otherwise, "NoWind" should be entered.

> Note: When a surface is specified with "NoWind", this has several implications. Within the heat balance code, this surface will default to using the simple ASHRAE exterior convection coefficient correlation with a zero wind speed. In addition, since the ASHRAE simple method does not have a separate value for equivalent long wavelength radiation to the sky and ground, using "NoWind" also eliminates long wavelength radiant exchange from the exterior of the surface to both the sky and the ground. Thus, only simple convection takes place at the exterior face of a surface specified with "NoWind".

1.10.34.1.9 Field: View Factor to Ground

The fraction of the ground plane (assumed horizontal) that is visible from a heat-transfer surface. It is used to calculate the diffuse solar radiation from the ground that is incident on the surface.

For example, if there are no obstructions, a vertical surface sees half of the ground plane and so View Factor to Ground = 0.5. A horizontal downward-facing surface sees the entire ground plane, so View Factor to Ground = 1.0. A horizontal upward-facing surface (horizontal roof) does not see the ground at all, so View Factor to Ground = 0.0.

Unused if reflections option in Solar Distribution field in Building object input unless a DaylightingDevice:Shelf or DaylightingDevice:Tubular has been specified.

If you do not use the reflections option in the Solar Distribution field in your Building object input, you are responsible for entering the View Factor to Ground for each heat-transfer surface. Typical values for a surface that is not shadowed are obtained by the simple equation:

View Factor to Ground = (1-cos(SurfTilt))/2

For example, this gives 0.5 for a wall of tilt 90°. If the tilt of the wall changes, then the View Factor to Ground must also change.

If you enter **autocalculate** in this field, EnergyPlus will automatically calculate the view factor to ground based on the tilt of the surface.

If **you do use the reflections option in the Solar Distribution field** in your Building object input, you do **not** have to enter View Factor to Ground values. In this case the program will automatically calculate the value to use for each exterior surface taking into account solar shadowing (including shadowing of the ground by the building) and reflections from obstructions (ref: Building, Field: Solar Distribution).

However, if you do use the reflections option AND you are modeling a DaylightingDevice:Shelf or DaylightingDevice:Tubular, then you still need to enter some values of View Factor to Ground. For DaylightingDevice:Shelf you need to enter View Factor to Ground for the window associated with the shelf. And for DaylightingDevice:Tubular you need to enter the View Factor to Ground for the FenestrationSurface:Detailed corresponding to the dome of the tubular device.

Note 1: The corresponding view factor to the sky for diffuse solar radiation is not a user input; it is calculated within EnergyPlus based on surface orientation, sky solar radiance distribution, and shadowing surfaces.

Note 2: The view factors to the sky and ground for thermal infrared (long-wave) radiation are not user inputs; they are calculated within EnergyPlus based on surface tilt and shadowing surfaces. Shadowing surfaces are considered to have the same emissivity and temperature as the ground, so they are lumped together with the ground in calculating the ground IR view factor.

1.10.34.1.10 Field: Number of Vertices

This field specifies the number of sides in the surface (number of X,Y,Z vertex groups). For further information, see the discussion on “Surface Vertices” above.

1.10.35 RoofCeiling:Detailed

The RoofCeiling:Detailed object is used to describe a roof or ceiling.

1.10.35.1 Inputs

1.10.35.1.1 Field: Name

This is a unique name associated with each building surface. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.35.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the “full” construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.35.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.35.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.35.1.5 Field: Outside Boundary Condition

This value can be one of several things depending on the actual kind of surface.

1. **Surface** – if this surface is an internal surface, then this is the choice. The value will either be a surface in the base zone or a surface in another zone. The heat balance between two zones can be accurately simulated by specifying a surface in an adjacent zone. EnergyPlus will simulate a group of zones simultaneously and will include the heat transfer between zones. However, as this increases the complexity of the calculations, it is not necessary to specify the other zone unless the two zones will have a significant temperature difference. If the two zones will not be very different (temperature wise), then the surface should use itself as the outside environment or specify this field as **Adiabatic**. The surface name on the “outside” of this surface (adjacent to) is placed in the next field.

2. **Adiabatic** – an internal surface in the same Zone. This surface will not transfer heat out of the zone, but will still store heat in thermal mass. Only the inside face of the surface will exchange heat with the zone (i.e. two adiabatic surfaces are required to model internal partitions where both sides of the surface are exchanging heat with the zone). The Outside Boundary Condition Object can be left blank.

3. **Zone** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent zone when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

4. **Space** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent space when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

5. **Outdoors** – if this surface is exposed to outside temperature conditions, then this is the choice. See Sun Exposure and Wind Exposure below for further specifications on this kind of surface.

6. **Ground** – The temperature on the outside of this surface will be the Ground Temperature.

7. **OtherSideCoefficients** – if this surface has a custom, user specified temperature or other parameters (See SurfaceProperty:OtherSideCoefficients specification), then this is the choice. The outside boundary condition will be the name of the SurfaceProperty:OtherSideCoefficients specification.

8. **OtherSideConditionsModel** – if this surface has a specially-modeled multi-skin component, such as a transpired collector or vented photovoltaic panel, attached to the outside (See SurfaceProperty:OtherSideConditionsModel specification), then this the choice. The outside face environment will be the name of the SurfaceProperty:OtherSideConditionsModel specification.

9. **GroundSlabPreprocessorAverage** – uses the average results from the Slab preprocessor calculations.

10. **GroundSlabPreprocessorCore** – uses the core results from the Slab preprocessor calculations.

11. **GroundSlabPreprocessorPerimeter** – uses the perimeter results from the Slab preprocessor calculations.

12. **GroundBasementPreprocessorAverageWall** – uses the average wall results from the Basement preprocessor calculations.

13. **GroundBasementPreprocessorAverageFloor** – uses the average floor results from the Basement preprocessor calculations.

14. **GroundBasementPreprocessorUpperWall** – uses the upper wall results from the Basement preprocessor calculations.

15. **GroundBasementPreprocessorLowerWall** – uses the lower wall results from the Basement preprocessor calculations.

1.10.35.1.6 Field: Outside Boundary Condition Object

If neither Surface, Zone, Space, Foundation, OtherSideCoefficients, or OtherSideConditionsModel are specified for the Outside Boundary Condition (previous field), then this field should be left blank.

As stated above, if the Outside Boundary Condition is “Surface”, then this field’s value must be the surface name whose inside face temperature will be forced on the outside face of the base surface. This permits heat exchange between adjacent zones (interzone heat transfer) when multiple zones are simulated, but can also be used to simulate middle zone behavior without modeling the adjacent zones. This is done by specifying a surface within the zone. For example, a middle floor zone can be modeled by making the floor the Outside Boundary Condition Object for the ceiling, and the ceiling the Outside Boundary Condition Object for the floor.

If the Outside Boundary Condition is Zone or Space, then this field should contain the zone or space name for the adjacent surface.

> Note: Zones with interzone heat transfer are not adiabatic and the internal surfaces contribute to gains or losses. Adiabatic surfaces are modeled by specifying the base surface itself in this field. Also, for interzone heat transfer, both surfaces must be represented – for example, if you want interzone heat transfer to an attic space, the ceiling in the lower zone must have a surface object with the outside face environment as the floor in the attic and, likewise, there must be a floor surface object in the attic that references the ceiling surface name in the lower zone.

Equally, if the Outside Boundary Condition is "OtherSideCoefficients", then this field's value must be the SurfaceProperty:OtherSideCoefficients name. Or if the Outside Boundary Condition is "OtherSideConditionsModel" then this field's value must be the SurfaceProperty:OtherSideConditionsModel name.

1.10.35.1.7 Field: Sun Exposure

If the surface is exposed to the sun, then "SunExposed" should be entered in this field. Otherwise, "NoSun" should be entered.

Note, a cantilevered floor could have "Outdoors" but "NoSun" exposure.

1.10.35.1.8 Field: Wind Exposure

If the surface is exposed to the Wind, then "WindExposed" should be entered in this field. Otherwise, "NoWind" should be entered.

> Note: When a surface is specified with "NoWind", this has several implications. Within the heat balance code, this surface will default to using the simple ASHRAE exterior convection coefficient correlation with a zero wind speed. In addition, since the ASHRAE simple method does not have a separate value for equivalent long wavelength radiation to the sky and ground, using "NoWind" also eliminates long wavelength radiant exchange from the exterior of the surface to both the sky and the ground. Thus, only simple convection takes place at the exterior face of a surface specified with "NoWind".

1.10.35.1.9 Field: View Factor to Ground

The fraction of the ground plane (assumed horizontal) that is visible from a heat-transfer surface. It is used to calculate the diffuse solar radiation from the ground that is incident on the surface.

For example, if there are no obstructions, a vertical surface sees half of the ground plane and so View Factor to Ground = 0.5. A horizontal downward-facing surface sees the entire ground plane, so View Factor to Ground = 1.0. A horizontal upward-facing surface (horizontal roof) does not see the ground at all, so View Factor to Ground = 0.0.

Unused if reflections option in Solar Distribution field in Building object input unless a DaylightingDevice:Shelf or DaylightingDevice:Tubular has been specified.

If you do not use the reflections option in the Solar Distribution field in your Building object input, you are responsible for entering the View Factor to Ground for each heat-transfer surface. Typical values for a surface that is not shadowed are obtained by the simple equation:

View Factor to Ground = (1-cos(SurfTilt))/2

For example, this gives 0.5 for a wall of tilt 90°. If the tilt of the wall changes, then the View Factor to Ground must also change.

If you enter **autocalculate** in this field, EnergyPlus will automatically calculate the view factor to ground based on the tilt of the surface.

If **you do use the reflections option in the Solar Distribution field** in your Building object input, you do **not** have to enter View Factor to Ground values. In this case the program will automatically calculate the value to use for each exterior surface taking into account solar shadowing (including shadowing of the ground by the building) and reflections from obstructions (ref: Building, Field: Solar Distribution).

However, if you do use the reflections option AND you are modeling a DaylightingDevice:Shelf or DaylightingDevice:Tubular, then you still need to enter some values of View Factor to Ground. For DaylightingDevice:Shelf you need to enter View Factor to Ground for the window associated with the shelf. And for DaylightingDevice:Tubular you need to enter the View Factor to Ground for the FenestrationSurface:Detailed corresponding to the dome of the tubular device.

Note 1: The corresponding view factor to the sky for diffuse solar radiation is not a user input; it is calculated within EnergyPlus based on surface orientation, sky solar radiance distribution, and shadowing surfaces.

Note 2: The view factors to the sky and ground for thermal infrared (long-wave) radiation are not user inputs; they are calculated within EnergyPlus based on surface tilt and shadowing surfaces. Shadowing surfaces are considered to have the same emissivity and temperature as the ground, so they are lumped together with the ground in calculating the ground IR view factor.

1.10.35.1.10 Field: Number of Vertices

This field specifies the number of sides in the surface (number of X,Y,Z vertex groups). For further information, see the discussion on "Surface Vertices" above.

1.10.36 Floor:Detailed

The Floor:Detailed object is used to describe floors.

1.10.36.1 Inputs

1.10.36.1.1 Field: Name

This is a unique name associated with each building surface. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.36.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.36.1.3 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.36.1.4 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space named <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.36.1.5 Field: Outside Boundary Condition

This value can be one of several things depending on the actual kind of surface.

1. **Surface** – if this surface is an internal surface, then this is the choice. The value will either be a surface in the base zone or a surface in another zone. The heat balance between two zones can be accurately simulated by specifying a surface in an adjacent zone. EnergyPlus will simulate a group of zones simultaneously and will include the heat transfer between zones. However, as this increases the complexity of the calculations, it is not necessary to specify the other zone unless the two zones will have a significant temperature difference. If the two zones will not be very different (temperature wise), then the surface should use itself as the outside environment or specify this field as **Adiabatic**. The surface name on the "outside" of this surface (adjacent to) is placed in the next field.

2. **Adiabatic** – an internal surface in the same Zone. This surface will not transfer heat out of the zone, but will still store heat in thermal mass. Only the inside face of the surface will exchange heat with the zone (i.e. two adiabatic surfaces are required to model internal partitions where both sides of the surface are exchanging heat with the zone). The Outside Boundary Condition Object can be left blank.

3. **Zone** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent zone when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

4. **Space** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent space when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

5. **Outdoors** – if this surface is exposed to outside temperature conditions, then this is the choice. See Sun Exposure and Wind Exposure below for further specifications on this kind of surface.

6. **Foundation** - uses an alternative model (currently only the Kiva™ model) to account for the multi-dimensional heat transfer of foundation surfaces. The Outside Boundary Condition Object will refer to the name of a Foundation:Kiva object (or be left blank to use the default foundation without extra insulation).

7. **Ground** - The temperature on the outside of this surface will be the Site:GroundTemperature:Surface value for the month..

8. **GroundFCfactorMethod** – if this surface is exposed to the ground and using the **Construction:CfactorUnd** then this is the choice. The temperature on the outside of this surface will be the Site:GroundTemperature:Fcfacto value for the month.

9. **OtherSideCoefficients** – if this surface has a custom, user specified temperature or other parameters (See SurfaceProperty:OtherSideCoefficients specification), then this is the choice. The outside boundary condition will be the name of the SurfaceProperty:OtherSideCoefficients specification.

10. **OtherSideConditionsModel** – if this surface has a specially-modeled multi-skin component, such as a transpired collector or vented photovoltaic panel, attached to the outside (See SurfaceProperty:OtherSideConditionsModel specification), then this the choice. The outside face environment will be the name of the SurfaceProperty:OtherSideConditionsModel specification.

11. **GroundSlabPreprocessorAverage** – uses the average results from the Slab preprocessor calculations.

12. **GroundSlabPreprocessorCore** – uses the core results from the Slab preprocessor calculations.

13. **GroundSlabPreprocessorPerimeter** – uses the perimeter results from the Slab preprocessor calculations.

14. **GroundBasementPreprocessorAverageWall** – uses the average wall results from the Basement preprocessor calculations.

15. **GroundBasementPreprocessorAverageFloor** – uses the average floor results from the Basement preprocessor calculations.

16. **GroundBasementPreprocessorUpperWall** – uses the upper wall results from the Basement preprocessor calculations.

17. **GroundBasementPreprocessorLowerWall** – uses the lower wall results from the Basement preprocessor calculations.

1.10.36.1.6 Field: Outside Boundary Condition Object

If neither Surface, Zone, Space, Foundation, OtherSideCoefficients, or OtherSideConditionsModel are specified for the Outside Boundary Condition (previous field), then this field should be left blank.

As stated above, if the Outside Boundary Condition is “Surface”, then this field’s value must be the surface name whose inside face temperature will be forced on the outside face of the base surface. This permits heat exchange between adjacent zones (interzone heat transfer) when multiple zones are simulated, but can also be used to simulate middle zone behavior without modeling the adjacent zones. This is done by specifying a surface within the zone. For example, a middle floor zone can be modeled by making the floor the Outside Boundary Condition Object for the ceiling, and the ceiling the Outside Boundary Condition Object for the floor.

If the Outside Boundary Condition is Zone or Space, then this field should contain the zone or space name for the adjacent surface.

> Note: Zones with interzone heat transfer are not adiabatic and the internal surfaces contribute to gains or losses. Adiabatic surfaces are modeled by specifying the base surface itself in this field. Also, for interzone heat transfer, both surfaces must be represented – for example, if you want interzone heat transfer to an attic space, the ceiling in the lower zone must have a surface object with the outside face environment as the floor in the attic and, likewise, there must be a floor surface object in the attic that references the ceiling surface name in the lower zone.

Equally, if the Outside Boundary Condition is “OtherSideCoefficients”, then this field’s value must be the SurfaceProperty:OtherSideCoefficients name. Or if the Outside Boundary Condition is “OtherSideConditionsModel” then this field’s value must be the SurfaceProperty:OtherSideConditionsModel name.

1.10.36.1.7 Field: Sun Exposure

If the surface is exposed to the sun, then “SunExposed” should be entered in this field. Otherwise, “NoSun” should be entered.

Note, a cantilevered floor could have “Outdoors” but “NoSun” exposure.

1.10.36.1.8 Field: Wind Exposure

If the surface is exposed to the Wind, then "WindExposed" should be entered in this field. Otherwise, "NoWind" should be entered.

> Note: When a surface is specified with "NoWind", this has several implications. Within the heat balance code, this surface will default to using the simple ASHRAE exterior convection coefficient correlation with a zero wind speed. In addition, since the ASHRAE simple method does not have a separate value for equivalent long wavelength radiation to the sky and ground, using "NoWind" also eliminates long wavelength radiant exchange from the exterior of the surface to both the sky and the ground. Thus, only simple convection takes place at the exterior face of a surface specified with "NoWind".

1.10.36.1.9 Field: View Factor to Ground

The fraction of the ground plane (assumed horizontal) that is visible from a heat-transfer surface. It is used to calculate the diffuse solar radiation from the ground that is incident on the surface.

For example, if there are no obstructions, a vertical surface sees half of the ground plane and so View Factor to Ground = 0.5. A horizontal downward-facing surface sees the entire ground plane, so View Factor to Ground = 1.0. A horizontal upward-facing surface (horizontal roof) does not see the ground at all, so View Factor to Ground = 0.0.

Unused if reflections option in Solar Distribution field in Building object input unless a DaylightingDevice:Shelf or DaylightingDevice:Tubular has been specified.

If you do not use the reflections option in the Solar Distribution field in your Building object input, you are responsible for entering the View Factor to Ground for each heat-transfer surface. Typical values for a surface that is not shadowed are obtained by the simple equation:

View Factor to Ground = (1-cos(SurfTilt))/2

For example, this gives 0.5 for a wall of tilt 90°. If the tilt of the wall changes, then the View Factor to Ground must also change.

If you enter **autocalculate** in this field, EnergyPlus will automatically calculate the view factor to ground based on the tilt of the surface.

If **you do use the reflections option in the Solar Distribution field** in your Building object input, you do **not** have to enter View Factor to Ground values. In this case the program will automatically calculate the value to use for each exterior surface taking into account solar shadowing (including shadowing of the ground by the building) and reflections from obstructions (ref: Building, Field: Solar Distribution).

However, if you do use the reflections option AND you are modeling a DaylightingDevice:Shelf or DaylightingDevice:Tubular, then you still need to enter some values of View Factor to Ground. For DaylightingDevice:Shelf you need to enter View Factor to Ground for the window associated with the shelf. And for DaylightingDevice:Tubular you need to enter the View Factor to Ground for the FenestrationSurface:Detailed corresponding to the dome of the tubular device.

Note 1: The corresponding view factor to the sky for diffuse solar radiation is not a user input; it is calculated within EnergyPlus based on surface orientation, sky solar radiance distribution, and shadowing surfaces.

Note 2: The view factors to the sky and ground for thermal infrared (long-wave) radiation are not user inputs; they are calculated within EnergyPlus based on surface tilt and shadowing surfaces. Shadowing surfaces are considered to have the same emissivity and temperature as the ground, so they are lumped together with the ground in calculating the ground IR view factor.

1.10.36.1.10 Field: Number of Vertices

This field specifies the number of sides in the surface (number of X,Y,Z vertex groups). For further information, see the discussion on "Surface Vertices" above.

Some examples of using these objects:

```
Floor:Detailed,
      Floor_NorthZone_1stFloor,!- Name
      FLOOR-SLAB-ASSEMBLY,     !- Construction Name
      NorthZone_1stFloor,      !- Zone Name
      Ground,                  !- Outside Boundary Condition
      ,                        !- Outside Boundary Condition Object
      NoSun,                   !- Sun Exposure
      NoWind,                  !- Wind Exposure
      0.0,                     !- View Factor to Ground
      4,                       !- Number of Vertices
      0, 11, 0,                          !- X,Y,Z  1 {m}
      25, 11, 0,                         !- X,Y,Z  2 {m}
      25, 5.5, 0,                        !- X,Y,Z  3 {m}
     0, 5.5, 0;                         !- X,Y,Z  4 {m}

  RoofCeiling:Detailed,
      Ceiling_SouthZone_1stFloor,  !- Name
      CEILING-FLOOR-ASSEMBLY,  !- Construction Name
      SouthZone_1stFloor,      !- Zone Name
      Surface,                 !- Outside Boundary Condition
      Floor_SouthZone_2ndFloor,!- Outside Boundary Condition Object
      NoSun,                   !- Sun Exposure
      NoWind,                  !- Wind Exposure
      0.0,                     !- View Factor to Ground
      4,                       !- Number of Vertices
      0, 0, 3.4,                         !- X,Y,Z  1 {m}
      25, 0, 3.4,                        !- X,Y,Z  2 {m}
      25, 5.5, 3.4,                      !- X,Y,Z  3 {m}
     0, 5.5, 3.4;                       !- X,Y,Z  4 {m}

  Wall:Detailed,
      InteriorWall_SouthZone_1stFloor,  !- Name
      INTERIOR-WALL-ASSEMBLY,  !- Construction Name
      SouthZone_1stFloor,      !- Zone Name
      Surface,                 !- Outside Boundary Condition
      InteriorWall_NorthZone_1stFloor,  !- Outside Boundary Condition Object
      NoSun,                   !- Sun Exposure
      NoWind,                  !- Wind Exposure
      0,                       !- View Factor to Ground
      4,                       !- Number of Vertices
      25, 5.5, 3.7,                      !- X,Y,Z  1 {m}
      25, 5.5, 0,                        !- X,Y,Z  2 {m}
      0, 5.5, 0,                         !- X,Y,Z  3 {m}
     0, 5.5, 3.7;                       !- X,Y,Z  4 {m}
```

1.10.37 BuildingSurface:Detailed

The BuildingSurface:Detailed object can more generally describe each of the surfaces.

1.10.37.1 Inputs

1.10.37.1.1 Field: Name

This is a unique name associated with each building surface. It is used in several other places as a reference (e.g. as the base surface name for a Window or Door).

1.10.37.1.2 Field: Surface Type

Used primarily for convenience, the surface type can be one of the choices illustrated above – Wall, Floor, Ceiling, Roof. Azimuth (facing) and Tilt are determined from the vertex coordinates. Note that "normal" floors will be tilted 180° whereas flat roofs/ceilings will be tilted 0°. EnergyPlus uses this field's designation, along with the calculated tilt of the surface, to issue warning messages when tilts are "out of range". Calculations in EnergyPlus use the actual calculated tilt values for the actual heat balance calculations. Note, however, that a floor tilted 0° is really facing "into" the zone and is not what you will desire for the calculations even though the coordinate may appear correct in the viewed DXF display.

"Normal" tilt for walls is 90° – here you may use the calculated Azimuth to make sure your walls are facing away from the zone's interior.

1.10.37.1.3 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface. Regardless of location in the building, the "full" construction (all layers) is used. For example, for an interior wall separating two zones, zone x would have the outside layer (e.g. drywall) as the material that shows in zone y and then the layers to the inside layer – the material that shows in zone x. For symmetric constructions, the same construction can be used in the surfaces described in both zones.

1.10.37.1.4 Field: Zone Name

This is the zone name to which the surface belongs. This field is required.

1.10.37.1.5 Field: Space Name

This is the space name to which the surface belongs. If this field is left blank, it will be automatically assigned to a space name <Zone Name> or <Zone Name>-Remainder. See Space for more details.

1.10.37.1.6 Field: Outside Boundary Condition

This value can be one of several things depending on the actual kind of surface.

1. **Surface** – if this surface is an internal surface, then this is the choice. The value will either be a surface in the base zone or a surface in another zone. The heat balance between two zones can be accurately simulated by specifying a surface in an adjacent zone. EnergyPlus will simulate a group of zones simultaneously and will include the heat transfer between zones. However, as this increases the complexity of the calculations, it is not necessary to specify the other zone unless the two zones will have a significant temperature difference. If the two zones will not be very different (temperature wise), then the surface should use itself as the outside environment or specify this field as **Adiabatic**. The surface name on the "outside" of this surface (adjacent to) is placed in the next field.

2. **Adiabatic** – an internal surface in the same Zone. This surface will not transfer heat out of the zone, but will still store heat in thermal mass. Only the inside face of the surface will exchange heat with the zone (i.e. two adiabatic surfaces are required to model internal partitions where both sides of the surface are exchanging heat with the zone). The Outside Boundary Condition Object can be left blank.

3. **Zone** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent zone when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

4. **Space** – this is similar to Surface but EnergyPlus will automatically create the required surface in the adjacent space when this is entered for the surface. If there are windows or doors on the surface, EnergyPlus automatically creates appropriate sub-surfaces as well.

5. **Outdoors** – if this surface is exposed to outside temperature conditions, then this is the choice. See Sun Exposure and Wind Exposure below for further specifications on this kind of surface.

6. **Foundation** - uses an alternative model (currently only the Kiva^{TM} model) to account for the multi-dimensional heat transfer of foundation surfaces. The Outside Boundary Condition Object will refer to the name of a Foundation:Kiva object (or be left blank to use the default foundation without extra insulation).

7. **Ground** - The temperature on the outside of this surface will be the Site:GroundTemperature:Surface value for the month.

8. **GroundFCfactorMethod** – if this surface is exposed to the ground and using the **Construction:CfactorUnd** then this is the choice. The temperature on the outside of this surface will be the Site:GroundTemperature:Fcfacto value for the month.

9. **OtherSideCoefficients** – if this surface has a custom, user specified temperature or other parameters (See SurfaceProperty:OtherSideCoefficients specification), then this is the choice. The outside boundary condition will be the name of the SurfaceProperty:OtherSideCoefficients specification.

10. **OtherSideConditionsModel** – if this surface has a specially-modeled multi-skin component, such as a transpired collector or vented photovoltaic panel, attached to the outside (See SurfaceProperty:OtherSideConditionsModel specification), then this the choice. The outside face environment will be the name of the SurfaceProperty:OtherSideConditionsModel specification.

11. **GroundSlabPreprocessorAverage** – uses the average results from the Slab preprocessor calculations.

12. **GroundSlabPreprocessorCore** – uses the core results from the Slab preprocessor calculations.

13. **GroundSlabPreprocessorPerimeter** – uses the perimeter results from the Slab preprocessor calculations.

14. **GroundBasementPreprocessorAverageWall** – uses the average wall results from the Basement preprocessor calculations.

15. **GroundBasementPreprocessorAverageFloor** – uses the average floor results from the Basement preprocessor calculations.

16. **GroundBasementPreprocessorUpperWall** – uses the upper wall results from the Basement preprocessor calculations.

17. **GroundBasementPreprocessorLowerWall** – uses the lower wall results from the Basement preprocessor calculations.

1.10.37.1.7 Field: Outside Boundary Condition Object

If neither Surface, Zone, Space, Foundation, OtherSideCoefficients, or OtherSideConditionsModel are specified for the Outside Boundary Condition (previous field), then this field should be left blank.

As stated above, if the Outside Boundary Condition is "Surface", then this field's value must be the surface name whose inside face temperature will be forced on the outside face of the base surface. This permits heat exchange between adjacent zones (interzone heat transfer) when multiple zones are simulated, but can also be used to simulate middle zone behavior without modeling the adjacent zones. This is done by specifying a surface within the zone. For example, a middle floor zone can be modeled by making the floor the Outside Boundary Condition Object for the ceiling, and the ceiling the Outside Boundary Condition Object for the floor.

If the Outside Boundary Condition is Zone or Space, then this field should contain the zone or space name for the adjacent surface.

> Note: Zones with interzone heat transfer are not adiabatic and the internal surfaces contribute to gains or losses. Adiabatic surfaces are modeled by specifying the base surface itself in this field. Also, for interzone heat transfer, both surfaces must be represented – for example, if you want interzone heat transfer to an attic space, the ceiling in the lower zone must have a surface object with the outside face environment as the floor in the attic and, likewise, there must be a floor surface object in the attic that references the ceiling surface name in the lower zone.

Equally, if the Outside Boundary Condition is "OtherSideCoefficients", then this field's value must be the SurfaceProperty:OtherSideCoefficients name. Or if the Outside Boundary Condition is "OtherSideConditionsModel" then this field's value must be the SurfaceProperty:OtherSideConditionsModel name.

1.10.37.1.8 Field: Sun Exposure

If the surface is exposed to the sun, then "SunExposed" should be entered in this field. Otherwise, "NoSun" should be entered.

Note, a cantilevered floor could have "Outdoors" but "NoSun" exposure.

1.10.37.1.9 Field: Wind Exposure

If the surface is exposed to the Wind, then "WindExposed" should be entered in this field. Otherwise, "NoWind" should be entered.

> Note: When a surface is specified with "NoWind", this has several implications. Within the heat balance code, this surface will default to using the simple ASHRAE exterior convection coefficient correlation with a zero wind speed. In addition, since the ASHRAE simple method does not have a separate value for equivalent long wavelength radiation to the sky and ground, using "NoWind" also eliminates long wavelength radiant exchange from the exterior of the surface to both the sky and the ground. Thus, only simple convection takes place at the exterior face of a surface specified with "NoWind".

1.10.37.1.10 Field: View Factor to Ground

The fraction of the ground plane (assumed horizontal) that is visible from a heat-transfer surface. It is used to calculate the diffuse solar radiation from the ground that is incident on the surface.

For example, if there are no obstructions, a vertical surface sees half of the ground plane and so View Factor to Ground = 0.5. A horizontal downward-facing surface sees the entire ground plane, so View Factor to Ground = 1.0. A horizontal upward-facing surface (horizontal roof) does not see the ground at all, so View Factor to Ground = 0.0.

Unused if reflections option in Solar Distribution field in Building object input unless a DaylightingDevice:Shelf or DaylightingDevice:Tubular has been specified.

If you do not use the reflections option in the Solar Distribution field in your Building object input, you are responsible for entering the View Factor to Ground for each heat-transfer surface. Typical values for a surface that is not shadowed are obtained by the simple equation:

View Factor to Ground = (1-cos(SurfTilt))/2

For example, this gives 0.5 for a wall of tilt 90°. If the tilt of the wall changes, then the View Factor to Ground must also change.

If you enter **autocalculate** in this field, EnergyPlus will automatically calculate the view factor to ground based on the tilt of the surface.

If **you do use the reflections option in the Solar Distribution field** in your Building object input, you do **not** have to enter View Factor to Ground values. In this case the program will automatically calculate the value to use for each exterior surface taking into account solar shadowing (including shadowing of the ground by the building) and reflections from obstructions (ref: Building, Field: Solar Distribution).

However, if you do use the reflections option AND you are modeling a DaylightingDevice:Shelf or DaylightingDevice:Tubular, then you still need to enter some values of View Factor to Ground. For DaylightingDevice:Shelf you need to enter View Factor to Ground for the window associated with the shelf. And for DaylightingDevice:Tubular you need to enter the View Factor to Ground for the FenestrationSurface:Detailed corresponding to the dome of the tubular device.

Note 1: The corresponding view factor to the sky for diffuse solar radiation is not a user input; it is calculated within EnergyPlus based on surface orientation, sky solar radiance distribution, and shadowing surfaces.

Note 2: The view factors to the sky and ground for thermal infrared (long-wave) radiation are not user inputs; they are calculated within EnergyPlus based on surface tilt and shadowing surfaces. Shadowing surfaces are considered to have the same emissivity and temperature as the ground, so they are lumped together with the ground in calculating the ground IR view factor.

1.10.37.1.11 Field: Number of Vertices

This field specifies the number of sides in the surface (number of X,Y,Z vertex groups). For further information, see the discussion on "Surface Vertices" above.

IDF example of three walls (first is an exterior wall, second and third are interzone partitions):

```
BuildingSurface:Detailed,Zn001:Wall001,  !- Base Surface Name
    Wall,EXTERIOR,  !- Class and Construction Name
    ZONE 1 @ 200 601 0 T,  !- Zone
    Outdoors,,  !- Outside Boundary Condition  and Target (if applicable)
     SunExposed,  !- Solar Exposure
     WindExposed,  !- Wind Exposure
    0.5000000    ,  !- VF to Ground
             4, !-Rectangle
    0.0000000E+00,  0.0000000E+00,   3.048000    ,
    0.0000000E+00,  0.0000000E+00,  0.0000000E+00,
     6.096000    ,  0.0000000E+00,  0.0000000E+00,
     6.096000    ,  0.0000000E+00,   3.048000    ;

  BuildingSurface:Detailed,Zn001:Wall006,  !-Base Surface Name
    Wall,INTERIOR,  !- Class and Construction Name
    HEARTLAND AREA,  !- Zone
    Surface,Zn004:Wall005,  !- Outside Boundary Conditions and Target (if applicable)
     NoSun,  !- Solar Exposure
     NoWind,  !- Wind Exposure
    0.5000000    ,  !- VF to Ground
             4, !-Rectangle
     38.01000    ,   28.25000    ,   10.00000    ,
     38.01000    ,   28.25000    ,  0.0000000E+00,
     38.01000    ,   18.25000    ,  0.0000000E+00,
     38.01000    ,   18.25000    ,   10.00000    ;

  BuildingSurface:Detailed,Zn001:Wall007,  !- Base Surface Name
    Wall,INTERIOR,  !- Class and Construction Name
    HEARTLAND AREA,  !- Zone
    Surface,Zn003:Wall006,  !- Outside Boundary Conditions and Target (if applicable)
     NoSun,  !- Solar Exposure
     NoWind,  !- Wind Exposure
    0.5000000    ,  !- VF to Ground
             4, !-Rectangle
     58.01000    ,   18.25000    ,   10.00000    ,
     58.01000    ,   18.25000    ,  0.0000000E+00,
     58.01000    ,   28.25000    ,  0.0000000E+00,
     58.01000    ,   28.25000    ,   10.00000    ;
```

1.10.38 FenestrationSurface:Detailed

This surface class is used for subsurfaces, which can be of five different types: Windows, Doors, GlassDoors, TubularDaylightDomes, and TubularDaylightDiffusers. A subsurface (such as a window) of a base surface (such as a wall) inherits several of the properties (such as Outside Boundary Condition, Sun Exposure, etc.) of the base surface. Windows, GlassDoors, TubularDaylightDomes, and TubularDaylightDiffusers are considered to have one or more glass layers and so transmit solar radiation. Doors are considered to be opaque. For Surface Type = Window and GlassDoor, shades and screens may be applied by referencing this subsurface in a window shading control (ref: WindowShadingControl object). To assign a shade to a window or glass door, see WindowMaterial: Shade. To assign a screen, see WindowMaterial:Screen. To assign a blind, see WindowMaterial:Blind. To assign switchable glazing, such as electrochromic glazing, see WindowShadingControl.

The surface can be of 3 sides or 4 sides. A 4-sided but non-rectangular fenestration surface is allowed and will be transferred to an equivalent rectangular surface with the same area and aspect ratio for the convection calculations at window air gaps.

1.10.38.1 Inputs

1.10.38.1.1 Field: Name

This is a unique name associated with the heat transfer subsurface. It may be used in other places as a reference (e.g. as the opposing subsurface of an interzone window or door).

1.10.38.1.2 Field: Surface Type

The choices for Surface Type are Window, Door, GlassDoor, TubularDaylightDome, and TubularDaylightDiffuser. Doors are assumed to be opaque (do not transmit solar radiation) whereas the other surface types do transmit solar radiation. Windows and Glass Doors are treated identically in the calculation of conduction heat transfer, solar gain, daylighting, etc. A Window or GlassDoor, but not a Door, can have a movable interior, exterior or between-glass shading device, such as blinds (ref: WindowMaterial:Blind object), and can have a frame and/or a divider (ref: WindowProperty:FrameAndDivider object). TubularDaylightDomes and TubularDaylightDomes are specialized subsurfaces for use with the DaylightingDevice:Tubular object to simulate Tubular Daylighting Devices (TDDs). TubularDaylightDomes and TubularDaylightDomes cannot have shades, screens or blinds. In the following, the term “window” applies to Window, GlassDoor, TubularDaylightDome, and TubularDaylightDome, if not otherwise explicitly mentioned.

As noted in the description of the BuildingSurface:Detailed, Azimuth (facing angle) and Tilt are calculated from the entered vertices. Tilts of subsurfaces will normally be the same as their base surface. If these are significantly beyond the “normals” for the base surface, warning messages may be issued. If the facing angles are not correct, you may have a window pointing “into” the zone rather than out of it – this would cause problems in the calculations. Note, too, that a “reveal” (inset or outset) may occur if the plane of the subsurface is not coincident with the base surface; the reveal has an effect on shading of the subsurface.

1.10.38.1.3 Field: Construction Name

This is the name of the subsurface’s construction (ref: Construction object [for Door] and Construction, Construction:ComplexFenestrationState, Construction:WindowDataFile objects [for Window and GlassDoor]).

For windows, if Construction Name is not found among the constructions on the input (.idf) file, the Window Data File (ref. Construction:WindowDataFile object) will be searched for that Construction Name (see “Importing Windows from WINDOW”). If that file is not present or if the Construction Name does not match the name of an entry on the file, an error will result. If there is a match, a window construction and its corresponding glass and gas materials will be created from the information read from the file.

1.10.38.1.4 Field: Building Surface Name

This is the name of a surface that contains this subsurface. Certain kinds of surfaces may not be allowed to have subsurfaces. For example, a surface in contact with the ground (Outside Boundary Condition = Ground) cannot contain a window.

1.10.38.1.5 Field: Outside Boundary Condition Object

If the base surface has Outside Boundary Condition = Surface or OtherSideCoefficients, then this field must also be specified for the subsurface. Otherwise, it can be left blank.

If the base surface has Outside Boundary Condition = Zone or Space, then this surface retains that characteristic and uses the same zone or space of the base surface. It can be entered here for clarity or it can be left blank.

If Outside Boundary Condition for the base surface is Surface, this field should specify the subsurface in the opposing zone that is the counterpart to this subsurface. The constructions of the subsurface and opposing subsurface must match, except that, for multi-layer constructions, the layer order of the opposing subsurface's construction must be the reverse of that of the subsurface.

If Outside Boundary Condition for the base surface is OtherSideCoefficients, this field could specify the set of SurfaceProperty:OtherSideCoefficients for this subsurface. If this is left blank, then the Other Side Coefficients of the base surface will be used for this subsurface. *Windows and GlassDoors are not allowed to have Other Side Coefficients.*

1.10.38.1.6 Field: View Factor to Ground

The fraction of the ground plane (assumed horizontal) that is visible from a heat-transfer surface. It is used to calculate the diffuse solar radiation from the ground that is incident on the surface.

For example, if there are no obstructions, a vertical surface sees half of the ground plane and so View Factor to Ground = 0.5. A horizontal downward-facing surface sees the entire ground plane, so View Factor to Ground = 1.0. A horizontal upward-facing surface (horizontal roof) does not see the ground at all, so View Factor to Ground = 0.0.

Unused if reflections option in Solar Distribution field in Building object input unless a Daylighting Device:Shelf or Daylighting Device:Tubular has been specified.

If you do not use the reflections option in the Solar Distribution field in your Building object input, you are responsible for entering the View Factor to Ground for each heat-transfer surface. Typical values for a surface that is not shadowed are obtained by the simple equation:

View Factor to Ground = (1-cos(SurfTilt))/2

For example, this gives 0.5 for a wall of tilt 90°. If the tilt of the wall changes, then the View Factor to Ground must also change.

If you enter **autocalculate** in this field, EnergyPlus will automatically calculate the view factor to ground based on the tilt of the surface.

If **you do use the reflections option in the Solar Distribution field** in your BUILDING object input, you do **not** have to enter View Factor to Ground values. In this case the program will automatically calculate the value to use for each exterior surface taking into account solar shadowing (including shadowing of the ground by the building) and reflections from obstructions (ref: Building, Field: Solar Distribution).

However, if you do use the reflections option AND you are modeling a DaylightingDevice:Shelf or DaylightingDevice:Tubular, then you should enter some values of View Factor to Ground. For DaylightingDevice:Shelf you need to enter View Factor to Ground for the window associated with the shelf. And for DaylightingDevice:Tubular you need to enter the View Factor to Ground for the FenestrationSurface:Detailed corresponding to the dome of the tubular device (ref: DaylightingDevice:Tubular).

Note 1: The corresponding view factor to the sky for diffuse solar radiation is not a user input; it is calculated within EnergyPlus based on surface orientation, sky solar radiance distribution, and shadowing surfaces.

Note 2: The view factors to the sky and ground for thermal infrared (long-wave) radiation are not user inputs; they are calculated within EnergyPlus based on surface tilt and shadowing surfaces. Shadowing surfaces are considered to have the same emissivity and temperature as the ground, so they are lumped together with the ground in calculating the ground infrared view factor.

Note 3: If the view factor to the ground is autosized and a DaylightingDevice:Shelf is present, EnergyPlus will adjust both the view factor to the ground first and then the sky so that the sum of all three view factors (to the ground, to the sky, and to the shelf) does not exceed 1.0.

1.10.38.1.7 Field: Frame and Divider Name

This field, if not blank, can be used to specify window frame, divider and reveal-surface data (ref: WindowProperty:FrameAndDivider object). It is used only for exterior GlassDoors and rectangular exterior Windows, i.e., those with OutsideFaceEnvironment = Outdoors.

This field should be blank for triangular windows.

1.10.38.1.8 Field: Multiplier

Used only for Surface Type = Window, Door or Glass Door. It is the number of identical items on the base surface. Using Multiplier can save input effort and calculation time. In the calculation the area (and area of frame and divider, if present and surface type is a window) is multiplied by Multiplier. The calculation of shadowing on the subsurfaces (and the calculation of the interior distribution of beam solar radiation transmitted by windows and glass doors) are done for the specified subsurface position and dimensions.

Multiplier should be used with caution. Multiplier > 1 can give inaccurate or nonsensical results in situations where the results are sensitive to window or glass door position. This includes shadowing on the window/glass door, daylighting from the window/glass door, and interior distribution of solar radiation from the window/glass door. In these cases, the results for the single input window/glass door, after multiplication, may not be representative of the results you would get if you entered each of the multiple subsurfaces separately.

If Multiplier > 1, you will get

–a *warning* if Solar Distribution = FullExterior or FullInteriorAndExterior (ref: Building - Field: Solar Distribution), indicating that the shadowing on the input window or the interior solar radiation distribution from the input window may not be representative of the actual group of windows. No warning is issued if Solar Distribution = MinimalShadowing.

–an *error* if the window is an exterior window/glass door in a zone that has a detailed daylighting calculation (Daylighting:Detailed specified for the zone). Since a single window with a multiplier can never give the same daylight illuminance as the actual set of windows, you are not allowed to use Multiplier in this situation.

1.10.38.1.9 Field: Number of Vertices

The number of sides the surface has (number of X,Y,Z vertex groups). For further information, see the discussion on "Surface Vertices" above. Door and GlassDoor subsurfaces are rectangular and therefore have four vertices. Window subsurfaces can be rectangular or triangular and therefore have four or three vertices, respectively.

1.10.38.1.10 Fields: Vertex Coordinates

This is a total of twelve fields giving the x,y,z coordinate values of the four vertices of rectangular subsurfaces [m], or a total of nine fields giving the x,y,z coordinate values of the three vertices of triangular windows.

For triangular windows the first vertex listed can be any of the three vertices, but the order of the vertices should be counter-clockwise if VertexEntry is CounterClockWise and clockwise if VertexEntry is ClockWise (ref: GlobalGeometryRules).

An IDF example of a rectangular subsurface (Window):

```
FenestrationSurface:Detailed,
   Zn001:Wall001:Win001,  !- SubSurface Name
    Window,SINGLE PANE HW WINDOW,  !- Class and Construction Name
    Zn001:Wall001,,         !- Base Surface Name and Target (if applicable)
    0.5000000    ,          !- VF to Ground
    WINDOW-CONTROL-DRAPES,  !- Window Shading Control
    TestFrameAndDivider,    !- Frame/Divider name
    5,                      !- Multiplier
    4,                      !- Rectangle (number of sides)
     1.524000    ,  0.1520000    ,   2.743000    ,
     1.524000    ,  0.1520000    ,  0.3050000    ,
     4.572000    ,  0.1520000    ,  0.3050000    ,
     4.572000    ,  0.1520000    ,   2.743000    ;
```

1.10.39 Window Modeling Options

The following table shows what input objects/fields to use to model different window options. It also gives the name of an example input, if available, that demonstrates the option.

Table 1.17: Window Modeling Options

Option	Object/Field or Output Variable	Input File (distributed with install)
Build up a window from **layers**	WindowMaterial:Glazing, WindowMaterial:-Gas, WindowMaterial:Shade, WindowMaterial:Screen, WindowMaterial:Blind, Construction	WindowTests.idf
Add an **overhang**	Shading:Zone:Detailed	5ZoneAirCooled.idf
Add a **shading device**	WindowMaterial:Shade, WindowMaterial:Screen or WindowMaterial:Blind; WindowShadingControl	WindowTests.idf, PurchAir-WindowBlind.idf
Control a shading device	Window:ShadingControl	PurchAirWindowBlind.idf
Determine when a shading device is on in a particular timestep	Print the variable "Surface Shading Device Is On Time Fraction"	PurchAirWindowBlind.idf
Control the **slat angle** of a blind	WindowShadingControl	PurchAirWindowBlind.idf
Add a **frame**	WindowProperty:-FrameAndDivider	PurchAirWithDaylighting.idf

Table 1.17: Window Modeling Options

Option	Object/Field or Output Variable	Input File (distributed with install)
Add a **divider**	WindowProperty:FrameAndDivider	PurchAirWithDaylighting.idf
Allow window to **daylight** a zone	Daylighting:Controls	PurchAirWithDaylighting.idf, DElight-Detailed-Comparison.idf
Find **solar reflected** onto window from neighboring buildings	Building/SolarDistribution field – uses “WithReflections”	ReflectiveAdjacent-Building.idf
Model **switchable glazing** (e.g., electrochromic glass)	WindowShadingControl	PurchAirWithDaylighting.idf
Add an **interior window**	Define two Fenestration-Surface:Detailed’s, one for each associated interior wall	PurchAirWithDoubleFacade-Daylighting.idf
Track how beam solar falls on interior surfaces	Building/Solar Distribution = FullInteriorAndExterior	PurchAirWithDoubleFacade-Daylighting.idf
Track beam solar transmitted through interior windows	Building/Solar Distribution = FullInteriorAndExterior	PurchAirWithDoubleFacade-Daylighting.idf
Add a **shading device on an interior window**	Not allowed	
Model an **airflow window** (aka, heat extract window)	WindowProperty:Airflow-Control	AirflowWindowsAndBetween-GlassBlinds.idf
Add a **storm window** glass layer	WindowProperty:Storm-Window	StormWindow.idf
Add **natural ventilation** through an open window	Ventilation or Airflow-Network objects (Airflow-Network:Multizone:Surface, AirflowNetwork:MultiZone:-Component:DetailedOpening, etc.)	AirflowNetwork3zvent.idf
Add **diffusing glass**	WindowMaterial:Glazing/-Solar Diffusing = Yes	
Add **dirt on window**	WindowMaterial:Glazing/-Dirt Correction Factor for Solar and Visible Transmittance	
Import window (and frame/divider if present) from **WINDOW** program	See “Importing Windows from WINDOW program”	
Find **daylighting through interior windows**	See “Double Facades: Daylighting through Interior Windows”	PurchAirWithDoubleFacade-Daylighting.idf

Table 1.17: Window Modeling Options

Option	Object/Field or Output Variable	Input File (distributed with install)
Determine when **condensation** occurs	Print the variables "Surface Window Inside Face Glazing Condensation Status," "Surface Window Inside Face Frame Condensation Status," "Surface Window Inside Face Divider Condensation Status"	

1.10.40 InternalMass

Any surface that would logically be described as an interior wall, floor or ceiling can just as easily be described as Internal Mass. Internal Mass surface types only exchange energy with the zone in which they are described; they do not see any other zones. There are two approaches to using internal mass. The first approach is to have several pieces of internal mass with each piece having a different construction type. The other approach is to choose an average construction type and combine all of the interior surfaces into a single internal mass. Similar to internal surfaces with an adiabatic boundary condition, the zone will only exchange energy with the inside of the Internal Mass construction. If both sides of the surface exchange energy with the zone then the user should input twice the area when defining the Internal Mass object. Note that furniture and other large objects within a zone can be described using internal mass. However, simplifying calculations using internal mass must be used with caution when the "FullInteriorAndExterior" or "FullInteriorAndExteriorWithReflections" Solar Distribution model (see Building parameters) is chosen. A single Internal Mass object can be applied to a set of zones or spaces by specifying a ZoneList or SpaceList name.

1.10.40.1 Inputs

Example

When zoning an office building, five west-facing offices have been combined into one zone. All of the offices have interior walls made of the same materials. As shown in the figure below, this zone may be described with 5 exterior walls and 11 internal walls or 1 exterior wall and 1 internal mass. Note that fewer surfaces will speed up the EnergyPlus calculations.

Example

A five-story building has the same ceiling/floor construction separating each of the levels. Zones that are on floors 2 through 4 may be described using a single piece of internal mass to represent both the floor and ceiling. The construction for this internal mass would be identical to the ceiling/floor construction that would be used to describe separate surfaces and the area of the internal mass surface would be the total surface area of the combined ceilings/floors (i.e. twice the total floor area).

1.10.40.1.1 Field: Name

This is a unique character string associated with the internal mass surface. Though it must be unique from other surface names, it is used primarily for convenience with internal mass surfaces.

1.10.40.1.2 Field: Construction Name

This is the name of the construction (ref: Construction object) used in the surface.

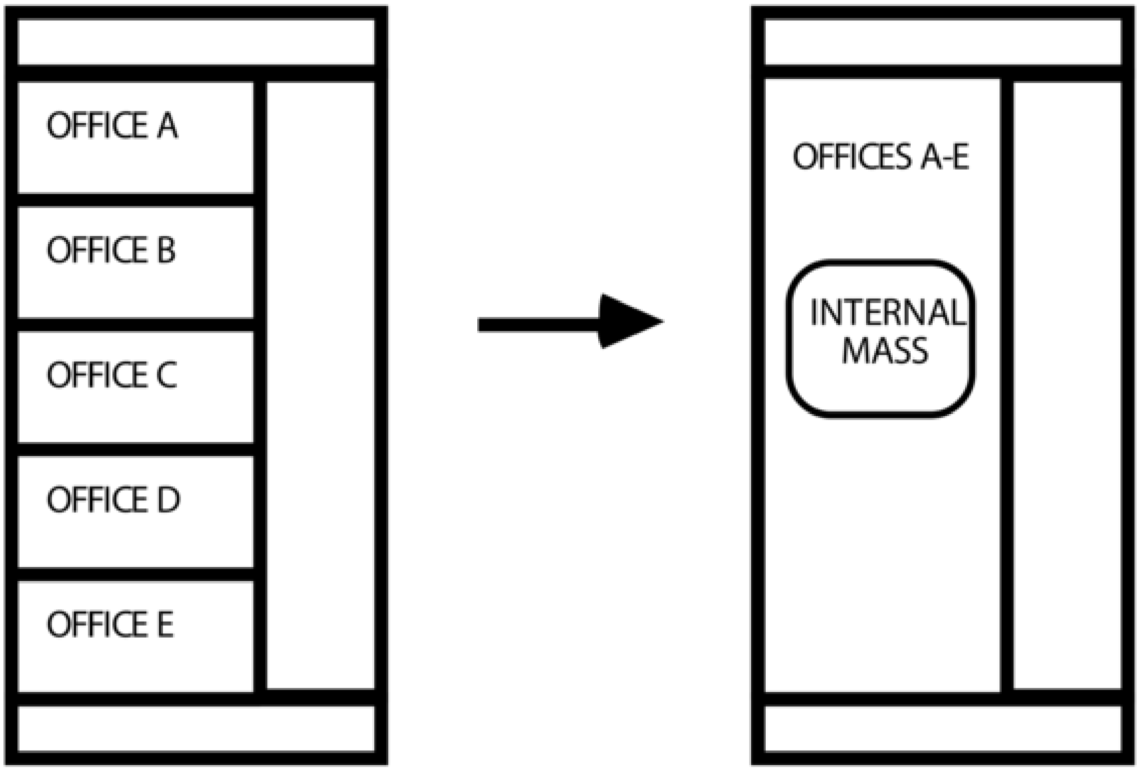

Figure 1.32: Representing 11 internal walls as internal mass

1.10.40.1.3 Field: Zone or ZoneList Name

This field is the name of the Zone or ZoneList in which the internal mass will be added. When the ZoneList option is used then this internal mass object is applied to each zone in the list. The name of the actual internal mass object in each zone becomes <Zone Name> <InternalMass Object Name> and should be less than the standard length (100 characters) for a name field. If it is greater than this standard length, it may be difficult to specify in output reporting as it will be truncated. A warning will be shown if the generated name is greater than 100 characters. If it duplicates another such concatenated name, there will be a severe error and terminate the run. This field is ignored when a Space or SpaceList Name is specified.

1.10.40.1.4 Field: Space or SpaceList Name

This field is the name of the Space or SpaceList in which the internal mass will be added. If this field is blank (or a ZoneList Name is specified), the internal mass will be added to the last space attached to the zone. When the SpaceList option is used then this internal mass object is applied to each space in the list. The name of the actual internal mass object in each space becomes <Space Name> <InternalMass Object Name> and should be less than the standard length (100 characters) for a name field. If it is greater than this standard length, it may be difficult to specify in output reporting as it will be truncated. A warning will be shown if the generated name is greater than 100 characters. If it duplicates another such concatenated name, there will be a severe error and terminate the run. This field is ignored when a ZoneList Name is specified for Zone or ZoneList Name. If this field is entered, then the Zone or ZoneList Name is ignored, and the surface is assigned to the corresponding Zone(s) for the Space(s).

1.10.40.1.5 Field: Surface Area

This field is the surface area of the internal mass. The area that is specified must be the entire surface area that is exposed to the zone. If both sides of a wall are completely within the same zone, then the area of both sides must be included when describing that internal wall.

IDF examples of Internal Mass surfaces:

```
InternalMass,
  Zn002:IntM001,                !- Surface Name
  INTERIOR,                     !- Construction Name
  DORM ROOMS AND COMMON AREAS,  !- Zone or ZoneList Name
  ,                             !- Space or SpaceList Name
  408.7734;                     !- Surface Area {m2}

InternalMass,
  Zn002:IntM002,                !- Surface Name
  PARTITION02,                  !- Construction Name
  DORM ROOMS AND COMMON AREAS,  !- Zone or ZoneList Name
  ,                             !- Space or SpaceList Name
  371.6122;                     !- Surface Area {m2}
```

1.10.41 Surface Output Variables/Reports

Note that Surface Outputs from specialized algorithms (such as Effective Moisture Penetration Depth (EMPD), Combined Heat and Moisture Transport (HAMT) and Conduction Finite Difference (CondFD) are discussed under the objects that describe the specialized inputs for these algorithms). You can access them via these links:

- Moisture Penetration Depth (EMPD) Outputs
- Conduction Finite Difference (CondFD) Outputs
- Heat and Moisture (HAMT) Outputs

Additionally, the output variables applicable to all heat transfer surfaces:

```
Zone,Sum,Surface Inside Face Heat Balance Calculation Iteration Count []
Zone,Average,Surface Inside Face Temperature [C]
Zone,Average,Surface Inside Face Interior Movable Insulation Temperature [C]
Zone,Average,Surface Outside Face Temperature [C]
Zone,Average,Surface Inside Face Adjacent Air Temperature [C]
Zone,Average,Surface Inside Face Convection Heat Transfer Coefficient [W/m2-K]
Zone,Average,Surface Inside Face Convection Heat Gain Rate [W]
Zone,Average,Surface Inside Face Convection Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Convection Heat Gain Energy [J]
Zone,Average,Surface Inside Face Net Surface Thermal Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face Net Surface Thermal Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Net Surface Thermal Radiation Heat Gain Energy [J]
Zone,Average,Surface Inside Face Solar Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face Solar Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Solar Radiation Heat Gain Energy [J]
Zone,Average,Surface Inside Face Lights Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face Lights Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Lights Radiation Heat Gain Energy [J]
Zone,Average,Surface Inside Face Internal Gains Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face Internal Gains Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Internal Gains Radiation Heat Gain Energy [J]
Zone,Average,Surface Inside Face System Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face System Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face System Radiation Heat Gain Energy [J]
Zone,Average,Surface Outside Face Convection Heat Transfer Coefficient [W/m2-K]
Zone,Average,Surface Outside Face Convection Heat Gain Rate [W]
Zone,Average,Surface Outside Face Convection Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Outside Face Convection Heat Gain Energy [J]
Zone,Average,Surface Outside Face Net Thermal Radiation Heat Gain Rate [W]
Zone,Average,Surface Outside Face Net Thermal Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Outside Face Net Thermal Radiation Heat Gain Energy [J]
```

```
Zone,Average,Surface Outside Face Thermal Radiation to Air Heat Transfer Coefficient [W/m2-K]
Zone,Average,Surface Outside Face Thermal Radiation to Sky Heat Transfer Coefficient [W/m2-K]
Zone,Average,Surface Outside Face Thermal Radiation to Ground Heat Transfer Coefficient [W/m2-K]
Zone,Average,Surface Outside Face Thermal Radiation to Surrounding Surfaces Heat Transfer Coefficient [W/
    m2-K]
Zone,Average,Surface Outside Face Surrounding Surfaces Average Temperature [C]
Zone,Average,Surface Inside Face Exterior Windows Incident Beam Solar Radiation Rate [W]
Zone,Sum,Surface Inside Face Exterior Windows Incident Beam Solar Radiation Energy [J]
Zone,Average,Surface Inside Face Exterior Windows Incident Beam Solar Radiation Rate per Area[W/m2]
Zone,Average,Surface Inside Face Interior Windows Incident Beam Solar Radiation Rate [W]
Zone,Average,Surface Inside Face Interior Windows Incident Beam Solar Radiation Rate per Area[W/m2]
Zone,Sum,Surface Inside Face Interior Windows Incident Beam Solar Radiation Energy [J]
Zone,Average,Surface Inside Face Initial Transmitted Diffuse Absorbed Solar Radiation Rate [W]
Zone,Average,Surface Inside Face Initial Transmitted Diffuse Transmitted Out Window Solar Radiation Rate
    [W]
Zone,Average,Surface Inside Face Absorbed Shortwave Radiation Rate [W]
Zone,Average,Surface Construction Index []
```

Output variables applicable to all exterior heat transfer surfaces:

```
Zone,Average,Surface Outside Face Outdoor Air Drybulb Temperature [C]
Zone,Average,Surface Outside Face Outdoor Air Wetbulb Temperature [C]
Zone,Average,Surface Outside Face Outdoor Air Wind Speed [m/s]
Zone,Average,Surface Outside Face Outdoor Air Wind Direction [deg]
Zone,Average,Surface Outside Face Sunlit Area [m2]
Zone,Average,Surface Outside Face Sunlit Fraction \protect\hyperlinksection-1
Zone,Average,Surface Outside Face Incident Solar Radiation Rate per Area[W/m2]
Zone,Average,Surface Outside Face Solar Radiation Heat Gain Rate [W]
Zone,Average,Surface Outside Face Solar Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Outside Face Solar Radiation Heat Gain Energy [J]
Zone,Average,Surface Outside Face Incident Beam Solar Radiation Rate per Area[W/m2]
Zone,Average,Surface Outside Face Incident Sky Diffuse Solar Radiation Rate per Area[W/m2]
Zone,Average,Surface Outside Face Incident Ground Diffuse Solar Radiation Rate per Area[W/m2]
Zone,Average,Surface Ext Diff Sol From Bm-To-Diff Refl From Ground[W/m2]
Zone,Average,Surface Outside Face Incident Sky Diffuse Ground Reflected Solar Radiation Rate per Area[W/
    m2]
Zone,Average,Surface Outside Face Incident Sky Diffuse Surface Reflected Solar Radiation Rate per Area[W/
    m2]
Zone,Average,Surface Outside Face Incident Beam To Beam Surface Reflected Solar Radiation Rate per Area[W
    /m2]
Zone,Average,Surface Outside Face Incident Beam To Diffuse Surface Reflected Solar Radiation Rate per
    Area[W/m2]
Zone,Average,Surface Outside Face Beam Solar Incident Angle Cosine Value[]
Zone,Average,Surface Anisotropic Sky Multiplier []
Zone,Average,Surface Window BSDF Beam Direction Number []
Zone,Average,Surface Window BSDF Beam Theta Angle [rad]
Zone,Average,Surface Window BSDF Beam Phi Angle [rad]
Zone,Average,Surface Outside Face Thermal Radiation to Air Heat Transfer Rate [W]
Zone,Average,Surface Outside Face Heat Emission to Air Rate [W]
```

Output variables applicable to opaque heat transfer surfaces (FLOOR, WALL, ROOF, DOOR). **Note – these are advanced variables – you must read the descriptions and understand before use – then you must use the Diagnostics object to allow reporting.**

```
Zone,Average,Surface Inside Face Solar Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face Solar Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Solar Radiation Heat Gain Energy [J]
Zone,Average,Surface Inside Face Lights Radiation Heat Gain Rate [W]
Zone,Average,Surface Inside Face Lights Radiation Heat Gain Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Lights Radiation Heat Gain Energy [J]
Zone,Average,Surface Inside Face Conduction Heat Transfer Rate [W]
Zone,Average,Surface Inside Face Conduction Heat Gain Rate [W]
Zone,Average,Surface Inside Face Conduction Heat Loss Rate [W]
Zone,Average,Surface Inside Face Conduction Heat Transfer Rate per Area [W/m2]
Zone,Sum,Surface Inside Face Conduction Heat Transfer Energy [J]
Zone,Average,Surface Outside Face Conduction Heat Transfer Rate [W]
Zone,Average,Surface Outside Face Conduction Heat Gain Rate [W]
Zone,Average,Surface Outside Face Conduction Heat Loss Rate [W]
Zone,Average,Surface Outside Face Conduction Heat Transfer Rate per Area [W/m2]
```

```
Zone,Sum,Surface Outside Face Conduction Heat Transfer Energy [J]
Zone,Average,Surface Average Face Conduction Heat Transfer Rate [W]
Zone,Average,Surface Average Face Conduction Heat Gain Rate [W]
Zone,Average,Surface Average Face Conduction Heat Loss Rate [W]
Zone,Average,Surface Average Face Conduction Heat Transfer Rate per Area [W/m2]
Zone,Sum,Surface Average Face Conduction Heat Transfer Energy [J]
Zone,Average,Surface Heat Storage Rate [W]
Zone,Average,Surface Heat Storage Gain Rate [W]
Zone,Average,Surface Heat Storage Loss Rate [W]
Zone,Average,Surface Heat Storage Rate per Area [W/m2]
Zone,Sum,Surface Heat Storage Energy [J]
Zone,Average,Zone Opaque Surface Inside Face Conduction [W]
Zone,Average,Zone Opaque Surface Inside Faces Total Conduction Heat Gain Rate [W]
Zone,Average,Zone Opaque Surface Inside Faces Total Conduction Heat Loss Rate [W]
Zone,Sum,Zone Opaque Surface Inside Faces Total Conduction Heat Gain Energy [J]
Zone,Sum,Zone Opaque Surface Inside Faces Total Conduction Heat Loss Energy [J]
Zone,Average,Zone Opaque Surface Outside Face Conduction [W]
Zone,Average,Zone Opaque Surface Outside Face Conduction Gain[W]
Zone,Average,Zone Opaque Surface Outside Face Conduction Loss[W]
Zone,Average, Surface Inside Face Beam Solar Radiation Heat Gain Rate [W]
```

1.10.42 Window Output Variables

Output variables applicable to exterior windows and glass doors:

```
Zone,Average,Enclosure Windows Total Transmitted Solar Radiation Rate [W]
Zone,Sum,Enclosure Windows Total Transmitted Solar Radiation Energy [J]
Zone,Sum,Zone Transmitted Solar Energy [J]
Zone,Average,Zone Windows Total Heat Gain Rate [W]
Zone,Sum,Zone Windows Total Heat Gain Energy [J]
Zone,Average,Zone Windows Total Heat Loss Rate [W]
Zone,Sum,Zone Windows Total Heat Loss Energy [J]
Zone,Average,Enclosure Exterior Windows Total Transmitted Beam Solar Radiation Rate [W]
Zone,Sum,Enclosure Exterior Windows Total Transmitted Beam Solar Radiation Energy [J]
Zone,Average,Enclosure Interior Windows Total Transmitted Beam Solar Radiation Rate [W]
Zone,Sum,Enclosure Interior Windows Total Transmitted Beam Solar Radiation Energy [J]
Zone,Average,Enclosure Exterior Windows Total Transmitted Diffuse Solar Radiation Rate [W]
Zone,Sum,Enclosure Exterior Windows Total Transmitted Diffuse Solar Radiation Energy [J]
Zone,Average,Enclosure Interior Windows Total Transmitted Diffuse Solar Radiation Rate [W]
Zone,Sum,Enclosure Interior Windows Total Transmitted Diffuse Solar Radiation Energy [J]
Zone,Average,Surface Window Total Glazing Layers Absorbed Solar Radiation Rate [W]
Zone,Average,Surface Window Total Glazing Layers Absorbed Shortwave Radiation Rate [W]
Zone,Sum,Surface Window Total Glazing Layers Absorbed Solar Radiation Energy [J]
Zone,Average,Surface Window Shading Device Absorbed Solar Radiation Rate [W]
Zone,Sum,Surface Window Shading Device Absorbed Solar Radiation Energy [J]
Zone,Average, Surface Window Transmitted Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Solar Radiation Energy [J]
Zone,Average,Surface Window Transmitted Beam Solar Radiation Rate [W]
Zone,Average,Surface Window Transmitted Beam To Beam Solar Radiation Rate [W]
Zone,Average,Surface Window Transmitted Beam To Diffuse Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Beam Solar Radiation Energy [J]
Zone,Sum,Surface Window Transmitted Beam To Beam Solar Radiation Energy [J]
Zone,Sum,Surface Window Transmitted Beam To Diffuse Solar Radiation Energy [J]
Zone,Average,Surface Window Transmitted Diffuse Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Diffuse Solar Radiation Energy [J]
Zone,Average,Surface Window System Solar Transmittance []
Zone,Average,Surface Window System Solar Absorptance []
Zone,Average,Surface Window System Solar Reflectance []
Zone,Average,Surface Window Gap Convective Heat Transfer Rate [W]
Zone,Sum,Surface Window Gap Convective Heat Transfer Energy [J]
Zone,Average,Surface Window Heat Gain Rate [W]
Zone,Sum,Surface Window Heat Gain Energy [J]
Zone,Average,Surface Window Heat Loss Rate [W]
Zone,Sum,Surface Window Heat Loss Energy [J]
Zone,Average,Surface Window Net Heat Transfer Rate [W]
Zone,Sum,Surface Window Net Heat Transfer Energy [J]
Zone,Average,Surface Window Glazing Beam to Beam Solar Transmittance[]
Zone,Average,Surface Window Glazing Beam to Diffuse Solar Transmittance []
```

```
Zone,Average,Surface Window Glazing Diffuse to Diffuse Solar Transmittance[]
Zone,Average,Surface Window Model Solver Iteration Count []
Zone,Average,Surface Window Solar Horizontal Profile Angle[deg]
Zone,Average,Surface Window Solar Vertical Profile Angle[deg]
Zone,Average,Surface Window Outside Reveal Reflected Beam Solar Radiation Rate [W]
Zone,Sum,Surface Window Outside Reveal Reflected Beam Solar Radiation Energy
Zone,Average,Surface Window Inside Reveal Reflected Beam Solar Radiation Rate [W]
Zone,Sum,Surface Window Inside Reveal Reflected Beam Solar Radiation Energy [J]
Zone,Average,Surface Window Inside Reveal Absorbed Beam Solar Radiation Rate [W]
Zone,Average,Surface Window Inside Face Glazing Condensation Status []
Zone,Average,Surface Window Inside Face Frame Condensation Status []
Zone,Average,Surface Window Inside Face Divider Condensation Status []
Zone,Average,Surface Shading Device Is On Time Fraction[]
Zone,Average,Surface Window Blind Slat Angle [deg]
Zone,Average,Surface Window Blind Beam to Beam Solar Transmittance[]
Zone,Average,Surface Window Blind Beam to Diffuse Solar Transmittance[]
Zone,Average,Surface Window Blind Diffuse to Diffuse Solar Transmittance[]
Zone,Average,Surface Window Blind and Glazing System Beam Solar Transmittance[]
Zone,Average,Surface Window Blind and Glazing System Diffuse Solar Transmittance[]
Zone,Average,Surface Window Screen Beam to Beam Solar Transmittance []
Zone,Average,Surface Window Screen Beam to Diffuse Solar Transmittance []
Zone,Average,Surface Window Screen Diffuse to Diffuse Solar Transmittance []
Zone,Average,Surface Window Screen and Glazing System Beam Solar Transmittance []
Zone,Average,Surface Window Screen and Glazing System Diffuse Solar Transmittance []
Zone,State,Surface Storm Window On Off Status []
Zone,Average,Surface Window Inside Face Frame and Divider Zone Heat Gain Rate [W]
Zone,Average,Surface Window Frame Heat Gain Rate [W]
Zone,Average,Surface Window Frame Heat Loss Rate [W]
Zone,Average,Surface Window Divider Heat Gain Rate [W]
Zone,Average,Surface Window Divider Heat Loss Rate [W]
Zone,Average,Surface Window Frame Inside Temperature [C]
Zone,Average,Surface Window Frame Outside Temperature [C]
Zone,Average,Surface Window Divider Inside Temperature [C]
Zone,Average,Surface Window Divider Outside Temperature [C]
```

If the user requests to display advanced report/output variables (e.g. see Output:Diagnostics keyword DisplayAdvancedReportVariables) the the following additional output variables are available for exterior windows and glass doors

```
Zone,Average,Surface Window Inside Face Glazing Zone Convection Heat Gain Rate [W]
Zone,Average,Surface Window Inside Face Glazing Net Infrared Heat Transfer Rate [W]
Zone,Average,Surface Window Shortwave from Zone Back Out Window Heat Transfer Rate [W]
Zone,Average,Surface Window Inside Face Frame and Divider Zone Heat Gain Rate [W]
Zone,Average,Surface Window Inside Face Gap between Shade and Glazing Zone Convection Heat Gain Rate [W]
Zone,Average,Surface Window Inside Face Shade Zone Convection Heat Gain Rate [W]
Zone,Average,Surface Window Inside Face Shade Net Infrared Heat Transfer Rate [W]
```

Output variables applicable to interior windows and glass doors:

```
Zone,Average,Surface Window Transmitted Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Solar Radiation Energy [J]
Zone,Average,Surface Window Transmitted Beam Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Beam Solar Radiation Energy [J]
Zone,Average,Surface Window Transmitted Beam to Beam Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Beam to Beam Solar Radiation Energy [J]
Zone,Average,Surface Window Transmitted Diffuse Solar Radiation Rate [W]
Zone,Sum,Surface Window Transmitted Diffuse Solar Radiation Energy [J]
Zone,Average,Surface Window Heat Gain Rate [W]
Zone,Sum,Surface Window Heat Gain Energy [J]
Zone,Average,Surface Window Heat Loss Rate [W]
Zone,Sum,Surface Window Heat Loss Energy [J]
```

If the user requests to display advanced report/output variables (e.g. see Output:Diagnostics keyword DisplayAdvancedReportVariables) the the following additional output variable is available for Equivalent Layer Window;

```
Zone,Average, Surface Window Inside Face Other Convection Heat Gain Rate [W]
```

Output variables applicable to interior and exterior windows and doors under certain conditions (see next three subsections for more information) are:

```
Zone,Average,Surface Window Total Absorbed Shortwave Radiation Rate Layer <x> [W]
Zone,Average,Surface Window Front Face Temperature Layer <x> [C]
Zone,Average,Surface Window Back Face Temperature Layer <x> [C]
```

1.10.42.1 Surface Window Total Absorbed Shortwave Radiation Rate Layer <x> [W]

This will output shortwave radiation absorbed in a window layer. The key values for this output variable are the surface name. Layers are numbered from the outside to the inside of the surface. The full listing will appear in the RDD file. Note that this variable is only defined for constructions defined by a Construction:ComplexFenestrationState.

1.10.42.2 Surface Window Front Face Temperature Layer <x> [C]

This will output a temperature for the front face of the layer. The layer front face is considered to be the face closest to the outside environment. The full listing will appear in the RDD file. Note that this variable is only defined for constructions defined by a Construction:ComplexFenestrationState. For other window constructions, this variable is also defined for the outer layer (Layer 1) only. The value will be identical to the Surface Outside Face Temperature.

1.10.42.3 Surface Window Back Face Temperature Layer <x> [C]

This will output a temperature for the back face of the layer. The layer back face is considered to be the face closest to the inside environment. The full listing will appear in the RDD file. Note that this variable is only defined for constructions defined by a Construction:ComplexFenestrationState. For other window constructions, this variable is also defined for the inner layer only. The value will be identical to the Surface Inside Face Temperature.

1.10.43 Surface Output Variables (all heat transfer surfaces)

The various output variables related to surface heat transfer are organized around the inside and outside face of each surface. The zone heat balance model draws energy balances at each side, or face, of a surface and so each surface essentially has two sets of results. The inside face is the side of a heat transfer surface that faces toward the thermal zone. The outside face is the side of a heat transfer surface that faces away from the thermal zone, typically facing outdoors. The Key Value for these is generally the user-defined name of the surface.

1.10.43.1 Surface Inside Face Heat Balance Calculation Iteration Count []

This output is the number of iterations used in a part of the solution for surface heat transfer that accounts for thermal radiation heat transfer between zone surfaces. This is simply a counter on the iteration loop for inside face surface modeling. There is only one instance of this output in a given run and the Key Value is “Simulation.”

1.10.43.2 Surface Inside Face Temperature [C]

This is the temperature of the surface’s inside face, in degrees Celsius. Former Name: Prior to version 7.1 this output was called Surface Inside Temperature.

1.10.43.3 Surface Inside Face Interior Movable Insulation Temperature [C]

This is the temperature of movable insulation installed on the inside of the construction at the movable insulation's inside face (the one facing the zone), in degrees Celsius. This variable is only valid when the surface has movable insulation and the movable insulation is actually scheduled to be present. For surfaces that have no movable insulation or the movable insulation is not scheduled to be present, the value of this variable is set to the Surface Inside Face Temperature. It should be noted that users can limit how often this output variable is generating by using a schedule to control when this output is produced. For example, this user could add the following syntax to an input file that includes movable insulation:

```
Output:Variable,Zone1:WestWall,Surface Inside Face Interior Movable Insulation Temperature,timestep,
  MovableInsulationSchedule;
```

In this example, interior movable insulation is used on the west wall of zone 1 and is only present when the schedule (MovableInsulationSchedule) is greater than zero. Adding this output variable syntax to the input file reports the temperature at the inside face of the interior movable insulation only when the movable insulation is present. The use of the movable insulation schedule in the output variable designation limits when this value shows up in the EnergyPlus output file to when it is actually scheduled to be present. At other times, no value will be reported. The limiting of when the output is generated allows the user to generate useful statistics instead of having those statistics influenced by values for when the movable insulation is not present. If the user does not use the output variable schedule feature, the output for this variable will equal the surface inside face temperature when the movable insulation is not present.

1.10.43.4 Surface Outside Face Temperature [C]

This is the temperature of the surface's outside face, in degrees Celsius. Former Name: Prior to version 7.1, this output was called Surface Outside Temperature.

1.10.43.5 Surface Inside Face Adjacent Air Temperature [C]

This is the effective bulk air temperature used for modeling the inside surface convection heat transfer. This is the same as the zone mean air temperature when using the mixing model for roomair. However, if more advanced roomair models are used, this variable will report the air temperature predicted by the roomair model as it was used in the surface heat balance model calculations. Former Name: Prior to version 7.1, this output was called Surface Int Adjacent Air Temperature.

1.10.43.6 Surface Inside Face Convection Heat Gain Rate [W]

1.10.43.7 Surface Inside Face Convection Heat Gain Rate per Area [W/m2]

1.10.43.8 Surface Inside Face Convection Heat Gain Energy [J]

These "inside face convection heat gain" output variables describe the heat transferred by convection between the inside face and the zone air. The values can be positive or negative with positive indicating heat is being added to the surface's face by convection. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

Former Name: Prior to version 7.1, these outputs were called "Surface Int Convection Heat *" and had used the opposite sign convention.

1.10.43.9 Surface Inside Face Convection Heat Transfer Coefficient [W/m2-K]

This is the coefficient that describes the convection heat transfer. It is the value of "Hc" in the classic convection model Q = Hc* A* (T – T). This is the result of the surface convection algorithm used for the inside face. Former Name: Prior to version 7.1, this output was called "Surface Int Convection Coeff."

1.10.43.10 Surface Inside Face Net Surface Thermal Radiation Heat Gain Rate [W]

1.10.43.11 Surface Inside Face Net Surface Thermal Radiation Heat Gain Rate per Area [W/m2]

1.10.43.12 Surface Inside Face Net Surface Thermal Radiation Heat Gain Energy [J]

These "inside face net surface thermal radiation heat gain" output variables describe the heat transferred by longwave infrared thermal radiation exchanges between the inside faces of other surfaces in the zone. The values can be positive or negative with positive indicating heat is being added to the surface's face by thermal radiation. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

1.10.43.13 Surface Inside Face Solar Radiation Heat Gain Rate [W]

1.10.43.14 Surface Inside Face Solar Radiation Heat Gain Rate per Area [W/m2]

1.10.43.15 Surface Inside Face Solar Radiation Heat Gain Energy [J]

These "inside face solar radiation heat gain" output variables describe the heat transferred by solar radiation onto the inside face. The values are always positive and indicate heat is being added to the surface's face by solar radiation. This is sunlight that has entered the zone through a window and been absorbed on the inside face of the surface. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

1.10.43.16 Surface Inside Face Lights Radiation Heat Gain Rate [W]

1.10.43.17 Surface Inside Face Lights Radiation Heat Gain Rate per Area [W/m2]

1.10.43.18 Surface Inside Face Lights Radiation Heat Gain Energy [J]

These "inside face lights radiation heat gain" output variables describe the heat transferred by shortwave radiation onto the inside face. The values are always positive and indicate heat is being added to the surface's face by shortwave radiation that emanated from electric lighting equipment and was absorbed by the surface. Note that this includes light from other zones that are transmitted through interzone windows. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

1.10.43.19 Surface Inside Face Internal Gains Radiation Heat Gain Rate [W]

1.10.43.20 Surface Inside Face Internal Gains Radiation Heat Gain Rate per Area [W/m2]

1.10.43.21 Surface Inside Face Internal Gains Radiation Heat Gain Energy [J]

These "inside face internal gains radiation heat gain" output variables describe the heat transferred by longwave infrared thermal radiation onto the inside face that emanated from internal gains such as lights, electric equipment, and people. The values are always positive and indicate heat is being added to the surface's face by the absorption of longwave thermal radiation. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

1.10.43.22 Surface Inside Face System Radiation Heat Gain Rate [W]

1.10.43.23 Surface Inside Face System Radiation Heat Gain Rate per Area [W/m2]

1.10.43.24 Surface Inside Face System Radiation Heat Gain Energy [J]

These "inside face system radiation heat gain" output variables describe the heat transferred by infrared thermal radiation onto the inside face that emanated from HVAC equipment such as baseboard heaters or high-temperature radiant heating panels. The values are always positive and indicate heat is being added to the surface's face by the absorption of thermal radiation. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

1.10.43.25 Surface Outside Face Convection Heat Gain Rate [W]

1.10.43.26 Surface Outside Face Convection Heat Gain Rate per Area [W/m2]

1.10.43.27 Surface Outside Face Convection Heat Gain Energy [J]

These "outside face convection" output variables describe heat transferred by convection between the outside face and the surrounding air. The values can be positive or negative with positive values indicating heat is added to the surface face by convection heat transfer. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m^2), and an energy version (J).

Former Name: Prior to version 7.1, these outputs were called "Surface Ext Convection Heat *" and used the opposite sign convention.

1.10.43.28 Surface Outside Face Convection Heat Transfer Coefficient [W/m2-K]

This is the coefficient that describes the convection heat transfer. It is the value of "Hc" in the classic convection model Q = Hc* A* (T – T). This is the result of the surface convection algorithm used for the outside face. Former Name: Prior to Version 7.1, this output was called "Surface Ext Convection Coeff."

1.10.43.29 Surface Outside Face Net Thermal Radiation Heat Gain Rate [W]

1.10.43.30 Surface Outside Face Net Thermal Radiation Heat Gain Rate per Area [W/m2]

1.10.43.31 Surface Outside Face Net Thermal Radiation Heat Gain Energy [J]

These "outside face net thermal radiation" output variables describe the heat transferred by longwave infrared thermal radiation exchanges between the surface and the surroundings of the outside face. This is the net of all forms of longwave thermal infrared radiation heat transfer. The values can be positive or negative with positive indicating the net addition of heat to the outside face. Different versions of the report are available including the basic heat gain rate (W), and a per unit area flux (W/m2), and an energy version (J).

Former Name: Prior to version 7.1, these outputs were called "Surface Ext Thermal Radiation Heat *" and used the opposite sign convention.

1.10.43.32 Surface Inside Face Exterior Windows Incident Beam Solar Radiation Rate [W]

1.10.43.33 Surface Inside Face Exterior Windows Incident Beam Solar Radiation Rate per Area [W/m2]

1.10.43.34 Surface Inside Face Exterior Windows Incident Beam Solar Radiation Energy [J]

Beam solar radiation from the exterior windows in a zone incident on the inside face of a surface in the zone. If Solar Distribution in the BUILDING object is equal to MinimalShadowing or FullExterior, it is assumed that all beam solar from exterior windows falls on the floor. In this case the value of this output variable can be greater than zero only for floor surfaces. If Solar Distribution equals FullInteriorExterior the program tracks where beam solar from exterior windows falls inside the zone, in which case the value of this variable can be greater than zero for floor as well as wall surfaces. Different versions of the report are available including the basic incident rate (W), a per unit area flux (W/m2), and an energy version (J).

1.10.43.35 Surface Inside Face Interior Windows Incident Beam Solar Radiation Rate [W]

1.10.43.36 Surface Inside Face Interior Windows Incident Beam Solar Radiation Rate per Area [W/m2]

1.10.43.37 Surface Inside Face Interior Windows Incident Beam Solar Radiation Energy [J]

Beam solar radiation from the interior (i.e., interzone) windows in a zone incident on the inside face of a surface in the zone. This value is calculated only if Solar Distribution in the BUILDING object is equal to FullInteriorExterior. However, the program does not track where this radiation falls. Instead, it is treated by the program as though it were diffuse radiation uniformly distributed over all of the zone surfaces. See **Figure 1.33**. Different versions of the report are available including the basic incident rate (W), a per unit area flux (W/m2), and an energy version (J).

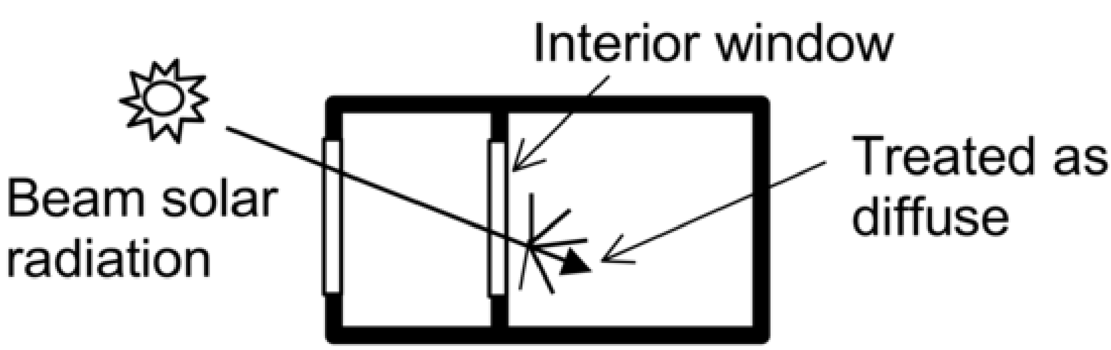

Figure 1.33: Beam solar radiation entering a zone through an interior window is distributed inside the zone as though it were diffuse radiation.

1.10.43.38 Surface Inside Face Initial Transmitted Diffuse Absorbed Solar Radiation Rate [W]

As of Version 2.1, diffuse solar transmitted through exterior and interior windows is no longer uniformly distributed. Instead, it is distributed according to the approximate view factors between the transmitting window and all other heat transfer surfaces in the zone. This variable is the amount of transmitted diffuse solar that is initially absorbed on the inside of each heat transfer surface. The portion of this diffuse solar that is reflected by all surfaces in the zone is subsequently redistributed uniformly to all heat transfer surfaces in the zone, along with interior reflected beam solar and shortwave radiation from lights. The total absorbed shortwave radiation is given by the next variable.

1.10.43.39 Surface Inside Face Absorbed Shortwave Radiation Rate [W]

As of Version 2.1, the previous variable plus absorbed shortwave radiation from uniformly distributed initially-reflected diffuse solar, reflected beam solar, and shortwave radiation from lights. This sum is the power of all sources of solar and visible radiation absorbed by the surface at the inside face.

1.10.43.40 Surface Construction Index []

The variable reports an index that points to specific Construction. The variable is an additional output variable when Output:Diagnostics, DisplayAdvancedReportVariables; is given in an input file.

In order to find a corresponding construction using the index, following steps are suggested:

1. Find a surface name from Surface Construction Index in the .csv file
2. Search the surface name in lines starting with HeatTransfer Surface in the .eio file
3. The seventh item is the construction name with matched surface name.

1.10.44 Surface Output Variables (exterior heat transfer surfaces)

1.10.44.1 Surface Outside Face Outdoor Air Drybulb Temperature [C]

The outdoor air dry-bulb temperature calculated at the height above ground of the surface centroid. Former Name: Prior to version 7.1, this output was called "Surface Ext Outdoor Dry Bulb."

1.10.44.2 Surface Outside Face Outdoor Air Wetbulb Temperature [C]

The outdoor air wet-bulb temperature calculated at the height above ground of the surface centroid. Former Name: Prior to version 7.1, this output was called "Surface Ext Outdoor Wet Bulb."

1.10.44.3 Surface Outside Face Outdoor Air Wind Speed [m/s]

The outdoor wind speed calculated at the height above ground of the surface centroid. This value may differ from site wind speed depending on the height of the surface (ref. Site:HeightVariation). In addition, the value may be different from the site wind speed when using OutdoorAir:Node with SurfaceProperty:LocalEnvironment or ZoneProperty:LocalEnvironment. It can also be modified using EMS. Former Name: Prior to version 7.1, this output was called "Surface Ext Wind Speed."

1.10.44.4 Surface Outside Face Outdoor Air Wind Direction [deg]

This output reports the actual wind direction at the outside face of a surface. As with the wind speed variable above, this value may be different from the site wind direction when using OutdoorAir:Node with

SurfaceProperty:LocalEnvironment or ZoneProperty:LocalEnvironment. It can also be modified using EMS.

1.10.44.5 Surface Outside Face Sunlit Area [m2]

The outside area of an exterior surface that is illuminated by (unreflected) beam solar radiation.

1.10.44.6 Surface Outside Face Sunlit Fraction []

The fraction of the outside area of an exterior surface that is illuminated by (unreflected) beam solar radiation. Equals Surface Outside Face Sunlit Area divided by total surface area.

1.10.44.7 Surface Outside Face Thermal Radiation to Air Heat Transfer Coefficient [W/m2-K]

This is the coefficient that describes thermal radiation heat transfer between the outside face and the air mass surrounding the surface. It is the value of "Hr" in the classic linearized model for thermal radiation Q = Hr * A * (T__surf – T__surfodb) when applied to the ambient air. Where T__surf = Surface Outside Face Temperature, and T__surfodb = Surface Outside Face Outdoor Air Drybulb Temperature. Former Name: Prior to version 7.1, this output was called "Surface Ext Rad to Air Coeff."

1.10.44.8 Surface Outside Face Thermal Radiation to Sky Heat Transfer Coefficient [W/m2-K]

This is the coefficient that describes thermal radiation heat transfer between the outside face and the sky surrounding the surface. It is the value of "Hr" in the classic linearized model for thermal radiation Q = Hr * A * (T__surf – T__sky) when applied to the sky. Where T__surf = Surface Outside Face Temperature, and T__sky = Site Sky Temperature. Former Name: Prior to version 7.1, this output was called "Surface Ext Rad to Sky Coeff."

1.10.44.9 Surface Outside Face Thermal Radiation to Ground Heat Transfer Coefficient [W/m2-K]

This is the coefficient that describes thermal radiation heat transfer between the outside face and the ground surrounding the surface. It is the value of "Hr" in the classic linearized model for thermal radiation Q = Hr * A * (T__surf – T__odb) when applied to the ground. Where T__surf = Surface Outside Face Temperature, and T__odb = Site Outdoor Air Drybulb Temperature (used as an approximation for the ground surface temperature). Former Name: Prior to version 7.1, this output was called "Surface Ext Rad to Ground Coeff."

1.10.44.10 Surface Outside Face Thermal Radiation to Air Heat Transfer Rate [W]

This is thermal radiation heat transfer rate between the outside face and the air mass surrounding the surface.

1.10.44.11 Surface Outside Face Heat Emission to Air Rate [W]

This is total heat transfer rate between the outside face and the air mass surrounding the surface by convection and thermal radiation.

1.10.44.12 Surface Outside Face Thermal Radiation to Surrounding Surfaces Heat Transfer Coefficient [W/m2-K]

This is the coefficient that describes thermal radiation heat transfer between the outside face of an exterior surface and the surrounding surfaces it views. It is the value of "Hr" in the classic linearized model for thermal radiation exchange Q = Hr * A * (T_surf – T_srdsurfs) when applied to the surrounding surfaces. Where T_surf = Surface Outside Face Temperature, and T_srdsurf = Average temperature of the surrounding surfaces viewed by an exterior surface.

1.10.44.13 Surface Outside Face Surrounding Surfaces Average Temperature [C]

This is the average surface temperature of the surrounding surfaces viewed by an exterior surface, in degrees Celsius. The surrounding surfaces average temperature is a view factor weighed surface temperature of multiple surrounding surfaces seen by an exterior surface. If an exterior surface views a single surrounding surface then the average temperature is the same as the user specified surrounding surface temperature.

1.10.44.14 Surface Outside Face Solar Radiation Heat Gain Rate [W]

1.10.44.15 Surface Outside Face Solar Radiation Heat Gain Rate per Area [W/m2]

1.10.44.16 Surface Outside Face Solar Radiation Heat Gain Energy [J]

These "outside face solar radiation" output variables describe the heat transferred by the absorption of solar radiation at the outside face. This is the result of incident solar radiation being absorbed at the surface face. The values are always positive.

1.10.44.17 Surface Outside Face Incident Solar Radiation Rate per Area [W/m2]

The total solar radiation incident on the outside of an exterior surface. It is the sum of:

- Surface Outside Face Incident Beam Solar Radiation Rate per Area
- Surface Outside Face Incident Sky Diffuse Solar Radiation Rate per Area
- Surface Outside Face Incident Ground Diffuse Solar Radiation Rate per Area
- Surface Outside Face Incident Sky Diffuse Surface Reflected Solar Radiation Rate per Area
- Surface Outside Face Incident Beam To Beam Surface Reflected Solar Radiation Rate per Area
- Surface Outside Face Incident Sky Diffuse Surface Reflected Solar Radiation Rate per Area

1.10.44.18 Surface Outside Face Incident Beam Solar Radiation Rate per Area [W/m2]

The solar beam radiation incident on the outside of an exterior surface, including the effects of shadowing, if present. The beam here is that directly from the sun; it excludes beam specularly reflected from obstructions.

1.10.44.19 Surface Outside Face Incident Sky Diffuse Solar Radiation Rate per Area [W/m2]

The solar diffuse radiation from the sky incident on the outside of an exterior surface, including the effects of shadowing, if present.

1.10.44.20 Surface Outside Face Incident Ground Diffuse Solar Radiation Rate per Area [W/m2]

The solar diffuse radiation incident on the outside of an exterior surface that arises from reflection of beam solar and sky diffuse solar from the ground. This is the sum of the next two output variables, "Surface Outside Face Incident Beam To Diffuse Ground Reflected Solar Radiation Rate per Area" and "Surface Outside Face Incident Sky Diffuse Ground Reflected Solar Radiation Rate per Area." The reflected solar radiation from the ground is assumed to be diffuse and isotropic (there is no specular component).

If "Reflections" option is not chosen in the Solar Distribution Field in the BUILDING object, the effects of shadowing are accounted for by the user-specified value of View Factor to Ground for the surface. If "Reflections" option is chosen, the program determines the effects of shadowing, including time-varying shadowing of the ground plane by the building itself.

1.10.44.21 Surface Outside Face Incident Beam To Diffuse Ground Reflected Solar Radiation Rate per Area [W/m2]

The solar diffuse radiation incident on the outside of an exterior surface that arises from beam-to-diffuse reflection from the ground. It is assumed that there is no beam-to-beam (specular) component. The beam here is that directly from the sun; it excludes beam specularly reflected from obstructions.

1.10.44.22 Surface Outside Face Incident Sky Diffuse Ground Reflected Solar Radiation Rate per Area [W/m2]

The solar diffuse radiation incident on the outside of an exterior surface that arises from sky diffuse solar reflection from the ground. The sky diffuse here is that directly from the sky; it excludes reflection of sky diffuse from obstructions.

1.10.44.23 Surface Outside Face Incident Sky Diffuse Surface Reflected Solar Radiation Rate per Area [W/m2]

The solar diffuse radiation incident on the outside of an exterior surface that arises from sky diffuse reflection from one or more obstructions. This value will be non-zero only if "Reflections" option is chosen in the BUILDING object.

1.10.44.24 Surface Outside Face Incident Beam To Beam Surface Reflected Solar Radiation Rate per Area [W/m2]

The solar beam radiation incident on the outside of an exterior surface that arises from beam-to-beam (specular) reflection from one or more obstructions. This value will be non-zero only if "Reflections" option is chosen in the BUILDING object. For windows, the program treats this beam radiation as diffuse radiation in calculating its transmission and absorption.

1.10.44.25 Surface Outside Face Incident Beam To Diffuse Surface Reflected Solar Radiation Rate per Area [W/m2]

The solar diffuse radiation incident on the outside of an exterior surface that arises from beam-to-diffuse reflection from building shades or building surfaces. This value will be non-zero only if "Reflections" option is chosen in the BUILDING object.

1.10.44.26 Surface Outside Face Beam Solar Incident Angle Cosine Value []

The cosine of the angle of incidence of (unreflected) beam solar radiation on the outside of an exterior surface. The value varies from 0.0 for beam parallel to the surface (incidence angle = 90^O) to 1.0 for beam perpendicular to the surface (incidence angle = 0^O). Negative values indicate the sun is behind the surface, i.e the surface does not see the sun.

1.10.44.27 Surface Anisotropic Sky Multiplier []

This is the view factor multiplier for diffuse sky irradiance on exterior surfaces taking into account the anisotropic radiance of the sky. The diffuse sky irradiance on a surface is given by Anisotropic Sky Multiplier * Diffuse Solar Irradiance.

1.10.44.28 Surface Window BSDF Beam Direction Number []

1.10.44.29 Surface Window BSDF Beam Phi Angle [rad]

1.10.44.30 Surface Window BSDF Beam Theta Angle [rad]

1.10.45 Opaque Surface Output Variables

The following variables apply only to opaque surfaces, where an opaque surface is considered here to be an exterior or interzone heat transfer surface of class FLOOR, WALL, ROOF or DOOR. **Note – these are advanced variables – you must read the descriptions and understand before use – then you must use the Output:Diagnostics object to allow reporting.**

1.10.45.1 Surface Inside Face Conduction Heat Transfer Rate [W]

1.10.45.2 Surface Inside Face Conduction Heat Transfer Rate per Area [W/m2]

1.10.45.3 Surface Inside Face Conduction Heat Gain Rate [W]

1.10.45.4 Surface Inside Face Conduction Heat Loss Rate [W]

These “inside face conduction” output variables describe heat flow by conduction right at the inside face of an opaque heat transfer surface. A positive value means that the conduction is from just inside the inside face toward the inside face. A negative value means that the conduction is from the inside face into the core of the heat transfer surface.

Note that Inside Face Conduction, when positive, does **not** necessarily indicate the heat flow from the surface to the zone air, which is governed by the inside face convection coefficient, the difference in temperature between the inside face and the zone air, and various radiation terms due to solar, internal gains, and radiant exchange with other surfaces in the zone.

Different versions of the reports are available. The basic heat gain rate (W) and a per unit area flux (W/m^2) can have positive or negative values with the sign convention that positive indicates heat flowing toward the face itself. There are also directed “gain” and “loss” versions that have only positive values or zero when the heat flow direction opposes.

Former Name: Prior to version 7.1, these outputs were called “Opaque Surface Inside Face Conduction *.”

Former Name: For Conduction Finite Difference simulations (CondFD), CondFD Inside Surface Heat Flux is replaced with Surface Inside Face Conduction Heat Transfer Rate Per Area. Likewise for CondFD Inside Heat Flux to Surface.

1.10.45.5 Surface Outside Face Conduction Heat Transfer Rate [W]

1.10.45.6 Surface Outside Face Conduction Heat Transfer Rate per Area [W/m2]

1.10.45.7 Surface Outside Face Conduction Heat Gain Rate [W]

1.10.45.8 Surface Outside Face Conduction Heat Loss Rate [W]

These “outside face conduction” output variables describe heat flow by conduction right at the outside face of an opaque heat transfer surface. A positive value means that the conduction is from just inside the outside face toward the outside face. A negative value means that the conduction is from the outside face into the core of the heat transfer surface.

Note that outside face conduction, when positive, does **not** necessarily indicate the heat flow from the surface to the surrounding air, due to the fact that there could be various terms such as convection and/or radiation terms based on whatever is the ”outside” environment for this surface.

When the surface in question is a partition, the output for this variable is set to zero because there is no outside face because the surface is fully exposed to the zone. When the surface is an interzone partition, the value will be non-zero because there will potentially be conduction into or out of the zone on the other side at this surface.

Different versions of the reports are available. The basic heat transfer rate (W) and a per unit area flux (W/m^2) can have positive or negative values with the sign convention that positive indicates heat flowing toward the face itself. There are also directed “gain” and “loss” versions that have only positive values or zero when the heat flow direction opposes.

Former Name: For Conduction Finite Difference simulations (CondFD), CondFD Outside Surface Heat Flux is replaced with Surface Outside Face Conduction Heat Transfer Rate Per Area. Likewise for CondFD Outside Heat Flux to Surface.

1.10.45.9 Surface Average Face Conduction Heat Transfer Rate [W]

1.10.45.10 Surface Average Face Conduction Heat Transfer Rate per Area [W/m2]

1.10.45.11 Surface Average Face Conduction Heat Gain Rate [W]

1.10.45.12 Surface Average Face Conduction Heat Loss Rate [W]

1.10.45.13 Surface Average Face Conduction Heat Transfer Energy [J]

These “average face conduction” output variables combine the inside face conduction and outside face conduction reports together to describe the conduction situation in a heat transfer surface in a nominal way. This is simply the average of the inside and outside face conduction rates, but with the sign convention for the outside face switched to match the inside face so that positive values here indicate heat flowing into the thermal zone.

Different versions of the reports are available. The basic heat conduction rate (W) and a per unit area flux (W/m^2) can have positive or negative values with the sign convention that positive indicates heat flowing toward the thermal zone. There are also directed “gain” and “loss” versions that have only positive values or zero when the heat flow direction opposes (W). Finally there is a version for total energy transfer (J).

1.10.45.14 Surface Heat Storage Rate [W]

1.10.45.15 Surface Heat Storage Rate per Area [W/m2]

1.10.45.16 Surface Heat Storage Gain Rate [W]

1.10.45.17 Surface Heat Storage Loss Rate [W]

1.10.45.18 Surface Heat Storage Energy [J]

These "heat storage" output variables combine the inside face conduction and outside face conduction reports together to describe the thermal storage situation in a heat transfer surface in a nominal way. This is simply the difference between the inside and outside face conduction, but with the sign convention arranged so that positive values indicate heat being added to the core of the surface.

Different versions of the reports are available. The basic heat storage rate (W) and a per unit area flux (W/m^2) can have positive or negative values with the sign convention that positive indicates heat being added to the surface's mass. There are also directed "gain" and "loss" versions that have only positive values or zero when the heat storage direction opposes (W). Finally there is a version for total energy stored (J).

1.10.45.19 Zone Opaque Surface Inside Face Conduction [W]

The sum of the Opaque Surface Inside Face Conduction values for all opaque surfaces in a zone for both positive and negative sums. For example, assume a zone has six opaque surfaces with Opaque Surface Inside Face Conduction values of 100, -200, 400, 50, 150 and –300 W. Then Zone Opaque Surface Inside Face Conduction = 700 - 500 = 200 W. Or if a zone has six opaque surfaces with Opaque Surface Inside Face Conduction values of -100, -200, 400, -50, 150 and –300W. Then Zone Opaque Surface Inside Face Conduction = 550 – 650 = -100 W.

1.10.45.20 Zone Opaque Surface Inside Faces Total Conduction Heat Gain Rate [W]

1.10.45.21 Zone Opaque Surface Inside Faces Total Conduction Heat Gain Energy [J]

These are the power and energy sums for the Opaque Surface Inside Face Conduction values for all opaque surfaces in a zone when that sum is positive. For example, assume a zone has six opaque surfaces with Opaque Surface Inside Face Conduction values of 100, -200, 400, 50, 150 and –300 W. Then Zone Opaque Surface Inside Faces Total Conduction Heat Gain Rate = 700 - 500 = 200 W.

1.10.45.22 Zone Opaque Surface Inside Faces Total Conduction Heat Loss Rate [W]

1.10.45.23 Zone Opaque Surface Inside Faces Total Conduction Heat Loss Energy [J]

These are the power and energy absolute value for the sums of the Opaque Surface Inside Face Conduction values for all opaque surfaces in a zone when that sum is negative. For example, assume a zone has six opaque surfaces with Opaque Surface Inside Face Conduction values of -100, -200, 400, -50, 150 and –300W. Then Zone Opaque Surface Inside Faces Total Conduction Heat Loss Rate = |550 – 650| = |-100| = 100 W.

1.10.45.24 Zone Opaque Surface Outside Face Conduction [W]

The sum of the Opaque Surface Outside Face Conduction values for all opaque surfaces in a zone for both positive and negative sums. For example, assume a zone has six opaque surfaces with Opaque Surface Outside Face Conduction values of 100, -200, 400, 50, 150 and –300 W. Then Zone Opaque Surface Outside

Face Conduction = 700 - 500 = 200 W. Or if a zone has six opaque surfaces with Opaque Surface Outside Face Conduction values of -100, -200, 400, -50, 150 and –300W. Then Zone Opaque Surface Outside Face Conduction = 550 – 650 = -100 W.

1.10.45.25 Zone Opaque Surface Outside Face Conduction Gain [W]

1.10.45.26 Zone Opaque Surface Outside Face Conduction Gain Energy [J]

These are the power and energy sums for the Opaque Surface Outside Face Conduction values for all opaque surfaces in a zone when that sum is positive. For example, assume a zone has six opaque surfaces with Opaque Surface Outside Face Conduction values of 100, -200, 400, 50, 150 and –300 W. Then Zone Opaque Surface Outside Face Conduction Gain = 700 - 500 = 200 W.

1.10.45.27 Zone Opaque Surface Outside Face Conduction Loss [W]

1.10.45.28 Zone Opaque Surface Outside Face Conduction Loss Energy [J]

These are the power and energy absolute value for the sums of the Opaque Surface Outside Face Conduction values for all opaque surfaces in a zone when that sum is negative. For example, assume a zone has six opaque surfaces with Opaque Surface Outside Face Conduction values of -100, -200, 400, -50, 150 and -300W. Then Zone Opaque Surface Outside Face Conduction Loss = |550 – 650| = |-100| = 100 W.

1.10.45.29 Surface Inside Face Beam Solar Radiation Heat Gain Rate [W]

Beam solar radiation from exterior windows absorbed on the inside face of an opaque heat transfer surface. For Solar Distribution = FullInteriorAndExterior, this quantity can be non-zero for both floor and wall surfaces. Otherwise, for Solar Distribution = FullExterior or MinimalShadowing, it can be non-zero only for floor surfaces since in this case all entering beam solar is assumed to fall on the floor. Note that this variable will not be operational (have a real value) unless there are exterior windows in the zone.

1.10.46 Window Output Variables

The following output variables apply to subsurfaces that are windows or glass doors. These two subsurface types are called “window” here. “Exterior window” means that the base surface of the window is an exterior wall, floor, roof or ceiling (i.e., the base surface is a BuildingSurface:Detailed with OutsideFaceEnvironment = ExteriorEnvironment). “Interior window” means that the base surface of the window is an inter-zone wall, floor or ceiling. “Glass” means a transparent solid layer, usually glass, but possibly plastic or other transparent material. “Shading device” means an interior, exterior or between-glass shade or blind, or an exterior screen (only exterior windows can have a shading device).

1.10.46.1 Enclosure Windows Total Transmitted Solar Radiation Rate [W]

1.10.46.2 Enclosure Windows Total Transmitted Solar Radiation Energy [J]

The total Surface Window Transmitted Solar Radiation Rate of all the exterior windows in an enclosure. If the user wishes to report these variables only for a specific enclosure and is uncertain as to what the name of the enclosure is, the enclosure name can be obtained from the the ”Input Verification and Results Summary” output report in the ”Space Summary” subtable.

1.10.46.3 Zone Windows Total Heat Gain Rate [W]

1.10.46.4 Zone Windows Total Heat Gain Energy [J]

The sum of the heat flow from all of the exterior windows in a zone when that sum is positive. (See definition of "heat flow" under "Window Heat Gain," below.)

1.10.46.5 Zone Windows Total Heat Loss Rate [W]

1.10.46.6 Zone Windows Total Heat Loss Energy [J]

The absolute value of the sum of the heat flow from all of the exterior windows in a zone when that sum is negative.

1.10.46.7 Surface Window Total Glazing Layers Absorbed Shortwave Radiation Rate [W]

1.10.46.8 Surface Window Total Glazing Layers Absorbed Solar Radiation Rate [W]

1.10.46.9 Surface Window Total Glazing Layers Absorbed Solar Radiation Energy [J]

The total exterior beam and diffuse solar radiation absorbed in all of the glass layers of an exterior window.

1.10.46.10 Surface Window Shading Device Absorbed Solar Radiation Rate [W]

Surface Window Shading Device Absorbed Solar Radiation Energy [J]

The exterior beam and diffuse solar radiation absorbed in the shading device, if present, of an exterior window.

1.10.46.11 Surface Window Transmitted Solar Radiation Rate [W]

1.10.46.12 Surface Window Transmitted Solar Radiation Energy [J]

The amount of beam and diffuse solar radiation entering a zone through an exterior window. It is the sum of the following two variables, "Surface Window Transmitted Beam Solar Radiation Rate" and "Surface Window Transmitted Diffuse Solar Radiation Rate."

1.10.46.13 Surface Window Transmitted Beam Solar Radiation Rate [W]

1.10.46.14 Surface Window Transmitted Beam Solar Radiation Energy [J]

The solar radiation transmitted by an exterior window whose source is beam solar incident on the outside of the window. For a bare window, this transmitted radiation consists of beam radiation passing through the glass (assumed transparent) and diffuse radiation from beam reflected from the outside window reveal, if present. For a window with a shade, this transmitted radiation is totally diffuse (shades are assumed to be perfect diffusers). For a window with a blind, this transmitted radiation consists of beam radiation that passes between the slats and diffuse radiation from beam-to-diffuse reflection from the slats. For a window with a screen, this value consists of direct beam radiation that is transmitted through the screen (gaps between the screen material) and diffuse radiation from beam-to-diffuse reflection from the screen material.

For each zone time step,

Surface Window Transmitted Beam Solar Radiation Rate = Surface Window Transmitted Beam To Beam Solar Radiation Rate + Surface Window Transmitted Beam To Diffuse Solar Radiation Rate

Surface Window Transmitted Beam Solar Radiation Energy = Surface Window Transmitted Beam To Beam Solar Radiation Energy + Surface Window Transmitted Beam To Diffuse Solar Radiation Energy

1.10.46.15 Surface Window Transmitted Beam To Beam Solar Radiation Rate [W]

1.10.46.16 Surface Window Transmitted Beam To Beam Solar Radiation Energy [J]

For a window with a blind, this transmitted radiation consists of beam radiation that passes between the slats. For a window with a screen, this value consists of direct beam radiation that is transmitted through the screen (gaps between the screen material).

1.10.46.17 Surface Window Transmitted Beam To Diffuse Solar Radiation Rate [W]

1.10.46.18 Surface Window Transmitted Beam To Diffuse Solar Radiation Energy [J]

For a window with a blind, this transmitted radiation consists of diffuse radiation reflected from beam by the slats. For a window with a screen, this value consists of diffuse radiation reflected by the screen material.

1.10.46.19 Enclosure Exterior Windows Total Transmitted Beam Solar Radiation Rate [W]

1.10.46.20 Enclosure Exterior Windows Total Transmitted Beam Solar Radiation Energy [J]

The sum of the Surface Window Transmitted Beam Solar Radiation Rate (see definition above) from all exterior windows in an enclosure. If the user wishes to report these variables only for a specific enclosure and is uncertain as to what the name of the enclosure is, the enclosure name can be obtained from the the ”Input Verification and Results Summary” output report in the ”Space Summary” subtable.

1.10.46.21 Enclosure Interior Windows Total Transmitted Beam Solar Radiation Rate [W]

1.10.46.22 Enclosure Interior Windows Total Transmitted Beam Solar Radiation Energy [J]

The sum of the Surface Window Transmitted Beam Solar Radiation Rate (see definition above) from all interior windows in an enclosure. If the user wishes to report these variables only for a specific enclosure and is uncertain as to what the name of the enclosure is, the enclosure name can be obtained from the the ”Input Verification and Results Summary” output report in the ”Space Summary” subtable.

1.10.46.23 Surface Window Transmitted Diffuse Solar Radiation Rate [W]

1.10.46.24 Surface Window Transmitted Diffuse Solar Radiation Energy [J]

The solar radiation transmitted by an exterior window whose source is diffuse solar incident on the outside of the window. For a bare window, this transmitted radiation consists of diffuse radiation passing through the glass. For a window with a shade, this transmitted radiation is totally diffuse (shades are assumed to be perfect diffusers). For a window with a blind, this transmitted radiation consists of diffuse radiation that passes between the slats and diffuse radiation from diffuse-to-diffuse reflection from the slats. For a window with a screen, this value consists of diffuse radiation transmitted through the screen (gaps between the screen material) and diffuse radiation from diffuse-to-diffuse reflection from the screen material.

1.10.46.25 Enclosure Exterior Windows Total Transmitted Diffuse Solar Radiation Rate [W]

1.10.46.26 Enclosure Exterior Windows Total Transmitted Diffuse Solar Radiation Energy [J]

The combined beam and diffuse solar that first entered adjacent zones through exterior windows in the adjacent zones, was subsequently reflected from interior surfaces in those zones (becoming diffuse through that reflection), and was then transmitted through interior windows into the current enclosure. If the user wishes to report these variables only for a specific enclosure and is uncertain as to what the name of the enclosure is, the enclosure name can be obtained from the the ”Input Verification and Results Summary” output report in the ”Space Summary” subtable.

1.10.46.27 Enclosure Interior Windows Total Transmitted Diffuse Solar Radiation Rate [W]

1.10.46.28 Enclosure Interior Windows Total Transmitted Diffuse Solar Radiation Energy [J]

The sum of the Surface Window Transmitted Diffuse Solar Radiation Rate (see definition above) from all interior windows in an enclosure. If the user wishes to report these variables only for a specific enclosure and is uncertain as to what the name of the enclosure is, the enclosure name can be obtained from the the ”Input Verification and Results Summary” output report in the ”Space Summary” subtable.

1.10.46.29 Surface Window System Solar Transmittance []

Effective solar transmittance of an exterior window, including effect of shading device, if present. Equal to “Surface Window Transmitted Solar Radiation Rate” divided by total exterior beam plus diffuse solar radiation incident on the window (excluding frame, if present).

1.10.46.30 Surface Window System Solar Absorptance []

Effective solar absorptance of an exterior window, including effect of shading device, if present. Equal to “Window Solar Absorbed: All Glass Layers” plus “Window Solar Absorbed: Shading Device” divided by total exterior beam plus diffuse solar radiation incident on window (excluding frame, if present)

1.10.46.31 Surface Window System Solar Reflectance []

Effective solar reflectance of an exterior window, including effect of shading device, if present. Equal to: [1.0 – “Surface Window System Solar Transmittance” – “Surface Window System Solar Absorptance”].

1.10.46.32 Surface Window Gap Convective Heat Transfer Rate [W]

1.10.46.33 Surface Window Gap Convective Heat Transfer Energy [J]

For an airflow window, the forced convective heat flow from the gap through which airflow occurs. This is the heat gained (or lost) by the air from the glass surfaces (and between-glass shading device surfaces, if present) that the air comes in contact with as it flows through the gap. If the gap airflow goes to the zone indoor air, the gap convective heat flow is added to the zone load. Applicable to exterior windows only.

1.10.46.34 Surface Window Net Heat Transfer Rate [W]

1.10.46.35 Surface Window Net Heat Transfer Energy [J]

The net heat flow to the zone through the glazing, frame and divider of an exterior window, including net solar gains. Negative values imply heat flow to the exterior. It is important to note that the boundary volume is at the inside face of the window surface (or shading device). Beam and diffuse solar which is absorbed in the window (and/or shading) layers will impact the window surface temperature which impacts the convective and IR heat flow components.

For a window *without an interior shading device*, this heat flow is equal to:

[Surface Window Transmitted Solar Radiation Rate]

+ [Surface Window Inside Face Glazing Zone Convection Heat Gain Rate]

+ [Surface Window Inside Face Glazing Net Infrared Heat Transfer Rate]

+ [Surface Window Inside Face Frame and Divider Zone Heat Gain Rate] (if present)

- [Surface Window Shortwave from Zone Back Out Window Heat Transfer Rate]

- [Surface Inside Face Initial Transmitted Diffuse Transmitted Out Window Solar Radiation Rate]

+ [Surface Window Gap Convective Heat Transfer Rate] (for WindowProperty:AirflowControl with destination IndoorAir)

Here, short-wave radiation is that from lights and interior solar radiation (beam and diffuse) incident on this window.

For a window *with an interior shading device*, this heat flow is equal to:

[Surface Window Transmitted Solar Radiation Rate]

+ [Surface Window Inside Face Gap between Shade and Glazing Zone Convection Heat Gain Rate]

+ [Surface Window Inside Face Shade Zone Convection Heat Gain Rate]

+ [Surface Window Inside Face Glazing Net Infrared Heat Transfer Rate]

+ [Surface Window Inside Face Shade Net Infrared Heat Transfer Rate]

+ [Surface Window Inside Face Frame and Divider Zone Heat Gain Rate] (if present)

- [Surface Window Shortwave from Zone Back Out Window Heat Transfer Rate]

- [Surface Inside Face Initial Transmitted Diffuse Transmitted Out Window Solar Radiation Rate]

+ [Surface Window Gap Convective Heat Transfer Rate] (for WindowProperty:AirflowControl with destination IndoorAir)

The total window heat flow can also be thought of as the sum of the solar and conductive gain *to the zone* from the window.

1.10.46.36 Surface Window Heat Gain Rate [W]

1.10.46.37 Surface Window Heat Gain Energy [J]

The total heat flow *to the zone* from the glazing, frame and divider of an exterior window when the Surface Window Net Heat Transfer (see definition above) is positive.

1.10.46.38 Surface Window Heat Loss Rate [W]

1.10.46.39 Surface Window Heat Loss Energy [J]

The total heat flow *from the zone* from the glazing, frame and divider of an exterior window when the Surface Window Net Heat Transfer (see definition above) is negative.

1.10.46.40 Surface Window Glazing Beam to Beam Solar Transmittance []

The fraction of exterior beam solar radiation incident on the glass of an exterior window that is transmitted through the glazing as beam radiation. This is for the base window without shading. Takes into account the angle of incidence of beam solar radiation on the glass.

1.10.46.41 Surface Window Glazing Beam to Diffuse Solar Transmittance []

The fraction of exterior beam solar radiation incident on the glazing of an exterior window that is transmitted through the glazing as diffuse radiation.

1.10.46.42 Surface Window Glazing Diffuse to Diffuse Solar Transmittance []

The fraction of exterior diffuse solar radiation incident on the glass of an exterior window that is transmitted through the glass assuming that the window has no shading device. It is assumed that incident diffuse solar is transmitted only as diffuse with no beam component.

1.10.46.43 Surface Window Model Solver Iteration Count []

The number of iterations needed by the window-layer heat balance solution to converge.

1.10.46.44 Surface Window Solar Horizontal Profile Angle [deg]

For a vertical exterior window, this is an angle appropriate for calculating beam solar quantities appropriate to horizontal window elements such as horizontal reveal surfaces, horizontal frame and divider elements and horizontal slats of window blinds. It is defined as the angle between the window outward normal and the projection of the sun's ray on the vertical plane containing the outward normal. See Figure 1.34.

For an exterior window of arbitrary tilt, it is defined as the angle between the window outward normal the projection of the sun's ray on the plane that contains the outward normal and is perpendicular to the ground.

If the sun is behind the window, the horizontal profile angle is not defined and is reported as 0.0.

Note that in most texts and in window equivalent layer model what we call "horizontal profile angle" is called "vertical profile angle."

1.10.46.45 Surface Window Solar Vertical Profile Angle [deg]

For a vertical exterior window, this is an angle appropriate for calculating beam solar quantities appropriate to vertical window elements such as vertical reveal surfaces, vertical frame and divider elements and vertical slats of window blinds. It is defined as the angle between the window outward normal and the projection of the sun's ray on the horizontal plane containing the outward normal. See **Figure 1.34.**

For an exterior window of arbitrary tilt, it is defined as the angle between the window outward normal the projection of the sun's ray on the plane that contains the outward normal and is perpendicular to the plane defined above for Surface Window Solar Horizontal Profile Angle for a window of arbitrary tilt.

If the sun is behind the window, the vertical profile angle is not defined and is reported as 0.0.

Note that in most texts and in window equivalent layer model what we call "vertical profile angle" is called "horizontal profile angle."

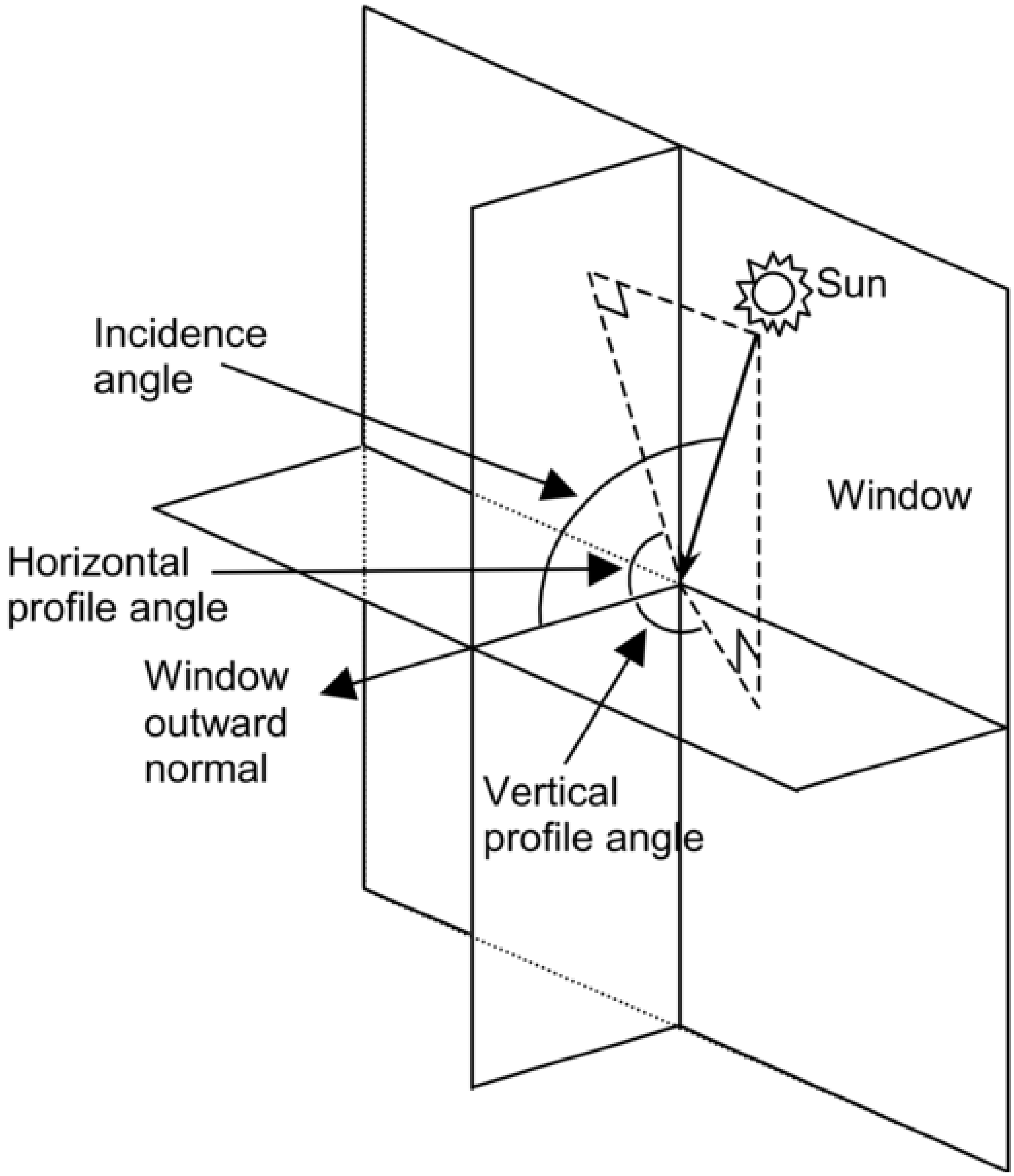

Figure 1.34: Vertical exterior window showing solar horizontal profile angle, solar vertical profile angle and solar incidence angle.

1.10.46.46 Surface Window Outside Reveal Reflected Beam Solar Radiation Rate [W]

1.10.46.47 Surface Window Outside Reveal Reflected Beam Solar Radiation Energy [J]

Beam solar radiation reflected from the outside reveal surfaces of a window (ref: Reveal Surfaces under WindowProperty:FrameAndDivider object). There are both rate and energy versions.

1.10.46.48 Surface Window Inside Reveal Reflected Beam Solar Radiation Rate [W]

1.10.46.49 Surface Window Inside Reveal Reflected Beam Solar Radiation Energy [J]

Beam solar radiation reflected from the inside reveal surfaces of a window (ref: Reveal Surfaces under WindowProperty:FrameAndDivider object). There are both rate and energy versions.

1.10.46.50 Surface Window Inside Reveal Absorbed Beam Solar Radiation Rate [W]

Beam solar radiation absorbed at the inside reveal surfaces of a window, in Watts.

1.10.46.51 Surface Window Inside Reveal Reflected Diffuse Zone Solar Radiation Rate [W]

Diffuse solar radiation reflected from inside reveal surfaces of a window into the zone, in Watts.

1.10.46.52 Surface Window Inside Reveal Reflected Diffuse Frame Solar Radiation Rate [W]

Diffuse solar radiation reflected from inside reveal surfaces onto the frame surfaces of a window, in Watts.

1.10.46.53 Surface Window Inside Reveal Reflected Diffuse Glazing Solar Radiation Rate [W]

Diffuse solar radiation reflected from inside reveal surfaces onto the glazing surfaces of a window, in Watts.

1.10.46.54 Surface Window Inside Face Glazing Condensation Status []

A value of 1 means that moisture condensation will occur on the innermost glass face of an exterior window (i.e., on the glass face in contact with the zone air). Otherwise the value is 0. The condition for condensation is glass inside face temperature < zone air dewpoint temperature.

For airflow exterior windows, in which forced air passes between adjacent glass faces in double- and triple-pane windows, a value of 1 means that condensation will occur on one or both of the glass faces in contact with the airflow. In this case the condition for condensation is:

- *For airflow source = indoorair*, temperature of either face in contact with airflow < zone air dewpoint temperature.
- *For airflow source = outdoorair*, temperature of either face in contact with airflow < outside air dewpoint temperature.

As for regular windows, the value will also be 1 if condensation occurs on the innermost glass face.

1.10.46.55 Surface Window Inside Face Frame Condensation Status []

If an exterior window has a frame and the value of this flag is 1, condensation will occur on the inside surface of the frame. The condition for condensation is frame inside surface temperature < zone air dewpoint temperature.

1.10.46.56 Surface Window Inside Face Divider Condensation Status []

If an exterior window has a divider and the value of this flag is 1, condensation will occur on the inside surface of the divider. The condition for condensation is divider inside surface temperature < zone air dewpoint temperature.

1.10.46.57 Surface Shading Device Is On Time Fraction []

The fraction of time that a shading device is on an exterior window. For a particular simulation timestep, the value is 0.0 if the shading device is off (or there is no shading device) and the value is 1.0 if the shading device is on. (It is assumed that the shading device, if present, is either on or off for the entire timestep.) If the shading device is switchable glazing, a value of 0.0 means that the glazing is in the unswitched (light colored) state, and a value of 1.0 means that the glazing is in the switched (dark colored) state.

For a time interval longer a timestep, this is the fraction of the time interval that the shading device is on. For example, take the case where the time interval is one hour and the timestep is 10 minutes. Then if the shading device is on for two timesteps in the hour and off for the other four timesteps, then the fraction of time that the shading device is on = 2/6 = 0.3333.

1.10.46.58 Surface Window Blind Slat Angle [deg]

For an exterior window with a blind, this is the angle in degrees between the glazing outward normal and the blind slat angle outward normal, where the outward normal points away from the front face of the slat. The slat angle varies from 0 to 180 deg. If the slat angle is 0 deg or 180 deg, the slats are parallel to the glazing and the slats are said to be "closed". If the slat angle is 90 deg, the slats are perpendicular to the glazing and the slats are said to be "fully open". See illustrations under WindowMaterial:Blind. For blinds with a fixed slat angle, the value reported here will be constant.

For WindowMaterial:Blind:EquivalentLayer blinds the slat angle varies from -90 to 90 deg. If the slat angle is -90 deg or 90 deg, the slats are parallel to the glazing and the slats are said to be "closed". If the slat angle is 0 deg, the slats are perpendicular to the glazing and the slats are said to be "fully open". For blinds with a fixed slat angle, the value reported here will be constant.

1.10.46.59 Surface Window Blind Beam to Beam Solar Transmittance []

For an exterior window with a blind, this is the fraction of exterior beam solar radiation incident on the blind that is transmitted through the blind as beam solar radiation when the blind is isolated (i.e., as though the window glass were not present). Depends on various factors, including slat angle, width, separation, and thickness, and horizontal solar profile angle (for blinds with horizontal slats) or vertical solar profile angle (for blinds with vertical slats). The transmittance value reported here will be non-zero only when some beam solar can pass through the blind without hitting the slats.

1.10.46.60 Surface Window Blind Beam to Diffuse Solar Transmittance []

For an exterior window with a blind, the fraction of exterior beam solar radiation incident on the blind that is transmitted through the blind as diffuse solar radiation when the blind is isolated (i.e., as though the window glass were not present). Depends on various factors, including slat angle, width, separation, thickness and reflectance, and horizontal solar profile angle (for blinds with horizontal slats) or vertical solar profile angle (for blinds with vertical slats).

1.10.46.61 Surface Window Blind Diffuse to Diffuse Solar Transmittance []

For an exterior window with a blind, the fraction of exterior diffuse solar radiation incident on the blind that is transmitted through the blind as diffuse solar radiation when the blind is isolated (i.e., as though the window glass were not present). Depends on various factors, including slat angle, width, separation, thickness and reflectance. For blinds with a fixed slat angle the transmittance value reported here will be constant.

1.10.46.62 Surface Window Blind and Glazing System Beam Solar Transmittance []

The fraction of exterior beam solar radiation incident on an exterior window with a blind (excluding window frame, if present) that is transmitted through the blind/glass system as beam solar radiation. Depends on various factors, including type of glass; solar incidence angle; slat angle, width, separation, and thickness; and horizontal solar profile angle (for blinds with horizontal slats) or vertical solar profile angle (for blinds with vertical slats).

1.10.46.63 Surface Window Blind and Glazing System Diffuse Solar Transmittance []

The fraction of exterior diffuse solar radiation incident on an exterior window with a blind (excluding window frame, if present) that is transmitted through the blind/glass system as diffuse solar radiation. Depends on various factors, including type of glass and slat angle, width, separation, thickness and reflectance. For blinds with a fixed slat angle the transmittance value reported here will be constant.

1.10.46.64 Surface Window Screen Beam to Beam Solar Transmittance []

For an exterior window with a screen, this is the fraction of exterior beam solar radiation incident on the screen that is transmitted through the screen as beam solar radiation when the screen is isolated (i.e., as though the window glass were not present). Depends on various factors, including the screen reflectance and the relative angle of the incident beam with respect to the screen. This value will include the amount of inward reflection of solar beam off the screen material surface if the user specifies this modeling option (i.e., Material: WindowScreen, field Reflected Beam Transmittance Accounting Method = Model as Direct Beam).

1.10.46.65 Surface Window Screen Beam to Diffuse Solar Transmittance[]

For an exterior window with a screen, the fraction of exterior beam solar radiation incident on the screen that is transmitted through the screen as diffuse solar radiation when the screen is isolated (i.e., as though the window glass were not present). Depends on various factors, including the screen reflectance and the relative angle of the incident beam with respect to the screen. This value is the amount of inward reflection of solar beam off the screen material surface if the user specifies this modeling option (i.e., Material: WindowScreen, field Reflected Beam Transmittance Accounting Method = Model as Diffuse); otherwise, this value will be zero.

1.10.46.66 Surface Window Screen Diffuse to Diffuse Solar Transmittance[]

For an exterior window with a screen, the fraction of exterior diffuse solar radiation incident on the screen that is transmitted through the screen as diffuse solar radiation when the screen is isolated (i.e., as though the window glass were not present). Depends on various factors including screen material geometry and reflectance. This value is calculated as an average, constant For a window with a screen, this value consists of diffuse radiation transmitted through the screen (gaps between the screen material) and diffuse radiation from diffuse-to-diffuse reflection from the screen material. For a window with a screen, this value consists of diffuse radiation transmitted through the screen (gaps between the screen material) and diffuse radiation from diffuse-to-diffuse reflection from the screen material.

1.10.46.67 Surface Window Screen and Glazing System Beam Solar Transmittance[]

The fraction of exterior beam solar radiation incident on an exterior window with a screen (excluding window frame, if present) that is transmitted through the screen/glass system as beam solar radiation. Depends on various factors, including the screen reflectance and the relative angle of the incident beam with respect to the screen. This value will include the amount of inward reflection of solar beam off the screen material surface if the user specifies this modeling option (i.e., Material: WindowScreen, field Reflected Beam Transmittance Accounting Method = Model as Direct Beam).

1.10.46.68 Surface Window Screen and Glazing System Diffuse Solar Transmittance []

The fraction of exterior diffuse solar radiation incident on an exterior window with a screen (excluding window frame, if present) that is transmitted through the screen/glass system as diffuse solar radiation. Depends on various factors including screen material geometry and reflectance.

1.10.46.69 Surface Window Transmitted Beam Solar Radiation Rate [W]

1.10.46.70 Surface Window Transmitted Beam Solar Radiation Energy [J]

The beam solar radiation transmitted through an interior window. Calculated only if Solar Distribution = FullInteriorAndExterior in your Building input. The origin of this radiation is beam solar that enters through an exterior window in a zone and then passes through an interior window into the adjacent zone. The amount of this radiation depends on several factors, including sun position, intensity of beam solar incident on the exterior window (including effects of shadowing, if present), relative position of the exterior and interior window, and the size and transmittance of the windows. Note that if there are two or more exterior windows in a zone, then beam solar from one or more of them may pass through the same interior window. Likewise, if there are more than two or more interior windows in a zone then beam solar from a single exterior window may pass through one or more of the interior windows. There are both rate and energy versions of the output.

1.10.46.71 Surface Storm Window On Off Status []

Indicates whether a storm window glass layer is present (ref: StormWindow object). The value is **0** if the storm window glass layer is off, **1** if it is on, and **–1** if the window does not have an associated storm window. Applicable only to exterior windows and glass doors.

1.10.46.72 Surface Inside Face Initial Transmitted Diffuse Transmitted Out Window Solar Radiation Rate [W]

As of Version 2.1, the diffuse solar transmitted through exterior windows that is initially distributed to another window in the zone and transmitted out of the zone through the inside face of the window. Some of this will be absorbed in the window layers and some will be transmitted through. For exterior windows, transmitted diffuse solar is "lost" to the exterior environment. For interior windows, this transmitted diffuse solar is distributed to heat transfer surfaces in the adjacent zone, and is part of the Surface Inside Face Initial Transmitted Diffuse Absorbed Solar Radiation Rate for these adjacent zone surfaces.

1.10.46.73 Additional Window Outputs (Advanced)

The following output variables for windows or glass doors are available when the user requests to display advanced output variables. These seven reports show the individual components that are combined to determine overall Surface Window Heat Gain Rate and/or Surface Window Heat Loss Rate (described above).

1.10.46.74 Surface Window Inside Face Glazing Zone Convection Heat Gain Rate [W]

The surface convection heat transfer from the glazing to the zone in watts. This output variable is the term called "[Convective heat flow to the zone from the zone side of the glazing]" under the description above for Surface Window Heat Gain Rate output variable. If the window has an interior shade or blind, then this is zero and the glazing's convection is included in the report called "Surface Window Inside Face Gap between Shade and Glazing Zone Convection Heat Gain Rate".

1.10.46.75 Surface Window Inside Face Glazing Net Infrared Heat Transfer Rate [W]

The net exchange of infrared radiation heat transfer from the glazing to the zone in watts. This output variable is the term called "[Net IR heat flow to the zone from zone side of the glazing]" under the description above for Surface Window Heat Gain Rate output variable.

1.10.46.76 Surface Window Shortwave from Zone Back Out Window Heat Transfer Rate [W]

This is the short-wave radiation heat transfer from the zone back out the inside face of the window in watts. This is a measure of the diffuse short-wave light (from beam and diffuse solar and electric lighting) that leaves the zone through the window.

1.10.46.77 Surface Window Inside Face Frame and Divider Zone Heat Gain Rate [W]

This is the convective heat transfer from any frames and/or dividers to the zone in watts. Negative values imply heat flow to the exterior. The Surface Window Frame Inside Temperature and Surface Window Divider Inside Temperature are computed from a heat balance on the frame/divider that includes solar radiation, long-wave radiation, convection, and conduction. The frame/divider is then included in the zone heat balance via convection at this inside temperature. See Engineering Reference "Window Frame and Divider Calculation" for more details.

1.10.46.78 Surface Window Frame Heat Gain Rate [W]

This is the positive heat flow from window frames to the zone in watts. This is part of the Surface Window Inside Face Frame and Divider Zone Heat Gain Rate.

1.10.46.79 Surface Window Frame Heat Loss Rate [W]

This is the negative heat flow from window frames to the zone in watts. This is part of the Surface Window Inside Face Frame and Divider Zone Heat Gain Rate.

1.10.46.80 Surface Window Frame Inside Temperature [C]

This is the temperature of the inside surface of the window frames.

1.10.46.81 Surface Window Frame Outside Temperature [C]

This is the temperature of the outside surface of the window frames.

1.10.46.82 Surface Window Divider Heat Gain Rate [W]

This is the positive heat flow from window dividers to the zone in watts. This is part of the Surface Window Inside Face Frame and Divider Zone Heat Gain Rate.

1.10.46.83 Surface Window Divider Heat Loss Rate [W]

This is the negative heat flow from window dividers to the zone in watts. This is part of the Surface Window Inside Face Frame and Divider Zone Heat Gain Rate.

1.10.46.84 Surface Window Divider Inside Temperature [C]

This is the temperature of the inside surface of the window dividers.

1.10.46.85 Surface Window Divider Outside Temperature [C]

This is the temperature of the outside surface of the window dividers.

1.10.46.86 Surface Window Inside Face Gap between Shade and Glazing Zone Convection Heat Gain Rate [W]

This is the convection surface heat transfer from the both the glazing and the shade's back face to the zone in Watts. This output variable is the term called "[Convective heat flow to the zone from the air flowing through the gap between glazing and shading device]" under the description above for Surface Window Heat Gain Rate output variable. For Equivalent Layer window this output variable is the convection heat gain from vented interior air gap to the zone in Watts.

1.10.46.87 Surface Window Inside Face Shade Zone Convection Heat Gain Rate [W]

This is the convection surface heat transfer from the front side of any interior shade or blind to the zone in Watts. This output variable is the term called "[Convective heat flow to the zone from the zone side of the shading device]" under the description above for Surface Window Heat Gain Rate output variable. For equivalent Layer window this output variable is the convection heat gain rate from the inside face of a glazing or a shade to the zone in Watts.

1.10.46.88 Surface Window Inside Face Shade Net Infrared Heat Transfer Rate [W]

The net exchange of infrared radiation heat transfer from the shade or blind to the zone in watts. This output variable is the term called "[Net IR heat flow to the zone from the zone side of the shading device]" under the description above for Surface Window Heat Gain Rate output variable.

1.10.46.89 Surface Window Inside Face Other Convection Heat Gain Rate [W]

The other (extra) convection heat transfer rate from the inside face of a an equivalent layer window. This output is computed from the difference in convection flux when using equivalent inside surface temperature of a window instead of the inside surface temperature from the standard surface heat balance calculation.

1.10.47 Thermochromic Window Outputs

1.10.47.1 Window Thermochromic Layer Temperature [C]

The temperature of the TC glass layer of a TC window at each time step.

1.10.47.2 Surface Window Thermochromic Layer Property Specification Temperature [C]

The temperature under which the optical data of the TC glass layer are specified.

The overall properties (U-factor/SHGC/VT) of the thermochromic windows at different specification temperatures are reported in the **.eio** file. These window constructions are created by EnergyPlus during run time. They have similar names with suffix "_TC_XX" where XX represents a specification temperature.

1.10.47.3 Switchable Window Outputs

1.10.47.4 Surface Window Switchable Glazing Switching Factor[]

The switching factor (tint level) of the switchable window: 0 means no switching – clear state; 1 means fully switched – dark state.

1.10.47.5 Surface Window Switchable Glazing Visible Transmittance[]

The visible transmittance of the switchable window.

1.10.48 Other Surface Outputs/Reports

Several reports can be selected for Surfaces. (See Group – Report for details on how to specify). Examples are:

1.10.48.1 DXF

This report produces a special file (**eplusout.dxf**) in the industry standard DXF (Drawing Interchange Format) for drawings. It is produced and accepted by many popular, commercial CAD programs. Detailed reference can be found on the AutoCAD™ website at: http://www.autodesk.com/techpubs/autocad/acadr14/dxf/index.htm.

EnergyPlus produces this file from the Report command:

```
Output:Reports, Surfaces, DXF;
```

1.10.48.2 Details, Vertices, DetailsWithVertices

This version of the report creates lines in the eplusout.eio file for each surface (except internal mass surfaces). Details of this reporting is shown in the Output Details and Examples document.

1.10.49 Shading Surfaces

Shading surfaces are entities outside of the building that may cast shadows on the building's heat transfer surfaces. These entities do not typically have enough thermal mass to be described as part of the building's thermal makeup.

The most important effect of shading surfaces is to reduce solar gain in windows that are shadowed. (However, in some cases, shading surfaces can reflect solar onto a wall or window and increase solar gain.)

There are two kinds of shading surfaces in EnergyPlus: **detached** and **attached**. A **detached** shading surface, such as a tree or neighboring building, is not connected to the building. An **attached** shading surface is typically an overhang or fin that is attached to a particular base surface of the building, usually a wall; attached shading surfaces are usually designed to shade specific windows.

Objects for detached shading surfaces:

- Shading:Site
- Shading:Building
- Shading:Site:Detailed
- Shading:Building:Detailed

Similarly to the surfaces, the detailed objects use vertex entry whereas the other objects are limited to rectangular representation.

Objects for attached shading surfaces:

- Shading:Overhang
- Shading:Overhang:Projection
- Shading:Fin
- Shading:Fin:Projection
- Shading:Zone:Detailed

EnergyPlus creates "bi-directional" shades from each shading surface entered. This means that the shade you input will cast a shadow no matter which side of the shade the sun is on. For example, a vertical fin will cast a shadow whether the sun is on the left side or right side of the fin. Prior to V1.0.2, a shading surface cast a shadow only in the hemisphere toward which the surface faced. This hemisphere is the one pointed to by the shading surface's outward normal vector, which is the cross product V23 x V21, where V23 is the vector from vertex 2 to vertex 3 of the shading surface and V21 is the vector from vertex 2 to vertex 1. Beginning with V1.0.2, the shades in EnergyPlus are "bi-directional" so that they can cast shadows in both hemispheres depending on the time-varying position of the sun relative to the shading surface.

It is important to note that EnergyPlus will automatically account for "self-shading" effects—such as in L-shaped buildings—in which some of the building's wall and roof surfaces shade other parts of the building, especially windows. This means that you only need to describe shading elements that aren't building heat-transfer surfaces.

Shading surfaces can also **reflect** solar radiation onto the building. This feature is simulated if you choose FullExteriorWithReflections or FullInteriorAndExteriorWithReflections in the Building object (ref: Building - Field: Solar Distribution). In this case, you may specify the reflectance properties of a shading surface with the ShadingProperty:Reflectance object. Note: If no ShadingProperty:Reflectance object is defined, then the shading surface reflectance values are assumed to be 0.2 and the fraction of shading surface that is glazed is assumed to be 0.0.

Shading surfaces also automatically shade diffuse solar radiation (and long-wave radiation) from the sky. And they will automatically shade diffuse solar radiation from the ground if Solar Distribution Field = FullExteriorWithReflections or FullInteriorAndExteriorWithReflections in the Building object. (If the reflections option for Solar Distribution is used, the program also takes into account the reduction of solar radiation reflected from the ground due to shading of the ground by shading surfaces and by the building itself.) Otherwise, shading surfaces will not shade diffuse radiation from the ground unless you enter a reduced value for View Factor to Ground for those building surfaces that are shaded (ref: BuildingSurface:Detailed - Field: View Factor to Ground and FenestrationSurface:Detailed - Field: View Factor to Ground).

1.10.50 Detached Shading Surfaces

1.10.51 Shading:Site, Shading:Building

These objects are used to describe rectangular shading elements that are external to the building. Examples are trees, high fences, near-by hills, and neighboring buildings.

If relative coordinates are used (ref: Field: Coordinate System in GlobalGeometryRules), shading surfaces entered with Shading:Site remain stationary if the building is rotated, whereas those entered with Shading:Building rotate with the building. If world coordinates are used Shading:Site and Shading:Building are equivalent.

These shading elements are always opaque.

1.10.51.1 Field: Name

This is a unique character string associated with the detached shading surface. Though it must be unique from other surface names, it is used primarily for convenience with detached shading surfaces.

1.10.51.2 Field: Azimuth Angle

Theoretically, this should face to the surface it is shading (i.e. if a south wall, this should be a north facing shade) but since EnergyPlus automatically generates the mirror image, the facing angle per se' is not so important.

1.10.51.3 Field: Tilt Angle

The tilt angle is the angle (in degrees) that the shade is tilted from horizontal (or the ground). Default for this field is 90 degrees.

1.10.51.4 Starting Corner for the surface

The rectangular surfaces specify the lower left corner of the surface for their starting coordinate. See the introductory paragraph for rules on this entry.

1.10.51.5 Field: Starting X Coordinate

This field is the X coordinate (in meters).

1.10.51.6 Field: Starting Y Coordinate

This field is the Y coordinate (in meters).

1.10.51.7 Field: Starting Z Coordinate

This field is the Z coordinate (in meters).

1.10.51.8 Field: Length

This field is the length of the shade in meters.

1.10.51.9 Field: Height

This field is the width of the shade in meters.

Examples of these (can be found in example files 4ZoneWithShading_Simple_1.idf and 4ZoneWithShading_Simple_2.idf)

```
Shading:Building,
    Bushes-East,             !- Name
    90,                      !- Azimuth Angle {deg}
    90,                      !- Tilt Angle {deg}
    45,                      !- Starting X Coordinate {m}
    0,                       !- Starting Y Coordinate {m}
    0,                       !- Starting Z Coordinate {m}
    50,                      !- Length {m}
    1;                       !- Height {m}

  Shading:Site,
    Bushes-North,            !- Name
    0,                       !- Azimuth Angle {deg}
    90,                      !- Tilt Angle {deg}
    45,                      !- Starting X Coordinate {m}
    50,                      !- Starting Y Coordinate {m}
    0,                       !- Starting Z Coordinate {m}
    50,                      !- Length {m}
    1;                       !- Height {m}
```

1.10.52 Shading:Site:Detailed, Shading:Building:Detailed

These objects are used to describe shading elements that are external to the building. Examples are trees, high fences, near-by hills, and neighboring buildings.

If relative coordinates are used (ref: Field: Coordinate System in GlobalGeometryRules), shading surfaces entered with Shading:Site:Detailed remain stationary if the building is rotated, whereas those entered with Shading:Building:Detailed rotate with the building. If world coordinates are used Shading:Site:Detailed and Shading:Building:Detailed are equivalent.

While "detached" implies that shading surfaces are not part of the building, the detached shading sequence can be used to describe attached shading surfaces that may shade heat transfer surfaces in more than one zone. For example, wing A of a building might shade several zones of wing B but wing A (for whatever reason) is not described in the geometry for the simulation so it is represented by a detached shade to get its shadowing effect.

1.10.52.1 Field: Name

This is a unique character string associated with the detached shading surface. Though it must be unique from other surface names, it is used primarily for convenience with detached shading surfaces.

1.10.52.2 Field: Transmittance Schedule Name

The name of a schedule of solar transmittance values from 0.0 to 1.0 for the shading surface. If a blank is entered in this field, the transmittance value defaults to 0.0, i.e., the shading surface is opaque at all times. This scheduling can be used to allow for seasonal transmittance change, such as for deciduous trees that have a higher transmittance in winter than in summer. Transmittance based on time of day can also be used—a movable awning, for example, where the transmittance is some value less than 1.0 when the awning is in place and is 1.0 when the awning is retracted.

The following assumptions are made in the shading surface transmittance calculation:

- Both sides of the shading surface have the same transmittance properties.
- The transmittance is the same for both beam and diffuse solar radiation.
- Beam solar transmittance is independent of angle of incidence on the shading surface.
- Beam radiation incident on a shading surface is transmitted as beam radiation with no change in direction, i.e., there is no beam-to-diffuse component.
- If two shading surfaces with non-zero transmittance overlap, the net transmittance is the product of the individual transmittances. Inter-reflection between the shading surfaces (and between the shading surfaces and the building) is ignored.
- For the daylighting calculation (ref: Group – Daylighting) the shading surface's visible transmittance is assumed to be the same as its solar transmittance.
- Shading devices are assumed to be opaque to long-wave radiation no matter what the solar transmittance value is.

Note that shading devices only shade solar radiation when the sun is up, which is automatically determined by EnergyPlus from latitude, time of year, etc. The user need only account for the time-varying transmittance of the shading device in the transmittance schedule, not whether the sun is up or not.

1.10.52.3 Field: Number Vertices

The number of sides in the surface (number of X,Y,Z vertex groups). For further information, see the discussion on "Surface Vertices" above.

IDF example of Detached Shading Surfaces:

```
Shading:Building:Detailed,
     EAST SIDE TREE,  !- Detached Shading
     ShadingTransmittance:0002,  !- Shadowing Transmittance & Schedule
     3, !-Triangle
     33.52800    ,   10.66800    ,   10.05800    ,
     33.52800    ,   13.71600    ,  0.9140000    ,
     33.52800    ,   4.572000    ,  0.9140000    ;

  Shading:Building:Detailed,
     WEST SIDE TREE,  !- Detached Shading
     ShadingTransmittance:0002,  !- Shadowing Transmittance & Schedule
     3, !-Triangle
    -3.048000    ,   7.620000    ,   10.05800    ,
    -3.048000    ,   4.572000    ,  0.9140000    ,
    -3.048000    ,   13.71600    ,  0.9140000    ;
```

1.10.53 Attached Shading Surfaces

Overhangs are usually horizontal devices that are used to shade windows. Fins are usually vertical devices that similarly shade windows.

1.10.54 Shading:Overhang

An overhang typically is used to shade a window in a building.

1.10.54.1 Inputs

1.10.54.1.1 Field: Name

This is the name of the overhang. It must be different from other surface names.

1.10.54.1.2 Field: Window or Door Name

The name of a window or door that this overhang shades.

1.10.54.1.3 Field: Height above Window or Door

This field is the height (meters) above the top of the door for the overhang.

1.10.54.1.4 Field Tilt Angle from Window/Door

This field is the tilt angle from the Window/Door. For a flat overhang, this would be 90 (degrees).

1.10.54.1.5 Field: Left Extension from Window/Door Width

This field is the width from the left edge of the window/door to the start of the overhang (meters).

1.10.54.1.6 Field: Right Extension from Window/Door Width

This field is the width from the right edge of the window/door to the start of the overhang (meters).

1.10.54.1.7 Field: Depth

This field is the depth of the overhang (meters) projecting out from the wall.

1.10.55 Shading:Overhang:Projection

An overhang typically is used to shade a window in a building. This object allows for specifying the depth of the overhang as a fraction of the window or door's height.

1.10.55.1 Inputs

1.10.55.1.1 Field: Name

This is the name of the overhang. It must be different from other surface names.

1.10.55.1.2 Field: Window or Door Name

The name of a window or door that this overhang shades.

1.10.55.1.3 Field: Height above Window or Door

This field is the height (meters) above the top of the door for the overhang.

1.10.55.1.4 Field Tilt Angle from Window/Door

This field is the tilt angle from the Window/Door. For a flat overhang, this would be 90 (degrees).

1.10.55.1.5 Field: Left Extension from Window/Door Width

This field is the width from the left edge of the window/door to the start of the overhang (meters).

1.10.55.1.6 Field: Right Extension from Window/Door Width

This field is the width from the right edge of the window/door to the start of the overhang (meters).

1.10.55.1.7 Field: Depth as Fraction of Window/Door Height

This field is the fraction of the window/door height to specify as the depth of the overhang (meters) projecting out from the wall.

1.10.56 Shading:Fin

Fins shade either side of windows/doors in a building. This object allows for specification of both fins for the window. Fin placement is relative to the edge of the glass and user must include the frame width when a frame is present.

1.10.56.1 Inputs

1.10.56.1.1 Field: Name

This is the name of the fin. It must be different from other surface names.

1.10.56.1.2 Field: Window or Door Name

The name of a window or door that this fin shades.

1.10.56.1.3 Field: Left Fin Extension from Window/Door

This field is the width from the left edge of the window/door to the plane of the left fin (meters). The extension width is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.56.1.4 Field: Left Fin Distance Above Top of Window

This field is the distance from the top of the window to the top of the left fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.56.1.5 Field: Left Fin Distance Below Bottom of Window

This field is the distance from the bottom of the window to the bottom of the left fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.56.1.6 Field: Left Fin Tilt Angle from Window/Door

This field is the tilt angle from the window / door for the left fin. Typically, a fin is 90 degrees (default) from its associated window/door.

1.10.56.1.7 Field: Left Fin Depth

This field is the depth (meters) of the left fin (projecting out from the wall).

1.10.56.1.8 Field: Right Fin Extension from Window/Door

This field is the width from the right edge of the window/door to the plane of the right fin (meters). The extension width is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.56.1.9 Field: Right Fin Distance Above Top of Window

This field is the distance from the top of the window to the top of the right fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.56.1.10 Field: Right Fin Distance Below Bottom of Window

This field is the distance from the bottom of the window to the bottom of the right fin (meters) and is relative to the edge of the glass and includes the frame width.

1.10.56.1.11 Field: Right Fin Tilt Angle from Window/Door

This field is the tilt angle from the window / door for the right fin. Typically, a fin is 90 degrees (default) from its associated window/door.

1.10.56.1.12 Field: Right Fin Depth

This field is the depth (meters) of the right fin (projecting out from the wall).

1.10.57 Shading:Fin:Projection

Fins shade either side of windows/doors in a building. This object allows for specification of both fins for the window. This object allows for specifying the depth of the fin as a fraction of the window or door's width. Fin placement is relative to the edge of the glass and user must include the frame width when a frame is present.

1.10.57.1 Inputs

1.10.57.1.1 Field: Name

This is the name of the fin. It must be different from other surface names.

1.10.57.1.2 Field: Window or Door Name

The name of a window or door that this fin shades.

1.10.57.1.3 Field: Left Fin Extension from Window/Door

This field is the width from the left edge of the window/door to the plane of the left fin (meters). The extension width is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.57.1.4 Field: Left Fin Distance Above Top of Window

This field is the distance from the top of the window to the top of the left fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.57.1.5 Field: Left Fin Distance Below Bottom of Window

This field is the distance from the bottom of the window to the bottom of the left fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.57.1.6 Field: Left Fin Tilt Angle from Window/Door

This field is the tilt angle from the window / door for the left fin. Typically, a fin is 90 degrees (default) from its associated window/door.

1.10.57.1.7 Field: Left Fin Depth as Fraction of Window/Door Width

This field is the fraction of the window/door width to specify as the depth of the left fin (meters) projecting out from the wall.

1.10.57.1.8 Field: Right Fin Extension from Window/Door

This field is the width from the right edge of the window/door to the plane of the right fin (meters).. The extension width is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.57.1.9 Field: Right Fin Distance Above Top of Window

This field is the distance from the top of the window to the top of the right fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.57.1.10 Field: Right Fin Distance Below Bottom of Window

This field is the distance from the bottom of the window to the bottom of the right fin (meters) and is relative to the edge of the glass and includes the frame width when a frame is present.

1.10.57.1.11 Field: Right Fin Tilt Angle from Window/Door

This field is the tilt angle from the window / door for the right fin. Typically, a fin is 90 degrees (default) from its associated window/door.

1.10.57.1.12 Field: Right Fin Depth as Fraction of Window/Door Width

This field is the fraction of the window/door width to specify as the depth of the right fin (meters) projecting out from the wall.

Examples of these (can be found in example files 4ZoneWithShading_Simple_1.idf and 4ZoneWithShading_Simple_2.idf)

```
Shading:Overhang:Projection,
    Zn001:Wall001:Win001:Shade001,  !- Name
    Zn001:Wall001:Win001,    !- Window or Door Name
    .7,                      !- Height above Window or Door {m}
    90,                      !- Tilt Angle from Window/Door {deg}
    .2,                      !- Left extension from Window/Door Width {m}
    .2,                      !- Right extension from Window/Door Width {m}
    .6;                      !- Depth as Fraction of Window/Door Height {m}

  Shading:Overhang,
    Zn001:Wall001:Door001:Shade001,  !- Name
    Zn001:Wall001:Door001,   !- Window or Door Name
    .6,                      !- Height above Window or Door {m}
    90,                      !- Tilt Angle from Window/Door {deg}
    0,                       !- Left extension from Window/Door Width {m}
    0,                       !- Right extension from Window/Door Width {m}
    3;                       !- Depth {m}

  Shading:Fin:Projection,
    Zn001:Wall001:Shade003,  !- Name
    Zn001:Wall001:Win001,    !- Window or Door Name
    .1,                      !- Left Extension from Window/Door {m}
    .1,                   !- Left Distance Above Top of Window {m}
    .1,                   !- Left Distance Below Bottom of Window {m}
    90,                   !- Left Tilt Angle from Window/Door {deg}
```

```
    .6,                     !- Left Depth as Fraction of Window/Door Width {m}
    .1,                     !- Right Extension from Window/Door {m}
    .1,                     !- Right Distance Above Top of Window {m}
    .1,                     !- Right Distance Below Bottom of Window {m}
    90,                     !- Right Tilt Angle from Window/Door {deg}
    .6;                     !- Right Depth as Fraction of Window/Door Width {m}
```

1.10.58 Shading:Zone:Detailed

This object is used to describe attached "subsurfaces" such as overhangs, wings or fins that project outward from a base surface. This classification is used for convenience; actually, a device of this type can cast shadows on the surface to which it is attached as well as on adjacent surfaces. For example, a fin may shade its parent wall as well as adjacent walls.

Note that a zone surface can cast shadows on other zone surfaces. However, you don't have to worry about such effects—for example, one wall of an L-shaped building shading another wall–because EnergyPlus will automatically check for this kind of "self shadowing" and do the proper calculations.

Unlike attached (or detached) shading surfaces, building surfaces can only cast shadows in the hemisphere towards which they face. This means, for example, that a roof that faces *upward* will not cast a shadow *downward*. (Thus, specifying an oversized roof in an attempt to account for the shading effects of overhangs will *not* work). Interior surfaces do not cast shadows of any kind.

1.10.58.1 Inputs

1.10.58.1.1 Field: Name

This is the name of the attached shading surface. It must be different from other surface names.

1.10.58.1.2 Field: Base Surface Name

This is the name of the surface to which this shading device is attached. This surface can be a wall (or roof) but not a window or door.

1.10.58.1.3 Field: Transmittance Schedule Name

The name of a schedule of solar transmittance values from 0.0 to 1.0 for the shading surface. If a blank is entered in this field, the transmittance value defaults to 0.0, i.e., the shading surface is opaque at all times. This scheduling can be used to allow for seasonal transmittance change, such as for deciduous trees that have a higher transmittance in winter than in summer. Transmittance based on time of day can also be used—a movable awning, for example, where the transmittance is some value less than 1.0 when the awning is in place and is 1.0 when the awning is retracted.

The following assumptions are made in the shading surface transmittance calculation:

- Both sides of the shading surface have the same transmittance properties.
- The transmittance is the same for both beam and diffuse solar radiation.
- Beam solar transmittance is independent of angle of incidence on the shading surface.
- Beam radiation incident on a shading surface is transmitted as beam radiation with no change in direction, i.e., there is no beam-to-diffuse component.
- If two shading surfaces with non-zero transmittance overlap, the net transmittance is the product of the individual transmittances. Inter-reflection between the shading surfaces (and between the shading surfaces and the building) is ignored.

- For the daylighting calculation (ref: Group – Daylighting) the shading surface's visible transmittance is assumed to be the same as its solar transmittance.

- Shading devices are assumed to be opaque to long-wave radiation no matter what the solar transmittance value is.

Note that shading devices only shade solar radiation when the sun is up, which is automatically determined by EnergyPlus from latitude, time of year, etc. The user need only account for the time-varying transmittance of the shading device in the transmittance schedule, not whether the sun is up or not.

1.10.58.1.4 Field: Number Vertices

The number of sides in the surface (number of X,Y,Z vertex groups). For further information, see the discussion on "Surface Vertices" above. The example below shows the correct input for an overhang (to shade the appropriate portion of the base wall and window).

Proper specification for this overhang (facing up) is:

4,(C,0,D),(C,-B,D),(C+A,-B,D),(C+A,0,D); () used to illustrate each vertex.

> Note that for horizontal surfaces, any corner may be chosen as the starting corner. The order of vertices determines whether the surface is facing up or down. Shading surfaces are mirrored automatically unless the user specifies "DoNotMirrorDetachedShading", so each shading surface need only be described once.

Thus, another shading surface will be created (facing down):

4,(C+A,-B,D),(C+A,0,D),(C,0,D),(C,-B,D);

IDF example of attached shading surfaces (overhang, fin):

```
Shading:Zone:Detailed,
  Zn001:Wall001:Shade001,        !- Surface Name
  Zn001:Wall001,                 !- Base Surface Name
  ShadingTransmittance:0001,     !- Shadowing Transmittance Schedule
  4,                             !-RectangularOverhang
  1.524000 , -0.3050000    , 2.865000,
  1.524000 ,  0.0000000E+00, 2.865000,
  4.572000 ,  0.0000000E+00, 2.865000,
  4.572000 , -0.3050000    , 2.865000;

Shading:Zone:Detailed,
  Zn003:Wall001:Shade001,        !- Surface Name
  Zn003:Wall001,                 !- Base Surface Name
  ShadingTransmittance:0001,     !- Shadowing Transmittance Schedule
  4,                             !-RectangularLeftFin
  57.97000 ,   8.450000    ,10.00000      ,
  57.97000 ,   8.450000    , 0.0000000E+00,
  57.97000 ,   6.450000    , 0.0000000E+00,
  57.97000 ,   6.450000    ,10.00000      ;

Shading:Zone:Detailed,
  Zn003:Wall001:Shade002,        !- Surface Name
  Zn003:Wall001,                 !- Base Surface Name
  ShadingTransmittance:0003,     !- Shadowing Transmittance Schedule
  4,                             !-RectangularRightFin
  77.97000 ,   6.450000    ,10.00000      ,
  77.97000 ,   6.450000    , 0.0000000E+00,
  77.97000 ,   8.450000    , 0.0000000E+00,
  77.97000 ,   8.450000    ,10.00000      ;
```

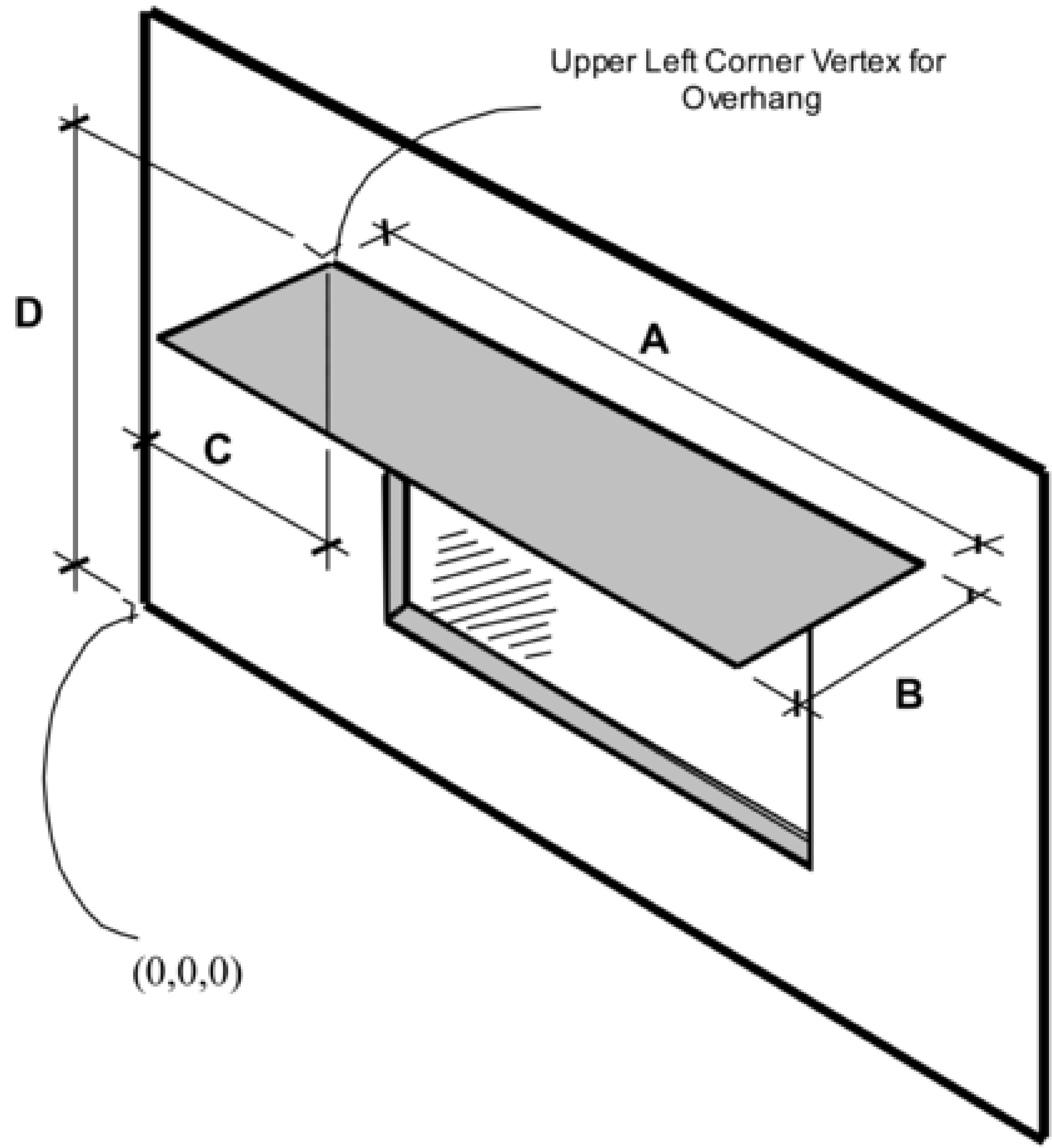

Figure 1.35: Illustration for Attached Shading Surface

1.10.59 ShadingProperty:Reflectance

Specifies the reflectance properties of a shading surface when the solar reflection calculation has requested, i.e., when if "WithReflections" option is chosen in the Building object (ref: Building - Field: Solar Distribution). It is assumed that shading surfaces are divided into an unglazed, diffusely reflecting portion and a glazed, specularly-reflecting portion, either of which may be zero. The reflectance properties are assumed to be the same on both sides of the shading surface.

Note that a shadowing transmittance schedule (ref: Shading Surfaces, Field: Transmittance Schedule Name) can be used with a reflective shading surface. However, EnergyPlus assumes that the reflectance properties of the shading surface are constant even if the transmittance varies.

If no ShadingProperty:Reflectance objects are entered, the default values shown here will be used for shading surfaces. Other surfaces have their reflectance properties defined by the materials in the outer layers of the constructions.

1.10.59.1 Inputs

1.10.59.1.1 Field: Shading Surface Name

The name of the Shading:Site, Shading:Building, Shading:Site:Detailed, Shading:Building:Detailed, Shading:Overhang, Shading:Overhang:Projection, Shading:Fin, Shading:Fin:Projection or Shading:Zone:Detailed object to which the following fields apply.

If this ShadingProperty:Reflectance object is not defined for a shading surface the default values listed in each of the following fields will be used in the solar reflection calculation.

1.10.59.1.2 Field: Diffuse Solar Reflectance of Unglazed Part of Shading Surface

The diffuse solar reflectance of the unglazed part of the shading surface (default = 0.2). This reflectance is assumed to be the same for beam-to-diffuse and diffuse-to-diffuse reflection. Beam-to-diffuse reflection is assumed to be independent of angle of incidence of beam radiation. Diffuse-to-diffuse reflection is assumed to be independent of angular distribution of the incident of diffuse radiation. The outgoing diffuse radiation is assumed to be isotropic (hemispherically uniform).

The sum of this reflectance and the shading surface transmittance should be less than or equal to 1.0.

1.10.59.1.3 Field: Diffuse Visible Reflectance of Unglazed Part of Shading Surface

The diffuse visible reflectance of the unglazed part of the shading surface (default = 0.2). This reflectance is assumed to be the same for beam-to-diffuse and diffuse-to-diffuse reflection. Beam-to-diffuse reflection is assumed to be independent of angle of incidence of beam radiation. Diffuse-to-diffuse reflection is assumed to be independent of angular distribution of the incident of diffuse radiation. The outgoing diffuse radiation is assumed to be isotropic (hemispherically uniform).

This value if used only for the daylighting calculation (ref: Daylighting:Controls). The sum of this reflectance and the shading surface transmittance should be less than or equal to 1.0.

1.10.59.1.4 Field: Fraction of Shading Surface That Is Glazed

The fraction of the area of the shading surface that consists of windows (default = 0.0). It is assumed that the windows are evenly distributed over the surface and have the same glazing construction (see following "Name of Glazing Construction"). This might be the case, for example, for reflection from the façade of a neighboring, highly-glazed building. For the reflection calculation the possible presence of shades, screens or blinds on the windows of the shading surface is ignored. Beam-to-beam (specular) reflection is assumed to occur only from the glazed portion of the shading surface. This reflection depends on angle of incidence as determined by the program from the glazing construction. Beam-to-diffuse reflection from the glazed portion is assumed to be zero. The diffuse-to-diffuse reflectance of the glazed portion is determined by the program from the glazing construction.

1.10.59.1.5 Field: Glazing Construction Name

The name of the construction of the windows on the shading surface. Required if Fraction of Shading Surface That Is Glazed is greater than 0.0.

IDF example of Shading Surface Reflectance for shading surface with specular reflection

```
Shading:Site:Detailed,
  Adjacent Glazed Facade,  !- User Supplied Surface Name
  ,   !- Shadowing Transmittance Schedule
  4,  !- Number of Surface Vertex Groups -- Number of (X,Y,Z) groups
  0,-24,30, !- Vertex 1 X,Y,Z coordinates
  0,-24,0,  !- Vertex 2 X,Y,Z coordinates
  0,0,0,    !- Vertex 3 X,Y,Z coordinates
  0,0,30;   !-Vertex 3 X,Y,Z coordinates


  ShadingProperty:Reflectance,
  Adjacent Glazed Facade, !- Name of Surface:Shading Object
  0.3,  !- Diffuse Solar Reflectance of Unglazed Part of Shading Surface
  0.3,  !- Diffuse Visible Reflectance of Unglazed Part of Shading Surface
  0.7,  !- Fraction of Shading Surface That Is Glazed
  GlassCon-1; !- Name of Glazing Construction
```

IDF example of Shading Surface Reflectance for shading surface without specular reflection

```
Shading:Site:Detailed,
  Adjacent Blank Facade,  !- User Supplied Surface Name
  ,   !- Shadowing Transmittance Schedule
  4,  !- Number of Surface Vertex Groups -- Number of (X,Y,Z) groups
  0,-24,30,
  0,-24,0,
  0,0,0,
  0,0,30;


  ShadingProperty:Reflectance,
  Adjacent Blank Facade, !- Name of Surface:Shading Object
  0.4,  !- Diffuse Solar Reflectance of Unglazed Part of Shading Surface
  0.4,  !- Diffuse Visible Reflectance of Unglazed Part of Shading Surface
  0.0,  !- Fraction of Shading Surface That Is Glazed
  ;     !- Name of glazing construction
```

1.10.60 WindowShadingControl

Window shading with coverings like drapes, blinds, screens or pull-down shades can be used to reduce the amount of solar radiation entering the window or reduce daylighting glare. It can also be used to reduce heat loss through the window (movable insulation). Leaving the window covering open in the winter can maximize solar heat gain and thereby reduce heating loads.

With WindowShadingControl you specify the type, location, and timing of the shading device, what variable or combination of variables controls deployment of the shading device, and what the control setpoint is. If the shading device is a blind, you also specify how the slat angle is controlled. Each WindowShadingControl object is associated with a zone and has a list of one or more windows and glass doors to which the shading control is applied (ref: FenestrationSurface:Detailed with Type = Window or GlassDoor, Window, and GlazedDoor).

NOTE: WindowShadingControl does not work with complex fenestration systems. Controlled complex fenestration systems can be made only with Energy Management Systems objects. Referencing a

FenestrationSurface:Detailed in a WindowShadingControl while using complex fenestration systems will be ignored by program.

As shown in Figure 1.36, a shading device can be inside the window (Shading Type = InteriorShade or InteriorBlind), outside the window (Shading Type = ExteriorShade or ExteriorBlind), or between panes of glass (Shading Type = BetweenGlassShade or BetweenGlassBlind). The exception is window screens which can only be outside the window (Shading Type = ExteriorScreen).

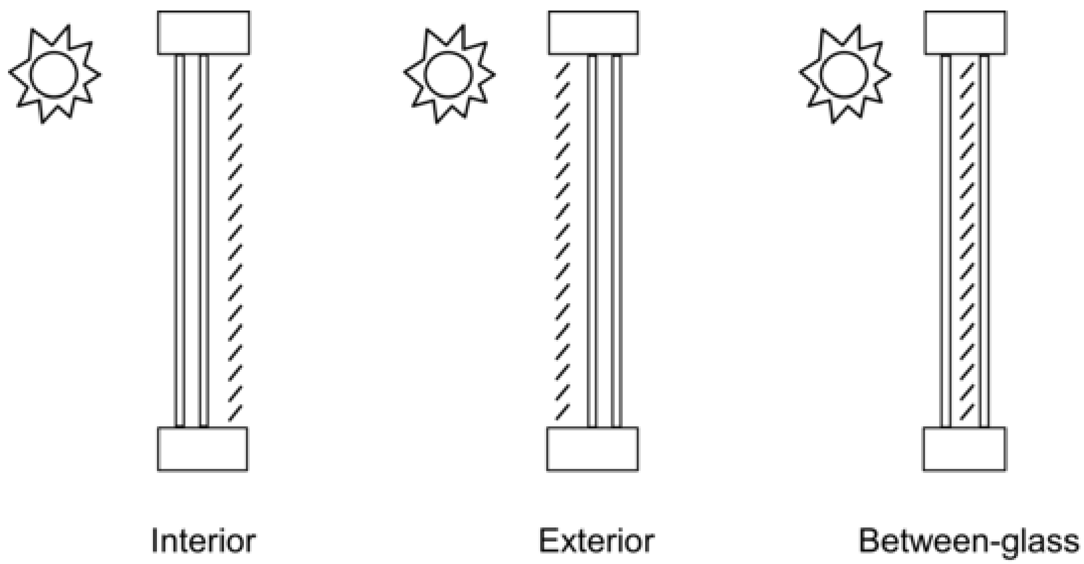

Figure 1.36: Allowed locations of a window shading device.

When a shading device is present it is either retracted or activated. When it is retracted it covers none of the window. When it is activated it covers the entire glazed part of the window (but not the frame). Whether the shading device is retracted or activated in a particular timestep depends on the control mechanism: see "Shading Control Type," below. To model a case in which the shading device, when activated, covers only **part** of the window you will have to divide the window into two separate windows, one with the shading device and one without the shading device.

A shading device can also be of a kind in which the optical properties of the glazing switch from one set of values to another in order to increase or decrease solar or visible transmittance (Shading Type = SwitchableGlazing).

There are two ways of specifying the actual shading device:

- **Specify "Name of Construction with Shading"**

  This is the name of a window Construction that has the shading device as one of its layers. The thermal and solar-optical properties of the shading device are given by the shading material referenced in that Construction (ref: Construction, WindowMaterial:Shade, WindowMaterial:Screen and WindowMaterial:Blind). To use this method you have to define two Constructions for the window, one without the shading device and one with it. See Example 1, below.

  The Construction without the shading device is referenced in the FenestrationSurface:Detailed input for the window (see IDF example, below). The Construction with the shading device is referenced by the window's WindowShadingControl.

  For Shading Type = InteriorShade, InteriorBlind, ExteriorShade, ExteriorScreen and ExteriorBlind these two Constructions must be identical expect for the presence of the shading layer in the shaded

Construction, otherwise you will get an error message. You will also get an error message if the Construction referenced by the window has a shading layer.

- **Specify the "Material Name of the Shading Device"**

  This is the name of a WindowMaterial:Shade, WindowMaterial:Screen or WindowMaterial:Blind. This method can be used with Shading Type = InteriorShade, InteriorBlind, ExteriorShade and ExteriorBlind. It cannot be used with Shading Type = BetweenGlassShade, BetweenGlassBlind, or SwitchableGlazing. If Shading Type = InteriorShade or ExteriorShade, then you specify the name of a WindowMaterial:Shade. If Shading Type = InteriorBlind or ExteriorBlind, then you specify the name of a WindowMaterial:Blind. If Shading Type = ExteriorScreen, then you specify the name of a WindowMaterial:Screen. See Example 2, below. This method is simpler to use since you don't have to specify two Constructions that differ only by the shading layer.

  When this method is used, the program will automatically create a shaded window construction by adding a shading layer to the outside or inside of the construction corresponding to the windows referenced by the WindowShadingControl. The name, created by the program, of this shaded construction is composed as follows: if the name of the window construction is CCC and the material name of the shading device is DDD, then the shaded construction name is CCC:DDD:INT for an interior shading device and CCC:DDD:EXT for an exterior shading device.

  This method is the required if you want to add a shading device to a construction brought in from a WINDOW Data File (ref:Construction:WindowDataFile).

  Note that if both "Name of Construction with Shading" and "Material Name of Shading Device" are specified, the former takes precedence.

Most Shading Control Types allow you to specify a schedule that determines when the control is active. One example is a control that is active seasonally. For example, to deploy shading only in the summer when the incident solar is high enough, use Shading Control Type = OnIfHighSolarOnWindow with a schedule that is 1 during the summer months and 0 otherwise and specify Shading Control Is Scheduled = YES.

To deploy one shading device during one time period and another during another time period for the same window, specify two different WindowShadingControl objects that both reference the same fenestration. Each of the WindowShadingControl input objects would include the same window on their lists of fenestration objects, but would reference different schedule input objects. The referenced schedule input objects should have non-zero values specified at different times for each WindowShadingControl. This approach can be used to add shading that is different for different seasons. When more than one WindowShadingControl is used for the same window based on schedule, the type of shading, control type, setpoints, and several other parameters must be the same in the different WindowShadingControl objects, but the shading device and construction can be different.

In addition, most Shading Control Types also allow you to specify that glare control is active in addition to the specified Control Type. For example, you might want to deploy shading when the solar incident on a window is too high OR the glare from the window is too high. This type of joint control requires that the window be in a daylit zone, that the maximum allowed glare be specified in the Daylighting object for the zone, and that Glare Control Is Active = YES in WindowShadingControl.

If Shading Type = InteriorBlind, ExteriorBlind or BetweenGlassBlind you can use WindowShadingControl to specify how the slat angle of the blind is controlled when the blind is in place.

A special type of WindowShadingControl is SwitchableGlazing. An example is electrochromic glazing in which the transmittance and reflectance of the glass is controlled electronically. For example, you could have electrochromic glazing switch from clear (high transmittance) to dark (low transmittance) to control solar gain. If you choose the Shading Type = SwitchableGlazing option for WindowShadingControl, the unswitched (clear) state is specified by the Construction referenced by the window and the switched (dark) state is specified by the Construction referenced by WindowShadingControl for that window. For example, if you specify Shading Type = SwitchableGlazing and Shading Control Type = OnIfHighSolarOnWindow, then the glazing will switch to the dark state whenever the solar radiation striking the window exceeds the Setpoint value.

For Shading Type = SwitchableGlazing the state of the window is either clear (unswitched) or dark (fully switched) for all Shading Control Types except MeetDaylightIlluminanceSetpoint. In this case, the transmittance of the glazing is adjusted to just meet the daylight illuminance set point at the first daylighting reference point (see Daylighting). This type of control assures that there is just enough solar gain to meet the daylighting requirements in a zone, and no more, thus reducing the cooling load.

To specify the order of when shades are deployed, the two different approaches can be used:

- If shades for each window are independently controlled, than a single WindowShadingControl object should be used and the Multiple Surface Control Type field should be set to Sequential. The windows should be specified in the Fenestration Surface N Name fields in the order that they should be deployed. The Shading Control Sequence Number field is not used being used since only one WindowShadingControl is used so it can be set to any value or left blank.

- If shades for a group of windows are deployed together and shades for another group of windows are deployed after that, than multiple WindowShadingControl objects should be used. Each WindowShadingControl should have Multiple Surface Control Type field should be set to Group and the names of each window in the group should be specified in the Fenestration Surface N Name fields. The Shading Control Sequence Number in the WindowShadingControl object that controls the first group of shades should be set to 1. The Shading Control Sequence Number in the WindowShadingControl object that controls the second group of shades should be set to 2. Any number of additional groups of shades can be added and Shading Control Sequence Number can be incremented by one for each group.

It is possible to mix these two approaches. For example, if first a group of shades is controlled together followed by a sequence of individual windows than these approaches can be combined with the Multiple Surface Control Type field should be set to Sequential in the second WindowShadingControl object.

1.10.60.1 Inputs

1.10.60.1.1 Field: Name

Name of the window shading control.

1.10.60.1.2 Field: Zone Name

Name of the zone where this shading control is used.

1.10.60.1.3 Field: Shading Control Sequence Number

If multiple WindowShadingControl objects are used in the same zone, then the order that they deploy the window shades can be set with this field. The first WindowShadingControl should be 1 and subsequent WindowShadingControl should 2, 3, etc. This is usually used when the Multiple Surface Control Type field is set to Group to control groups of windows in a certain order.

1.10.60.1.4 Field: Shading Type

The type of shading device. The choices are:

InteriorShade: A diffusing shade is on the inside of the window. (In the shaded Construction the shading layer must be a WindowMaterial:Shade.)

ExteriorShade: A diffusing shade is on the outside of the window. (In the shaded Construction the shading layer must be a WindowMaterial:Shade.)

BetweenGlassShade: A diffusing shade is between two glass layers. (In the shaded Construction the shading layer must be a WindowMaterial:Shade.) This shading type is allowed only for double- and triple-glazing. For triple-glazing the shade must be between the two inner glass layers.

ExteriorScreen: An insect screen is on the outside of the window. (In the shaded Construction the shading layer must be a WindowMaterial:Screen.)

InteriorBlind: A slat-type shading device, such as a Venetian blind, is on the inside of the window. (In the shaded Construction the shading layer must be a WindowMaterial:Blind.)

ExteriorBlind: A slat-type shading device is on the outside of the window. (In the shaded Construction the shading layer must be a WindowMaterial:Blind.)

BetweenGlassBlind: A slat-type shading device is between two glass layers. (In the shaded Construction the shading layer must be a WindowMaterial:Blind.) This shading type is allowed only for double- and triple-glazing. For triple-glazing the blind must be between the two inner glass layers.

SwitchableGlazing: Shading is achieved by changing the characteristics of the window glass, such as by darkening it.

1.10.60.1.5 Field: Construction with Shading Name

Name of the window Construction that has the shading in place. The properties of the shading device are given by the shading material referenced in that Construction (ref: Construction, WindowMaterial:Shade, WindowMaterial:Screen and WindowMaterial:Blind). For Shading Type = SwitchableGlazing, this is the name of the Construction that corresponds to the window in its fully-switched (darkest) state.

Specifying "Name of Construction with Shading" is required if Shading Type = BetweenGlassShade, BetweenGlassBlind, or SwitchableGlazing. For other Shading Types, you may alternatively specify "Material Name of Shading Device" (see below).

1.10.60.1.6 Field: Shading Control Type

Specifies how the shading device is controlled, i.e., it determines whether the shading device is "on" or "off." For blinds, screens and shades, when the device is "on" it is assumed to cover all of the window except its frame; when the device is "off" it is assumed to cover none of the window (whether "on" or "off" the shading device is assumed to cover none of the wall that the window is on).

For switchable glazing, "on" means that the glazing is in the fully-switched state and "off" means that it is in the unswitched state; for example, for electrochromic glazing, "on" means the glazing is in its darkest state and "off" means it is in its lightest state.

The choices for Shading Control Type are the following. If SetPoint is applicable its units are shown in parentheses.

AlwaysOn: Shading is always on.

AlwaysOff: Shading is always off.

The following six control types are used primarily to reduce zone cooling load due to window solar gain.

OnIfScheduleAllows: Shading is on if schedule value is non-zero. Requires that Schedule Name be specified and Shading Control Is Scheduled = Yes.

Note: For exterior window screens *AlwaysOn, AlwaysOff, and OnIfScheduleAllows* are the only valid shading control types.

OnIfHighSolarOnWindow: Shading is on if beam plus diffuse solar radiation incident on the window exceeds SetPoint (W/m^2) and schedule, if specified, allows shading.

OnIfHighHorizontalSolar: Shading is on if total (beam plus diffuse) horizontal solar irradiance exceeds SetPoint (W/m^2) and schedule, if specified, allows shading.

OnIfHighOutdoorAirTemperature: Shading is on if outside air temperature exceeds SetPoint (C) and schedule, if specified, allows shading.

OnIfHighZoneAirTemperature: Shading is on if zone air temperature in the previous timestep exceeds SetPoint (C) and schedule, if specified, allows shading.

OnIfHighZoneCooling: Shading is on if zone cooling rate in the previous timestep exceeds SetPoint (W) and schedule, if specified, allows shading.

OnIfHighGlare: Shading is on if the total daylight glare index at the zone's first daylighting reference point from all of the exterior windows in the zone exceeds the maximum glare index specified in the daylighting input for zone (ref: Group – Daylighting). Applicable only to windows in zones with daylighting.

Note: Unlike other Shading Control Types, glare control is active whether or not a schedule is specified.

MeetDaylightIlluminanceSetpoint: Used only with ShadingType = SwitchableGlazing in zones with daylighting controls. In this case the transmittance of the glazing is adjusted to just meet the daylight illuminance set point at the first daylighting reference point. Note that the daylight illuminance set point is specified in the Daylighting:Controls object for the Zone; it is not specified as a WindowShadingControl SetPoint. When the glare control is active, if meeting the daylight illuminance set point at the first daylighting reference point results in higher discomfort glare index (DGI) than the specified zone's maximum allowable DGI for either of the daylight reference points, the glazing will be further dimmed until the DGI equals the specified maximum allowable value.

The following three control types can be used to reduce zone heating load during the winter by reducing window conductive heat loss at night and leaving the window unshaded during the day to maximize solar gain. They are applicable to any Shading Type except ExteriorScreen but are most appropriate for interior or exterior shades with high insulating value ("movable insulation"). "Night" means the sun is down and "day" means the sun is up.

OnNightIfLowOutdoorTempAndOffDay: Shading is on at night if the outside air temperature is less than SetPoint (C) and schedule, if specified, allows shading. Shading is off during the day.

OnNightIfLowInsideTempAndOffDay: Shading is on at night if the zone air temperature in the previous timestep is less than SetPoint (C) and schedule, if specified, allows shading. Shading is off during the day.

OnNightIfHeatingAndOffDay: Shading is on at night if the zone heating rate in the previous timestep exceeds SetPoint (W) and schedule, if specified, allows shading. Shading is off during the day.

The following two control types can be used to reduce zone heating and cooling load. They are applicable to any Shading Type except ExteriorScreen but are most appropriate for translucent interior or exterior shades with high insulating value ("translucent movable insulation").

OnNightIfLowOutdoorTempAndOnDayIfCooling: Shading is on at night if the outside air temperature is less than SetPoint (C). Shading is on during the day if the zone cooling rate in the previous timestep is non-zero. Night and day shading is subject to schedule, if specified.

OnNightIfHeatingAndOnDayIfCooling: Shading is on at night if the zone heating rate in the previous timestep exceeds SetPoint (W). Shading is on during the day if the zone cooling rate in the previous timestep is non-zero. Night and day shading is subject to schedule, if specified.

The following control types can be used to reduce zone cooling load. They are applicable to any Shading Type except ExteriorScreen but are most appropriate for interior or exterior blinds, interior or exterior shades with low insulating value, or switchable glazing.

OffNightAndOnDayIfCoolingAndHighSolarOnWindow: Shading is off at night. Shading is on during the day if the solar radiation incident on the window exceeds SetPoint (W/m^2) and if the zone cooling rate in the previous timestep is non-zero. Daytime shading is subject to schedule, if specified.

OnNightAndOnDayIfCoolingAndHighSolarOnWindow: Shading is on at night. Shading is on during the day if the solar radiation incident on the window exceeds SetPoint (W/m^2) and if the zone cooling rate in the previous timestep is non-zero. Day and night shading is subject to schedule, if specified. (This Shading Control Type is the same as the previous one, except the shading is on at night rather than off.)

OnIfHighOutdoorAirTempAndHighSolarOnWindow: Shading is on if the outside air temperature exceeds the Setpoint (C) and if the solar radiation incident on the window exceeds SetPoint 2 (W/m^2).

OnIfHighOutdoorAirTempAndHighHorizontalSolar: Shading is on if the outside air temperature exceeds the Setpoint (C) and if the horizontal solar radiation exceeds SetPoint 2 (W/m^2).

OnIfHighZoneAirTempAndHighSolarOnWindow: Shading is on if the zone air temperature exceeds the Setpoint (C) and if the solar radiation incident on the window exceeds SetPoint 2 (W/m^2).

OnIfHighZoneAirTempAndHighHorizontalSolar: Shading is on if the zone air temperature exceeds the Setpoint (C) and if the horizontal solar radiation exceeds SetPoint 2 (W/m^2).

The following three control types can be used to model human behavior of turning on shading during the day to reduce view luminance or solar gain and leaving shading on until a certain time point. They can only be used in zones with daylighting controls. Window view luminance at the first daylighting reference point is used to compare with luminance setpoint. They are applicable to any Shading Type except ExteriorScreen.

OnIfHighSolarOrHighLuminanceTillMidnight: Shading is on if the solar radiation incident on the window exceeds SetPoint (W/m^2) or the luminance from the window exceeds SetPoint 2 (cd/m^2). When shading is on, it remains on till midnight.

OnIfHighSolarOrHighLuminanceTillSunset: Shading is on if the solar radiation incident on the window exceeds SetPoint (W/m^2) or the luminance from the window exceeds SetPoint 2 (cd/m^2). When shading is on, it remains on till sunset.

OnIfHighSolarOrHighLuminanceTillNextMorning: Shading is on if the solar radiation incident on the window exceeds SetPoint (W/m^2) or the luminance from the window exceeds SetPoint 2 (cd/m^2). When shading is on, it remains on till sunrise next day.

1.10.60.1.7 Field: Schedule Name

Required if Shading Control Is Scheduled = Yes. If schedule value > 0 , shading control is active, i.e., shading can be on only if the shading control test passes. If schedule value = 0, shading is off whether or not the control test passes. If Schedule Name is not specified, shading control is assumed to be active at all times.

1.10.60.1.8 Field: Setpoint

The setpoint for activating window shading. The units depend on the type of trigger:

- W/m^2 for solar-based controls
- W for cooling- or heating-based controls
- Degrees C for temperature-based controls

SetPoint is unused for Shading Control Type = OnIfScheduleAllows, OnIfHighGlare and DaylightIlluminance.

1.10.60.1.9 Field: Shading Control Is Scheduled

Accepts values YES and NO. The default is NO. Not applicable for Shading Control Type = OnIfHighGlare and should be blank in that case.

If YES, Schedule Name is required and that schedule determines whether the shading control specified by Shading Control Type is active or inactive (see Schedule Name, above).

If NO, Schedule Name is not applicable (should be blank) and the shading control is unscheduled.

Shading Control Is Scheduled = YES is required if Shading Control Type = OnIfScheduleAllows.

1.10.60.1.10 Field: Glare Control Is Active

Accepts values YES and NO. The default is NO.

If YES and the window is in a daylit zone, shading is on if the zone's discomfort glare index exceeds the maximum discomfort glare index specified in the Daylighting object referenced by the zone. For switchable windows with *MeetDaylightIlluminanceSetpoint* shading control, if Glare Control is active, the windows are always continuously dimmed as necessary to meet the zone's maximum allowable DGI while providing appropriate amount of daylight for the zone.

The glare test is OR'ed with the test specified by Shading Control Type. For example, if Glare Control Is Active = YES and Shading Control Type = OnIfHighZoneAirTemp, then shading is on if glare is too high OR if the zone air temperature is too high.

Glare Control Is Active = YES is required if Shading Control Type = OnIfHighGlare.

1.10.60.1.11 Field: Shading Device Material Name

The name of a WindowMaterial:Shade, WindowMaterial:Screen or WindowMaterial:Blind. Required if "Name of Construction with Shading" is not specified. Not applicable if Shading Type = BetweenGlassShade, BetweenGlassBlind or SwitchableGlazing and should be blank in this case. If both "Name of Construction with Shading" and "Material Name of Shading Device" are entered the former takes precedence.

1.10.60.1.12 Field: Type of Slat Angle Control for Blinds

Applies only to Shading Type = InteriorBlind, ExteriorBlind or BetweenGlassBlind. Specifies how the slat angle is controlled. The choices are FixedSlatAngle, ScheduledSlatAngle and BlockBeamSolar.

If FixedSlatAngle (the default), the angle of the slat is fixed at the value input for the WindowMaterial:Blind that is contained in the construction specified by Name of Construction with Shading or is specified by Material Name of Shading Device.

If ScheduledSlatAngle, the slat angle varies according to the schedule specified by Slat Angle Schedule Name, below.

If BlockBeamSolar, the slat angle is set each timestep to just block beam solar radiation. If there is no beam solar on the window the slat angle is set to the value input for the WindowMaterial:Blind that is contained in the construction specified by Name of Construction with Shading or is specified by Material Name of Shading Device. The BlockBeamSolar option prevents beam solar from entering the window and causing possible unwanted glare if the beam falls on work surfaces while at the same time allowing near-optimal indirect radiation for daylighting.

1.10.60.1.13 Field: Slat Angle Schedule Name

This is the name of a schedule of slat angles that is used when Type of Slat Angle Control for Blinds = ScheduledSlatAngle. You should be sure that the schedule values fall within the range given by the Minimum Slat Angle and Maximum Slat Angle values entered in the corresponding WindowMaterial:Blind. If not, the program will force them into this range.

1.10.60.1.14 Field: Setpoint 2

Used only as the second setpoint for the following setpoint control types: OnIfHighOutdoorAirTempAndHighSolarC OnIfHighOutdoorAirTempAndHighHorizontalSolar, OnIfHighZoneAirTempAndHighSolarOnWindow, OnIfHighZoneAirTempAndHighHorizontalSolar, OnIfHighSolarOrHighLuminanceTillMidnight, OnIfHighSolarOrHigh and OnIfHighSolarOrHighLuminanceTillNextMorning.

1.10.60.1.15 Field: Daylighting Controls Object Name

Name of the Daylighting:Controls object that provides the glare and illuminance control to the zone.

1.10.60.1.16 Field: Multiple Surface Control Type

The field can have one of two options:

Sequential: The following list of fenestration surfaces are controlled individually in the order specified.

Group: The entire list of fenestration surfaces is controlled simultaneously, and if glare control is needed, the entire group of window shades are deployed together a the same time.

1.10.60.1.17 Field: Fenestration Surface <n> Name

The name of a FenestrationSurface:Detailed, Window, or GlazedDoor object controlled by this WindowShadingControl. This field can be repeated to apply the same shading control to more than one fenestration surface. When Multiple Surface Control Type is set to Sequential, the order of the Fenestration Surface Names is the order that the shades will be deployed. The object is extensible so additional fields of fenestration surface names can be added to the object. All of the fenestration surfaces must be either in the zone specified in the Zone Name field or in an adjacent zone connected by an interior window.

An IDF example: window with interior roll shade that is deployed when solar incident on the window exceeds 50 W/m^2.

```
! Example 1: Interior movable shade specified by giving name of shaded construction
! in WindowShadingControl

WindowMaterial:Glazing, GLASS - CLEAR SHEET 1 / 8 IN,  !- Material Name
       SpectralAverage,! Optical data type {SpectralAverage or Spectral}
       ,                ! Name of spectral data set when Optical Data Type = Spectral
       0.003        ,  !- Thickness {m}
       0.837        ,  !- Solar Transmittance at Normal Incidence
       0.075        ,  !- Solar Reflectance at Normal Incidence: Front Side
       0.075        ,  !- Solar Reflectance at Normal Incidence: Back Side
       0.898        ,  !- Visible Transmittance at Normal Incidence
       0.081        ,  !- Visible Reflectance at Normal Incidence: Front Side
       0.081        ,  !- Visible Reflectance at Normal Incidence: Back Side
       0.0          ,  !- IR Transmittance
       0.8400000    ,  !- IR Emissivity: Front Side
       0.8400000    ,  !- IR Emissivity: Back Side
       0.9000000    ;  !- Conductivity {W/m-K}

WindowMaterial:Shade, ROLL SHADE,  !- Material Name
       0.3          ,   !- Solar Transmittance at normal incidence
       0.5000000    ,   !- Solar Reflectance (same for front and back side)
       0.3          ,   !- Visible Transmittance at normal incidence
       0.5000000    ,   !- Visible reflectance (same for front and back side)
       0.9000000    ,   !- IR Emissivity (same for front and back side)
       0.05         ,   !- IR Transmittance
       0.003        ,   !- Thickness
       0.1          ,   !- Conductivity {W/m-K}
       0.0          ,   !- Top Opening Multiplier
       0.0          ,   !- Bottom Opening Multiplier
       0.5          ,   !- Left-Side Opening Multiplier
       0.5          ,   !- Right-Side Opening Multiplier
       0.0          ;   !- Air-Flow Permeability

Construction, SINGLE PANE WITH NO SHADE,  ! Name of construction without shade
       GLASS - CLEAR SHEET 1 / 8 IN;  !- First material layer

Construction, SINGLE PANE WITH INT SHADE, ! Name of construction with shade
       GLASS - CLEAR SHEET 1 / 8 IN,  !- First material layer
       ROLL SHADE                  ;  !- Second material layer

WindowShadingControl,
       CONTROL ON INCIDENT SOLAR,    !- Name
       West Zone,                    !- Zone Name
       1,                            !- Shading Control Sequence Number
       InteriorShade,                !- Shading Type
       SINGLE PANE WITH INT SHADE,   !- Name of construction with shading device
       OnIfHighSolarOnWindow,        !- Shading Control Type
```

```
       ,                             !- Schedule name
       50.0,                         !- Setpoint {W/m2}
       NO,                           !- Shading Control Is Scheduled
       NO,                           !- Glare Control Is Active
       ,                             !- Material Name of Shading Device
       ,                             !- Type of Slat Angle Control for Blinds
       ,                             !- Slat Angle Schedule Name
       ,                             !- Setpoint 2 {W/m2, deg C or cd/m2}
       West Zone_DaylCtrl,           !- Daylighting Control Object Name
       Sequential,                   !- Multiple Surface Control Type
       Zn001:Wall001:Win001;         !- Fenestration Surface 1 Name

FenestrationSurface:Detailed,
       Zn001:Wall001:Win001,         !- SubSurface Name
       Window                  ,     !- Class
       SINGLE PANE WITH NO SHADE,    !- Name of construction without shading device
       Zn001:Wall001           ,     !- Base Surface Name
       ,                             !- Target
       0.5000000               ,     !- VF to Ground
       ,                             !- Frame/Divider name
       1.0                     ,     !- Multiplier
       4                       ,     !- Number of vertices (assumed rectangular)
       0.548 ,  0.0 ,   2.5    ,     !- x,y,z of vertices {m}
       0.548 ,  0.0 ,   0.5    ,
       5.548 ,  0.0 ,   0.5    ,
       5.548 ,  0.0 ,   2.5    ;
```

```
! Example 2: Interior movable shade specified by giving name of shading device in WindowShadingControl

WindowMaterial:Glazing, GLASS - CLEAR SHEET 1 / 8 IN,  !- Material Name
       SpectralAverage,! Optical data type {SpectralAverage or Spectral}
       ,              ! Name of spectral data set when Optical Data Type = Spectral
       0.003        ,  !- Thickness {m}
       0.837        ,  !- Solar Transmittance at Normal Incidence
       0.075        ,  !- Solar Reflectance at Normal Incidence: Front Side
       0.075        ,  !- Solar Reflectance at Normal Incidence: Back Side
       0.898        ,  !- Visible Transmittance at Normal Incidence
       0.081        ,  !- Visible Reflectance at Normal Incidence: Front Side
       0.081        ,  !- Visible Reflectance at Normal Incidence: Back Side
       0.0          ,  !- IR Transmittance
       0.8400000    ,  !- IR Emissivity: Front Side
       0.8400000    ,  !- IR Emissivity: Back Side
       0.9000000    ;  !- Conductivity {W/m-K}

WindowMaterial:Shade, ROLL SHADE,  !- Material Name
       0.3          ,   !- Solar Transmittance at normal incidence
       0.5000000    ,   !- Solar Reflectance (same for front and back side)
       0.3          ,   !- Visible Transmittance at normal incidence
       0.5000000    ,   !- Visible reflectance (same for front and back side)
       0.9000000    ,   !- IR Emissivity (same for front and back side)
       0.05         ,   !- IR Transmittance
       0.003        ,   !- Thickness
       0.1          ,   !- Conductivity {W/m-K}
       0.0          ,   !- Top Opening Multiplier
       0.0          ,   !- Bottom Opening Multiplier
       0.5          ,   !- Left-Side Opening Multiplier
       0.5          ,   !- Right-Side Opening Multiplier
       0.0          ;   !- Air-Flow Permeability

Construction, SINGLE PANE WITH NO SHADE,  ! Name of construction without shade
       GLASS - CLEAR SHEET 1 / 8 IN;  !- First material layer

WindowShadingControl,
       CONTROL ON INCIDENT SOLAR,    !- Name
       West Zone,                    !- Zone Name
       1,                            !- Shading Control Sequence Number
       InteriorShade,                !- Shading Type
       ,                             !- Name of shaded construction
       OnIfHighSolarOnWindow,        !- Shading Control Type
```

```
       ,                             !- Schedule name
       50.0,                         !- Setpoint {W/m2}
       NO,                           !- Shading Control Is Scheduled
       NO,                           !- Glare Control Is Active
       ROLL SHADE,                   !- Material Name of Shading Device
       ,                             !- Type of Slat Angle Control for Blinds
       ,                             !- Slat Angle Schedule Name
       ,                             !- Setpoint 2 {W/m2, deg C or cd/m2}
       West Zone_DaylCtrl,           !- Daylighting Control Object Name
       Sequential,                   !- Multiple Surface Control Type
       Zn001:Wall001:Win001;         !- Fenestration Surface 1 Name

FenestrationSurface:Detailed,
       Zn001:Wall001:Win001,         !- SubSurface Name
       Window                   ,    !- Class
       SINGLE PANE WITH NO SHADE,    !- Name of construction without shade
       Zn001:Wall001            ,    !- Base Surface Name
       ,                             !- Target
       0.5000000                ,    !- VF to Ground
       ,                             !- Frame/Divider name
       1.0                      ,    !- Multiplier
       4                        ,    !- Number of vertices (assumed rectangular)
       0.548 ,  0.0 ,   2.5     ,    !- x,y,z of vertices {m}
       0.548 ,  0.0 ,   0.5     ,
       5.548 ,  0.0 ,   0.5     ,
       5.548 ,  0.0 ,   2.5     ;
```

1.10.61 WindowProperty:FrameAndDivider

The WindowProperty:FrameAndDivider object is referenced by exterior windows that have

- a frame, and/or
- a divider, and/or
- reveal surfaces that reflect beam solar radiation.

A ***frame*** surrounds the glazing in a window (see Figure 1.37 and Figure 1.38). It is assumed that all frame characteristics—such as width, conductance and solar absorptance—are the same for the top, bottom and side elements of the frame. If the frame elements are not the same then you should enter area-weighted average values for the frame characteristics.

The window vertices that you specify in the FenestrationSurface:Detailed object are those of the glazed part of the window, not the frame. EnergyPlus automatically subtracts the area of the frame—determined from the glazing dimensions and the frame width—from the area of the wall containing the window.

A ***divider***, as shown in Figure 1.37, Figure 1.38 and Figure 1.39, divides the glazing up into separate lites. It is assumed that all divider elements have the same characteristics. If not, area-weighted average values should be used. EnergyPlus automatically subtracts the divider area from the glazed area of the window.

Reveal surfaces, as shown in Figure 1.40, are associated with the setback of the glazing from the outside and/or inside surface of the parent wall. If the depth and solar absorptance of these surfaces are specified, the program will calculate the reflection of beam solar radiation from these surfaces. The program also calculates the shadowing (onto the window) of beam and diffuse solar radiation by outside reveal surfaces.

In EnergyPlus, a window can have any combination of frame, divider and reveal surfaces, or none of these.

The best source of frame and divider characteristics is the WINDOW program, which will calculate the values required by EnergyPlus for different frame and divider types. In particular, the THERM program within the WINDOW program will calculate the effective conductance of frames and dividers; this is the conductance taking 2-D heat transfer effects into account.

Note that a window's frame and divider characteristics, along with other window information, can be read in from the Window Data File (see "Importing Windows from the WINDOW program" and "Construction:WindowDataFile object"). In this case the WindowProperty:FrameAndDivider referenced by the window is not applicable and should be blank unless you want to specify reveal surfaces for beam solar reflection.

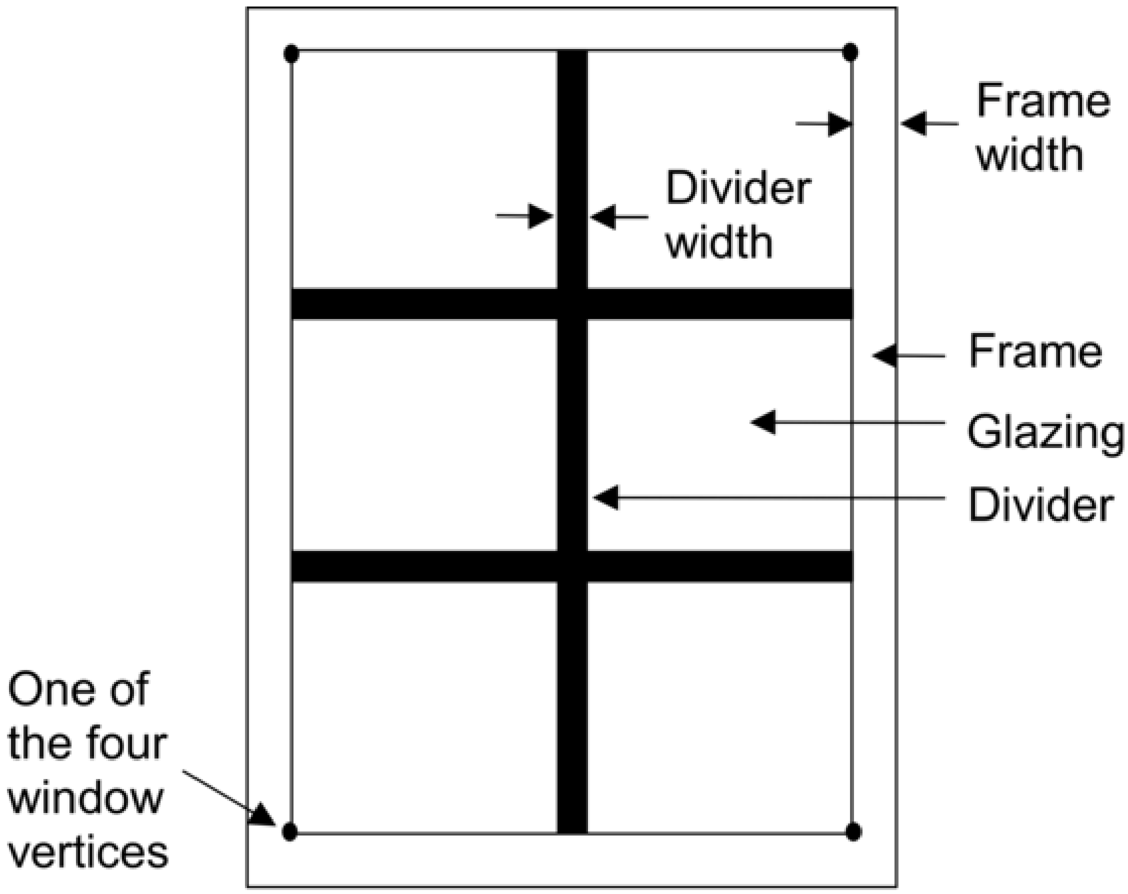

Figure 1.37: A window with a frame and divider.

In the illustration above, the divider has two horizontal elements and one vertical element.

1.10.61.1 Inputs

1.10.61.1.1 Field: Name

The name of the frame/divider object. It is referenced by WindowProperty:FrameAndDivider Name in FenestrationSurface:Detailed.

Frame Fields

1.10.61.1.2 Field: Frame Width

The width of the frame elements when projected onto the plane of the window. It is assumed that the top, bottom and side elements of the frame have the same width. If not, an average frame width should be entered such that the projected frame area calculated using the average value equals the sum of the areas of the frame elements.

1.10.61.1.3 Field: Frame Outside Projection

The amount by which the frame projects outward from the outside surface of the window glazing. If the outer surface of the frame is flush with the glazing, Frame Outside Projection = 0.0. Used to calculate shadowing of frame onto glass, solar absorbed by frame, IR emitted and absorbed by frame, and convection from frame.

1.10.61.1.4 Field: Frame Inside Projection

The amount by which the frame projects inward from the inside surface of the window glazing. If the inner surface of the frame is flush with the glazing, Frame Inside Projection = 0.0. Used to calculate solar absorbed by frame, IR emitted and absorbed by frame, and convection from frame.

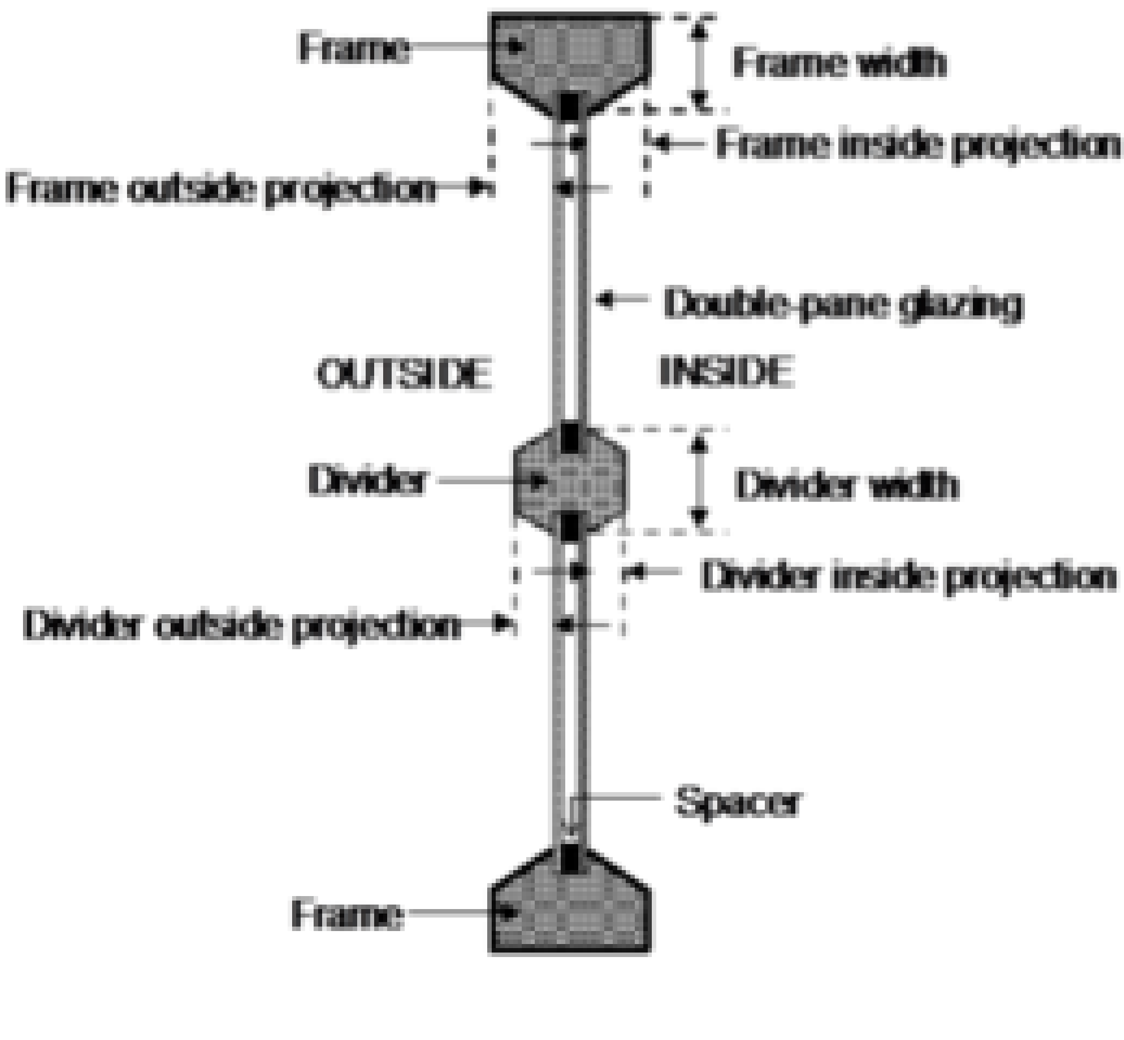

Figure 1.38: Illustration showing frame and divider dimensioning.

1.10.61.1.5 Field: Frame Conductance

The effective thermal conductance of the frame measured from inside to outside frame surface (no air films) and taking 2-D conduction effects into account. Obtained from the WINDOW program or other 2-D calculation.

1.10.61.1.6 Field: Ratio of Frame-Edge Glass Conductance to Center-Of-Glass Conductance

The glass conductance near the frame (excluding air films) divided by the glass conductance at the center of the glazing (excluding air films). Used only for multi-pane glazing constructions. This ratio is greater than 1.0 because of thermal bridging from the glazing across the frame and across the spacer that separates the glass panes. Values can be obtained from the WINDOW program the user-selected glazing construction and frame characteristics.

1.10.61.1.7 Field: Frame Solar Absorptance

The solar absorptance of the frame. The value is assumed to be the same on the inside and outside of the frame and to be independent of angle of incidence of solar radiation. If solar reflectance (or reflectivity) data is available, then absorptance is equal to 1.0 minus reflectance (for opaque materials).

1.10.61.1.8 Field: Frame Visible Absorptance

The visible absorptance of the frame. The value is assumed to be the same on the inside and outside of the frame and to be independent of angle of incidence of solar radiation. If visible reflectance (or reflectivity) data is available, then absorptance is equal to 1.0 minus reflectance (for opaque materials).

1.10.61.1.9 Field: Frame Thermal Hemispherical Emissivity

The thermal emissivity of the frame, assumed the same on the inside and outside.

Divider Fields

1.10.61.1.10 Field: Divider Type

The type of divider (see figure below). Divider Type = Suspended is applicable only to multi-pane glazing. It means that the divider is suspended between the panes. (If there are more than two glass layers, the divider is assumed to be placed between the two outermost layers.)

Divider Type = DividedLite means that the divider elements project out from the outside and inside surfaces of the glazing and divide the glazing into individual lites. For multi-pane glazing, this type of divider also has between-glass elements that separate the panes.

1.10.61.1.11 Field: Divider Width

The width of the divider elements when projected onto the plane of the window. It is assumed that the horizontal and vertical divider elements have the same width. If not, an average divider width should be entered such that the projected divider area calculated using the average value equals the sum of the areas of the divider elements.

1.10.61.1.12 Field: Number of Horizontal Dividers

The number of divider elements parallel to the top and bottom of the window.

1.10.61.1.13 Field: Number of Vertical Dividers

The number of divider elements parallel to the sides of the window.

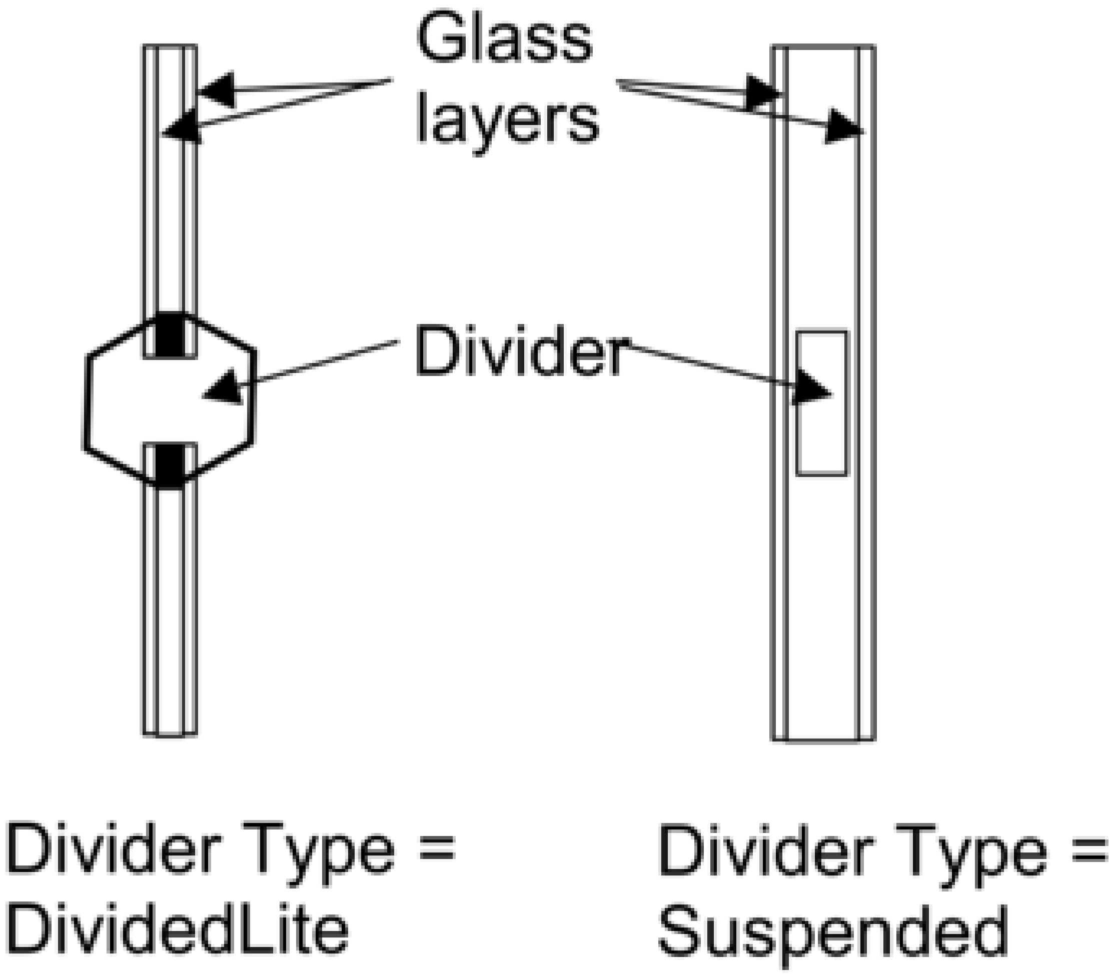

Figure 1.39: Illustration showing divider types.

1.10.61.1.14 Field: Divider Outside Projection

The amount by which the divider projects out from the outside surface of the window glazing. For Divider Type = Suspended, Divider Projection = 0.0. Used to calculate shadowing of divider onto glass, solar absorbed by divider, IR emitted and absorbed by divider, and convection from divider.

1.10.61.1.15 Field: Divider Inside Projection

The amount by which the divider projects inward from the inside surface of the window glazing. If the inner surface of the divider is flush with the glazing, Divider Inside Projection = 0.0. Used to calculate solar absorbed by divider, IR emitted and absorbed by divider, and convection from divider.

1.10.61.1.16 Field: Divider Conductance

The effective thermal conductance of the divider measured from inside to outside divider surface (no air films) and taking 2-D conduction effects into account. Obtained from the WINDOW program or other 2-D calculation.

1.10.61.1.17 Field: Ratio of Divider-Edge Glass Conductance to Center-Of-Glass Conductance

The glass conductance near the divider (excluding air films) divided by the glass conductance at the center of the glazing (excluding air films). Used only for multi-pane glazing constructions. This ratio is greater than 1.0 because of thermal bridging from the glazing across the divider and across the spacer that separates the glass panes. Values can be obtained from the WINDOW program for the user-selected glazing construction and divider characteristics.

1.10.61.1.18 Field: Divider Solar Absorptance

The solar absorptance of the divider. The value is assumed to be the same on the inside and outside of the divider and to be independent of angle of incidence of solar radiation. If solar reflectance (or reflectivity) data is available, then absorptance is equal to 1.0 minus reflectance (for opaque materials).

1.10.61.1.19 Field: Divider Visible Absorptance

The visible absorptance of the divider. The value is assumed to be the same on the inside and outside of the divider and to be independent of angle of incidence of solar radiation. If visible reflectance (or reflectivity) data is available, then absorptance is equal to 1.0 minus reflectance (for opaque materials).

1.10.61.1.20 Field: Divider Thermal Hemispherical Emissivity

The thermal emissivity of the divider, assumed the same on the inside and outside.

Reveal Surface Fields

The following fields specify the properties of the window reveal surfaces (reveals occur when the window is not in the same plane as the base surface). From this information and from the geometry of the window and the sun position, the program calculates beam solar radiation absorbed and reflected by the top, bottom, right and left sides of outside and inside window reveal surfaces. In doing this calculation, the shadowing on a reveal surface by other reveal surfaces is determined using the orientation of the reveal surfaces and the sun position.

It is assumed that:

- The window is an exterior window.
- The reveal surfaces are perpendicular to the window plane.

- If an exterior shade, screen or blind is in place it shades exterior and interior reveal surfaces so that in this case there is no beam solar on these surfaces.
- If an interior shade or blind is in place it shades the interior reveal surfaces so that in this case there is no beam solar on these surfaces.
- The possible shadowing on inside reveal surfaces by a window divider is ignored.
- The outside reveal surfaces (top, bottom, left, right) have the same solar absorptance and depth. This depth is not input here but is automatically determined by the program—from window and wall vertices–as the distance between the plane of the outside face of the glazing and plane of the outside face of the parent wall.
- The inside reveal surfaces are divided into two categories: (1) the bottom reveal surface, called here the "inside sill;" and (2) the other reveal surfaces (left, right and top).
- The left, right and top inside reveal surfaces have the same depth and solar absorptance. The inside sill is allowed to have depth and solar absorptance values that are different from the corresponding values for the other inside reveal surfaces.
- The inside sill depth is required to be greater than or equal to the depth of the other inside reveal surfaces. If the inside sill depth is greater than zero the depth of the other inside reveal surfaces is required to be greater than zero.
- The reflection of beam solar radiation from all reveal surfaces is assumed to be isotropic diffuse; there is no specular component.
- Half of the beam solar reflected from outside reveal surfaces is goes towards the window; the other half goes back to the exterior environment (i.e., reflection of this outward-going component from other outside reveal surfaces is not considered).
- The half that goes towards the window is added to the other solar radiation incident on the window. Correspondingly, half of the beam solar reflected from inside reveal surfaces goes towards the window, with the other half going into the zone. The portion going towards the window that is not reflected is absorbed in the glazing or is transmitted back out into the exterior environment.
- The beam solar that is absorbed by outside reveal surfaces is added to the solar absorbed by the outside surface of the window's parent wall; similarly, the beam solar absorbed by the inside reveal surfaces is added to the solar absorbed by the inside surface of the parent wall.

The net effect of beam solar reflected from outside reveal surfaces is to increase the heat gain to the zone, whereas the effect of beam solar reflected from inside reveal surfaces is to decrease the heat gain to the zone since part of this reflected solar is transmitted back out the window.

If the window has a frame, the absorption of reflected beam solar by the inside and outside surfaces of the frame is considered. The shadowing of the frame onto interior reveal surfaces is also considered.

1.10.61.1.21 Field: Outside Reveal Solar Absorptance

The solar absorptance of outside reveal surfaces.

1.10.61.1.22 Field: Inside Sill Depth

The depth of the inside sill, measured from the inside surface of the glazing to the edge of the sill (see Figure 1.40).

1.10.61.1.23 Field: Inside Sill Solar Absorptance

The solar absorptance of the inside sill.

Field: Inside Reveal Depth

The depth of the inside reveal surfaces other than the sill, measured from the inside surface of the glazing to the edge of the reveal surface (see Figure 1.40).

1.10.61.1.24 Field: Inside Reveal Solar Absorptance

The solar absorptance of the inside reveal surfaces other than the sill.

1.10.61.1.25 Field: NFRC Product Type for Assembly Calculations

The selection made for this field corresponds to NFRC 100 "Procedure for Determining Fenestration Product U-factors" product types which are used when computing the overall u-factor, SHGC, and visual transmittance. The default is CurtainWall. The options are: CasementDouble, CasementSingle, DualAction, Fixed, Garage, Greenhouse, HingedEscape, HorizontalSlider, Jal, Pivoted, Projecting, DoorSidelite, Skylight, SlidingPatioDoor, CurtainWall, SpandrelPanel, SideHingedDoor, DoorTransom, TropicalAwning, TubularDaylightingDevice, and VerticalSlider. The sizes used in the calculation of the overall u-factor, overall SHGC, and overall VT are based on NFRC 100 Table 4-3 (see Figure 1.41).

An IDF example:

```
WindowProperty:FrameAndDivider,
      TestFrameAndDivider, ! Frame/Divider Name
      0.05, ! Frame Width
      0.04, ! Frame Outside Projection
      0.03, ! Frame Inside Projection
      5.0,  ! Frame Conductance
      1.3,  ! Ratio of Frame-Edge Glass Conductance to Center-Of-Glass Conductance
      0.8,  ! Frame Solar Absorptance
      0.8,  ! Frame Visible Absorptance
      0.9,  ! Frame Thermal Emissivity
      DividedLite, ! Divider Type
      0.03, ! Divider Width
      2,    ! Number of Horizontal Dividers
      2,    ! Number of Vertical Dividers
      0.03, ! Divider Outside Projection
      0.03, ! Divider Inside Projection
      5.0,  ! Divider Conductance
      1.3,  ! Ratio of Divider-Edge Glass Conductance to Center-Of-Glass Conductance
      0.8,  ! Divider Solar Absorptance
      0.8,  ! Divider Visible Absorptance
      0.9,  ! Divider Thermal Emissivity
      0.7,  ! Outside Reveal Solar Absorptance
      0.25, ! Inside Sill Depth (m)
      0.6,  ! Inside Sill Solar Absorptance
      0.2,  ! Inside Reveal Depth (m)
      0.5,  ! Inside Reveal Solar Absorptance
      ProjectingSingle; ! NFRC Product Type for Assembly Calculations
```

1.10.62 WindowProperty:AirflowControl

This object is used to specify the control mechanism for windows in which forced air flows in the gap between adjacent layers of glass. Such windows are called "airflow windows." They are also known as "heat-extract windows" or "climate windows."

A common application is to reduce the zone load by exhausting indoor air through the window. In the cooling season this picks up and expels some of the solar heat absorbed by the window glass (and by the between-glass shade or blind, if present). In the heating season this warms the window, reducing the heat

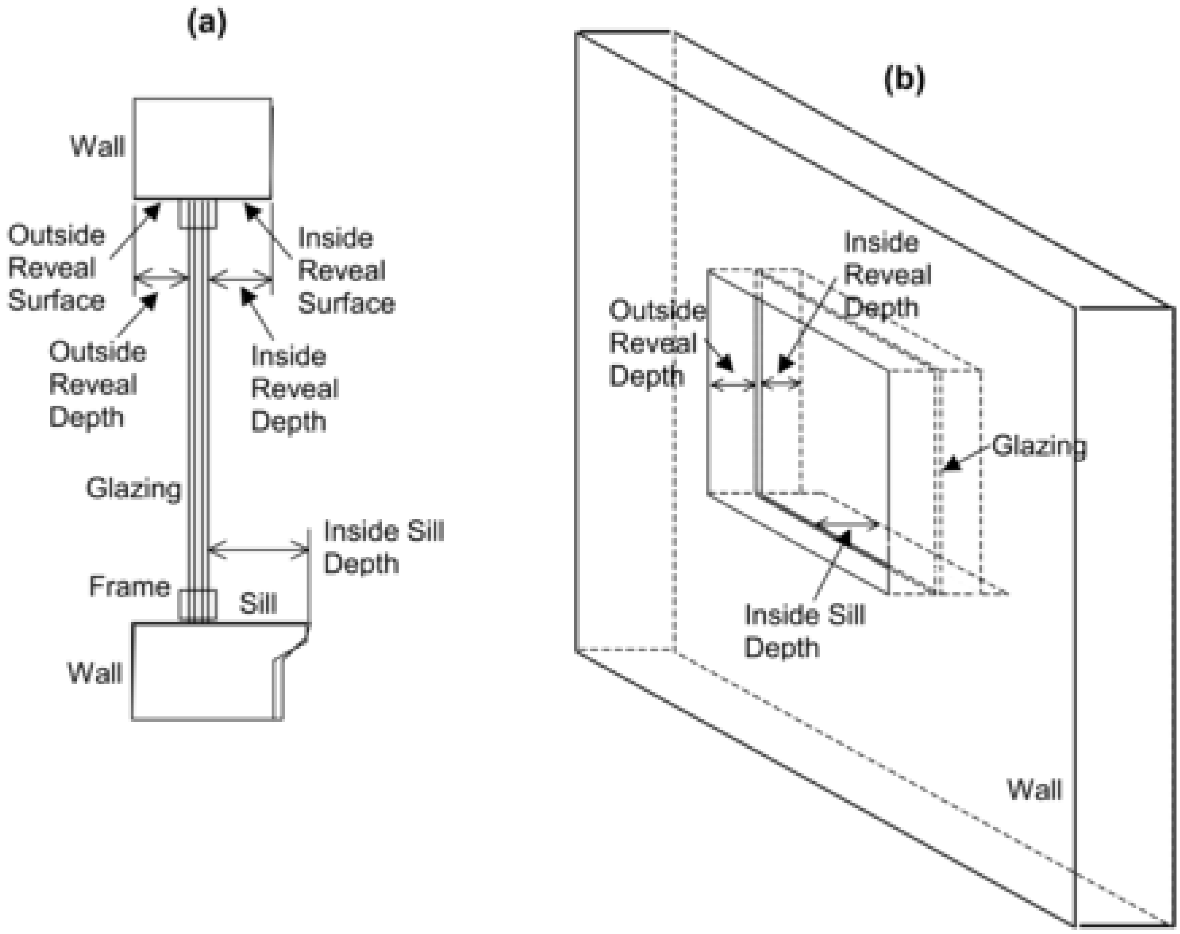

Figure 1.40: (a) Vertical section through a window (with frame) showing outside and inside reveal surfaces and inside sill. (b) Perspective view looking from the outside of a window (without frame) showing reveal surfaces. Note that "Outside Reveal Depth" is not a user input; it is calculated by the program from the window and wall vertices.

Table 4-3 – Product Types and Model Sizes

Product Type	Opening (X) Non-operating (O)	Model Size (width by height) SI (IP)
Casement – Double[1]	XX, XO, OO	1200 mm x 1500 mm (47 in x 59 in)
Casement – Single	X[6]	600 mm x 1500 mm (24 in x 59 in)
Dual Action	X[6]	1200mm x 1500 mm (47 in x 59 in)
Fixed (includes non-standard shapes)	O	1200 mm x 1500 mm (47 in x 59 in)
Garage (Vehicular Access)/Rolling Door	X[6]	2134 mm x 2134 mm (84 in x 84 in)
Greenhouse/Garden[2]	X[6]	1500 mm x 1200 mm (59 in x 47 in)
Hinged Escape	X[6]	1500 mm x 1200 mm (59 in x 47 in)
Horizontal Slider	XO or XX	1500 mm x 1200 mm (59 in x 47 in)
Jal/Jal Awning	X[6]	1200 mm x 1500 mm (47 in x 59.in)
Pivoted	X[6]	1200 mm x 1500 mm (47 in x 59 in)
Projecting (Awning, Dual)	XX[6]	1500 mm x 1200 mm (59 in x 47 in)
Projecting (Awning – Single)	X[6]	1500 mm x 600 mm (59 in x 24 in)
Door Sidelite[5]	X or O	600 mm x 2090 mm (24 in x 82 3/8 in)
Skylight/Roof Window	X[2]	1200 mm x 1200 mm (47 in x 47 in)
Sliding Patio Door with Frame	XO or XX[8]	2000 mm x 2000 mm (79 in x 79 in)
Curtain Wall/Window Wall/Storefront/Sloped Glazing	OO[3]	2000 mm x 2000 mm (79 in x 79 in)
Spandrel Panel	OO	2000 mm x 1200 mm (79 in x 47 in)
Side-Hinged Exterior Door	O, X, XO or XX[4]	960 mm x 2090 mm (37 3/4 in x 82 3/8 in); or 1920 mm x 2090 mm (75 1/2 in x 82 3/8 in)
Door Transom[5,7]	O	2000 mm x 600 mm (79 in x 24 in)
Tropical Awning	X[6]	1500 mm x 1200 mm (59 in x 47 in)
Tubular Daylighting Device	O	350 mm Dia. (14 in Dia.)
Vertical Slider	XO or XX	1200 mm by 1500 mm (47 in by 59 in)

Figure 1.41: NFRC 100 Product Types and Model Sizes.

loss from the window. A side benefit is increased thermal comfort. This is because the inside surface of the window will generally be cooler in summer and warmer in winter.

The surface output variable “Surface Window Gap Convective Heat Transfer Rate” gives the heat picked up (or lost) by the gap airflow.

1.10.62.1 Inputs

1.10.62.1.1 Field: Name

Name of the window that this WindowProperty:AirflowControl refers to. It must be a window with two or three glass layers, i.e., double- or triple-glazing. For triple-glazing the airflow is assumed to be between the two inner glass layers.

An error will result if the gas in the airflow gap is other than air. If an airflow window has a between-glass shade or blind, the gas in the gap on either side of the shade or blind must be air.

1.10.62.1.2 Field: Airflow Source

The source of the gap airflow. The choices are:

IndoorAir: Indoor air from the window’s zone is passed through the window.

OutdoorAir: Outdoor air is passed through the window.

1.10.62.1.3 Field: Airflow Destination

This is where the gap air goes after passing through the window. The choices are:

IndoorAir: The gap air goes to the indoor air of the window’s zone.

OutdoorAir: The gap air goes to the outside air.

ReturnAir. The gap air goes to the return air for the window’s zone. This choice is allowed only if Airflow Source = InsideAir. If the return air flow is zero, the gap air goes to the indoor air of the window’s zone. If the sum of the gap airflow for all of the windows in a zone with Airflow Destination = ReturnAir exceeds the return airflow, then the difference between this sum and the return airflow goes to the indoor air. The name of the Return Air Node may be specified below.

Figure 1.42 shows the allowed combinations of Airflow Source and Airflow Destination. The allowed combinations of Airflow Source and Airflow Destination are:

IndoorAir -> OutdoorAir

IndoorAir -> IndoorAir

IndoorAir -> ReturnAir

OutdoorAir -> IndoorAir

OutdoorAir -> OutdoorAir

1.10.62.1.4 Field: Maximum Flow Rate

The maximum value of the airflow, in m^3/s per m of glazing width. The value is typically 0.006 to 0.009 m^3/s-m (4 to 6 cfm/ft).

The airflow can be modulated by specifying Airflow Has Multiplier Schedule = Yes and giving the name of the Airflow Multiplier Schedule (see below).

The fan energy used to move the air through the gap is generally very small and so is ignored.

1.10.62.1.5 Field: Airflow Control Type

Specifies how the airflow is controlled. The choices are:

AlwaysOnAtMaximumFlow. The airflow is always equal to Maximum Airflow.

AlwaysOff. The airflow is always zero.

ScheduledOnly. The airflow in a particular timestep equals Maximum Airflow times the value of the Airflow Multiplier Schedule for that timestep.

1.10.62.1.6 Field: Airflow Is Scheduled

Specifies if the airflow is scheduled. The choices are:

Yes. The airflow is scheduled.

No. The airflow is not scheduled.

If Yes, Airflow Multiplier Schedule Name is required.

1.10.62.1.7 Field: Airflow Multiplier Schedule Name

The name of a schedule with values between 0.0 and 1.0. The timestep value of the airflow is Maximum Airflow times the schedule value. Required if Airflow Is Scheduled = Yes. Unused if Airflow Is Scheduled = No. This schedule should have a ScheduleType with Numeric Type = Continuous and Range = 0.0 : 1.0.

1.10.62.1.8 Field: Airflow Return Air Node Name

The name of the return air node for this airflow window if the Airflow Destination is ReturnAir. If left blank, this defaults to the first return air node for the zone containing the window surface.

An IDF example: window with a constant airflow from inside to outside at 0.008 m^3/s-m.

```
WindowProperty:AirflowControl,
      Zn001:Wall001:Win002,   !- Name
      InsideAir,              !- Airflow Source
      OutsideAir,             !- Airflow Destination
      0.008,                  !- Maximum Flow Rate {m3/s-m}
      AlwaysOnAtMaxFlow,      !- Airflow Control Type
      No,                     !- Airflow Is Scheduled
      ;                       !- Airflow Multiplier Schedule Name
      ,                       !- Airflow Return Air Node Name
```

1.10.63 WindowProperty:StormWindow

This object allows you to assign a movable exterior glass layer ("storm window" or "storm glass") that is usually applied to a window in the winter to reduce heat loss and removed in the summer. A WindowProperty:StormWindow object is required for each window that has an associated storm window. It is assumed that:

- When the storm glass is in place it is the outermost layer of the window, it covers only the glazed part of the window and not the frame, and it forms a tight seal. See Figure 1.43.

- When the storm glass is not in place it is completely removed and has no effect on window heat transfer.

- The gap between the storm glass and rest of the glazing is filled with air.

With the addition of a storm window, single glazing effectively becomes double glazing, double glazing becomes triple glazing, etc.

The presence of a storm window is indicated by the output variable "Surface Storm Window On Off Status" (see "Window Output Variables"). This flag is **0** if the storm window is off, **1** if it is on, and **−1** if the window does not have an associated storm window.

The program automatically creates a window construction (ref: Construction) that consists of the storm window glass layer and its adjacent air layer added to the original (unshaded, or "bare") window construction. In the eplusout.eio file this construction is called BARECONSTRUCTIONWITHSTORMWIN:*n*, where *n* is the number of the associated StormWin object. If the window has a shaded construction, the program creates a construction called SHADEDCONSTRUCTIONWITHSTORMWIN:*n* that consists of the storm window glass layer and its adjacent air layer added to the original shaded window construction.

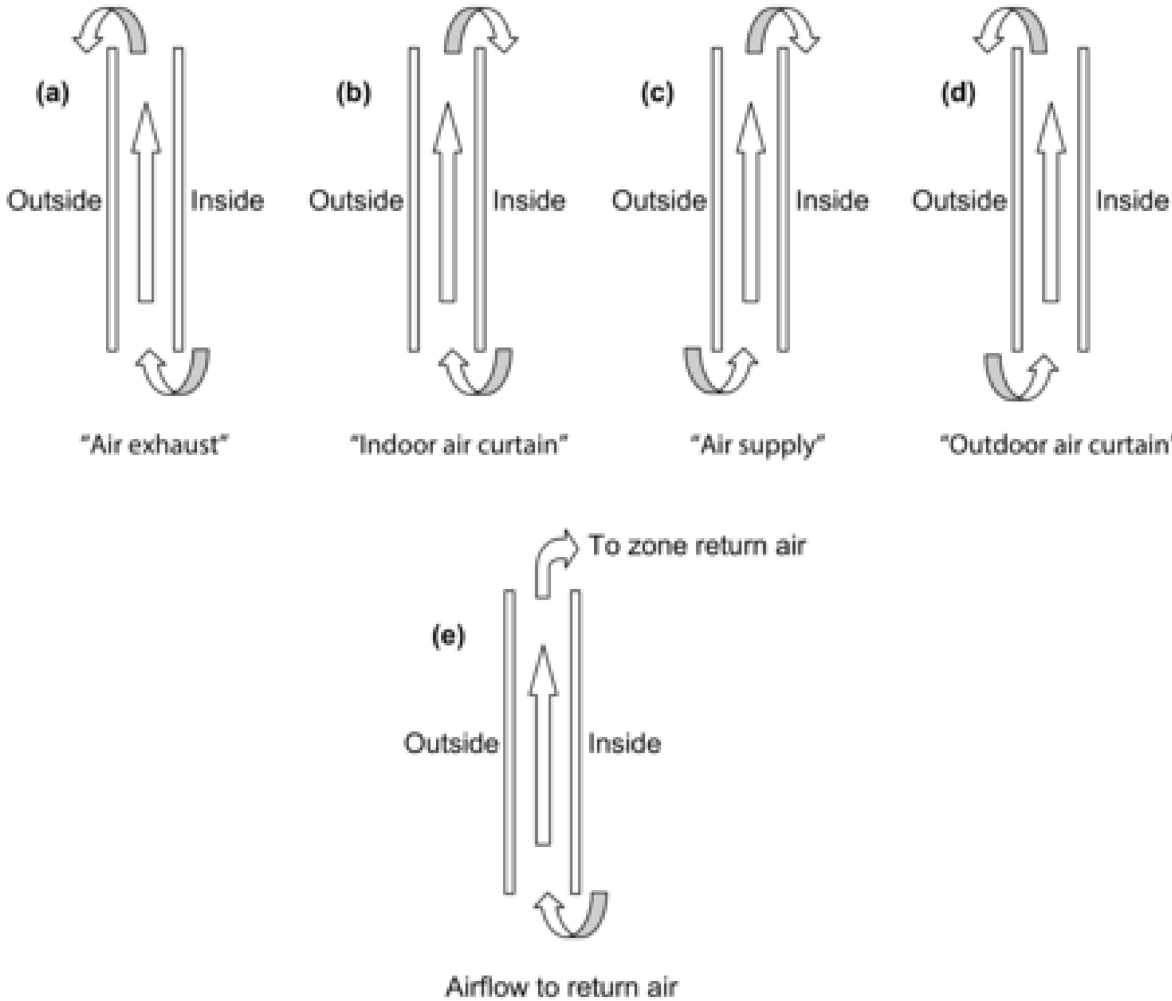

Figure 1.42: Gap airflow configurations for airflow windows. (a) **Air exhaust window**: Airflow Source = InsideAir, Airflow Destination = OutsideAir; (b) **Indoor air curtain window**: Airflow Source = InsideAir, Airflow Destination = InsideAir; (c) **Air supply window**: Airflow Source = OutsideAir, Airflow Destination = InsideAir; (d) **Outdoor air curtain window**: Airflow Source = OutsideAir, Airflow Destination = OutsideAir; (e) **Airflow to Return Air**: Airflow Source = InsideAir, Airflow Destination = ReturnAir. Based on "Active facades," Version no. 1, Belgian Building Research Institute, June 2002.

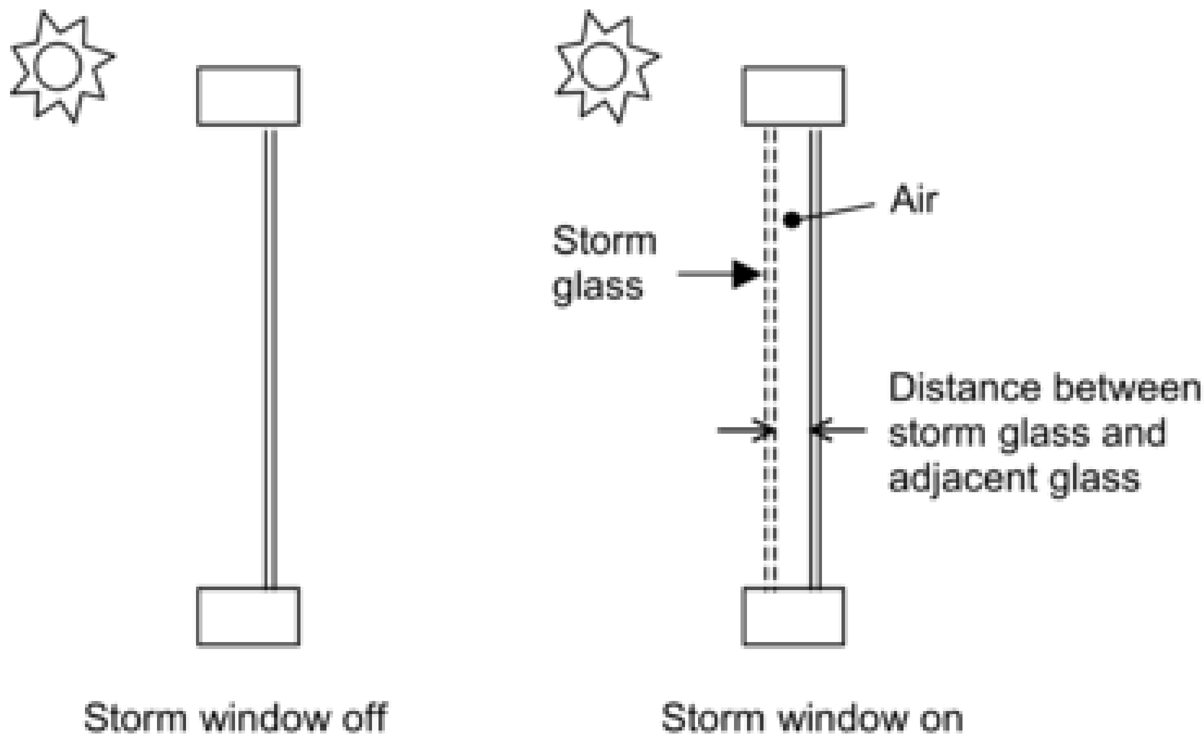

Figure 1.43: Section through a single-glazed window without (left) and with (right) a storm glass layer. Not to scale.

The program also creates a WindowMaterial:Gas layer corresponding to the air layer adjacent to the storm glass. In the eplusout.eio file this layer is called AIR:STORMWIN:*k*MM, where *k* is the thickness of the air layer expressed as an integer number of millimeters.

1.10.63.1 Inputs

1.10.63.1.1 Field: Window Name

This is the name of a window (or glass door) to which the storm glass is applied. Not all windows can accept WindowProperty:StormWindow. The rules are:

- The window must be an exterior window. WindowProperty:StormWindow is not applicable to interior (interzone) windows.
- The window construction (without the storm glass layer) can have up to three glass layers.
- If the window has an associated shaded construction (ref: WindowShadingControl), that construction can have an interior shade or blind and up to three glass layers, or a between-glass shade or blind and two glass layers. The shaded construction cannot have an exterior shade or blind, cannot have a between-glass shade or blind and three glass layers, and cannot be switchable glazing.
- The window cannot be an airflow window, i.e., a window that has an associated WindowProperty:AirflowControl.

1.10.63.1.2 Field: Storm Glass Layer Name

This is the name of a window glass material. Storm windows are assumed to consist of a single layer of glass. A storm window frame, if present, is ignored.

1.10.63.1.3 Field: Distance Between Storm Glass Layer and Adjacent Glass

The separation between the storm glass and the rest of the window (Figure 1.43). It is measured from the inside of the storm glass layer to the outside of the adjacent glass layer.

1.10.63.1.4 Field: Month that Storm Glass Layer Is Put On

The number of the month (January = 1, February = 2, etc.) during which the storm window is put in place.

1.10.63.1.5 Field: Day of Month that Storm Glass Layer Is Put On

The day of the month that the storm window is put in place. It is assumed that the storm window is put in place at the beginning of this day, i.e., during the first simulation timestep of the day, and remains in place until that month and day given by the following two fields.

1.10.63.1.6 Field: Month that Storm Glass Layer Is Taken Off

The number of the month (January = 1, February = 2, etc.) during which the storm window is removed.

1.10.63.1.7 Field: Day of Month that Storm Glass Layer Is Taken Off

The day of the month that the storm window is removed. It is assumed that the storm window is removed at the beginning of this day, i.e., during the first simulation timestep of the day, and stays off until the month and day given by Month that Storm Glass Layer Is Put On, Day of Month that Storm Glass Layer Is Put On.

In the northern hemisphere, the month the storm window is put on is generally greater than the month it is taken off (for example put on in month 10, when it starts to get cold, and taken off in month 5, when it starts to warm up). In the southern hemisphere this is reversed: month on is less than month off.

An IDF example of WindowProperty:StormWindow. The storm window is put in place on October 15 and removed on May 1.

```
WindowProperty:StormWindow,
   Window1, !- Name of Window to Which Storm Window Glass Layer is Applied
   GlassA,  !- Name of Material:WindowGlass or MATERIAL:WindowGlass:AltInput that is the storm window
    layer
   0.060,   !- Distance from storm window to adjacent glass (m)
   10,      !- Month that Storm Window Is Put On
   15,      !- Day of Month that Storm Window Is Put On
   5,       !- Month that Storm Window Is Taken Off
   1;       !- Day of Month that Storm Window Is Taken Off
```

1.10.64 Importing Windows from WINDOW program

WINDOW v6.3 and later is capable of writing IDF excerpts for Window data. This is the preferred method as no external file is necessary. See the Tips document for details on obtaining the IDF excerpt.

The WINDOW program calculates the U-value, Solar Heat Gain Coefficient, solar transmission/absorption characteristics, visible transmission characteristics and other properties of a window under standard indoor and outdoor conditions. WINDOW treats the whole window system—glazing, frame and divider. A sub-program of WINDOW called THERM uses a 2-D finite element calculation to determine the effective conductance of frame, divider and edge-of-glass elements. Another sub-program, OPTICS, determines the solar-optical properties of glazing, including laminates and coated glass.

WINDOW can write a data file containing a description of the window that was analyzed. An example of this file (which is no longer the preferred method) is shown in the Tips document under WINDOW generated files. is shown below. This file, which can be named by the user, can be read by EnergyPlus. For more complete description and examples, see the object description – Construction:WindowDataFile.

In this way, the same window that was created in WINDOW can be imported into EnergyPlus for annual energy analysis without having to re-input the window data. To obtain WINDOW, THERM, or OPTICS go to http://windows.lbl.gov and choose the software link. A major advantage of using WINDOW to create window input for EnergyPlus is that you will have direct access to WINDOW's expanding database of over 1000 different glass types; and you will be able to browse through this database according to different criteria (color, transmittance, solar heat gain coefficient, etc.) to help you select the best glass type for your application.

Although WINDOW writes only one window entry on the WINDOW data file, EnergyPlus users can combine two or more of these files to end up with a single data file with multiple window entries of different types. In this way a library of windows from WINDOW can be built up if so desired. If you combine files like this you should be sure not to leave out or change any of lines from the original files.

There are four methods for inputting window constructions in EnergyPlus:

1. input full spectral data for each layer in the IDF,
2. input spectral average data for each layer in the IDF,
3. items 1 and 2 can be accomplished by reporting the IDF excerpt method from WINDOW
4. import WINDOW report containing layer-by-layer calculated values and overall glazing system angular values.

Note: When using method 4, the overall glazing system angular dependent properties, including Tsol, Abs, Rfsol, Rbsol, Tvis, Rfvis, and Rbvis, are not used by EnergyPlus. Therefore, methods 1 and 2 and preferably 3 are recommended.

- The SHGC calculations in EnergyPlus for window layers input using full spectral data use a spectral weighting data set (derived from Optics5 data file ISO-9845GlobalNorm.std) that is different from the WINDOW default spectral weighting data set (W5_NFRC_2003.std). This difference accounts for most of the variation in SHGC values reported by EnergyPlus and WINDOW for full spectral data window layer input. This variation is more pronounced for window constructions of three glass layers or more.

- Users intending to select a window construction based on SHGC value for energy code compliance should base their selection on the value reported by WINDOW since this is the officially recognized value.

In EnergyPlus, the Window data file is searched for each "Construction:WindowDataFile" object in the EnergyPlus input. This object has a very simple form:

```
Construction:WindowDataFile,
  ConstructionName,
  FileName; ! Default is Window5DataFile.dat in the ``run'' folder.
```

If there is a window called ConstructionName on the Window data file, the data for that window is read from the file and the following EnergyPlus objects and their names are created. The "W5" prefixed to these names indicates that the object originated in the Window data file.

1.10.65 Using LBNLs Windows Calculation Engine

This is new windows simulation engine that will be used in both EnergyPlus and Berkeley Lab WINDOW. The intent is for this engine to replace all other window calculation modules in EnergyPlus. For the time being, other modules will still be available. To use the new Window Calculation Engine, insert the following into IDF:

```
WindowsCalculationEngine,
ExternalWindowsModel;
```

1.11 Group – Advanced Surface Concepts

This group of objects describe concepts applied to heat transfer surfaces that are of an advanced nature. Careful consideration must be given before using these.

1.11.1 SurfaceProperty:HeatTransferAlgorithm

This object, and three other related objects, can be used to control which surface heat transfer model is used on specific surfaces. The separate object called HeatBalanceAlgorithm is used to control the heat transfer model in an overall way while this object can be used to revise the algorithm selections for specific surfaces. This object allows selectively overriding the global setting in HeatBalanceAlgorithm to choose one of the following models for a particular surface:

- CTF (Conduction Transfer Functions),
- EMPD (Effective Moisture Penetration Depth with Conduction Transfer Functions).
- CondFD (Conduction Finite Difference)
- HAMT (Combined Heat And Moisture Finite Element)

1.11.1.1 Inputs

1.11.1.1.1 Field: Surface Name

This is the name of the surface that will be assigned to use the heat transfer algorithm selected in the next field. This should be a name of a surface defined elsewhere.

1.11.1.1.2 Field: Algorithm

This field is used to determine the heat transfer algorithm that is to be applied to the surface named in the previous field. The allowable choices are:

- ConductionTransferFunction
- MoisturePenetrationDepthConductionTransferFunction
- ConductionFiniteDifference
- CombinedHeatAndMoistureFiniteElement

1.13.19.3.3 RoomAirflowNetwork Node Relative Humidity [%]

The relative humidity ratio at a Room Air Node in a zone in percent.

1.13.19.3.4 HVAC,Average,RoomAirflowNetwork Node SumIntSensibleGain [W]

A sum of internal sensible gains assigned to a Room Air node in a zone.

1.13.19.3.5 HVAC,Average,RoomAirflowNetwork Node SumIntLatentGain [W]

A sum of internal latent gains assigned to a Room Air node in a zone.

1.13.19.3.6 HVAC,Average,RoomAirflowNetwork Node NonAirSystemResponse [W]

A sum of system convective gains, collected via NonAirSystemResponse, and assigned to a Room Air node in a zone.

1.13.19.3.7 HVAC,Average,RoomAirflowNetwork Node SysDepZoneLoadsLagged [W]

A sum of SysDepZoneLoads saved to be added to zone heat balance at the next HVAC time step. A typical system is a zone dehumidifier.

1.14 Group – Internal Gains (People, Lights, Other internal zone equipment)

Not all the influence for energy consumption in the building is due to envelope and ambient conditions. This group of objects describes other internal gains that may come into play (People, Lights, Various Equipment Types).

1.14.1 Specifying Applicable Zone(s) or Space(s)

Each internal gains object has a field for "Zone or ZoneList or Space or SpaceList Name". All internal gains are modeled at the Space level, so each internal gain input object may be expanded to multiple instances of that gain according to the following rules with names constructed using the <Object Name>, <Zone Name>, or <Space Name>. If the resulting concatenated name is greater than 100 characters a warning will be shown and it will be truncated. If the resulting name duplicates another such concatenated name, there will be a severe error and terminate the run. The concatenated name is used when referring to specific instances of internal gains for Output:Variables, Energy Management System (EMS) controls, and Demand Limiting Controls.

Space Name will result in one instance of the internal gain, named <Object Name>. The full magnitude of the gain will be applied to the Space using the design level, space floor area, or space occupancy as appropriate.

SpaceList Name will result in one instance of the internal gain for each Space in the SpaceList, named <Space Name> <Object Name>. The full magnitude of the gain will be applied to each Space in the SpaceList using the design level, space floor area, or space occupancy as appropriate.

Zone Name will result in one instance of the internal gain for each Space in the Zone. If there is only one Space in the Zone, then the single instance will be named <Object Name>. If there is more than one Space in the Zone, then each instance will be named <Space Name> <Object Name>. The full magnitude of the gain will be split between the Spaces in the Zone apportioned by the Space floor area or occupancy depending on the input method.

ZoneList Name will result in one instance of the internal gain for each Space in each Zone in the ZoneList. Each instance will be named <Space Name> <Object Name>. The full magnitude of the gain will be applied to each Zone in the ZoneList then split between the Spaces in each Zone apportioned by the Space floor area or occupancy depending on the input method.

One exception to this is ElectricEquipment:ITE:AirCooled which only allows Zone or Space Name. If a Zone Name is specified which contains more than one Space, the same rules apply.

As an example, assume that Zone 1 contains Space 1A and Space 1B, and Zone 2 contains only Space 2 (objects are abbreviated for clarity):

```
Space,
Space 1A, !- Name
Zone 1,   !- Zone Name
75.0,     !- Floor Area

Space,
Space 1B, !- Name
Zone 1;   !- Zone Name
25.0,     !- Floor Area

Space,
Space 2,  !- Name
Zone 2,   !- Zone Name

SpaceList,
All Spaces, !- Name
Space 1A,   !- Space 1 Name
Space 1B,   !- Space 2 Name
Space 2,    !- Space 3 Name

People,
Office People, !- Name
Zone 1,     !- Zone or ZoneList or Space or SpaceList Name
100.0,      !- Number of People

People,
Lunchroom People, !- Name
Zone 2,     !- Zone or ZoneList or Space or SpaceList Name
50.0,       !- Number of People

People,
Cleaning Crew, !- Name
All Spaces, !- Zone or ZoneList or Space or SpaceList Name
2.0,        !- Number of People

Resulting instances of People:

Space 1A Office People, 75 people. (concatenated name, apportioned by space floor area)
Space 1B Office People, 25 people. (concatenated name, apportioned by space floor area)
Lunchroom People, 50 people.       (simple object name for a single space)
Space 1A Cleaning Crew, 2 people.  (full load applied to each space in SpaceList)
Space 1B Cleaning Crew, 2 people.
Space 2 Cleaning Crew, 2 people.
```

1.14.2 People

The people statement is used to model the occupant's effect on the space conditions. The following definition addresses the basic affects as well as providing information that can be used to report the thermal comfort of a group of occupants. The Fanger, Pierce Two-Node, Kansas State University Two-Node, ASHRAE Standard 55 Elevated Air Cooling Effect model, and ASHRAE Standard 55 Ankle Draft

Risk thermal comfort models are available in EnergyPlus. A user may select any of these models for each People statement by simply adding the appropriate choice keyword after the air velocity schedule name. Thermal comfort calculations will only be made for people statements that include specific requests for these thermal comfort models. This object also requires input of carbon dioxide generation rate based on people activity level for zone carbon dioxide simulations.

1.14.2.1 Inputs

1.14.2.1.1 Field: Name

The name of the People object. Must be unique across all People objects.

1.14.2.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this People object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the people/area and area/person options of the Number of People Calculation Method to place a varying number of people at the same density in each zone or space. The names of the actual people objects may be concatenated as <Space Name> <People Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

When multiple people objects are defined for the same zone, and they have different SET, PMV, Heat/Cold Stress Temperature Threshold, a warning message will be produced to caution users that zone-level resilience metrics results will only be meaningful if each zone has at most one People object defined. This is especially true for SET Degree-Hours and Discomfort-Weighted Exceedance Hours tables. In the SET Degree-Hour case, this is because in the calculation of zone level Heating/Cooling SET Degree-Hours, if multiple People objects are defined for one zone, EnergyPlus will only use the SET of one People object to compute the zone level SET Degree-Hours, and neglecting the SET of other people object defined in the same zone. As a result, when multiple people objects with different SET, PMV, or Heat/Cold Stress Temperature are linked to one zone, the corresponding zone-level resilience metrics results might be invalid.

1.14.2.1.3 Field: Number of People Schedule Name

This field is the name of the schedule (ref: Schedules) that modifies the number of people parameter (see Number of People Calculation Method and related fields). The schedule values can be any positive number. The actual number of people in a zone as defined by this statement is the product of the number of people field and the value of the schedule specified by name in this field.

1.14.2.1.4 Field: Number of People Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal number of occupants (people) in the Zone. The key/choices are:

People With this choice, the method used will be a straight insertion of the number of occupants (people). (The Number of People field should be filled.)

People/Area With this choice, the method used will be a factor per floor area of the zone or space. (The People per Zone Floor Area field should be filled).

Area/Person With this choice, the method used will be a factor of floor area per person. (The Zone Floor Area per Person field should be filled).

1.14.2.1.5 Field: Number of People

This field is used to represent the maximum number of people in a zone that is then multiplied by a schedule fraction (see schedule field). In EnergyPlus, this is slightly more flexible in that the number of people could be a “diversity factor” applied to a schedule of real numbers. Note that while the schedule value can vary from hour to hour, the number of people field is constant for all simulation environments.

1.14.2.1.6 Field: People per Floor Area

This factor (person/m^2) is used, along with the Zone or Space Floor Area to determine the maximum number of people as described in the Number of People field. The choice from the method field should be “people/area”.

1.14.2.1.7 Field: Floor Area per Person

This factor (m^2/person) is used, along with the Zone or Space Floor Area to determine the maximum number of people as described in the Number of People field. The choice from the method field should be “area/person”.

1.14.2.1.8 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the type of heat being given off by people in a zone. The number specified in this field will be multiplied by the total sensible energy emitted by people to give the amount of long wavelength radiation gain from human beings in a zone. The remainder of the sensible load is assumed to be convective heat gain. Note that latent gains from people are not included in either the radiant or convective heat gains. See the Engineering Reference document for more details. Default value is 0.30.

1.14.2.1.9 Field: Sensible Heat Fraction

The user can use this field to specify a fixed sensible fraction for the heat gain due to this PEOPLE object. Normally the program calculates the sensible/latent split; this field gives the user control over this split. This field is autocalculated: if the field is blank or **autocalculate**, the program will calculate the sensible/latent split; if a value is entered, it will be used as the sensible fraction of the current total heat gain.

1.14.2.1.10 Field: Activity Level Schedule Name

This field is the name of the schedule that determines the amount of heat gain per person in the zone under design conditions. This heat gain impacts the basic zone heat balance as well as the modeling of thermal comfort. This value is modified somewhat based on a correlation to account for variations in space temperature. The schedule values may be any positive number and the units for this parameter is Watts per person. This schedule represents the total heat gain per person including convective, radiant, and latent. An internal algorithm is used to determine what fraction of the total is sensible and what fraction is latent. Then, the sensible portion is divided into radiant and convective portions using the value specified for Fraction Radiant (above). See the Engineering Reference document for more details.

Values for activity level can range anywhere from approximately 100-150 Watts per person for most office activities up to over 900 Watts per person for strenuous physical activities such as competitive wrestling. The following table (Table 1.29) is based on Table 1.10 from the 2005 ASHRAE Handbook of Fundamentals, page 8.6. In addition to the information from the ASHRAE HOF, there is an added column of values in W/Person such as necessary for the activity level schedule values. This column uses the standard adult body surface area of 1.8 m^2 to multiply the activity levels in W/m^2 that are used in the table. Warnings are produced when the activity level schedule values fall outside normal ranges. Having too low or too high values can also skew thermal comfort reporting values.

1.14.2.1.11 Field: Carbon Dioxide Generation Rate

This numeric input field specifies carbon dioxide generation rate per person with units of m3/s-W. The total carbon dioxide generation rate from this object is:

Number of People * People Schedule * People Activity * Carbon Dioxide Generation Rate. The default value is 3.82E-8 m3/s-W (obtained from ASHRAE Standard 62.1-2007 value at 0.0084 cfm/met/person over the general adult population). The maximum value can be 10 times the default value.

1.14.2.1.12 Field: Cold Stress Temperature Threshold

This field is used in the calculation of hours of safety for cold events, which is defined as the longest duration (number of hours), starting from the beginning time of the risk period (e.g., the start time of a power outage), to not below a certain temperature threshold (specified in this field). If not specified, 60F (or 15.56C) will be used as the default temperature threshold.

1.14.2.1.13 Field: Heat Stress Temperature Threshold

This field is used in the calculation of hours of safety for hot events, which is defined as the longest duration (number of hours), starting from the beginning time of the risk period (e.g., the start time of a power outage), to not above a certain temperature threshold (specified in this field). If not specified, 86F (or 30C) will be used as the default temperature threshold.

1.14.2.1.14 Field: Enable ASHRAE 55 comfort warnings

This field accepts either "Yes" or "No" as values. When "Yes" is specified, warnings are generated when the space conditions are outside of the ASHRAE 55 comfort range as discussed in the sections that follow titled "Simplified ASHRAE 55-2004 Graph Related Outputs" and "Simplified ASHRAE 55 Warnings." The default is not to provide these warnings so if you want to know if your space is outside this comfort range you must set this field to Yes.

Table 1.29: Metabolic Rates for Various Activities

Activity	Activity Level Schedule Value (W/Person)	Activity Level (W/m^2)	met*
Resting			
Sleeping	72	40	0.7
Reclining	81	45	0.8
Seated, quiet	108	60	1
Standing, relaxed	126	70	1.2
Walking (on level surface)			
3.2 km/h (0.9 m/s)	207	115	2
4.3 km/h (1.2 m/s)	270	150	2.6
6.4 km/h (1.8 m/s)	396	220	3.8
Office Activities			
Reading, seated	99	55	1
Writing	108	60	1
Typing	117	65	1.1
Filing, seated	126	70	1.2

Table 1.29: Metabolic Rates for Various Activities

Activity	Activity Level Schedule Value (W/Person)	Activity Level (W/m^2)	met*
Filing, standing	144	80	1.4
Walking about	180	100	1.7
Lifting/packing	216	120	2.1
Miscellaneous Occupational Activities			
Cooking	171 to 207	95 to 115	1.6 to 2.0
Housecleaning	207 to 360	115 to 200	2.0 to 3.4
Seated, heavy limb movement	234	130	2.2
Machine work	189	105	1.8
sawing (table saw)	207 to 252	115 to 140	2.0 to 2.4
light (electrical industry)	423	235	4
Handling 50 kg bags	423	235	4
Pick and shovel work	423 to 504	235 to 280	4.0 to 4.8
Miscellaneous Leisure Activities			
Dancing, social	252 to 459	140 to 255	2.4 to 4.4
Calisthenics/exercise	315 to 423	175 to 235	3.0 to 4.0
Tennis, singles	378 to 486	210 to 270	3.6 to 4.0
Basketball, competitive	522 to 792	290 to 440	5.0 to 7.6
Wrestling, competitive	738 to 909	410 to 505	7.0 to 8.7

***Note that one met = 58.1 W/m^2**

1.14.2.1.15 Field: Mean Radiant Temperature Calculation Type

This field specifies the type of Mean Radiant Temperature (MRT) calculation the user wishes to use for the thermal comfort model. At the present time, there are three options for MRT calculation type: enclosure averaged, surface weighted, and a list of angle factors. The default calculation is “EnclosureAveraged” and is used if field is left blank. In the enclosure averaged MRT calculation, the MRT used for the thermal comfort calculations is for an “average” point in the zone. MRT is calculated based on an area-emissivity weighted average of all of the surfaces in the same radiant enclosure (one or more Spaces). In cases where the emissivity of all of the surfaces are sufficiently small (near zero), the mean radiant temperature will be set to the mean air temperature of the space to avoid divide by zero errors. The other MRT calculation type is “SurfaceWeighted”. The goal of this calculation type is to estimate a person in the space close to a particular surface without having to define exact view factors for all of the surfaces and the location of the person in the space. The MRT used in the thermal comfort calculations when the “surface weighted” calculation type is selected is actually the average of the temperature of the surface to which the person is closest (defined by the next field “Surface Name”) and the enclosure averaged MRT (defined above). The surface temperature alone is not used because in theory the maximum view factor from a person to any flat surface is roughly 0.5. In the “surfaceweighted” calculation, the surface in question actually gets slightly more weighting than 50% since the surface selected is still a part of the zone average MRT calculation. Again, this simplification was made to avoid the specification of view factors and the exact location of the person.

A third option is to use "AngleFactor". This option allows for more explicit positioning of the person within the space by defining the angle factors from the person to the various surfaces in the enclosure. This option requires the user to list the surfaces that the person can see from a radiation standpoint and also define the angle (or view) factor for each surface. The ComfortViewFactorAngles object (see next object description) is intended to give the user this opportunity.

1.14.2.1.16 Field: Surface Name/Angle Factor List Name

This field is only valid when the user selects "SurfaceWeighted" or "AngleFactor" for the MRT calculation type (see the previous input field description). In the case of "SurfaceWeighted", the field is the name of a surface within the enclosure the people are residing. This surface will be used in the MRT calculation as defined above to come up with a more representative MRT for a person near a particular surface. The MRT used for thermal comfort calculations using the "SurfaceWeighted" MRT calculation method is the average of the temperature of the surface specified in this field and the "zone averaged" MRT (see the Mean Radiant Temperature calculation type field above). In the case of "AngleFactor", the field is the name of a ComfortViewFactorAngles input object defined elsewhere. This field is required when the previous field is set to "SurfaceWeighted" or "AngleFactor" and is set to run one of the following thermal comfort models: Fanger, Pierce, KSU, CoolingEffectASH55 or AnkleDraftASH55.

1.14.2.1.17 Field: Work Efficiency Schedule Name

This field is the name of the schedule that determines the efficiency of energy usage within the human body that will be used for thermal comfort calculations. Note that all energy produced by the body is assumed to be converted to heat for the zone heat balance calculation. A value of zero corresponds to all of the energy produced in the body being converted to heat. A value of unity corresponds to all of the energy produced in the body being converted to mechanical energy. The values for this parameter defined in the schedule must be between 0.0 and 1.0. Any value greater than zero will result in a reduction of heat that impacts the thermal comfort energy balance of a person within the space, resulting in PMV results appearing lower than expected. Ensure that if this value is non-zero, the base activity level is chosen to ensure that the net activity converted to heat and zone conditions are sufficient to maintain thermal comfort. This field is required to run one of the following thermal comfort models: Fanger, Pierce, KSU, CoolingEffectASH55 or AnkleDraftASH55. If a schedule is listed here but no thermal comfort model is selected, then a warning message will be produced and this schedule will be listed as unused in the error file.

1.14.2.1.18 Field: Clothing Insulation Calculation Method

This field is a key/choice field that tells which of the next two fields are filled and is descriptive of the method for calculating the clothing insulation value of occupants (people) in the Zone. The key/choices are:

- ClothingInsulationSchedule

With this choice, the method used will be a straight insertion of the scheduled clothing insulation values of occupants (people). (The Clothing Insulation Schedule Name field should be filled.)

- DynamicClothingModelASHRAE55

With this choice, the method used will be the dynamic predictive clothing insulation model developed by Schiavon and Lee (2013) based on 6,333 selected observations taken from ASHRAE RP-884 and RP-921 databases. It varies the clothing insulation as a function of outdoor air temperature measured at 6am as illustrated below.

- CalculationMethodSchedule

With this choice, the method used can be either the ClothingInsulationSchedule or the DynamicClothingModelASHRAE55, depending on a schedule (to be entered as the next field) that determines which method to use in different time of a day. When this option is chosen, the next field "Clothing Insulation Calculation Method Schedule Name" is a required input.

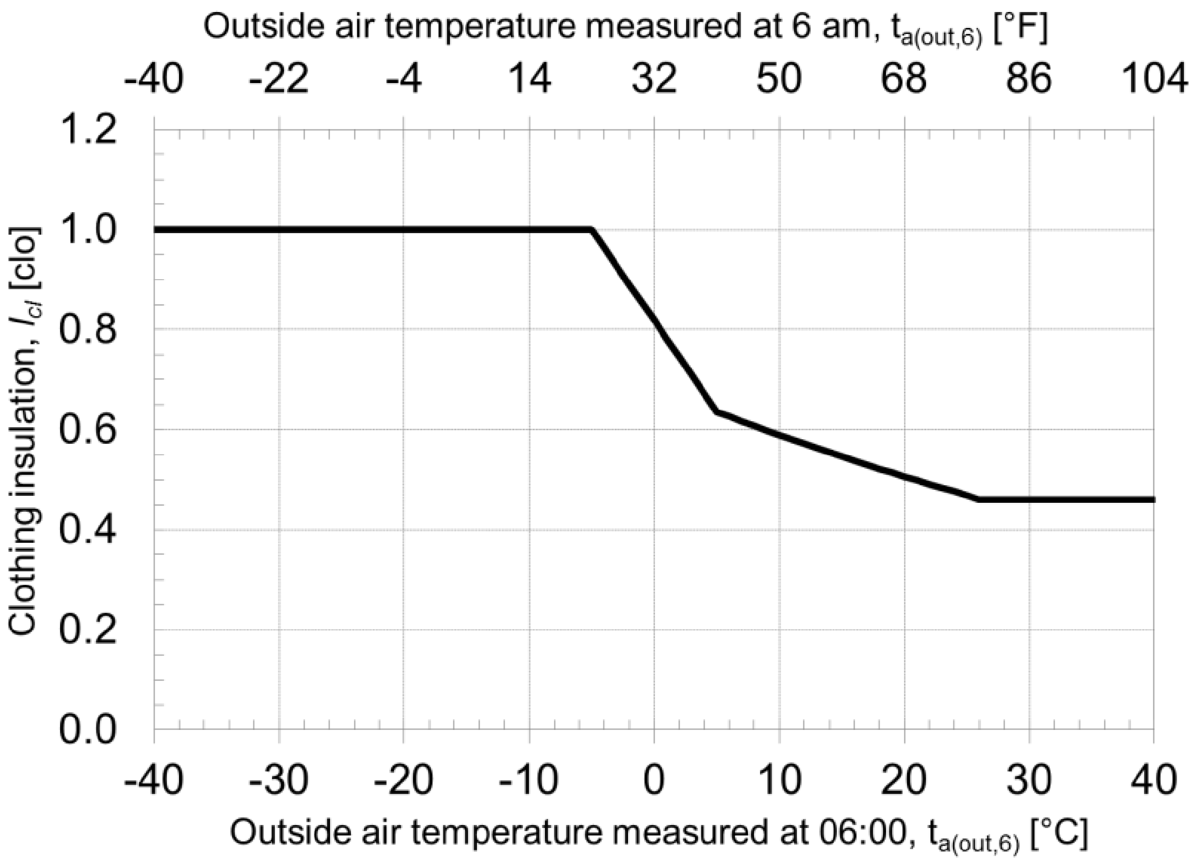

Figure 1.69: Graphical representation of the dynamic predictive clothing insulation model

1.14.2.1.19 Field: Clothing Insulation Calculation Method Schedule Name

This field specifies which clothing insulation method (ClothingInsulationSchedule or DynamicClothingModelASHR to use at a particular time of the day. A schedule value of 1 means the ClothingInsulationSchedule method, and 2 means the DynamicClothingModelASHRAE55 method. This field is only required when the "Clothing Insulation Calculation Method" field is set to **CalculationMethodSchedule**. If this field is left blank, the specified clothing insulation calculation method will be used and not changed during the simulation.

1.14.2.1.20 Field: Clothing Insulation Schedule Name

This field is the name of the schedule that defines the amount of clothing being worn by a typical zone occupant during various times in the simulation period. The choice from the Clothing Insulation Calculation Method field should be "**ClothingInsulationSchedule**". This parameter must be a positive real number and has units of Clo. Typical values for Clo can be seen in the ASHRAE 2009 HOF Table 7, page 9.8 (for clothing ensembles) and Table 1.17, page 9.9 (for garment values)) or https://comfort.cbe.

berkeley.edu/. This field is required to run one of the following thermal comfort models: Fanger, Pierce, KSU, CoolingEffectASH55 or AnkleDraftASH55. If a schedule is listed here but no thermal comfort model is selected, then a warning message will be produced and this schedule will be listed as unused in the error file.

1.14.2.1.21 Field: Air Velocity Schedule Name

This field is the name of the schedule that approximates the amount of air movement in the space as a function of time throughout the simulation period. The user has control over this parameter through the schedule that requires the individual numbers in the schedule to be positive real numbers having units of meters per second. This field is required to run one of the following thermal comfort models: Fanger, Pierce, KSU, CoolingEffectASH55 or AnkleDraftASH55. If a schedule is listed here but no thermal comfort model is selected, then a warning message will be produced and this schedule will be listed as unused in the error file.

1.14.2.1.22 Field: Thermal Comfort Model Type (up to 7 allowed)

The final one to five fields are optional and are intended to trigger various thermal comfort models within EnergyPlus. By entering the keywords Fanger, Pierce, KSU, AdaptiveASH55, AdaptiveCEN15251, CoolingEffectASH55, and AnkleDraftASH55, the user can request the Fanger, Pierce Two-Node, Kansas State UniversityTwo-Node, the adaptive comfort models of the ASHRAE Standard 55 and CEN Standard 15251, ASHRAE Standard 55 Elevated Air Cooling Effect model, and ASHRAE Standard 55 Ankle Draft Risk model results for this particular people statement. Detailed descriptions and requirements of the seven models as listed below.

Fanger Fanger's Comfort model is applied to calculate related thermal comfort metrics. Fanger Model PMV, PPD, and Clothing Surface Temperature are calculated and reported as each time step. Apart from existing required fields in People object, extra fields required for this model include Surface Name/Angle Factor List Name, Work Efficiency Schedule Name, Clothing Insulation Schedule Name, and Air Velocity Schedule Name.

Pierce The Pierce Two-Node model is applied to calculate related thermal comfort metrics. Pierce Model Effective Temperature PMV, Standard Effective Temperature PMV, Discomfort Index, Thermal Sensation Index, and Standard Effective Temperature are calculated and reported as each time step.Apart from existing required fields in People object, extra fields required for this model include Surface Name/Angle Factor List Name, Work Efficiency Schedule Name, Clothing Insulation Schedule Name, and Air Velocity Schedule Name.

KSU The KSU Two-Node Model is applied to calculate related thermal comfort metrics. KSU Model Thermal Sensation Vote is calculated and reported as each time step. Note that the KSU model is computationally intensive and may noticeably increase the execution time of the simulation. Apart from existing required fields in People object, extra fields required for this model include Surface Name/Angle Factor List Name, Work Efficiency Schedule Name, Clothing Insulation Schedule Name, and Air Velocity Schedule Name.

AdaptiveASH55 Adaptive Comfort Model Based on ASHRAE Standard 55-2010 is applied to calculate related thermal comfort metrics. ASHRAE 55 Adaptive Model 90% Acceptability Status, 80% Acceptability Status, Running Average Outdoor Air Temperature, and the Adaptive Model Temperature are calculated and reported as each time step. AdaptiveASH55 is only applicable when the running average outdoor air temperature for the past 7 days is between 10.0 and 33.5C.

AdaptiveCEN15251 Adaptive Comfort Model Based on European Standard EN15251-2007 is applied to calculate related thermal comfort metrics. CEN 15251 Adaptive Model Category I/II/II Status, Running Average Outdoor Air Temperature, and the Adaptive Model Temperature are calculated and reported as each time step. AdaptiveCEN15251 is only applicable when the running average outdoor air temperature for the past 30 days is between 10.0 and 30.0C.

CoolingEffectASH55 ASHRAE 55-2017 Elevated Air Speed Cooling Effect Model is applied to calculate related thermal comfort metrics. Elevated Air Speed Cooling Effect, Cooling Effect Adjusted PMV, and Cooling Effect Adjusted PPD are calculated and reported as each time step. Apart from existing required fields in People object, extra fields required for this model include Surface Name/Angle Factor List Name, Work Efficiency Schedule Name, Clothing Insulation Schedule Name, and Air Velocity Schedule Name.

AnkleDraftASH55 ASHRAE 55-2017 Ankle Draft Risk Model is applied to calculate related thermal comfort metrics. Zone Thermal Comfort ASHRAE 55 Ankle Draft PPD is calculated and reported as each time step. Apart from existing required fields in People object, extra fields required for this model include Surface Name/Angle Factor List Name, Work Efficiency Schedule Name, Clothing Insulation Schedule Name, Air Velocity Schedule Name, and Ankle Level Air Velocity Schedule Name. Ankle draft PPD calculations are only applicable for relative air velocity is below 0.2 m/s, and the subject's metabolic rate and clothing level should be kept below 1.3 met and 0.7 clo. PPD at ankle draft will be set to -1.0 if if these conditions are not met.

For descriptions of the thermal comfort calculations, see the Engineering Reference document.

Note that since up to seven models may be specified, the user may opt to have EnergyPlus calculate the thermal comfort for people identified with this people statement using all seven models if desired.

1.14.2.1.23 Field: Ankle Level Air Velocity Schedule Name

This field is the name of the schedule that approximates the amount of air movement at the occupants' ankle level (0.1 m above floor level) as a function of time throughout the simulation period. The user has control over this parameter through the schedule that requires the individual numbers in the schedule to be positive real numbers having units of meters per second. This field is required to run the AnkleDraftASH55 thermal comfort model. If a schedule is listed here but no thermal comfort model is selected, then a warning message will be produced and this schedule will be listed as unused in the error file.

The following IDF example allows for a maximum of 31 people with scheduled occupancy of "Office Occupancy", 60% radiant using an Activity Schedule of "Activity Sch". The example allows for thermal comfort reporting.

```
People,
  Kitchen_ZN_1_FLR_1,   !- Name
  Kitchen_ZN_1_FLR_1,   !- Zone or ZoneList or Space or SpaceList Name
  BLDG_OCC_SCH,         !- Number of People Schedule Name
  People,               !- Number of People Calculation Method
  25.2000,,,            !- Number of People, People per Zone Floor Area, Zone Floor Area per Person
  0.3000,               !- Fraction Radiant
  AUTOCALCULATE,        !- Sensible Heat Fraction
  ACTIVITY_SCH,         !- Activity Level Schedule Name
  3.82E-8,              !- Carbon Dioxide Generation Rate {m3/s-W}
  No,                   !- Enable ASHRAE 55 Comfort Warnings
  EnclosureAveraged,    !- Mean Radiant Temperature Calculation Type
  ,                     !- Surface Name/Angle Factor List Name
  WORK_EFF_SCH,         !- Work Efficiency Schedule Name
  ClothingInsulationSchedule, !- Clothing Insulation Calculation Method
  ,                     !- Clothing Insulation Calculation Method Schedule Name
  CLOTHING_SCH,         !- Clothing Insulation Schedule Name
  AIR_VELO_SCH,         !- Air Velocity Schedule Name
  Fanger;               !- Thermal Comfort Model 1 Type
```

A simpler example, without using the thermal comfort reporting option:

```
People,
  RIGHT FORK,     !- Name
  RIGHT FORK,     !- Zone or ZoneListName
  Dorm Occupancy, !- Number of People Schedule Name
  people,         !- Number of People Calculation Method
  8.00000,        !- Number of People,
  ,               !- People per Zone Floor Area
  ,               !- Zone Floor Area per Person
  0.6000000,      !- Fraction Radiant
  Autocalculate,  !- Sensible Heat Fraction
  Activity Sch,   !- Activity level Schedule Name
```

And with the sensible fraction specified:

```
People,
  SPACE1-1 People 1,    !- Name
  SPACE1-1,             !- Zone or ZoneListName
  OCCUPY-1,             !- Number of People Schedule Name
  people,               !- Number of People Calculation Method
  11,                   !- Number of People
  ,                     !- People per Zone Floor Area
  ,                     !- Zone Floor Area per Person
  0.3,                  !- Fraction Radiant
  0.55,                 !- Sensible Heat Fraction
  ActSchd;              !- Activity level Schedule Name
```

Global People Object:

```
ZoneList,AllOccupiedZones,SPACE1-1,SPACE2-1,SPACE3-1,SPACE4-1,SPACE5-1;

People,
  AllZones with People, !- Name
  AllOccupiedZones,     !- Zone or ZoneList or Space or SpaceList Name
  OCCUPY-1,             !- Number of People Schedule Name
  People/Area,          !- Number of People Calculation Method
  ,                     !- Number of People
  .11,                  !- People per Floor Area {person/m2}
  ,                     !- Floor Area per Person {m2/person}
  0.3,                  !- Fraction Radiant
  ,                     !- Sensible Heat Fraction
  ActSchd,              !- Activity Level Schedule Name
  3.82E-8,              !- Carbon Dioxide Generation Rate {m3/s-W}
  No,                   !- Enable ASHRAE 55 Comfort Warnings
  surfaceweighted,      !- Mean Radiant Temperature Calculation Type
  Zn001:Wall001,        !- Surface Name/Angle Factor List Name
  Work Eff Sch,         !- Work Efficiency Schedule Name
  ClothingInsulationSchedule,  !- Clothing Insulation Calculation Method
  ,                     !- Clothing Insulation Calculation Method Schedule Name
  Clothing Sch,         !- Clothing Insulation Schedule Name
  Air Velo Sch,         !- Air Velocity Schedule Name
  FANGER,               !- Thermal Comfort Model 1 Type
  PIERCE,               !- Thermal Comfort Model 2 Type
  KSU;                  !- Thermal Comfort Model 3 Type
```

1.14.2.2 Outputs

People objects have output variables for individual objects and for space and zone totals.

People specific outputs include:

- Zone,Average,People Occupant Count []
- Zone,Sum,People Radiant Heating Energy [J]
- Zone,Average,People Radiant Heating Rate [W]
- Zone,Sum,People Convective Heating Energy [J]

- Zone,Average,People Convective Heating Rate [W]
- Zone,Sum,People Sensible Heating Energy [J]
- Zone,Average,People Sensible Heating Rate [W]
- Zone,Sum,People Latent Gain Energy [J]
- Zone,Average,People Latent Gain Rate [W]
- Zone,Sum,People Total Heating Energy [J]
- Zone,Average,People Total Heating Rate [W]
- Zone,Average,Space People Occupant Count []
- Zone,Sum,Space People Radiant Heating Energy [J]
- Zone,Average,Space People Radiant Heating Rate [W]
- Zone,Sum,Space People Convective Heating Energy [J]
- Zone,Average,Space People Convective Heating Rate [W]
- Zone,Sum,Space People Sensible Heating Energy [J]
- Zone,Average,Space People Sensible Heating Rate [W]
- Zone,Sum,Space People Latent Gain Energy [J]
- Zone,Average,Space People Latent Gain Rate [W]
- Zone,Sum,Space People Total Heating Energy [J]
- Zone,Average,Space People Total Heating Rate [W]
- Zone,Average,Zone People Occupant Count []
- Zone,Sum,Zone People Radiant Heating Energy [J]
- Zone,Average,Zone People Radiant Heating Rate [W]
- Zone,Sum,Zone People Convective Heating Energy [J]
- Zone,Average,Zone People Convective Heating Rate [W]
- Zone,Sum,Zone People Sensible Heating Energy [J]
- Zone,Average,Zone People Sensible Heating Rate [W]
- Zone,Sum,Zone People Latent Gain Energy [J]
- Zone,Average,Zone People Latent Gain Rate [W]
- Zone,Sum,Zone People Total Heating Energy [J]
- Zone,Average,Zone People Total Heating Rate [W]
- Zone,Average,People Air Temperature [C]

- Zone,Average,People Air Relative Humidity [%]
- Zone,Average,Zone Thermal Comfort Mean Radiant Temperature [C]
- Zone,Average,Zone Thermal Comfort Operative Temperature [C]
- Zone,Average,Zone Thermal Comfort Fanger Model PMV []
- Zone,Average,Zone Thermal Comfort Fanger Model PPD [%]
- Zone,Average,Zone Thermal Comfort Clothing Surface Temperature [C]
- Zone,Average,Zone Thermal Comfort Pierce Model Effective Temperature PMV []
- Zone,Average,Zone Thermal Comfort Pierce Model Standard Effective Temperature PMV []
- Zone,Average,Zone Thermal Comfort Pierce Model Discomfort Index []
- Zone,Average,Zone Thermal Comfort Pierce Model Thermal Sensation Index []
- Zone,Average,Zone Thermal Comfort Pierce Model Standard Effective Temperature [C]
- Zone,Average,Zone Thermal Comfort KSU Model Thermal Sensation Index []
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Adaptive Model 80% Acceptability Status []
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Adaptive Model 90% Acceptability Status []
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Adaptive Model Running Average Outdoor Air Temperature [C]
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Adaptive Model Temperature [C]
- Zone,Average,Zone Thermal Comfort CEN 15251 Adaptive Model Category I Status []
- Zone,Average,Zone Thermal Comfort CEN 15251 Adaptive Model Category II Status []
- Zone,Average,Zone Thermal Comfort CEN 15251 Adaptive Model Category III Status
- Zone,Average,Zone Thermal Comfort CEN 15251 Adaptive Model Running Average Outdoor Air Temperature [C]
- Zone,Average,Zone Thermal Comfort CEN 15251 Adaptive Model Temperature [C]
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Elevated Air Speed Cooling Effect [C]
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Elevated Air Speed Cooling Effect Adjusted PMV []
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Elevated Air Speed Cooling Effect Adjusted PPD []
- Zone,Average,Zone Thermal Comfort ASHRAE 55 Ankle Draft PPD []

It should be noted that if a user is trying to output the Standard Effective Temperature (SET) that the Pierce two-node model must be selected. This variable is calculated as part of the Pierce model and can be seen in the output by requesting Zone Thermal Comfort Pierce Model Standard Effective Temperature.

1.14.2.2.1 People Occupant Count []

This field is the number of people for this PEOPLE object during the timestep in question.

1.14.2.2.2 People Radiant Heating Rate [W]

1.14.2.2.3 People Radiant Heating Energy [J]

These output variables are the amount of radiant heat gain for this People object in Watts (for rate) or Joules. This is determined by the current sensible heat gain from people to the zone and the "Fraction Radiant" specified in the input. The radiant gains from people are distributed to the surfaces using an area weighting scheme.

1.14.2.2.4 People Convective Heating Rate [W]

1.14.2.2.5 People Convective Heating Energy [J]

These output variables are the amount of convective heat gain for this People object in Watts (for rate) or Joules. This is determined by the current sensible heat gain from people to the zone and the "Fraction Radiant" specified in input. Note that the radiant and convective gains should add up to the sensible heat gain from people. The convective heat gain from people is added to the zone air heat balance directly.

1.14.2.2.6 People Latent Gain Rate [W]

1.14.2.2.7 People Latent Gain Energy [J]

These output variables are the amount of latent heat gain for this People object in Watts (for rate) or Joules. This amount is based on the number of people in the space as well as the total amount of energy produced by a typical person defined by the activity schedule in the input. An internal algorithm is used to determine what fraction of the total is sensible and what fraction is latent. Details about this split are included in the Engineering Reference document.

1.14.2.2.8 People Sensible Heating Rate [W]

1.14.2.2.9 People Sensible Heating Energy [J]

These output variables are the amount of sensible heat gain for this People object in Watts (for rate) or Joules. This amount is based on the number of people in the space as well as the total amount of energy produced by a typical person defined by the activity schedule in the input. An internal algorithm (described in the Engineering Reference document) is used to determine what fraction of the total is sensible and what fraction is latent. The sensible plus the latent heat gain from people equals the total gain specified in the input.

1.14.2.2.10 People Total Heating Rate [W]

1.14.2.2.11 People Total Heating Energy [J]

These output variables are the total amount of heat gain for this People object in Watts (for rate) or Joules. This is derived from the activity level times the number of occupants.

1.14.2.2.12 People Air Temperature [C]

This output variable represents the zone air temperature based on the Fanger thermal comfort model. If there is a ZoneControl:Thermostat:ThermalComfort object specified and the thermal zone is occupied, then the value of "People Air Temperature" is determined based on the thermal comfort that satisfies the thermal comfort setpoint PMV value specified; otherwise, it is set to average zone air temperature.

1.14.2.2.13 People Air Relative Humidity [%]

This output variable represents the zone air relative humidity based on the Fanger thermal comfort model. If there is a ZoneControl:Thermostat:ThermalComfort object specified and the thermal zone is occupied, then the value of "People Air Relative Humidity" is determined from the mean zone air temperature and zone air humidity ratio that satisfies the thermal comfort setpoint PMV value specified; otherwise, it is calculated from the zone air temperature and humidity ratio averaged over the time step.

1.14.2.2.14 Space or Zone People Occupant Count []

This field is the total number of people within the space or zone during the timestep in question.

1.14.2.2.15 Space or Zone People Radiant Heating Rate [W]

1.14.2.2.16 Space or Zone People Radiant Heating Energy [J]

These output variables are the amount of radiant heat gain from people within the space or zone in Watts (for rate) or Joules. This is determined by the current sensible heat gain from people to the space or zone and the "Fraction Radiant" specified in the input. The radiant gains from people are distributed to the surfaces using an area weighting scheme.

1.14.2.2.17 Space or Zone People Convective Heating Rate [W]

1.14.2.2.18 Space or Zone People Convective Heating Energy [J]

These output variables are the amount of convective heat gain from people within the space or zone in Watts (for rate) or Joules. This is determined by the current sensible heat gain from people to the space or zone and the "Fraction Radiant" specified in input. Note that the radiant and convective gains should add up to the sensible heat gain from people. The convective heat gain from people is added to the space or zone air heat balance directly.

1.14.2.2.19 Space or Zone People Latent Gain Rate [W]

1.14.2.2.20 Space or Zone People Latent Gain Energy [J]

These output variables are the amount of latent heat gain from people within the space or zone in Watts (for rate) or Joules. This amount is based on the number of people in the space as well as the total amount of energy produced by a typical person defined by the activity schedule in the input. An internal algorithm is used to determine what fraction of the total is sensible and what fraction is latent. Details about this split are included in the Engineering Reference document.

1.14.2.2.21 Space or Zone People Sensible Heating Rate [W]

1.14.2.2.22 Space or Zone People Sensible Heating Energy [J]

These output variables are the amount of sensible heat gain from people within the space or zone in Watts (for rate) or Joules. This amount is based on the number of people in the space as well as the total amount of energy produced by a typical person defined by the activity schedule in the input. An internal algorithm (described in the Engineering Reference document) is used to determine what fraction of the total is sensible and what fraction is latent. The sensible plus the latent heat gain from people equals the total gain specified in the input.

1.14.2.2.23 Space or Zone People Total Heating Rate [W]

1.14.2.2.24 Space or Zone People Total Heating Energy [J]

These output variables are the total amount of heat gain from people within the space or zone in Watts (for rate) or Joules. Derived from the activity level times the number of occupants, this is summed for each people object within a zone.

1.14.2.2.25 Zone Thermal Comfort Mean Radiant Temperature [C]

This output variable is the mean radiant temperature used in the thermal comfort calculations. This value is computed according to the "MRT Calculation Type" specified in the PEOPLE object. If a high temperature radiant system is present in the zone, this value will be adjusted according to the current heater operation and the "Fraction of radiant energy incident on people" specified in the HIGH TEMP RADIANT SYSTEM object.

1.14.2.2.26 Zone Thermal Comfort Operative Temperature [C]

This output variable is the operative temperature as defined by the thermal comfort operations. Specifically, it is the average of the thermal comfort mean radiant temperature and the zone air temperature.

Note for all Thermal Comfort reporting: Though the published values for thermal comfort "vote" have a discrete scale (e.g. –3 to +3 or –4 to +4), the calculations in EnergyPlus are carried out on a continuous scale and, thus, reporting may be "off the scale" with specific conditions encountered in the space. This is not necessarily an error in EnergyPlus – rather a different approach that does not take the "limits" of the discrete scale values into account.

1.14.2.2.27 Zone Thermal Comfort Fanger Model PMV []

This field is the "predicted mean vote" (PMV) calculated using the Fanger thermal comfort model. Details on the equations used to calculate the Fanger PMV are shown in the EnergyPlus Engineering Reference. If the zone in question is currently being controlled using a thermostat object, then the value of the PMV is determined by using the air temperature and humidity that is calculated at the system time step; otherwise, if the zone is uncontrolled, the PMV is determined using the zone air temperature and humidity that is averaged over the zone time step.

1.14.2.2.28 Zone Thermal Comfort Fanger Model PPD [%]

This field is the "predicted percentage of dissatisfied" (PPD) calculated using the Fanger thermal comfort model. Details on the equations used to calculate the Fanger PPD are shown in the EnergyPlus Engineering Reference. If the zone in question is currently being controlled using a thermostat object, then the value of the PPD is determined by using the air temperature and humidity that is calculated at the system time step; otherwise, if the zone is uncontrolled, the PPD is determined using the zone air temperature and humidity that is averaged over the zone time step.

1.14.2.2.29 Zone Thermal Comfort Clothing Surface Temperature [C]

This output variable is the calculation of the clothing surface temperature using the Fanger thermal comfort model.

1.14.2.2.30 Zone Thermal Comfort Pierce Model Effective Temperature PMV []

This field is the "predicted mean vote" (PMV) calculated using the effective temperature and the Pierce two-node thermal comfort model.

1.14.2.2.31 Zone Thermal Comfort Pierce Model Standard Effective Temperature PMV []

This field is the "predicted mean vote" (PMV) calculated using the "standard" effective temperature and the Pierce two-node thermal comfort model.

1.14.2.2.32 Zone Thermal Comfort Pierce Model Discomfort Index []

This field is the "discomfort index" calculated using the the Pierce two-node thermal comfort model.

1.14.2.2.33 Zone Thermal Comfort Pierce Model Thermal Sensation Index []

This field is the "thermal sensation index" (PMV) calculated using the Pierce two-node thermal comfort model.

1.14.2.2.34 Zone Thermal Comfort Pierce Model Standard Effective Temperature [C]

This field is the "standard effective temperature" (SET) calculated using the Pierce two-node thermal comfort model. Note that if a user wishes to report the Pierce Model SET that it must be done using the Pierce two-node model and the user must select "Pierce" as one of the Thermal Comfort model types as shown above in the input syntax for the People statement.

1.14.2.2.35 Zone Thermal Comfort KSU Model Thermal Sensation Vote []

This field is the "thermal sensation vote" (TSV) calculated using the KSU two-node thermal comfort model.

1.14.2.2.36 Zone Thermal Comfort ASHRAE 55 Adaptive Model 90% Acceptability Status []

This field is to report whether the operative temperature falls into the 90% acceptability limits of the adaptive comfort in ASHRAE 55-2010. A value of 1 means within (inclusive) the limits, a value of 0 means outside the limits, and a value of -1 means not applicable (when unoccupied or running average outdoor temp is outside the range of 10.0 to 33.5C).

1.14.2.2.37 Zone Thermal Comfort ASHRAE 55 Adaptive Model 80% Acceptability Status []

This field is to report whether the operative temperature falls into the 80% acceptability limits of the adaptive comfort in ASHRAE 55-2010. A value of 1 means within (inclusive) the limits, a value of 0 means outside the limits, and a value of -1 means not applicable (when unoccupied or running average outdoor temp is outside the range of 10.0 to 33.5C).

1.14.2.2.38 Zone Thermal Comfort ASHRAE 55 Adaptive Model Running Average Outdoor Air Temperature [C]

This field reports the mean monthly outdoor air temperature, an input parameter for the ASHRAE-55 adaptive comfort model. This can be computed in two ways. If the .stat file is provided for the simulation, this field will reflect the monthly daily average temperature.

If the .epw file is used, the field reports the simple running average of the daily average outdoor dry-bulb temperatures of the previous 30 days.

1.14.2.2.39 Zone Thermal Comfort ASHRAE 55 Adaptive Model Temperature [C]

This field reports the ideal indoor operative temperature, or comfort temperature, as determined by the ASHRAE-55 adaptive comfort model. The 80% acceptability limits for indoor operative temperature are defined as no greater than 3.5°C from the adaptive comfort temperature. The 90% acceptability limits are defined as no greater than 2.5°C from the adaptive comfort temperature. A value of -1 means not applicable (when running average outdoor temp is outside the range of 10.0 to 33.5°C).

1.14.2.2.40 Zone Thermal Comfort CEN 15251 Adaptive Model Category I Status

This field is to report whether the operative temperature falls into the Category I (90% acceptability) limits of the adaptive comfort in the European Standard EN15251-2007. A value of 1 means within (inclusive) the limits, a value of 0 means outside the limits, and a value of -1 means not applicable (when unoccupied or running average outdoor temp is outside the range of 10.0 to 30.0°C).

1.14.2.2.41 Zone Thermal Comfort CEN 15251 Adaptive Model Category II Status

This field is to report whether the operative temperature falls into the Category II (80% acceptability) limits of the adaptive comfort in the European Standard EN15251-2007. A value of 1 means within (inclusive) the limits, a value of 0 means outside the limits, and a value of -1 means not applicable (when unoccupied or running average outdoor temp is outside the range of 10.0 to 30.0°C).

1.14.2.2.42 Zone Thermal Comfort CEN 15251 Adaptive Model Category III Status

This field is to report whether the operative temperature falls into the Category III (65% acceptability) limits of the adaptive comfort in the European Standard EN15251-2007. A value of 1 means within (inclusive) the limits, a value of 0 means outside the limits, and a value of -1 means not applicable (when unoccupied or running average outdoor temp is outside the range of 10.0 to 30.0°C).

1.14.2.2.43 Zone Thermal Comfort CEN 15251 Adaptive Model Running Average Outdoor Air Temperature

This field reports the weighted average of the outdoor air temperature of the previous seven days, an input parameter for the CEN-15251 adaptive comfort model.

1.14.2.2.44 Zone Thermal Comfort CEN 15251 Adaptive Model Temperature

This field reports the ideal indoor operative temperature, or comfort temperature, as determined by the CEN-15251 adaptive comfort model. Category I, II, and II limits for indoor operative temperature are defined as no greater than 2, 3, and 4 degrees C from this value respectively. A value of -1 means not applicable (when running average outdoor temp is outside the range of 10.0 to 30.0°C).

1.14.2.2.45 Zone Thermal Comfort ASHRAE 55 Elevated Air Speed Cooling Effect

This field is the calculated Cooling Effect of the elevated air speed in degree celsius. It is the value that, when subtracted equally from both the average air temperature and the mean radiant temperature, yields the same SET under still air as in the first SET calculation under elevated air speed.

1.14.2.2.46 Zone Thermal Comfort ASHRAE 55 Elevated Air Speed Cooling Effect Adjusted PMV

This field is the “predicted mean vote” (PMV) calculated using the Fanger PMV model, adjusted by the ASHRAE 55 Elevated Air Speed Cooling Effect. The Cooling Effect adjusted PMV for an environment with elevated average air speed is calculated using the adjusted average air temperature, the adjusted radiant temperature, and still air (0.1 m/s).

1.14.2.2.47 Zone Thermal Comfort ASHRAE 55 Elevated Air Speed Cooling Effect Adjusted PPD

This field is the “predicted percentage of dissatisfied” (PPD) calculated using the Fanger PMV-PPD model, adjusted by the ASHRAE 55 Elevated Air Speed Cooling Effect. The Cooling Effect adjusted PPD for an environment with elevated average air speed is calculated using the adjusted average air temperature, the adjusted radiant temperature, and still air (0.1 m/s).

1.14.2.2.48 Zone Thermal Comfort ASHRAE 55 Ankle Draft PPD

This field is the “predicted percentage of dissatisfied” (PPD) on draft at ankle level. It is used as the metric to evaluate the ankle draft risk as a function of PMV and air speed at the ankle level (0.1 m).

1.14.2.3 Outputs

The following output variables are all based on whether the humidity ratio and the operative temperature is within the region shown in ASHRAE Standard 55-2004 in Figure 5.2.1.1. For these outputs the operative temperature is simplified to be the average of the air temperature and the mean radiant temperature. For summer, the 0.5 Clo level is used and, for winter, the 1.0 Clo level is used. The graphs below are based on the following tables which extend the ASHRAE values to zero humidity ratio.

Table 1.30: Winter Clothes (1.0 Clo)

Operative Temperature (C)	Humidity Ratio (kgWater/kgDryAir)
19.6	0.012
23.9	0.012
26.3	0.000
21.7	0.000

Table 1.31: Summer Clothes (0.5 Clo)

Operative Temperature (C)	Humidity Ratio (kgWater/kgDryAir)
23.6	0.012
26.8	0.012
28.3	0.000

Table 1.31: Summer Clothes (0.5 Clo)

Operative Temperature (C)	Humidity Ratio (kgWater/kgDryAir)
25.1	0.000

1.14.2.3.1 Zone Thermal Comfort ASHRAE 55 Simple Model Summer Clothes Not Comfortable Time[hr]

The time when the zone is occupied that the combination of humidity ratio and operative temperature is not in the ASHRAE 55-2004 summer clothes region (see above)

1.14.2.3.2 Zone Thermal Comfort ASHRAE 55 Simple Model Winter Clothes Not Comfortable Time[hr]

The time when the zone is occupied that the combination of humidity ratio and operative temperature is not in the ASHRAE 55-2004 winter clothes region (see above)

1.14.2.3.3 Zone Thermal Comfort ASHRAE 55 Simple Model Summer or Winter Clothes Not Comfortable Time[hr]

The time when the zone is occupied that the combination of humidity ratio and operative temperature is not in the ASHRAE 55-2004 summer or winter clothes region (see above)

1.14.2.3.4 Facility Thermal Comfort ASHRAE 55 Simple Model Summer Clothes Not Comfortable Time[hr]

The time when any zone is occupied that the combination of humidity ratio and operative temperature is not in the ASHRAE 55-2004 summer clothes region (see above)

1.14.2.3.5 Facility Thermal Comfort ASHRAE 55 Simple Model Winter Clothes Not Comfortable Time [hr]

The time when any zone is occupied that the combination of humidity ratio and operative temperature is not in the ASHRAE 55-2004 winter clothes region (see above)

1.14.2.3.6 Facility Thermal Comfort ASHRAE 55 Simple Model Summer or Winter Clothes Not Comfortable Time [hr]

The time when any zone is occupied that the combination of humidity ratio and operative temperature is not in the ASHRAE 55-2004 summer or winter clothes region (see above)

1.14.3 Simplified ASHRAE 55 Warnings

The simplified ASHRAE 55 calculations may be computed for occupied zones and, possibly, warnings are shown on the .err file at the end of each simulated environment. To enable this option set the "Enable ASHRAE 55 comfort warnings" field of the People object to Yes. These warnings will not be generated by default.

If you enable the warnings, the simplified ASHRAE 55 calculations are done for occupied zones and, possibly, warnings are shown on the .err file at the end of each simulated environment.

```
** Warning ** More than 4% of time (350.4 hours) uncomfortable in zone ZSF1
** ~~~ ** 553.0 hours were uncomfortable based on ASHRAE 55-2004 graph (Section 5.2.1.1)
** ~~~ ** During Environment [10/01 - 09/30]: CHICAGO IL USA TMY2-94846 WMO\# = 725300
** Warning ** More than 4% of time (350.4 hours) uncomfortable in zone ZNF1
```

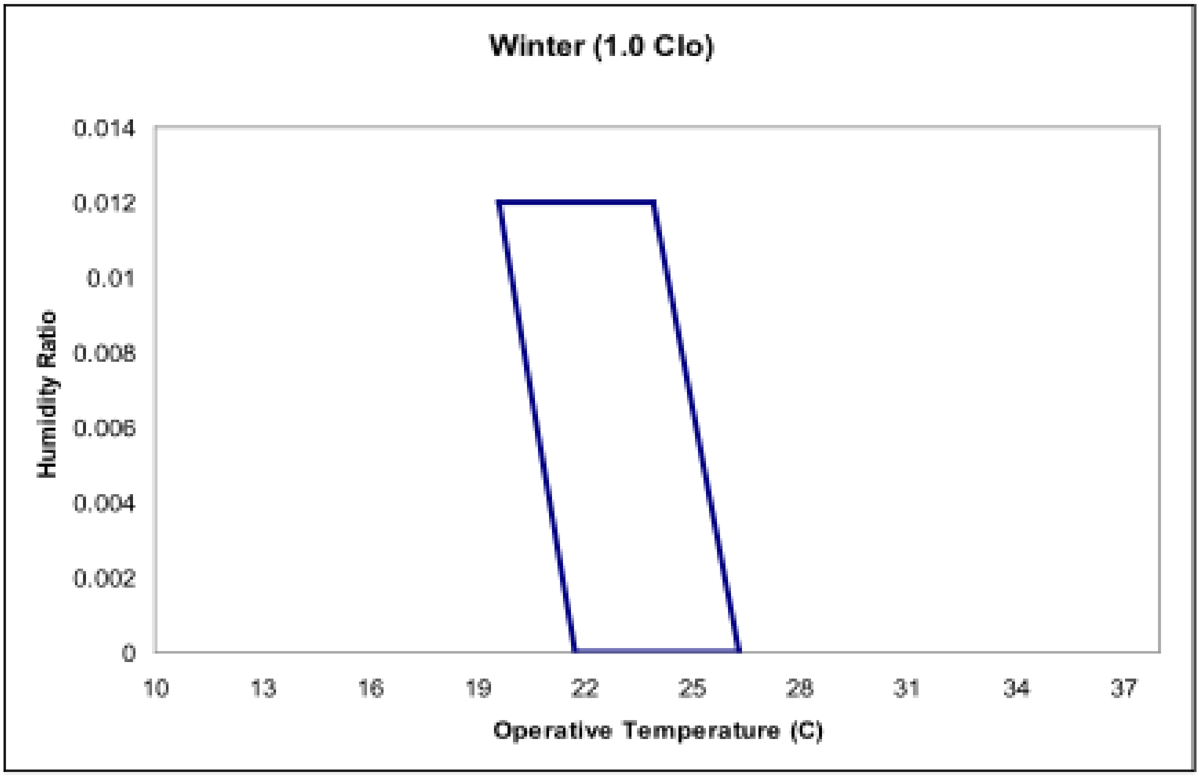

Figure 1.70: Winter Comfort Range

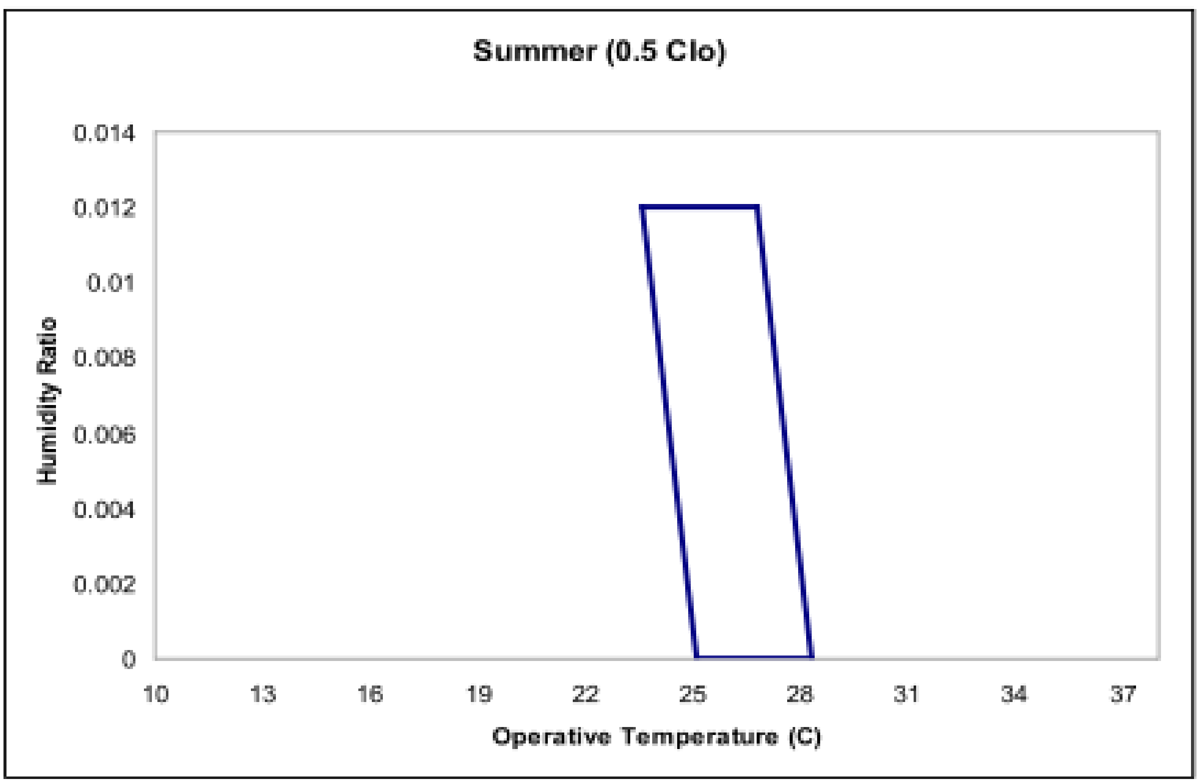

Figure 1.71: Summer Comfort Range

```
** ~~~ ** 827.8 hours were uncomfortable based on ASHRAE 55-2004 graph (Section 5.2.1.1)
** ~~~ ** During Environment [10/01 - 09/30]: CHICAGO IL USA TMY2-94846 WMO\# = 725300
** Warning ** More than 4% of time (350.4 hours) uncomfortable in zone ZSF2
** ~~~ ** 593.5 hours were uncomfortable based on ASHRAE 55-2004 graph (Section 5.2.1.1)
** ~~~ ** During Environment [10/01 - 09/30]: CHICAGO IL USA TMY2-94846 WMO\# = 725300
** Warning ** More than 4% of time (350.4 hours) uncomfortable in zone ZNF2
** ~~~ ** 875.8 hours were uncomfortable based on ASHRAE 55-2004 graph (Section 5.2.1.1)
** ~~~ ** During Environment [10/01 - 09/30]: CHICAGO IL USA TMY2-94846 WMO\# = 725300
```

You may decide if you need to change parameters to reduce these "uncomfortable" hours. The individual output variables shown previously may help you get more details on when these are occurring.

Following are some suggestions that might be applicable:

- Eliminate occupancy when conditioning equipment is off.
- Note that the ASHRAE graph lower limit is (19.6°C to 21.7°C)—heating setpoints may need to be nearer 22.2°C (72°F) than 21.1°C (70°F).
- Unoccupied heating setpoint should be nearer 16.7°C (62°F) rather than 12.8°C (55°F) to reduce the start up recovery.
- Start the occupied setpoint schedule, fan availability schedule, cooling pump availability schedule, reheat coil availability, one hour before occupancy. Seasonal turn on and off of equipment may cause more warnings (but potentially more energy consumption).
- Unoccupied cooling setpoint should be nearer 29.4°C (85°F) rather than 40.0°C (104°F) to reduce the start up recovery.

1.14.4 ComfortViewFactorAngles

When requesting EnergyPlus to do a thermal comfort calculation, the program user has three options for defining how the mean radiant temperature will be calculated. The user may select "EnclosureAveraged" which results in a mean radiant temperature that is characteristic of an "average" location near the center of the enclosure (one or more Spaces). The user may also elect to place the person near a particular surface by selecting "SurfaceWeighted" in the People statement. This takes the average of the enclosure mean radiant temperature and the temperature of the surface that the person is near and uses this value as the mean radiant temperature when calculating thermal comfort.

The third option is for the user to more explicitly position the person within the space by defining the angle factors from the person to the various surfaces in the same radiant enclosure. This option requires the user to list the surfaces that the person can see from a radiation standpoint and also define the angle (or view) factor for each surface.

1.14.4.1 Inputs

1.14.4.1.1 Field: Name

This field is an unique user assigned name for the list of surfaces that can be seen radiantly by the person for whom thermal comfort is to be evaluated. Any reference to this list by a People statement will use this name.

1.14.4.1.2 Field: Surface <#> Name

This field is the name of a surface in the radiant enclosure seen by the person. All surfaces listed should be in the same radiant enclosure (one or more Spaces). This should also be the same radiant enclosure as the People instance which references this ComfortViewFactorAngles object. If not, a warning will be issued, but the simulation will proceed with the specified surfaces.

1.14.4.1.3 Field: Angle Factor <#>

This field is the fraction that this surface contributes to the total mean radiant temperature. This can be thought of as a weighting factor for this surface and the actual mean radiant temperature used in the thermal comfort model is simply the sum of all angle factors multiplied by the corresponding inside surface temperature, weighted by the surface emissivity. The sum of all angle factors within any angle factor list must equal unity, otherwise the program will not accept the input as valid.

Note that the Surface Name/Angle Factor pair is extensible to accommodate as many surfaces as required.

An example IDF with an electric low temperature radiant system is shown below.

```
ComfortViewFactorAngles,
  South Zone Angle Factors, !- name of angle factor list
  Zn001:Flr001,          !- Surface name 1
  0.75,                  !- Angle factor for surface 1
  Zn001:Wall001,         !- Surface name 2
  0.15,                  !- Angle factor for surface 2
  Zn001:Roof001,         !- Surface name 3
  0.10;                  !- Angle factor for surface 3
```

1.14.5 Lights

The Lights statement allows you to specify information about a zone's electric lighting system, including design power level and operation schedule, and how the heat from lights is distributed thermally.

A zone may have multiple Lights statements. For example, one statement may describe the general lighting in the zone and another the task lighting. Or you can use multiple Lights statements for a zone that has two or more general lighting systems that differ in design level, schedule, etc.

1.14.5.1 Inputs

1.14.5.1.1 Field: Name

The name of the Lights object.

1.14.5.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this Lights object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the watts/area and watts/person options of the Design Level Calculation Method to place a varying lighting load at the same density in each zone or space. The names of the actual lights objects may be concatenated as <Space Name> <Lights Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.5.1.3 Field: Schedule Name

The name of the schedule that modifies the lighting power design level (see Design Level Calculation Method field and related subsequent fields). The schedule values can be any positive number. The electrical input for lighting in a particular timestep is the product of the design level and the value of this schedule in that timestep. If the design level is the maximum lighting power input the schedule should contain values between 0.0 and 1.0.

1.14.5.1.4 Field: Design Level Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal lighting level in the Zone. The key/choice options are:

LightingLevel With this choice, the method used will be a straight insertion of the lighting level (Watts) for the Zone. (The Lighting Level field should be filled.)

Watts/Area With this choice, the method used will be a factor per floor area of the zone or space. (The Watts per Floor Area field should be filled).

Watts/Person With this choice, the method used will be a factor of lighting level (watts) per person. (The Watts per person field should be filled).

1.14.5.1.5 Field: Lighting Level

This is typically the maximum electrical power input (in Watts) to lighting in a zone, including ballasts, if present. This value is multiplied by a schedule fraction (see previous field) to get the lighting power in a particular timestep. In EnergyPlus, this is slightly more flexible in that the lighting design level could be a "diversity factor" applied to a schedule of real numbers.

1.14.5.1.6 Field: Watts per Floor Area

This factor (watts/m^2) is used, along with the Zone or Space Floor Area to determine the maximum lighting level as described in the Lighting Level field. The choice from the method field should be "watts/area".

1.14.5.1.7 Field: Watts per Person

This factor (watts/person) is used, along with the number of occupants (people) to determine the maximum lighting level as described in the Lighting Level field. The choice from the method field should be "watts/person".

1.14.5.1.8 Heat Gains from Lights:

The electrical input to lighting ultimately appears as heat that contributes to zone loads or to return air heat gains. In EnergyPlus this heat is divided into four different fractions. Three of these are given by the input fields Return Air Fraction, Fraction Radiant and Fraction Visible. A fourth, defined as the fraction of the heat from lights convected to the zone air, is calculated by the program as:

$$f_{\text{convected}} = 1.0 - (\text{Return Air Fraction} + \text{Fraction Radiant} + \text{Fraction Visible})$$

You will get an error message if Return Air Fraction + Fraction Radiant + Fraction Visible exceeds 1.0.

These fractions depend on the type of lamp and luminaire, whether the luminaire is vented to the return air, etc.

1.14.5.1.9 Field: Return Air Fraction

The fraction of the heat from lights that goes into the zone return air (i.e., into the zone outlet node). If the return air flow is zero or the zone has no return air system, the program will put this fraction into the zone air. Return Air Fraction should be non-zero only for luminaires that are return-air ducted (see Table 1.32 and Figure 1.72). (However, see the field "Return Air Fraction Is Calculated from Plenum Temperature," below, for an approach to modeling the case where Return Air Fraction is caused by *conduction* between a luminaire that is in contact with a return-air plenum.)

1.14.5.1.10 Field: Fraction Radiant

The fraction of heat from lights that goes into the zone as long-wave (thermal) radiation. The program calculates how much of this radiation is absorbed by the inside surfaces of the zone according the area times thermal absorptance product of these surfaces.

1.14.5.1.11 Field: Fraction Visible

The fraction of heat from lights that goes into the zone as visible (short-wave) radiation. The program calculates how much of this radiation is absorbed by the inside surfaces of the zone according the area times solar absorptance product of these surfaces.

Approximate values of Return Air Fraction, Fraction Radiant and Fraction Visible are given in Table 1.32 for overhead fluorescent lighting for a variety of luminaire configurations. The data is based on ASHRAE 1282-RP "Lighting Heat Gain Distribution in Buildings" by Daniel E. Fisher and Chanvit Chantrasrisalai.

Table 1.32: Approximate values of Return Air Fraction, Fraction Radiant and Fraction Visible for overhead fluorescent lighting for different luminaire configurations.

Fixture No.	Luminaire Feature	Return Air Fraction	Fraction Radiant	Fraction Visible	fconvected
1	Recessed, Parabolic Louver, Non-Vented, T8	0.31	0.22	0.20	0.27
2	Recessed, Acrylic Lens, Non-Vented, T8	0.56	0.12	0.20	0.12
3	Recessed, Parabolic Louver, Vented, T8	0.28	0.19	0.20	0.33
4	Recessed, Acrylic Lens, Vented, T8	0.54	0.10	0.18	0.18
5	Recessed, Direct/Indirect, T8	0.34	0.17	0.16	0.33
6	Recessed, Volumetric, T5	0.54	0.13	0.20	0.13
7	Downlights, Compact Fluorescent, DTT	0.86	0.04	0.10	0.00
8	Downlights, Compact Fluorescent, TRT	0.78	0.09	0.13	0.00
9a	Downlights, Incandescent, A21	0.29	0.10	0.6	0.01
9b	Downlights, Incandescent, BR40	0.21	0.08	0.71	0.00
10	Surface Mounted, T5HO	0.00	0.27	0.23	0.50
11	Pendant, Direct/Indirect, T8	0.00	0.32	0.23	0.45
12	Pendant, Indirect, T5HO	0.00	0.32	0.25	0.43

Table 1.32: Approximate values of Return Air Fraction, Fraction Radiant and Fraction Visible for overhead fluorescent lighting for different luminaire configurations.

Fixture No.	Luminaire Feature	Return Air Fraction	Fraction Radiant	Fraction Visible	$f_{convected}$
1	Recessed, Parabolic Louver, Non-Vented, T8 - Ducted	0.27	0.27	0.21	0.25
5	Recessed, Direct/Indirect, T8 - Ducted	0.27	0.22	0.17	0.34
1	Recessed, Parabolic Louver, Non-Vented, T8 - Half Typical Supply Airflow Rate	0.45	0.30	0.22	0.03
3	Recessed, Parabolic Louver, Vented, T8 - Half Typical Supply Airflow Rate	0.43	0.25	0.21	0.11
5	Recessed, Direct/Indirect, T8 - Half Typical Supply Airflow Rate	0.43	0.27	0.18	0.12
1	Recessed, Parabolic Louver, Non-Vented, T8 - Half Typical Supply Airflow Rate	0.10	0.16	0.20	0.54
3	Recessed, Parabolic Louver, Vented, T8 - Half Typical Supply Airflow Rate	0.11	0.15	0.19	0.55
5	Recessed, Direct/Indirect, T8 - Half Typical Supply Airflow Rate	0.04	0.13	0.16	0.67

1.14.5.1.12 Field: Fraction Replaceable

This field defines the daylighting control for the LIGHTS object.

If **Daylighting:Controls** is specified for the space or zone, this field is used as an on/off flag for dimming controls. If set to 0.0, the lights are not dimmed by the daylighting controls. If set to 1.0, the lights are allowed to be dimmed by the applicable daylighting control. If daylighting controls are operating in the space or zone, all of the applicable Lights objects with a Fraction Replaceable greater than zero will be reduced by a multiplicative factor that accounts for how much the electric lighting is lowered due to daylighting (ref. Daylighting Lighting Power Multiplier).

1.14.5.1.13 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Task Lights", "Hall Lights", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter objects). Subcategories are also reported in the ABUPS table. If this field is omitted or blank, the lights will be assigned to the "General" end-use subcategory. Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and the LEED EAp2-4/5. Performance Rating Method Compliance table.

1.14.5.1.14 Field: Return Air Fraction Calculated from Plenum Temperature

Accepts values Yes or No (the default). Yes is for advanced used only. In this case the program will calculate the return air fraction by assuming that it is due to conduction of some of the light heat into the zone's return air plenum and that the amount of the conduction depends on the plenum air temperature. A Yes value should only be used for luminaires that are recessed and non-vented, as shown in Figure 1.72.

The value you enter for the Return Air Fraction field will be ignored and you can enter, for fluorescent lighting, Fraction Radiant = 0.37 and Fraction Visible = 0.18, as indicated in Table 1.32.

This feature requires that the coefficients described below be determined from measurements or detailed calculations since they are very sensitive to the luminaire type, lamp type, thermal resistance between fixture and plenum, etc.

If "Return Air Fraction Is Calculated from Plenum Temperature" = Yes, the return air fraction is calculated *each timestep* from the following empirical correlation:

$$(\mathrm{ReturnAirFraction})_{\mathrm{calculated}} = \mathrm{C}_1 - \mathrm{C}_2 \times \mathrm{T}_{\mathrm{plenum}} \tag{1.24}$$

where $\mathrm{T}_{\mathrm{plenum}}$ is the previous-time-step value of the return plenum air temperature (C),

and C_1 and C_2 are the values of the coefficients entered in the next two fields.

To compensate for the change in the return air fraction relative to its input value, the program modifies Fraction Radiant and $f_{\mathrm{convected}}$ by a scale factor such that

$$(\mathrm{ReturnAirFraction})_{\mathrm{calculated}} + (\mathrm{FractionRadiant})_{\mathrm{modified}} + (\mathrm{f}_{\mathrm{convected}})_{\mathrm{modified}} + (\mathrm{FractionVisible})_{\mathrm{input}} = 1.0 \tag{1.25}$$

It is assumed that Fraction Visible is a constant equal to its input value.

1.14.5.1.15 Field: Return Air Fraction Function of Plenum Temperature Coefficient 1

The coefficient C_1 in the equation for (Return Air Fraction)$_{calculated}$.

1.14.5.1.16 Field: Return Air Fraction Function of Plenum Temperature Coefficient 2

The coefficient C_2 in the equation for (Return Air Fraction)$_{calculated}$. Its units are 1/°C.

1.14.5.1.17 Field: Return Air Heat Gain Node Name

Name of the return air node for this heat gain. If left blank, it defaults to the first return air node for the zone containing this Lights object. Leave blank when using a ZoneList name.

1.14.5.1.18 Field:Exhaust Air Heat Gain Node Name

Name of the exhaust air node for this heat gain. If left blank, no heat gain from return air fraction will be added to the zone exhaust node. When the exhaust node name is entered, the return air heat gain will be shared by both return and exhaust nodes. The equipment can draw air from the exhaust node, but the inlet air properties is combined properties with mixing mass flow rates of both nodes and added lights heat gain.

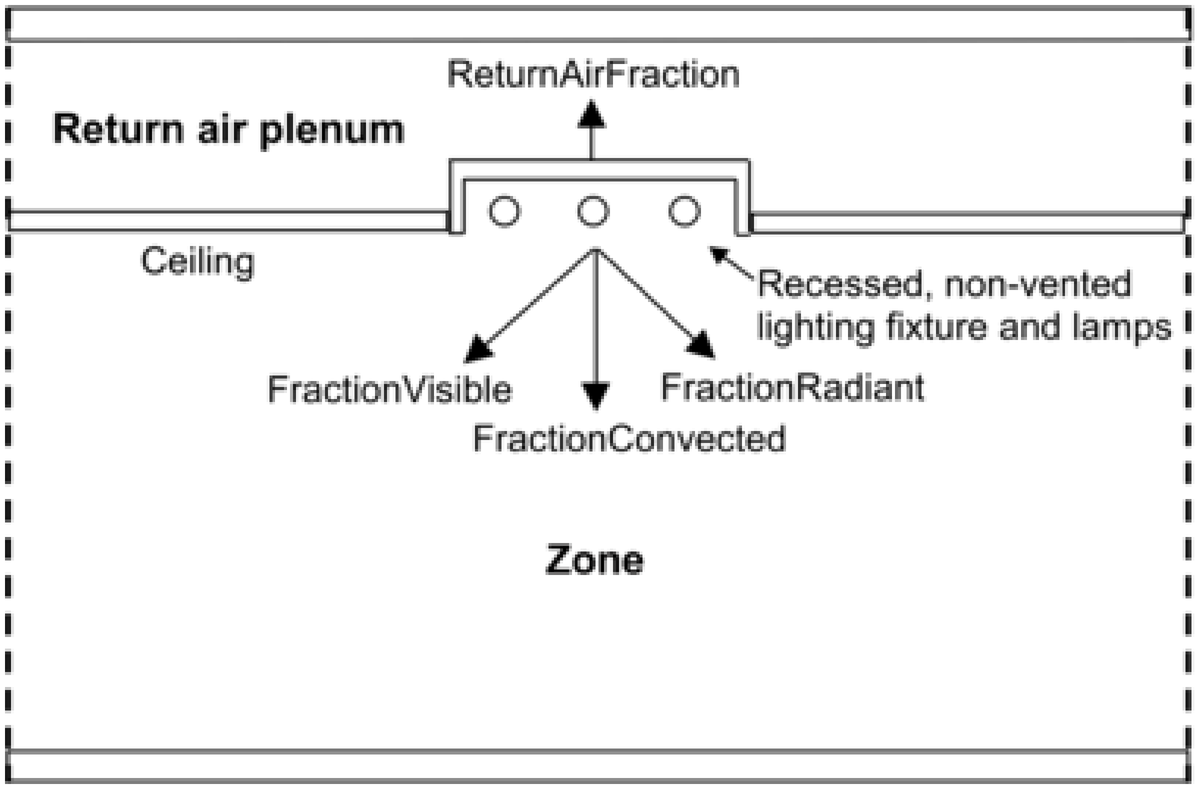

Figure 1.72: Vertical section through a zone and its return air plenum showing recessed lighting (not to scale). The heat from lights is divided into four fractions, three of which—ReturnAirFraction, FractionRadiant and FractionConvected—depend on plenum air temperature.

An IDF example:

```
Lights,
  RIGHT FORK Lights 1,  !- Name
  RIGHT FORK,           !- Zone Name
  Office Lighting,      !- SCHEDULE Name
  LightingLevel,        !- Design Level calculation method
  1039.706,             !- Lighting Level {W}
  ,                     !- Watts per Floor Area {W/m2}
  ,                     !- Watts per Person {W/person}
  0.0000000E+00,        !- Return Air Fraction
  0.4000000,            !- Fraction Radiant
  0.2000000,            !- Fraction Visible
  1.0,                  !- Fraction Replaceable
  GeneralLights;        !- End-Use Subcategory
```

Global Lights Object:

```
ZoneList,AllOccupiedZones,SPACE1-1,SPACE2-1,SPACE3-1,SPACE4-1,SPACE5-1;

Lights,
  AllZones with Lights, !- Name
```

```
AllOccupiedZones,        !- Zone or ZoneList or Space or SpaceList Name
LIGHTS-1,                !- Schedule Name
Watts/Area,              !- Design Level Calculation Method
,                        !- Lighting Level {W}
16,                      !- Watts per Zone Floor Area {W/m2}
,                        !- Watts per Person {W/person}
0.2,                     !- Return Air Fraction
0.59,                    !- Fraction Radiant
0.2,                     !- Fraction Visible
0,                       !- Fraction Replaceable
GeneralLights;           !- End-Use Subcategory
```

1.14.5.2 Outputs

Lights objects have output variables for individual objects and for space and zone totals.

A related output is Daylighting Lighting Power Multiplier.

- Zone,Average,Lights Electricity Rate [W]
- Zone,Sum,Lights Radiant Heat Gain [J]
- Zone,Average,Lights Radiant Heating Rate [W]
- Zone,Sum,Lights Visible Radiation Heating Energy [J]
- Zone,Average,Lights Visible Radiation Heating Rate [W]
- Zone,Sum,Lights Convective Heating Energy [J]
- Zone,Average,Lights Convective Heating Rate [W]
- Zone,Sum,Lights Return Air Heating Energy [J]
- Zone,Average,Lights Return Air Heating Rate [W]
- Zone,Sum,Lights Total Heating Energy [J]
- Zone,Average,Lights Total Heating Rate [W]
- Zone,Sum,Lights Electricity Energy [J]
- Zone,Average,Space Lights Electricity Rate [W]
- Zone,Average,Space Lights Electricity Energy [J]
- Zone,Sum,Space Lights Radiant Heating Energy [J]
- Zone,Average,Space Lights Radiant Heating Rate [W]
- Zone,Sum,Space Lights Visible Radiation Heating Energy [J]
- Zone,Average,Space Lights Visible Radiation Heating Rate [W]
- Zone,Sum,Space Lights Convective Heating Energy [J]
- Zone,Average,Space Lights Convective Heating Rate [W]
- Zone,Sum,Space Lights Return Air Heating Energy [J]

- Zone,Average,Space Lights Return Air Heating Rate [W]
- Zone,Sum,Space Lights Total Heating Energy [J]
- Zone,Average,Space Lights Total Heating Rate [W]
- Zone,Sum,Space Lights Electricity Energy [J]
- Zone,Average,Zone Lights Electricity Rate [W]
- Zone,Average,Zone Lights Electricity Energy [J]
- Zone,Sum,Zone Lights Radiant Heating Energy [J]
- Zone,Average,Zone Lights Radiant Heating Rate [W]
- Zone,Sum,Zone Lights Visible Radiation Heating Energy [J]
- Zone,Average,Zone Lights Visible Radiation Heating Rate [W]
- Zone,Sum,Zone Lights Convective Heating Energy [J]
- Zone,Average,Zone Lights Convective Heating Rate [W]
- Zone,Sum,Zone Lights Return Air Heating Energy [J]
- Zone,Average,Zone Lights Return Air Heating Rate [W]
- Zone,Sum,Zone Lights Total Heating Energy [J]
- Zone,Average,Zone Lights Total Heating Rate [W]
- Zone,Sum,Zone Lights Electricity Energy [J]

1.14.5.2.1 Lights Radiant Heating Rate [W]

1.14.5.2.2 Lights Radiant Heating Energy [J]

The amount of heat gain from lights that is in the form of long-wave (thermal) radiation entering the zone. This heat is absorbed by the inside surfaces of the zone according to an area times long-wave absorptance weighting scheme.

1.14.5.2.3 Lights Visible Radiation Heating Rate [W]

1.14.5.2.4 Lights Visible Radiation Heating Energy [J]

The amount of heat gain from lights that is in the form of visible (short-wave) radiation entering the zone. This heat is absorbed by the inside surfaces of the zone according to an area times short-wave absorptance weighting scheme.

1.14.5.2.5 Lights Convective Heating Rate [W]

1.14.5.2.6 Lights Convective Heating Energy [J]

The amount of heat gain from lights that is convected to the zone air.

1.14.5.2.7 Lights Return Air Heating Rate [W]

1.14.5.2.8 Lights Return Air Heating Energy [J]

The amount of heat gain from lights that goes into the zone's return air (and, therefore, does not directly contribute to the zone load). If the zone has no return air system or the zone's air system is off, this heat will be added to the zone air.

1.14.5.2.9 Lights Total Heating Rate [W]

1.14.5.2.10 Lights Total Heating Energy [J]

The total heat gain from lights. It is the sum of the following four outputs, i.e., Total Heat Gain = Return Air Heat Gain + Radiant Heat Gain + Visible Heat Gain + Convective Heat Gain. It is also equal to the electrical input to the lights.

1.14.5.2.11 Lights Electricity Rate [W]

The electric power input for the lights.

1.14.5.2.12 Lights Electricity Energy [J]

The lighting electrical consumption including ballasts, if present. This will have the same value as Lights Total Heating Energy (above).

The energy amount is also included in the following Electricity meters:

```
Electricity:Facility
Electricity:Building
Electricity:Zone:<Zone Name>
Electricity:SpaceType:<Space Type Name>
InteriorLights:Electricity
InteriorLights:Electricity:Zone:<Zone Name>
InteriorLights:Electricity:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorLights:Electricity
<End-Use Subcategory>:InteriorLights:Electricity:Zone:<Zone Name>
<End-Use Subcategory>:InteriorLights:Electricity:SpaceType:<Space Type Name>
```

1.14.5.2.13 Space or Zone Lights Radiant Heating Rate [W]

1.14.5.2.14 Space or Zone Lights Radiant Heating Energy [J]

The amount of heat gain from all lights in the space or zone that is in the form of long-wave (thermal) radiation entering the space or zone. This heat is absorbed by the inside surfaces of the space or zone according to an area times long-wave absorptance weighting scheme.

1.14.5.2.15 Space or Zone Lights Visible Radiation Heating Rate [W]

1.14.5.2.16 Space or Zone Lights Visible Radiation Heating Energy [J]

The amount of heat gain from all lights in the space or zone that is in the form of visible (short-wave) radiation entering the space or zone. This heat is absorbed by the inside surfaces of the space or zone according to an area times short-wave absorptance weighting scheme.

1.14.5.2.17 Space or Zone Lights Convective Heating Rate [W]

1.14.5.2.18 Space or Zone Lights Convective Heating Energy [J]

The amount of heat gain from all lights in the space or zone that is convected to the space or zone air.

1.14.5.2.19 Space or Zone Lights Return Air Heating Rate [W]

1.14.5.2.20 Space or Zone Lights Return Air Heating Energy [J]

The amount of heat gain from all lights in the space or zone that goes into the space or zone's return air (and, therefore, does not directly contribute to the space or zone load). If the space or zone has no return air system or the space or zone's air system is off, this heat will be added to the space or zone air.

1.14.5.2.21 Space or Zone Lights Total Heating Rate [W]

1.14.5.2.22 Space or Zone Lights Total Heating Energy [J]

The total heat gain from all lights in the space or zone. It is the sum of the following four outputs, i.e., Total Heat Gain = Return Air Heat Gain + Radiant Heat Gain + Visible Heat Gain + Convective Heat Gain. It is also equal to the electrical input to the lights.

1.14.5.2.23 Space or Zone Lights Electricity Rate [W]

1.14.5.2.24 Space or Zone Lights Electricity Energy [J]

The electric power input for all lights in the space or zone. This will have the same value as Zone Lights Total Heating Rate or Energy (above).

1.14.6 ElectricEquipment

The object models equipment in the zone which consumes electricity, such as computers, televisions, and cooking equipment, also known as "plug loads." All of the energy consumed by the equipment becomes a heat gain in the zone or is lost (exhausted) as specified below.

1.14.6.1 Inputs

1.14.6.1.1 Field: Name

The name of the ElectricEquipment object.

1.14.6.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this ElectricEquipment object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the watts/area and watts/person options of the Design Level Calculation Method to place a varying equipment load at the same density in each zone or space. The names of the actual electric equipment objects may be concatenated as <Space Name> <ElectricEquipment Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.6.1.3 Field: Schedule Name

This field is the name of the schedule that modifies the design level parameter for electric equipment (see Design Level Calculation Method field and related subsequent fields). The schedule values can be any positive number. The actual electrical input for equipment in a zone as defined by this statement is the product of the design level field and the value of the schedule specified by name in this field.

1.14.6.1.4 Field: Design Level Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal electric equipment level in the Zone or Space. The key/choices are:

- EquipmentLevel

With this choice, the method used will be a straight insertion of the electric equipment level (Watts) for the Zone. (The Design Level field should be filled.)

- Watts/Area

With this choice, the method used will be a factor per floor area of the zone or space. (The Watts per Floor Area field should be filled).

- Watts/Person

With this choice, the method used will be a factor of equipment level (watts) per person. (The Watts per Person field should be filled).

1.14.6.1.5 Field: Design Level

This field (in Watts) is typically used to represent the maximum electrical input to equipment in a zone that is then multiplied by a schedule fraction (see previous field). In EnergyPlus, this is slightly more flexible in that the electric equipment design level could be a "diversity factor" applied to a schedule of real numbers. Note that while the schedule value can vary from hour to hour, the design level field is constant for all simulation environments.

1.14.6.1.6 Field: Watts per Floor Area

This factor (watts/m^2) is used, along with the Zone or Space Floor Area to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "Watts/Area".

1.14.6.1.7 Field: Watts per Person

This factor (watts/person) is used, along with the number of occupants (people) to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "Watts/Person".

1.14.6.1.8 Heat Gains from Electric Equipment:

The electrical input to the equipment ultimately appears as heat that contributes to zone loads. In EnergyPlus this heat is divided into four different fractions. Three of these are given by the input fields Fraction Latent, Fraction Radiant and Fraction Lost. A fourth, defined as the fraction of the heat from electric equipment convected to the zone air, is calculated by the program as:

$$f_{\text{convected}} = 1.0 - (\text{FractionLatent} + \text{FractionRadiant} + \text{FractionLost}) \tag{1.26}$$

You will get an error message if Fraction Latent + Fraction Radiant + Fraction Lost exceeds 1.0.

1.14.6.1.9 Field: Fraction Latent

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of latent heat given off by electric equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by electric equipment to give the amount of latent energy produced by the electric equipment. This energy affects the moisture balance within the zone.

1.14.6.1.10 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of long-wave radiant heat being given off by electric equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by electric equipment to give the amount of long wavelength radiation gain from electric equipment in a zone.

1.14.6.1.11 Field: Fraction Lost

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of "lost" heat being given off by electric equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by electric equipment to give the amount of heat which is "lost" and does not impact the zone energy balances. This might correspond to electrical energy converted to mechanical work or heat that is vented to the atmosphere.

1.14.6.1.12 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Computers", "Copy Machines", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter objects). Subcategories are also reported in the ABUPS table. If this field is omitted or blank, the equipment will be assigned to the "General" end-use subcategory. Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and the LEED Summary table EAp2-4/5. Performance Rating Method Compliance.

An IDF example:

```
ElectricEquipment,
  DORM ROOMS AND COMMON AREAS ElecEq 1, !- Name
  DORM ROOMS AND COMMON AREAS,          !- Zone Name
  Residence Equipment,                  !- SCHEDULE Name
  EquipmentLevel,                       !- Design Level calculation method
  9210.921,                             !- Design Level {W}
  ,                                     !- Watts per Zone Floor Area {watts/m2}
  ,                                     !- Watts per Person {watts/person}
  0.0000000E+00,                        !- Fraction Latent
  0.3000000,                            !- Fraction Radiant
  0.0000000E+00,                        !- Fraction Lost
  Computers;                            !- End-use Subcategory
```

Global ElectricEquipment example:

```
ZoneList,AllOccupiedZones,SPACE1-1,SPACE2-1,SPACE3-1,SPACE4-1,SPACE5-1;

ElectricEquipment,
  AllZones with Electric Equipment,     !- Name
  AllOccupiedZones,                     !- Zone or ZoneList or Space or SpaceList Name
  EQUIP-1,                              !- Schedule Name
  Watts/Person,                         !- Design Level Calculation Method
  ,                                     !- Design Level {W}
  ,                                     !- Watts per Zone Floor Area {W/m2}
  96,                                   !- Watts per Person {W/person}
  0,                                    !- Fraction Latent
  0.3,                                  !- Fraction Radiant
  0;                                    !- Fraction Lost
```

1.14.7 GasEquipment

The object models equipment in the zone which consumes natural gas, such as cooking equipment or a gas fireplace. All of the energy consumed by the equipment becomes a heat gain in the zone or is lost (exhausted) as specified below.

1.14.7.1 Inputs

1.14.7.1.1 Field: Name

The name of the GasEquipment object.

1.14.7.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this GasEquipment object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the watts/area and watts/person options of the Design Level Calculation Method to place a varying equipment load at the same density in each zone or space. The names of the actual gas equipment objects may be concatenated as <Space Name> <GasEquipment Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.7.1.3 Field: Schedule Name

This field is the name of the schedule that modifies the design level parameter for gas equipment (see Design Level Calculation Method field and related subsequent fields). The schedule values can be any positive number. The actual energy input for gas equipment in a zone as defined by this statement is the product of the design level field and the value of the schedule specified by name in this field.

1.14.7.1.4 Field: Design Level Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal gas equipment level in the Zone. The key/choices are:

- EquipmentLevel

With this choice, the method used will be a straight insertion of the gas equipment level (Watts) for the Zone. (The Design Level field should be filled.)

- Watts/Area or Power/Area

With this choice, the method used will be a factor per floor area of the zone or space. (The Power per Floor Area field should be filled).

- Watts/Person or Power/Person

With this choice, the method used will be a factor of equipment level (watts) per person. (The Power per Person field should be filled).

1.14.7.1.5 Field: Design Level

This field (in Watts) is typically used to represent the maximum energy input to gas equipment in a zone that is then multiplied by a schedule fraction (see previous field). In EnergyPlus, this is slightly more flexible in that the gas equipment design level could be a "diversity factor" applied to a schedule of real numbers. Note that while the schedule value can vary from hour to hour, the design level field is constant for all simulation environments.

1.14.7.1.6 Field: Power per Floor Area

This factor (watts/m^2) is used, along with the Zone or Space Area to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "**Watts/Area**" or "**Power/Area**".

1.14.7.1.7 Field: Power per Person

This factor (watts/person) is used, along with the number of occupants (people) to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "**Watts/Person**" or "**Power/Person**".

1.14.7.1.8 Heat Gains from Gas Equipment:

The fuel input to the equipment ultimately appears as heat that contributes to zone loads. In EnergyPlus this heat is divided into four different fractions. Three of these are given by the input fields Fraction Latent, Fraction Radiant and Fraction Lost. A fourth, defined as the fraction of the heat from gas equipment convected to the zone air, is calculated by the program as:

$$f_{convected} = 1.0 - (\text{FractionLatent} + \text{FractionRadiant} + \text{FractionLost}) \tag{1.27}$$

You will get an error message if Fraction Latent + Fraction Radiant + Fraction Lost exceeds 1.0.

1.14.7.1.9 Field: Fraction Latent

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of latent heat given off by gas equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by gas equipment to give the amount of latent energy produced by the gas equipment. This energy affects the moisture balance within the zone.

1.14.7.1.10 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of long-wave radiant heat being given off by gas equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by gas equipment to give the amount of long wavelength radiation gain from gas equipment in a zone.

1.14.7.1.11 Field: Fraction Lost

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of "lost" heat being given off by gas equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by gas equipment to give the amount of heat which is "lost" and does not impact the zone energy balances. This might correspond to input energy converted to mechanical work or heat that is vented to the atmosphere.

1.14.7.1.12 Field: Carbon Dioxide Generation Rate

This numeric input field specifies carbon dioxide generation rate with units of m3/s-W. The default value of 0.0 assumes the equipment is fully vented to outdoors. In the absence of better information, the user might consider using a value of 3.45E-8 m3/s-W which assumes the equipment is not vented to outdoors. This value is converted from natural gas CO_2 emission rate at 11.7 lbs CO_2 per therm. The CO_2 emission rate is provided by U.S. Energy Information Administration, "Frequently Asked Questions - Environment, Questions About Environmental Emissions", http://tonto.eia.doe.gov/ask/environment_faqs.asp#CO2_quantity, January 2010. The maximum value for this input field is 3.45×10^{-7} m3/s-W.

1.14.7.1.13 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Cooking", "Clothes Drying", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter objects). Subcategories are also reported in the ABUPS table. If this field is omitted or blank, the equipment will be assigned to the "General" end-use subcategory. Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and the LEED Summary table EAp2-4/5 Performance Rating Method Compliance.

An IDF example:

```
GasEquipment,
  DORM ROOMS AND COMMON AREAS GasEq 1,  !- Name
  DORM ROOMS AND COMMON AREAS,          !- Zone or ZoneList or Space or SpaceList Name
  Gas Eq Sch,                           !-Schedule Name
```

```
  EquipmentLevel,                        !- Design Level Calculation Method
  29287.51,                              !- Design Level {W}
  ,                                      !- Power per Floor Area {W/m2}
  ,                                      !- Power per Person {W/Person}
  0,                                     !- Fraction Latent
  0.3,                                   !- Fraction Radiant
  0,                                     !- Fraction Lost
  0,                                     !- Carbon Dioxide Generation Rate {m3/s-W}
  Cooking;                               !- End-Use Subcategory
```

Global Gas Equipment example:

```
ZoneList,OfficeZones,Left Fork, Middle Fork, Right Fork;

GasEquipment,
  Office Zones with Gas,                 !- Name
  OfficeZones,                           !- Zone or ZoneList or Space or SpaceList Name
  Gas Eq Sch,                            !- Schedule Name
  Watts/Area,                            !- Design Level Calculation Method
  ,                                      !- Design Level {W}
  197,                                   !- Power per Floor Area {W/m2}
  ,                                      !- Power per Person {W/Person}
  0.0000000E+00,                         !- Fraction Latent
  0.3000000,                             !- Fraction Radiant
  0.0000000E+00;                         !- Fraction Lost
```

1.14.8 HotWaterEquipment

The object models hot water equipment in the zone which consumes district heating, such as cooking equipment or process loads. All of the energy consumed by the equipment becomes a heat gain in the zone or is lost (exhausted) as specified below. This object consumes district heating energy directly and does not cause a load on a hot water plant loop or water heater. For domestic hot water uses, such as sinks and showers, see WaterUse:Equipment.

1.14.8.1 Inputs

1.14.8.1.1 Field: Name

The name of the HotWaterEquipment object.

1.14.8.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this HotWaterEquipment object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the watts/area and watts/person options of the Design Level Calculation Method to place a varying equipment load at the same density in each zone or space. The names of the actual hot water equipment objects may be concatenated as <Space Name> <HotWaterEquipment Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.8.1.3 Field: Schedule Name

This field is the name of the schedule that modifies the design level parameter for hot water equipment (see Design Level Calculation Method field and related subsequent fields). The schedule values can be any positive number. The actual energy input for hot water equipment in a zone as defined by this statement is the product of the design level field and the value of the schedule specified by name in this field.

1.14.8.1.4 Field: Design Level Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal hot water equipment level in the Zone. The key/choices are:

- EquipmentLevel

With this choice, the method used will be a straight insertion of the hot water equipment level (Watts) for the Zone. (The Design Level field should be filled.)

- Watts/Area or Power/Area

With this choice, the method used will be a factor per floor area of the zone or space. (The Power per Floor Area field should be filled).

- Watts/Person or Power/Person

With this choice, the method used will be a factor of equipment level (watts) per person. (The Power per Person field should be filled).

1.14.8.1.5 Field: Design Level

This field (in Watts) is typically used to represent the maximum energy input to hot water equipment in a zone that is then multiplied by a schedule fraction (see previous field). In EnergyPlus, this is slightly more flexible in that the hot water equipment design level could be a "diversity factor" applied to a schedule of real numbers. Note that while the schedule value can vary from hour to hour, the design level field is constant for all simulation environments.

1.14.8.1.6 Field: Power per Floor Area

This factor (watts/m^2) is used, along with the Zone or Space Area to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "**Watts/Area**" or "**Power/Area**".

1.14.8.1.7 Field: Power per Person

This factor (watts/person) is used, along with the number of occupants (people) to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "**Watts/Person**" or "**Power/Person**".

1.14.8.1.8 Heat Gains from Hot Water Equipment:

The fuel input to the equipment ultimately appears as heat that contributes to zone loads. In EnergyPlus this heat is divided into four different fractions. Three of these are given by the input fields Fraction Latent, Fraction Radiant and Fraction Lost. A fourth, defined as the fraction of the heat from hot water equipment convected to the zone air, is calculated by the program as:

$$f_{\text{convected}} = 1.0 - (\text{FractionLatent} + \text{FractionRadiant} + \text{FractionLost}) \tag{1.28}$$

You will get an error message if Fraction Latent + Fraction Radiant + Fraction Lost exceeds 1.0.

1.14.8.1.9 Field: Fraction Latent

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of latent heat given off by hot water equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by hot water equipment to give the amount of latent energy produced by the hot water equipment. This energy affects the moisture balance within the zone.

1.14.8.1.10 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of long-wave radiant heat being given off by hot water equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by hot water equipment to give the amount of long wavelength radiation gain from hot water equipment in a zone.

1.14.8.1.11 Field: Fraction Lost

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of "lost" heat being given off by hot water equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by hot water equipment to give the amount of heat which is "lost" and does not impact the zone energy balances. This might correspond to input energy converted to mechanical work or heat that is vented to the atmosphere.

1.14.8.1.12 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Cooking", "Clothes Drying", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter object). Subcategories are also reported in the ABUPS table. If this field is omitted or blank, the equipment will be assigned to the "General" end-use subcategory. Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and the LEED Summary table EAp2-4/5 Performance Rating Method Compliance.

IDF Examples:

```
HotWaterEquipment,
  SPACE2-1 HWEq 1,    !- Name
  SPACE2-1,           !- Zone or ZoneList or Space or SpaceList Name
  EQUIP-1,            !- SCHEDULE Name
  EquipmentLevel,     !- Design Level calculation method
  300,                !- Design Level {W}
  ,                   !- Power per Floor Area {watts/m2}
  ,                   !- Power per Person {watts/person}
  0.2,                !- Fraction Latent
  0.1,                !- Fraction Radiant
  0.5,                !- Fraction Lost
  Dishwashing;        !- End-Use Subcategory
```

Global Hot Water Equipment example:

```
ZoneList,OfficeZones,Left Fork, Middle Fork, Right Fork;

HotWaterEquipment,
  Office Zones with HoWater Equipment,!- Name
  OfficeZones,                        !- Zone or ZoneList or Space or SpaceList Name
  HotWater Eq Sch,                    !- Schedule Name
  Watts/Area,                         !- Design Level Calculation Method
  ,                                   !- Design Level {W}
  50,                                 !- Power per Floor Area {W/m2}
  ,                                   !- Power per Person {W/Person}
  0.0000000E+00,                      !- Fraction Latent
  0.3000000,                          !- Fraction Radiant
  0.0000000E+00;                      !- Fraction Lost
```

1.14.9 SteamEquipment

The object models steam equipment in the zone which consumes district heating, such as cooking equipment or process loads. All of the energy consumed by the equipment becomes a heat gain in the zone or is lost (exhausted) as specified below. This object consumes district heating energy directly and does not cause a load on a steam plant loop.

1.14.9.1 Inputs

1.14.9.1.1 Field: Name

The name of the SteamEquipment object.

1.14.9.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this SteamEquipment object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the watts/area and watts/person options of the Design Level Calculation Method to place a varying equipment load at the same density in each zone or space. The names of the actual steam equipment objects may be concatenated as <Space Name> <SteamEquipment Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.9.1.3 Field: Schedule Name

This field is the name of the schedule that modifies the design level parameter for steam equipment (see Design Level Calculation Method field and related subsequent fields). The schedule values can be any positive number. The actual energy input for steam equipment in a zone as defined by this statement is the product of the design level field and the value of the schedule specified by name in this field.

1.14.9.1.4 Field: Design Level Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal steam equipment level in the Zone. The key/choices are:

- EquipmentLevel

With this choice, the method used will be a straight insertion of the steam equipment level (Watts) for the Zone. (The Design Level field should be filled.)

- Watts/Area or Power/Area

With this choice, the method used will be a factor per floor area of the zone. (The Power per Floor Area field should be filled).

- Watts/Person or Power/Person

With this choice, the method used will be a factor of equipment level (watts) per person. (The Power per Person field should be filled).

1.14.9.1.5 Field: Design Level

This field (in Watts) is typically used to represent the maximum energy input to steam equipment in a zone that is then multiplied by a schedule fraction (see previous field). In EnergyPlus, this is slightly more flexible in that the steam equipment design level could be a "diversity factor" applied to a schedule of real numbers. Note that while the schedule value can vary from hour to hour, the design level field is constant for all simulation environments.

1.14.9.1.6 Field: Power per Floor Area

This factor (watts/m^2) is used, along with the Zone or Space Area to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "**Watts/Area**" or "**Power/Area**".

1.14.9.1.7 Field: Power per Person

This factor (watts/person) is used, along with the number of occupants (people) to determine the maximum equipment level as described in the Design Level field. The choice from the method field should be "**Watts/Person**" or "**Power/Person**".

1.14.9.1.8 Heat Gains from Steam Equipment:

The fuel input to the equipment ultimately appears as heat that contributes to zone loads. In EnergyPlus this heat is divided into four different fractions. Three of these are given by the input fields Fraction Latent, Fraction Radiant and Fraction Lost. A fourth, defined as the fraction of the heat from steam equipment convected to the zone air, is calculated by the program as:

$$f_{\text{convected}} = 1.0 - (\text{FractionLatent} + \text{FractionRadiant} + \text{FractionLost}) \tag{1.29}$$

You will get an error message if Fraction Latent + Fraction Radiant + Fraction Lost exceeds 1.0.

1.14.9.1.9 Field: Fraction Latent

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of latent heat given off by steam equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by steam equipment to give the amount of latent energy produced by the steam equipment. This energy affects the moisture balance within the zone.

1.14.9.1.10 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of long-wave radiant heat being given off by steam equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by steam equipment to give the amount of long wavelength radiation gain from steam equipment in a zone.

1.14.9.1.11 Field: Fraction Lost

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of "lost" heat being given off by steam equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by steam equipment to give the amount of heat which is "lost" and does not impact the zone energy balances. This might correspond to input energy converted to mechanical work or heat that is vented to the atmosphere.

1.14.9.1.12 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Cooking", "Clothes Drying", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter objects). Subcategories are also reported in the ABUPS table. If this field is omitted or blank, the equipment will be assigned to the "General" end-use subcategory. Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and the LEED Summary table EAp2-4/5 Performance Rating Method Compliance.

IDF Examples:

```
SteamEquipment,
  SPACE4-1 ElecEq 1, !- Name
  SPACE4-1,          !- Zone or ZoneList or Space or SpaceList Name
  EQUIP-1,           !- SCHEDULE Name
  EquipmentLevel,    !- Design Level calculation method
  1050,              !- Design Level {W}
  ,                  !- Power per Floor Area {watts/m2}
  ,                  !- Power per Person {watts/person}
```

```
  0.5,                !- Fraction Latent
  0.3,                !- Fraction Radiant
  0,                  !- Fraction Lost
  Laundry;            !- End-Use Subcategory
```

1.14.10 OtherEquipment

Other Equipment object is provided as an additional source for heat gains or losses directly to the zone with a fuel type that is configurable. If a fuel type is specified, the energy is attributed to the appropriate end use. Otherwise, a loss can be entered by putting a negative value into the Design Level field and this object will not have an end-use component – gains or losses do not show up in the bottom energy lines (except as influencing overall zone gains or losses).

1.14.10.1 Inputs

1.14.10.1.1 Field: Name

The name of the OtherEquipment object.

1.14.10.1.2 Field: Fuel Type

This field designates the appropriate meter for the equipment. Valid fuel types are: None, Electricity, NaturalGas, Propane, FuelOilNo1, FuelOilNo2, Diesel, Gasoline, Coal, DistrictHeatingSteam, DistrictHeatingWater, DistrictCooling, OtherFuel1 and OtherFuel2. The fuel type triggers the application of consumption amounts to the appropriate energy meters. If the None fuel type is selected (the default if left blank), no end uses will be associated with the object, only the zone gains.

1.14.10.1.3 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this OtherEquipment object to one or more thermal zones or spaces in the building. The Zone, Zonelist or SpaceList options can be used effectively with the watts/area and watts/person options of the Design Level Calculation Method to place a varying equipment load at the same density in each zone or space. The names of the actual other equipment objects may be concatenated as <Space Name> <OtherEquipment Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.10.1.4 Field: Schedule Name

This field is the name of the schedule that modifies the design level parameter for other equipment (see Design Level Calculation Method field and related subsequent fields). The schedule values can be any positive number. The actual energy input for other equipment in a zone as defined by this statement is the product of the design level field and the value of the schedule specified by name in this field.

1.14.10.1.5 Field: Design Level Calculation Method

This field is a key/choice field that tells which of the next three fields are filled and is descriptive of the method for calculating the nominal other equipment level in the Zone. The key/choices are:

- EquipmentLevel

With this choice, the method used will be a straight insertion of the other equipment level (Watts) for the Zone. (The Design Level field should be filled.)

- Watts/Area or Power/Area

With this choice, the method used will be a factor per floor area of the zone. (The Power per Floor Area field should be filled).

- Watts/Person or Power/Person

With this choice, the method used will be a factor of equipment level (watts) per person. (The Power per Person field should be filled).

1.14.10.1.6 Field: Design Level

This field (in Watts) is typically used to represent the maximum energy input to other equipment in a zone that is then multiplied by a schedule fraction (see previous field). In EnergyPlus, this is slightly more flexible in that the other equipment design level could be a "diversity factor" applied to a schedule of real numbers. This value can be negative to denote a loss if the None fuel type is selected, otherwise this must be positive. Note that while the schedule value can vary from hour to hour, the design level field is constant for all simulation environments.

1.14.10.1.7 Field: Power per Floor Area

This factor (watts/m^2) is used, along with the Zone or Space Area to determine the maximum equipment level as described in the Design Level field. This value can be negative to denote a loss if the None fuel type is selected, otherwise this must be positive. The choice from the method field should be "**Watts/Area**" or "**Power/Area**".

1.14.10.1.8 Field: Power per Person

This factor (watts/person) is used, along with the number of occupants (people) to determine the maximum equipment level as described in the Design Level field. This value can be negative to denote a loss if the None fuel type is selected, otherwise this must be positive. The choice from the method field should be "**Watts/Person**" or "**Power/Person**".

1.14.10.1.9 Heat Gains/Losses from Other Equipment:

The fuel input to the equipment ultimately appears as heat that contributes to zone loads. In EnergyPlus this heat is divided into four different fractions. Three of these are given by the input fields Fraction Latent, Fraction Radiant and Fraction Lost. A fourth, defined as the fraction of the heat from other equipment convected to the zone air, is calculated by the program as:

$$f_{\text{convected}} = 1.0 - (\text{FractionLatent} + \text{FractionRadiant} + \text{FractionLost}) \tag{1.30}$$

You will get an error message if FractionLatent + FractionRadiant + FractionLost exceeds 1.0.

1.14.10.1.10 Field: Fraction Latent

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of latent heat given off by other equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by other equipment to give the amount of latent energy produced by the other equipment. This energy affects the moisture balance within the zone.

1.14.10.1.11 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of long-wave radiant heat being given off by other equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by other equipment to give the amount of long wavelength radiation gain from other equipment in a zone.

1.14.10.1.12 Field: Fraction Lost

This field is a decimal number between 0.0 and 1.0 and is used to characterize the amount of "lost" heat being given off by other equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by other equipment to give the amount of heat which is "lost" and does not impact the zone energy balances. This might correspond to input energy converted to mechanical work or heat that is vented to the atmosphere.

1.14.10.1.13 Field: Carbon Dioxide Generation Rate

This numeric input field specifies carbon dioxide generation rate with units of m3/s-W. The default value of 0.0 assumes the equipment is fully vented to outdoors. The maximum value for this input field is 3.45E-7 m3/s-W.

1.14.10.1.14 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Cooking", "Clothes Drying", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter objects). Subcategories are also reported in the ABUPS table. If this field is omitted or blank, the equipment will be assigned to the "General" end-use subcategory. Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and in the LEED Summary table EAp2-4/5 Performance Rating Method Compliance.

IDF Examples

```
OtherEquipment,
        BASE-1 OthEq 1, !- Name
        Propane,  !- Fuel Use Type
        BASE-1, !- Zone or ZoneList or Space or SpaceList Name
        ALWAYSON, !- SCHEDULE Name
        EquipmentLevel, !- Design Level calculation method
        6766., !- Design Level {W}
        , !- Power per Floor Area {watts/m2}
        , !- Power per Person {watts/person}
        0, !- Fraction Latent
        0.3, !- Fraction Radiant
        0, !- Fraction Lost
        1.2E-7, !- Carbon Dioxide Generation Rate
        SubCategory1; !- End-Use Subcategory
```

1.14.11 Internal Gain Equipment Outputs

Each type of equipment object has output variables for individual objects and for space and zone totals.

Electric Equipment

- Zone,Average,Electric Equipment Electricity Rate [W]
- Zone,Sum,Electric Equipment Electricity Energy [J]
- Zone,Sum,Electric Equipment Radiant Heating Energy [J]
- Zone,Average,Electric Equipment Radiant Heating Rate [W]
- Zone,Sum,Electric Equipment Convective Heating Energy [J]
- Zone,Average,Electric Equipment Convective Heating Rate [W]
- Zone,Sum,Electric Equipment Latent Gain Energy [J]
- Zone,Average,Electric Equipment Latent Gain Rate [W]

- Zone,Sum,Electric Equipment Lost Heat Energy [J]
- Zone,Average,Electric Equipment Lost Heat Rate [W]
- Zone,Sum,Electric Equipment Total Heating Energy [J]
- Zone,Average,Electric Equipment Total Heating Rate [W]
- Zone,Average,Space Electric Equipment Electricity Rate [W]
- Zone,Sum,Space Electric Equipment Electricity Energy [J]
- Zone,Sum,Space Electric Equipment Radiant Heating Energy [J]
- Zone,Average,Space Electric Equipment Radiant Heating Rate [W]
- Zone,Sum,Space Electric Equipment Convective Heating Energy [J]
- Zone,Average,Space Electric Equipment Convective Heating Rate [W]
- Zone,Sum,Space Electric Equipment Latent Gain Energy [J]
- Zone,Average,Space Electric Equipment Latent Gain Rate [W]
- Zone,Sum,Space Electric Equipment Lost Heat Energy [J]
- Zone,Average,Space Electric Equipment Lost Heat Rate [W]
- Zone,Sum,Space Electric Equipment Total Heating Energy [J]
- Zone,Average,Space Electric Equipment Total Heating Rate [W]
- Zone,Average,Zone Electric Equipment Electricity Rate [W]
- Zone,Sum,Zone Electric Equipment Electricity Energy [J]
- Zone,Sum,Zone Electric Equipment Radiant Heating Energy [J]
- Zone,Average,Zone Electric Equipment Radiant Heating Rate [W]
- Zone,Sum,Zone Electric Equipment Convective Heating Energy [J]
- Zone,Average,Zone Electric Equipment Convective Heating Rate [W]
- Zone,Sum,Zone Electric Equipment Latent Gain Energy [J]
- Zone,Average,Zone Electric Equipment Latent Gain Rate [W]
- Zone,Sum,Zone Electric Equipment Lost Heat Energy [J]
- Zone,Average,Zone Electric Equipment Lost Heat Rate [W]
- Zone,Sum,Zone Electric Equipment Total Heating Energy [J]
- Zone,Average,Zone Electric Equipment Total Heating Rate [W]

Gas Equipment

- Zone,Average,Gas Equipment NaturalGas Rate [W]

- Zone,Sum,Gas Equipment NaturalGas Energy [J]
- Zone,Sum,Gas Equipment Radiant Heating Energy [J]
- Zone,Sum,Gas Equipment Convective Heating Energy [J]
- Zone,Sum,Gas Equipment Latent Gain Energy [J]
- Zone,Sum,Gas Equipment Lost Heat Energy [J]
- Zone,Sum,Gas Equipment Total Heating Energy [J]
- Zone,Average,Gas Equipment Radiant Heating Rate [W]
- Zone,Average,Gas Equipment Convective Heating Rate [W]
- Zone,Average,Gas Equipment Latent Gain Rate [W]
- Zone,Average,Gas Equipment Lost Heat Rate [W]
- Zone,Average,Gas Equipment Total Heating Rate [W]
- Zone,Average,Space Gas Equipment NaturalGas Rate [W]
- Zone,Sum,Space Gas Equipment NaturalGas Energy [J]
- Zone,Sum,Space Gas Equipment Radiant Heating Energy [J]
- Zone,Average,Space Gas Equipment Radiant Heating Rate [W]
- Zone,Sum,Space Gas Equipment Convective Heating Energy [J]
- Zone,Average,Space Gas Equipment Convective Heating Rate [W]
- Zone,Sum,Space Gas Equipment Latent Gain Energy [J]
- Zone,Average,Space Gas Equipment Latent Gain Rate [W]
- Zone,Sum,Space Gas Equipment Lost Heat Energy [J]
- Zone,Average,Space Gas Equipment Lost Heat Rate [W]
- Zone,Sum,Space Gas Equipment Total Heating Energy [J]
- Zone,Average,Space Gas Equipment Total Heating Rate [W]
- Zone,Average,Zone Gas Equipment NaturalGas Rate [W]
- Zone,Sum,Zone Gas Equipment NaturalGas Energy [J]
- Zone,Sum,Zone Gas Equipment Radiant Heating Energy [J]
- Zone,Average,Zone Gas Equipment Radiant Heating Rate [W]
- Zone,Sum,Zone Gas Equipment Convective Heating Energy [J]
- Zone,Average,Zone Gas Equipment Convective Heating Rate [W]
- Zone,Sum,Zone Gas Equipment Latent Gain Energy [J]

- Zone,Average,Zone Gas Equipment Latent Gain Rate [W]
- Zone,Sum,Zone Gas Equipment Lost Heat Energy [J]
- Zone,Average,Zone Gas Equipment Lost Heat Rate [W]
- Zone,Sum,Zone Gas Equipment Total Heating Energy [J]
- Zone,Average,Zone Gas Equipment Total Heating Rate [W]

HotWater Equipment

- Zone,Average,Hot Water Equipment District Heating Rate [W]
- Zone,Sum,Hot Water Equipment District Heating Energy [J]
- Zone,Sum,Hot Water Equipment Radiant Heating Energy [J]
- Zone,Average,Hot Water Equipment Radiant Heating Rate [W]
- Zone,Sum,Hot Water Equipment Convective Heating Energy [J]
- Zone,Average,Hot Water Equipment Convective Heating Rate [W]
- Zone,Sum,Hot Water Equipment Latent Gain Energy [J]
- Zone,Average,Hot Water Equipment Latent Gain Rate [W]
- Zone,Sum,Hot Water Equipment Lost Heat Energy [J]
- Zone,Average,Hot Water Equipment Lost Heat Rate [W]
- Zone,Sum,Hot Water Equipment Total Heating Energy [J]
- Zone,Average,Hot Water Equipment Total Heating Rate [W]
- Zone,Average,Space Hot Water Equipment District Heating Rate [W]
- Zone,Sum,Space Hot Water Equipment District Heating Energy [J]
- Zone,Sum,Space Hot Water Equipment Radiant Heating Energy [J]
- Zone,Average,Space Hot Water Equipment Radiant Heating Rate [W]
- Zone,Sum,Space Hot Water Equipment Convective Heating Energy [J]
- Zone,Average,Space Hot Water Equipment Convective Heating Rate [W]
- Zone,Sum,Space Hot Water Equipment Latent Gain Energy [J]
- Zone,Average,Space Hot Water Equipment Latent Gain Rate [W]
- Zone,Sum,Space Hot Water Equipment Lost Heat Energy [J]
- Zone,Average,Space Hot Water Equipment Lost Heat Rate [W]
- Zone,Sum,Space Hot Water Equipment Total Heating Energy [J]
- Zone,Average,Space Hot Water Equipment Total Heating Rate [W]

- Zone,Average,Zone Hot Water Equipment District Heating Rate [W]
- Zone,Sum,Zone Hot Water Equipment District Heating Energy [J]
- Zone,Sum,Zone Hot Water Equipment Radiant Heating Energy [J]
- Zone,Average,Zone Hot Water Equipment Radiant Heating Rate [W]
- Zone,Sum,Zone Hot Water Equipment Convective Heating Energy [J]
- Zone,Average,Zone Hot Water Equipment Convective Heating Rate [W]
- Zone,Sum,Zone Hot Water Equipment Latent Gain Energy [J]
- Zone,Average,Zone Hot Water Equipment Latent Gain Rate [W]
- Zone,Sum,Zone Hot Water Equipment Lost Heat Energy [J]
- Zone,Average,Zone Hot Water Equipment Lost Heat Rate [W]
- Zone,Sum,Zone Hot Water Equipment Total Heating Energy [J]
- Zone,Average,Zone Hot Water Equipment Total Heating Rate [W]

Steam Equipment

- Zone,Average,Steam Equipment District Heating Rate [W]
- Zone,Sum,Steam Equipment District Heating Energy [J]
- Zone,Sum,Steam Equipment Radiant Heating Energy [J]
- Zone,Average,Steam Equipment Radiant Heating Rate [W]
- Zone,Sum,Steam Equipment Convective Heating Energy [J]
- Zone,Average,Steam Equipment Convective Heating Rate [W]
- Zone,Sum,Steam Equipment Latent Gain Energy [J]
- Zone,Average,Steam Equipment Latent Gain Rate [W]
- Zone,Sum,Steam Equipment Lost Heat Energy [J]
- Zone,Average,Steam Equipment Lost Heat Rate [W]
- Zone,Sum,Steam Equipment Total Heating Energy [J]
- Zone,Average,Steam Equipment Total Heating Rate [W]
- Zone,Average,Space Steam Equipment District Heating Rate [W]
- Zone,Sum,Space Steam Equipment District Heating Energy [J]
- Zone,Sum,Space Steam Equipment Radiant Heating Energy [J]
- Zone,Average,Space Steam Equipment Radiant Heating Rate [W]
- Zone,Sum,Space Steam Equipment Convective Heating Energy [J]

- Zone,Average,Space Steam Equipment Convective Heating Rate [W]
- Zone,Sum,Space Steam Equipment Latent Gain Energy [J]
- Zone,Average,Space Steam Equipment Latent Gain Rate [W]
- Zone,Sum,Space Steam Equipment Lost Heat Energy [J]
- Zone,Average,Space Steam Equipment Lost Heat Rate [W]
- Zone,Sum,Space Steam Equipment Total Heating Energy [J]
- Zone,Average,Space Steam Equipment Total Heating Rate [W]
- Zone,Average,Zone Steam Equipment District Heating Rate [W]
- Zone,Sum,Zone Steam Equipment District Heating Energy [J]
- Zone,Sum,Zone Steam Equipment Radiant Heating Energy [J]
- Zone,Average,Zone Steam Equipment Radiant Heating Rate [W]
- Zone,Sum,Zone Steam Equipment Convective Heating Energy [J]
- Zone,Average,Zone Steam Equipment Convective Heating Rate [W]
- Zone,Sum,Zone Steam Equipment Latent Gain Energy [J]
- Zone,Average,Zone Steam Equipment Latent Gain Rate [W]
- Zone,Sum,Zone Steam Equipment Lost Heat Energy [J]
- Zone,Average,Zone Steam Equipment Lost Heat Rate [W]
- Zone,Sum,Zone Steam Equipment Total Heating Energy [J]
- Zone,Average,Zone Steam Equipment Total Heating Rate [W]

Other Equipment

- Zone,Average,Other Equipment Fuel Rate [W]
- Zone,Sum,Other Equipment Fuel Energy [J]
- Zone,Sum,Other Equipment Radiant Heating Energy [J]
- Zone,Average,Other Equipment Radiant Heating Rate [W]
- Zone,Sum,Other Equipment Convective Heating Energy [J]
- Zone,Average,Other Equipment Convective Heating Rate [W]
- Zone,Sum,Other Equipment Latent Gain Energy [J]
- Zone,Average,Other Equipment Latent Gain Rate [W]
- Zone,Sum,Other Equipment Lost Heat Energy [J]
- Zone,Average,Other Equipment Lost Heat Rate [W]

- Zone,Sum,Other Equipment Total Heating Energy [J]
- Zone,Average,Other Equipment Total Heating Rate [W]
- Zone,Sum,Space Other Equipment Radiant Heating Energy [J]
- Zone,Average,Space Other Equipment Radiant Heating Rate [W]
- Zone,Sum,Space Other Equipment Convective Heating Energy [J]
- Zone,Average,Space Other Equipment Convective Heating Rate [W]
- Zone,Sum,Space Other Equipment Latent Gain Energy [J]
- Zone,Average,Space Other Equipment Latent Gain Rate [W]
- Zone,Sum,Space Other Equipment Lost Heat Energy [J]
- Zone,Average,Space Other Equipment Lost Heat Rate [W]
- Zone,Sum,Space Other Equipment Total Heating Energy [J]
- Zone,Average,Space Other Equipment Total Heating Rate [W]
- Zone,Sum,Zone Other Equipment Radiant Heating Energy [J]
- Zone,Average,Zone Other Equipment Radiant Heating Rate [W]
- Zone,Sum,Zone Other Equipment Convective Heating Energy [J]
- Zone,Average,Zone Other Equipment Convective Heating Rate [W]
- Zone,Sum,Zone Other Equipment Latent Gain Energy [J]
- Zone,Average,Zone Other Equipment Latent Gain Rate [W]
- Zone,Sum,Zone Other Equipment Lost Heat Energy [J]
- Zone,Average,Zone Other Equipment Lost Heat Rate [W]
- Zone,Sum,Zone Other Equipment Total Heating Energy [J]
- Zone,Average,Zone Other Equipment Total Heating Rate [W]

1.14.11.0.1 Electric Equipment Electricity Rate [W]

1.14.11.0.2 Electric Equipment Electricity Energy [J]

1.14.11.0.3 Space or Zone Electric Equipment Electricity Rate [W]

1.14.11.0.4 Space or Zone Electric Equipment Electricity Energy [J]

The electric equipment electric power consumption in Watts (for power) or Joules (for energy). It is the sum of the radiant, convective, latent and lost components. Electric Equipment Electricity Energy is added to the following electricity meters:

```
Electricity:Facility
Electricity:Building
Electricity:Zone:<Zone Name>
Electricity:SpaceType:<Space Type Name>
InteriorEquipment:Electricity
InteriorEquipment:Electricity:Zone:<Zone Name>
InteriorEquipment:Electricity:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:Electricity
<End-Use Subcategory>:InteriorEquipment:Electricity:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:Electricity:SpaceType:<Space Type Name>
```

1.14.11.0.5 Gas Equipment NaturalGas Rate [W]

1.14.11.0.6 Gas Equipment NaturalGas Energy [J]

1.14.11.0.7 Space or Zone Gas Equipment NaturalGas Rate [W]

1.14.11.0.8 Space or Zone Gas Equipment NaturalGas Energy [J]

The gas equipment natural gas consumption in Watts (for power) or Joules (for energy). It is the sum of the radiant, convective, latent and lost components. Gas Equipment NaturalGas Energy is added to the following NaturalGas meters:

```
NaturalGas:Facility
NaturalGas:Building
NaturalGas:Zone:<Zone Name>
NaturalGas:SpaceType:<Space Type Name>
InteriorEquipment:NaturalGas
InteriorEquipment:NaturalGas:Zone:<Zone Name>
InteriorEquipment:NaturalGas:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:NaturalGas
<End-Use Subcategory>:InteriorEquipment:NaturalGas:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:NaturalGas:SpaceType:<Space Type Name>
```

1.14.11.0.9 Hot Water Equipment District Heating Rate [W]

1.14.11.0.10 Hot Water Equipment District Heating Energy [J]

1.14.11.0.11 Space or Zone Hot Water Equipment District Heating Rate [W]

1.14.11.0.12 Space or Zone Hot Water Equipment District Heating Energy [J]

The hot water equipment district heating consumption in Watts (for power) or Joules (for energy). It is the sum of the radiant, convective, latent and lost components. Hot Water Equipment District Heating Energy is added to the following district heating meters:

```
DistrictHeatingWater:Facility
DistrictHeatingWater:Building
DistrictHeatingWater:Zone:<Zone Name>
DistrictHeatingWater:SpaceType:<Space Type Name>
InteriorEquipment:DistrictHeatingWater
InteriorEquipment:DistrictHeatingWater:Zone:<Zone Name>
InteriorEquipment:DistrictHeatingWater:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:DistrictHeatingWater
```

```
<End-Use Subcategory>:InteriorEquipment:DistrictHeatingWater:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:DistrictHeatingWater:SpaceType:<Space Type Name>
```

1.14.11.0.13 Steam Equipment District Heating Rate [W]

1.14.11.0.14 Steam Equipment District Heating Energy [J]

1.14.11.0.15 Space or Zone Steam Equipment District Heating Rate [W]

1.14.11.0.16 Space or Zone Steam Equipment District Heating Energy [J]

The steam equipment district heating consumption in Watts (for power) or Joules (for energy). It is the sum of the radiant, convective, latent and lost components. Steam Equipment District Heating Energy is added to the following:

```
DistrictHeatingWater:Facility
DistrictHeatingWater:Building
DistrictHeatingWater:Zone:<Zone Name>
DistrictHeatingWater:SpaceType:<Space Type Name>
InteriorEquipment:DistrictHeatingWater
InteriorEquipment:DistrictHeatingWater:Zone:<Zone Name>
InteriorEquipment:DistrictHeatingWater:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:DistrictHeatingWater
<End-Use Subcategory>:InteriorEquipment:DistrictHeatingWater:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:DistrictHeatingWater:SpaceType:<Space Type Name>
```

1.14.11.0.17 Other Equipment Fuel Rate [W]

1.14.11.0.18 Other Equipment Fuel Energy [J]

1.14.11.0.19 Space or Zone Other Equipment Fuel Rate [W]

1.14.11.0.20 Space or Zone Other Equipment Fuel Energy [J]

The other equipment fuel consumption in Watts (for power) or Joules (for energy). It is the sum of the radiant, convective, latent and lost components. Other Equipment Fuel Energy is added to the following fuel meters corresponding to the Fuel Type input. If Fuel Type = None, this energy is not metered.

```
<Fuel Type>:Facility
<Fuel Type>:Building
<Fuel Type>:Zone:<Zone Name>
<Fuel Type>:SpaceType:<Space Type Name>
InteriorEquipment:<Fuel Type>
InteriorEquipment:<Fuel Type>:Zone:<Zone Name>
InteriorEquipment:<Fuel Type>:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:<Fuel Type>
<End-Use Subcategory>:InteriorEquipment:<Fuel Type>:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:<Fuel Type>:SpaceType:<Space Type Name>
```

1.14.11.0.21 <Type> Equipment Radiant Heating Rate [W]

1.14.11.0.22 <Type> Equipment Radiant Heating Energy [J]

1.14.11.0.23 Space or Zone <Type> Equipment Radiant Heating Rate [W]

1.14.11.0.24 Space or Zone <Type> Equipment Radiant Heating Energy [J]

The amount of heat gain from equipment that is in the form of long-wave (thermal) radiation entering the zone. This heat is absorbed by the inside surfaces of the space or zone according to an area times long-wave absorptance weighting scheme.

1.14.11.0.25 <Type> Equipment Convective Heating Rate [W]

1.14.11.0.26 <Type> Equipment Convective Heating Energy [J]

1.14.11.0.27 Space or Zone <Type> Equipment Convective Heating Rate [W]

1.14.11.0.28 Space or Zone <Type> Equipment Convective Heating Energy [J]

The amount of heat gain from equipment that is convected to the space or zone air.

1.14.11.0.29 <Type> Latent Gain Rate [W]

1.14.11.0.30 <Type> Equipment Latent Gain Energy [J]

1.14.11.0.31 Space or Zone <Type> Latent Gain Rate [W]

1.14.11.0.32 Space or Zone <Type> Equipment Latent Gain Energy [J]

The amount of heat gain from equipment that goes added to the space or zone air as a latent (moisture) gain.

1.14.11.0.33 <Type> Lost Heat Rate [W]

1.14.11.0.34 <Type> Equipment Lost Heat Energy [J]

1.14.11.0.35 Space or Zone <Type> Lost Heat Rate [W]

1.14.11.0.36 Space or Zone <Type> Equipment Lost Heat Energy [J]

The amount of energy input to the equipment that is removed (exhausted) and does not add any heat gain to the space or zone.

1.14.11.0.37 <Type> Equipment Total Heating Rate [W]

1.14.11.0.38 <Type> Equipment Total Heating Energy [J]

1.14.11.0.39 Space or Zone <Type> Equipment Total Heating Rate [W]

1.14.11.0.40 Space or Zone <Type> Equipment Total Heating Energy [J]

The total heat gain from equipment. It is the sum of the following components: Total Heat Gain = Radiant Heat Gain + Convective Heat Gain + Latent Heat Gain. It is also equal to the power input to the equipment minus any Lost Heat.

1.14.12 IndoorLivingWall

Indoor greenery systems such as indoor living walls are panels of plants, which grow hydroponically or soil-based. The living wall structures can be either free-standing or attached to walls. This IndoorLivingWall object simulates the thermal performance of indoor living wall in the built environment. The IndoorLivingWall directly connects with the inside surface heat balance, zone air heat balance, and zone air moisture balance in EnergyPlus.

1.14.12.1 Inputs

1.14.12.1.1 Field: Name

The name of the IndoorLivingWall object, which must be unique across all IndoorLivingWall objects.

1.14.12.1.2 Field: Schedule Name

This field is the name of the schedule (ref: Schedules) that modifies the presence of indoor living walls. The schedule values can be zero or one. Zero represents the scenario that the indoor living wall was removed from the thermal zone while one represents the scenario that the indoor living wall is in the presence of the thermal zone.

1.14.12.1.3 Field: ET Calculation Method

This field lists two choices of calculation methods for evapotranspiration (ET) including Penman-Monteith model and Stanghellini model. Users can also develop a customize ET function to override ET rates of indoor living walls through Energy Management System (EMS) objects, Python PlugIns objects, and Python API.

1.14.12.1.4 Field: Lighting Method

This field lists three different methods to obtain net radiation for indoor living walls. They are artificial grow lights only (LED), daylighting only (Daylight), and artificial lights and daylight (LED-Daylight). If the LED lighting method is selected, the field LED Intensity Schedule Name should be used to define the LED lighting intensity level. If the Daylight or LED-Daylight method is selected, the field Daylighting Reference Point Name should be used to determine the daylighting sensor location. If the LED-Daylight method is selected, a targeted lighting intensity schedule should be defined. Based on the available daylighting, the required LED lighting level and power will be automatically adjusted to meet the targeted LED intensity level.

1.14.12.1.5 Field: LED Schedule Name

This field is the name of the schedule (ref: Schedules) that defines the fraction of LED nominal intensity level for indoor living walls. The scheduled value can be any fractional value between 0 and 1.

1.14.12.1.6 Field: Daylighting Control Name

If daylighting is used in the selected lighting methods (Daylight or LED-Daylight), users should define an object of Daylighting:Control to obtain the daylighting illuminance level and an object for Dayliging:ReferencePoint for the daylighting sensor location in the thermal zone. The name of the object of Daylighting:Controls should be specified in this field.

1.14.12.1.7 Field: LED-Daylight Targeted Lighting Intensity Schedule Name

This field is the name of the schedule for the targeted LED intensity levels. The schedule values can be any positive number representing the targeted photosynthetic photon flux density (PPFD) in the unit of $\mu mol/(m^2\,s)$.

1.14.12.1.8 Field: Total Leaf Area

This field is the estimated one-sided leaf area [m^2] of an indoor living wall. Based on the users' input, leaf area index (LAI) is calculated as the ratio of the total leaf area and the partition wall area. Typical LAIs are 1.0 for grass and 3.0 for bushes and shrubs. The maximum LAI is 2.0 for the IndoorLivingWall module in EnergyPlus. If the calculated LAI is greater than 2.0, the maximum value of 2.0 is used for LAI in the simulation.

1.14.12.1.9 Field: LED Nominal Intensity

This field defines the nominal LED intensity level for indoor living wall systems. The value represents the photosynthetic photon flux density (PPFD) of LED grow light. PPFD is measured in $\mu mol/(m^2\,s)$ which establishes exactly how many photosynthetically active radiation (PAR) photons are landing on a specific area.

1.14.12.1.10 Field: LED Nominal Power

This field defines the nominal total LED power [W] for the indoor living wall system.

1.14.12.1.11 Field: Radiant Fraction of LED Lights

This field defines the radiant fraction of the LED lighting power input. The remaining power input is dissipated by convection and contributes to the zone air heat balance.

An IDF example:

```
IndoorLivingWall,
  Space1-1IndoorLivingWall,   !-Name
  SPACE1-1SouthPartition,     !-Surface Name
  AlwaysOn,                   !-Schedule Name
  Penman-Monteith,            !-ET Calculation Method
  LED,                        !-Lighting Method
  AlwaysOff,                  !-LED Intensity Schedule Name
  ,                           !-Daylighting Control Name
  ,                           !-LED - Daylight Targeted Lighting Intensity Schedule Name
  30,                         !- Total Leaf Area {m2}
  32.5,                       !- LED Nominal Intensity {umol/m2-s}
  640,                        !- LED Nominal Power {W}
  0.6;                        !- Radiant Fraction of LED Lights
```

1.14.12.2 Outputs

IndoorLivingWall objects have output variables for individual objects. The following outputs are available:

- Zone,Average,Indoor Living Wall Plant Surface Temperature [C]
- Zone,Average,Indoor Living Wall Sensible Heat Gain Rate [W]
- Zone,Average,Indoor Living Wall Latent Heat Gain Rate [W]
- Zone,Average,Indoor Living Wall Evapotranspiration Rate [$kg/(m^2\,s)$]
- Zone,Average,Indoor Living Wall Energy Rate Required For Evapotranspiration Per Unit Area [W/m^2]
- Zone,Average,Indoor Living Wall LED Operational PPFD [$\mu mol/(m^2\,s)$]
- Zone,Average,Indoor Living Wall PPFD [$\mu mol/(m^2\,s)$]
- Zone,Average,Indoor Living Wall Vapor Pressure Deficit [Pa]
- Zone,Average,Indoor Living Wall LED Sensible Heat Gain Rate [W]
- Zone,Average,Indoor Living Wall LED Operational Power [W]
- Zone,Sum,Indoor Living Wall LED Electricity Energy [J]

1.14.12.2.1 Indoor Living Wall Plant Surface Temperature [C]

This output is the plant surface temperature in C. Plant surface temperature is determined using surface heat balance of indoor living walls.

1.14.12.2.2 Indoor Living Wall Sensible Heat Gain Rate [W]

This output is the sensible heat gain rate from indoor living walls in W and determined by surface heat balance. Positive sign represents heat gain of spaces; negative sign represents heat loss of spaces.

1.14.12.2.3 Indoor Living Wall Latent Heat Gain Rate [W]

This output is the latent heat gain rate from indoor living walls in W. Latent heat gain is determined based on the increase of enthalpy due to the increase of absolute humidity (or humidity ratio) from plant evapotranspiration.

1.14.12.2.4 Indoor Living Wall Evapotranspiration Rate [kg/(m^2 s)]

This output is evapotranspiration (ET) rate of indoor living walls in kg/(m^2 s). The ET rate is directly calculated by ET models(Penman-Monteith model, Stanghellini model or ET model from users).

1.14.12.2.5 Indoor Living Wall Energy Rate Required For Evapotranspiration Per Unit Area [W/m^2]

This output is energy rate per unit area required for plant evapotranspiration (ET) of indoor living walls in W/m^2. The energy rate per unit area for ET is determined by latent heat of vaporization for total ET over time period and surface area.

1.14.12.2.6 Indoor Living Wall LED Operational PPFD [µmol/(m^2 s)]

This output is operational photosynthetic photon flux density (PPFD) of LED light in µmol/(m^2 s). The operational PPFD of LED light is determined based on lighting method. If lighting method is Daylight, the operational LED PPFD is zero. If lighting method is LED, the operational LED PPFD is nominal LED PPFD. If lighting method is LED-Daylight, the operational LED PPFD is calculated based on targeted PPFD and daylighting level with the consideration of nominal LED PPFD.

1.14.12.2.7 Indoor Living Wall PPFD [µmol/(m^2 s)]

This output is the actual PPFD including LED and/or daylight for indoor living walls in µmol/(m^2 s).

1.14.12.2.8 Indoor Living Wall Vapor Pressure Deficit [Pa]

This output is the vapor pressure deficit in Pa. The output represents the difference between saturated vapor pressure and vapor pressure of the moist air of current condition.

1.14.12.2.9 Indoor Living Wall LED Sensible Heat Gain Rate [W]

This output is the LED sensible heat gain rate for indoor living walls in W. The output determines convective heat gain from LED lights when LED lights are on.

1.14.12.2.10 Indoor Living Wall LED Operational Power [W]

This output is LED operational power for indoor living walls in W. The LED operational power is determined by LED nominal power in proportion to the ratio of LED operational PPFD to LED nominal PPFD.

1.14.12.2.11 Indoor Living Wall LED Electricity Energy [J]

This output is LED electricity energy for indoor living walls in J. The LED electricity energy is calculated by LED operational power multiplied by time period.

1.14.13 ElectricEquipment:ITE:AirCooled

This object describes air-cooled electric information technology equipment (ITE) which has variable power consumption as a function of loading and temperature.

1.14.13.1 Inputs

1.14.13.1.1 Field: Name

The name of this object.

1.14.13.1.2 Field: Zone or Space Name

This field applies this ElectricEquipment:ITE:AirCooled object to a thermal zone or space in the building. If a Zone Name is specified and the zone contains more than one Space, the names of the actual IT equipment objects will be concatenated as <Space Name> <ITEquipment Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.13.1.3 Field: Air Flow Calculation Method

This field specifies the method used to calculate the IT inlet temperature and zone return air temperature.

If **FlowFromSystem** is chosen, the zone is assumed to be well-mixed.

If **FlowControlWithApproachTemperatures** is chosen, Supply and Return approach temperature should be defined to indicate the temperature difference due to the air distribution. The inputs of *Air Inlet Connection Type*, *Design Recirculation Fraction* and *Recirculation Function of Loading and Supply Temperature Curve Name* are ignored. For multiple ITE objects defined for one zone, the same calculation method should apply. The **FlowControlWithApproachTemperatures** only applies to ITE zones with single duct VAV terminal unit. Other return air heat gains from window or lights are not allowed when **FlowControlWithApproachTemperatures** is chosen.

The default method is FlowFromSystem.

1.14.13.1.4 Field: Design Power Input Calculation Method

This field is a key/choice field that tells which of the next two fields are filled and is descriptive of the method for calculating the nominal electric power input to the ITE. The key/choices are:

Watts/Unit: With this choice, the design power input will be the product of Design Level per Unit and Number of Units. (Both of these fields should be filled.) This is the default.

Watts/Area: With this choice, the design power input will be a factor per floor area of the zone. (The Watts per Floor Area field should be filled).

1.14.13.1.5 Field: Watts per Unit

This field (in Watts) is typically used to represent the design electrical power input to the ITE when fully loaded and the entering air temperatures is at the specified design value. This field is used if the choice from the method field is "EquipmentLevel".

1.14.13.1.6 Field: Number of Units

This field is multiplied times the Design Level per Unit to determine the design electrical power input to this ITE object when fully loaded and the entering air temperature is at the specified design value. This field is used if the choice from the method field is "EquipmentLevel". The default is 1.

1.14.13.1.7 Field: Watts per Floor Area

This factor (Watts/m2) is used, along with the Zone Area to determine the design electrical power input as described in the Design Level field above. This field is used if the choice from the method field is "Watts/Area".

1.14.13.1.8 Field: Design Power Input Schedule Name

This field is the name of the operating schedule that modifies the design level power input for this equipment This schedule specifies the fraction (typically 0.0 to 1.0) of this equipment which is available (powered up), regardless of CPU utilization. If this field is blank, the schedule is assumed to always be 1.0.

1.14.13.1.9 Field: CPU Loading Schedule Name

This field is the name of the schedule that specifies the CPU loading for this equipment as a fraction from 0.0 (idle) to 1.0 (full load). If this field is blank, the schedule is assumed to always be 1.0.

1.14.13.1.10 Field: CPU Power Input Function of Loading and Air Temperature Curve Name

The name of a two-variable curve or table lookup object which modifies the CPU power input as a function of CPU loading (x) and air inlet node temperature (y). This curve (table) should equal 1.0 at design conditions (CPU loading = 1.0 and Design Entering Air Temperature).

1.14.13.1.11 Field: Design Fan Power Input Fraction

This field is a decimal number between 0.0 and 1.0 and is used to specify the fraction of the total power input at design conditions which is for the cooling fan(s). If fan power data is not available, set this fraction to 0.0. The default is 0.0.

1.14.13.1.12 Field: Design Fan Air Flow Rate per Power Input

Specifies the cooling fan air flow rate in m^3/s per Watt of total electric power input at design conditions (CPU loading = 1.0 and Design Entering Air Temperature).

This is normalized by power input to allow the design power input to be changed without needing to change this value.

1.14.13.1.13 Field: Air Flow Function of Loading and Air Temperature Curve Name

The name of a two-variable curve or table lookup object which modifies the cooling air flow rate as a function of CPU loading (x) and air inlet node temperature (y). This curve (table) should equal 1.0 at design conditions (CPU loading = 1.0 and Design Entering Air Temperature).

1.14.13.1.14 Field: Fan Power Input Function of Flow Curve Name

The name of a single-variable curve or table lookup object which modifies the fan power input as a function of airflow fraction (x). This curve (table) should equal 1.0 at the design air flow rate (flow fraction = 1.0).

1.14.13.1.15 Field: Design Entering Air Temperature

Specifies the entering air temperature in °C at design conditions. The default is 15°C.

1.14.13.1.16 Field: Environmental Class

Specifies the allowable operating conditions for the air inlet conditions. The available inputs are A1, A2, A3, A4, B, C, H1, or None. This is used to report the "ITE Air Inlet Operating Range Exceeded Time." If None is specified (the default), then no reporting of time outside allowable conditions will be done.

The related reporting variables (such as "ITE Air Inlet Operating Range Exceeded Time") are based on the following limits[3] on the air inlet temperature and humidity conditions shown in Table 1.33:

Table 1.33: ElectricEquipment:ITE:AirCooled Air Inlet Limiting Temperature and Humidity Levels

Environmental Class →	A1	A2	A3	A4	B	C	H1
Dry-Bulb$_{min}$ [°C]	15	10	5	5	5	5	5
Dry-Bulb$_{max}$ [°C]	32	35	40	45	35	40	25
Dew-Point$_{min}$ [°C]	-12	-12	-12	-12	–	–	-12
Dew-Point$_{max}$ [°C]	17	21	24	24	28	28	17
Relative Humidity$_{min}$ [%]	8	8	8	8	8	8	8
Relative Humidity$_{max}$ [%]	80	80	85	90	80	80	80

1.14.13.1.17 Field: Air Inlet Connection Type

Specifies the type of connection between the zone and the ITE air inlet node. The choices are:

AdjustedSupply: This option is used to apply a recirculation adjustment to the ITE inlet conditions. If this option is specified, then the Supply Air Node Name is required and the air inlet temperature to the ITE will be the current supply air node temperature adjusted by the current recirculation fraction. All heat output is added to the zone air heat balance as a convective gain. AdjustedSupply is the default.

ZoneAirNode: This option is used if there is no containment and the ITE air inlet node is at the average zone condition. All heat output is added to the zone air heat balance as a convective gain.

RoomAirModel: This option connects the ITE air inlet and outlet nodes to a room air model (Ref. RoomAirModelType and RoomAir:Node). Currently in EnergyPlus, this option has not been fully implemented. If the user chooses this option, the program will issue a warning message and this field will be adjusted to ZoneAirNode.

This field is only used when Air Flow Calculation Method is **FlowFromSystem**.

1.14.13.1.18 Field: Air Inlet Room Air Model Node Name

Specifies the name of a room air model node (ref. RoomAir:Node) which is the air inlet to this equipment. This field is required if the Air Node Connection Type = RoomAirModel.

[3] Sources: [1]. ASHRAE, 2021. "Thermal Guidelines for Data Processing Environments (5th Edition)," Peachtree Corners: ASHRAE; [2]. Quirk, Davidson, and Schmidt, 2022. "ASHRAE's Data Center Thermal Guidelines—Air-Cooled Evolution," ASHRAE Journal, May 2022.

1.14.13.1.19 Field: Air Outlet Room Air Model Node Name

Specifies the node name of a room air model node (ref. RoomAir:Node) which is the air outlet from this equipment. This field is required if the Air Node Connection Type = RoomAirModel.

1.14.13.1.20 Field: Supply Air Node Name

Specifies the node name of the supply air inlet node service this ITE. If the Air Node Connection Type = AdjustedSupply ,then this field is required, and the conditions at this node will be used to determine the ITE air inlet conditions. This field is also required if reporting of the Supply Heat Index is desired. Also required if Calculation Method = **FlowControlWithApproachTemperatures**.

1.14.13.1.21 Field: Design Recirculation Fraction

Specifies the recirculation fraction for this equipment at design conditions. This field is used only if the Air Node Connection Type = AdjustedSupply. The recirculation fraction is defined as the ratio of recirculated air flow to total air flow entering the ITE. Recirculation is dependent upon many factors including rack and containment configuration. The default is 0.0 (no recirculation). This field is only used when Air Flow Calculation Method = **FlowFromSystem**.

1.14.13.1.22 Field: Recirculation Function of Loading and Supply Temperature Curve Name

The name of a two-variable curve or table lookup object which modifies the Design Recirculation Fraction as a function of CPU loading (x) and supply air node temperature (y). This curve (table) should equal 1.0 at design conditions (CPU loading = 1.0 and Design Entering Air Temperature). This field is used only if the Air Node Connection Type = AdjustedSupply. If this curve is left blank, then the curve is assumed to always equal 1.0. This field is only used when Air Flow Calculation Method = **FlowFromSystem**.

1.14.13.1.23 Field: Design Electric Power Supply Efficiency

This field is a decimal number used to specify the efficiency of the power supply system serving this ITE. The default is 1.0.

1.14.13.1.24 Field: Electric Power Supply Efficiency Function of Part Load Ratio Curve Name

The name of a single-variable curve or table lookup object which modifies the electric power supply efficiency as a function of part load ratio (x). This curve (table) should equal 1.0 at the design power consumption (part load ratio = 1.0). If this curve is left blank, then the curve is assumed to always equal 1.0.

1.14.13.1.25 Field: Fraction of Electric Power Supply Losses to Zone

This field is a decimal number between 0.0 and 1.0 and is used to specify the fraction of the electric power supply losses which are a heat gain to the zone containing the ITE. If this value is less than 1.0, the remainder of the losses are assumed to be lost to the outdoors. The default is 1.0.

1.14.13.1.26 Field: CPU End-Use Subcategory

This equipment is metered on the Interior Equipment end-use category for Electricity. This field allows you to specify a user-defined end-use subcategory for the CPU power consumption. A new meter for reporting is created for each unique subcategory (ref: Output:Meter object). Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and in the LEED Summary table EAp2-4/5 Performance Rating Method Compliance. The default is ITE-CPU.

1.14.13.1.27 Field: Fan End-Use Subcategory

This equipment is metered on the Interior Equipment end-use category for Electricity. This field allows you to specify a user-defined end-use subcategory for the fan power consumption. A new meter for reporting is created for each unique subcategory (ref: Output:Meter object). Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and in the LEED Summary table EAp2-4/5 Performance Rating Method Compliance. The default is ITE-Fans.

1.14.13.1.28 Field: Electric Power Supply End-Use Subcategory

This equipment is metered on the Interior Equipment end-use category for Electricity. This field allows you to specify a user-defined end-use subcategory for the electric power supply power consumption. A new meter for reporting is created for each unique subcategory (ref: Output:Meter object). Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and in the LEED Summary table EAp2-4/5 Performance Rating Method Compliance. The default is ITE-UPS

1.14.13.1.29 Field: Supply Temperature Difference

The difference of the IT inlet temperature from the AHU supply air temperature ($\Delta T_{\text{supply}} = T_{\text{in}} - T_{\text{supply}}$). Either Supply Temperature Difference or Supply Temperature Difference Schedule is required if Air Flow Calculation Method is set to **FlowControlWithApproachTemperatures**. This field is ignored when Air Flow Calculation Method is **FlowFromSystem**.

1.14.13.1.30 Field: Supply Temperature Difference Schedule

The difference schedule of the IT inlet temperature from the AHU supply air temperature ($\Delta T_{\text{supply}} = T_{\text{in}} - T_{\text{supply}}$). Either Supply Temperature Difference or Supply Temperature Difference Schedule is required if Air Flow Calculation Method is set to **FlowControlWithApproachTemperatures**. This field is ignored when Air Flow Calculation Method is **FlowFromSystem**.

1.14.13.1.31 Field: Return Temperature Difference

The difference of the actual AHU return air temperature to the IT equipment outlet temperature ($\Delta T_{\text{return}} = T_{\text{return}} - T_{\text{out}}$). Either Return Temperature Difference or Return Temperature Difference Schedule is required if Air Flow Calculation Method is set to **FlowControlWithApproachTemperatures**. This field is ignored when Air Flow Calculation Method is **FlowFromSystem**.

1.14.13.1.32 Field: Return Temperature Difference Schedule

The difference schedule of the actual AHU return air temperature to the IT equipment outlet temperature ($\Delta T_{\text{return}} = T_{\text{return}} - T_{\text{out}}$). Either Return Temperature Difference or Return Temperature Difference Schedule is required if Air Flow Calculation Method is set to **FlowControlWithApproachTemperatures** This field is ignored when Air Flow Calculation Method is **FlowFromSystem**.

An IDF example:

```
ElectricEquipment:ITE:AirCooled,
  Data Center Servers,      !- Name
  Main Zone,                !- Zone or Space Name
  FlowFromSystem,           !- Air Flow Calculation Method
  Watts/Unit,               !- Design Power Input Calculation Method
  500,                      !- Watts per Unit {W}
  200,                      !- Number of Units
  ,                         !- Watts per Floor Area {W/m2}
  Data Center Operation Schedule,  !- Design Power Input Schedule Name
  Data Center CPU Loading Schedule,  !- CPU Loading  Schedule Name
  Model 5250 Power fLoadTemp,  !- CPU Power Input Function of Loading and Air Temperature Curve Name
  0.4,                      !- Design Fan Power Input Fraction
  0.0001,                   !- Design Fan Air Flow Rate per Power Input {m3/s-W}
```

```
Model 5250 AifFlow fLoadTemp,  !- Air Flow Function of Loading and Air Temperature Curve Name
ECM FanPower fFlow,      !- Fan Power Input Function of Flow Curve Name
15,                      !- Design Entering Air Temperature {C}
A3,                      !- Environmental Class
AdjustedSupply,          !- Air Inlet Connection Type
,                        !- Air Inlet Room Air Model Node Name
,                        !- Air Outlet Room Air Model Node Name
Main Zone Inlet Node,    !- Supply Air Node Name
0.1,                     !- Design Recirculation Fraction
Data Center Recirculation fLoadTemp,  !- Recirculation Function of Loading and Supply Temperature Curve
   Name
0.9,                     !- Design Electric Power Supply Efficiency
UPS Efficiency fPLR,     !- Electric Power Supply Efficiency Function of Part Load Ratio Curve Name
1,                       !- Fraction of Electric Power Supply Losses to Zone
ITE-CPU,                 !- CPU End-Use Subcategory
ITE-Fans,                !- Fan End-Use Subcategory
ITE-UPS;                 !- Electric Power Supply End-Use Subcategory
```

Another IDF example when Air Flow Calculation Method = FlowControlWithApproachTemperatures:

```
ElectricEquipment:ITE:AirCooled,
  SmallDataCenter ComputerRoom Elec Equip,     !- Name
  ComputerRoom ZN,                             !- Zone or Space Name
  FlowControlWithApproachTemperatures,         !- Air Flow Calculation Method
  Watts/Area,                     !- Design Power Input Calculation Method
  ,                               !- Watts per Unit {W}
  ,                               !- Number of Units
  430.556416668389,               !- Watts per Floor Area {W/m2}
  Data Center Operation Schedule,  !- Design Power Input Schedule Name
  DataCenter Equipment_SCH,  !- CPU Loading  Schedule Name
  Data Center Servers Power fLoadTemp,  !- CPU Power Input Function of Loading and Air Temperature
  Curve Name
  0.4,                     !- Design Fan Power Input Fraction
  0.0001,                  !- Design Fan Air Flow Rate per Power Input {m3/s-W}
  Data Center Servers Airflow fLoadTemp,  !- Air Flow Function of Loading and Air Temperature Curve
  Name
  ECM FanPower fFlow,      !- Fan Power Input Function of Flow Curve Name
  15,                      !- Design Entering Air Temperature {C}
  A3,                      !- Environmental Class
  AdjustedSupply,          !- Air Inlet Connection Type
  ,                        !- Air Inlet Room Air Model Node Name
  ,                        !- Air Outlet Room Air Model Node Name
  Node 6,                          !- Supply Air Node Name
  0,                     !- Design Recirculation Fraction
  Data Center Recirculation fLoadTemp,  !- Recirculation Function of Loading and Supply Temperature
  Curve Name
  0.9,                     !- Design Electric Power Supply Efficiency
  UPS Efficiency fPLR,     !- Electric Power Supply Efficiency Function of Part Load Ratio Curve Name
  1,                       !- Fraction of Electric Power Supply Losses to Zone
  ITE-CPU,                 !- CPU End-Use Subcategory
  ITE-Fans,                !- Fan End-Use Subcategory
  ITE-UPS,                 !- Electric Power Supply End-Use Subcategory
  3,                       !- field Supply Approach Temperature
  ,                        !- field Supply Approach Temperature Schedule
  -3,                      !- field Return Approach Temperature
  ;                        !- field Return Approach Temperature Schedule
```

1.14.13.2 Outputs

ElectricEquipment:ITE:AirCooled Outputs:

- Zone,Average,ITE CPU Electricity Rate [W]
- Zone,Average,ITE Fan Electricity Rate [W]
- Zone,Average,ITE UPS Electricity Rate [W]
- Zone,Average,ITE CPU Electricity Rate at Design Inlet Conditions [W]
- Zone,Average,ITE Fan Electricity Rate at Design Inlet Conditions [W]

- Zone,Average,ITE UPS Heat Gain to Zone Rate [W]
- Zone,Average,ITE Total Heat Gain to Zone Rate [W]
- Zone,Sum,ITE CPU Electricity Energy [J]
- Zone,Sum,ITE Fan Electricity Energy [J]
- Zone,Sum,ITE UPS Electricity Energy [J]
- Zone,Sum,ITE CPU Electricity Energy at Design Inlet Conditions [J]
- Zone,Sum,ITE Fan Electricity Energy at Design Inlet Conditions [J]
- Zone,Sum,ITE UPS Heat Gain to Zone Energy [J]
- Zone,Sum,ITE Total Heat Gain to Zone Energy [J]
- Zone,Average,ITE Standard Density Air Volume Flow Rate [m3/s]
- Zone,Average,ITE Current Density Air Volume Flow Rate [m3/s]
- Zone,Average,ITE Air Mass Flow Rate [kg/s]
- Zone,Average,ITE Air Inlet Dry-Bulb Temperature [C]
- Zone,Average,ITE Air Inlet Dewpoint Temperature [C]
- Zone,Average,ITE Air Inlet Relative Humidity [%]
- Zone,Average,ITE Air Outlet Dry-Bulb Temperature [C]
- Zone,Average,ITE Supply Heat Index []
- Zone,Sum,ITE Air Inlet Operating Range Exceeded Time [hr]
- Zone,Sum,ITE Air Inlet Dry-Bulb Temperature Above Operating Range Time [hr]
- Zone,Sum,ITE Air Inlet Dry-Bulb Temperature Below Operating Range Time [hr]
- Zone,Sum,ITE Air Inlet Dewpoint Temperature Above Operating Range Time [hr]
- Zone,Sum,ITE Air Inlet Dewpoint Temperature Below Operating Range Time [hr]
- Zone,Sum,ITE Air Inlet Relative Humidity Above Operating Range Time [hr]
- Zone,Sum,ITE Air Inlet Relative Humidity Below Operating Range Time [hr]
- Zone,Average,ITE Air Inlet Dry-Bulb Temperature Difference Above Operating Range [deltaC]
- Zone,Average,ITE Air Inlet Dry-Bulb Temperature Difference Below Operating Range [deltaC]
- Zone,Average,ITE Air Inlet Dewpoint Temperature Difference Above Operating Range [deltaC]
- Zone,Average,ITE Air Inlet Dewpoint Temperature Difference Below Operating Range [deltaC]
- Zone,Average,ITE Air Inlet Relative Humidity Difference Above Operating Range [%]
- Zone,Average,ITE Air Inlet Relative Humidity Difference Below Operating Range [%]
- Zone,Average,Space ITE CPU Electricity Rate [W]
- Zone,Average,Space ITE Fan Electricity Rate [W]
- Zone,Average,Space ITE UPS Electricity Rate [W]
- Zone,Average,Space ITE CPU Electricity Rate at Design Inlet Conditions [W]
- Zone,Average,Space ITE Fan Electricity Rate at Design Inlet Conditions [W]
- Zone,Average,Space ITE UPS Heat Gain to Zone Rate [W]
- Zone,Average,Space ITE Total Heat Gain to Zone Rate [W]
- Zone,Sum,Space ITE CPU Electricity Energy [J]
- Zone,Sum,Space ITE Fan Electricity Energy [J]
- Zone,Sum,Space ITE UPS Electricity Energy [J]
- Zone,Sum,Space ITE CPU Electricity Energy at Design Inlet Conditions [J]
- Zone,Sum,Space ITE Fan Electricity Energy at Design Inlet Conditions [J]
- Zone,Sum,Space ITE UPS Heat Gain to Zone Energy [J]
- Zone,Sum,Space ITE Total Heat Gain to Zone Energy [J]
- Zone,Average,Space ITE Standard Density Air Volume Flow Rate [m3/s]
- Zone,Average,Space ITE Air Mass Flow Rate [kg/s]
- Zone,Average,Space ITE Average Supply Heat Index []
- Zone,Sum,Space ITE Any Air Inlet Operating Range Exceeded Time [hr]
- Zone,Sum,Space ITE Any Air Inlet Dry-Bulb Temperature Above Operating Range Time [hr]

- Zone,Sum,Space ITE Any Air Inlet Dry-Bulb Temperature Below Operating Range Time [hr]
- Zone,Sum,Space ITE Any Air Inlet Dewpoint Temperature Above Operating Range Time [hr]
- Zone,Sum,Space ITE Any Air Inlet Dewpoint Temperature Below Operating Range Time [hr]
- Zone,Sum,Space ITE Any Air Inlet Relative Humidity Above Operating Range Time [hr]
- Zone,Sum,Space ITE Any Air Inlet Relative Humidity Below Operating Range Time [hr]
- Zone,Average,Zone ITE Adjusted Return Air Temperature [C]
- Zone,Average,Zone ITE CPU Electricity Rate [W]
- Zone,Average,Zone ITE Fan Electricity Rate [W]
- Zone,Average,Zone ITE UPS Electricity Rate [W]
- Zone,Average,Zone ITE CPU Electricity Rate at Design Inlet Conditions [W]
- Zone,Average,Zone ITE Fan Electricity Rate at Design Inlet Conditions [W]
- Zone,Average,Zone ITE UPS Heat Gain to Zone Rate [W]
- Zone,Average,Zone ITE Total Heat Gain to Zone Rate [W]
- Zone,Sum,Zone ITE CPU Electricity Energy [J]
- Zone,Sum,Zone ITE Fan Electricity Energy [J]
- Zone,Sum,Zone ITE UPS Electricity Energy [J]
- Zone,Sum,Zone ITE CPU Electricity Energy at Design Inlet Conditions [J]
- Zone,Sum,Zone ITE Fan Electricity Energy at Design Inlet Conditions [J]
- Zone,Sum,Zone ITE UPS Heat Gain to Zone Energy [J]
- Zone,Sum,Zone ITE Total Heat Gain to Zone Energy [J]
- Zone,Average,Zone ITE Standard Density Air Volume Flow Rate [m3/s]
- Zone,Average,Zone ITE Air Mass Flow Rate [kg/s]
- Zone,Average,Zone ITE Average Supply Heat Index []
- Zone,Sum,Zone ITE Any Air Inlet Operating Range Exceeded Time [hr]
- Zone,Sum,Zone ITE Any Air Inlet Dry-Bulb Temperature Above Operating Range Time [hr]
- Zone,Sum,Zone ITE Any Air Inlet Dry-Bulb Temperature Below Operating Range Time [hr]
- Zone,Sum,Zone ITE Any Air Inlet Dewpoint Temperature Above Operating Range Time [hr]
- Zone,Sum,Zone ITE Any Air Inlet Dewpoint Temperature Below Operating Range Time [hr]
- Zone,Sum,Zone ITE Any Air Inlet Relative Humidity Above Operating Range Time [hr]
- Zone,Sum,Zone ITE Any Air Inlet Relative Humidity Below Operating Range Time [hr]

1.14.13.2.1 Space or Zone ITE CPU Electricity Rate [W]

1.14.13.2.2 ITE CPU Electricity Rate [J]

1.14.13.2.3 Space or Zone ITE CPU Electricity Energy [J]

1.14.13.2.4 ITE CPU Electricity Energy [J]

The electric power (or energy) input to the ITE equipment CPU (total power input less cooling fan power). The ITE CPU Electricity Energy output is also added to a meter object with Resource Type = Electricity, End Use Key = InteriorEquipment, Group Key = Building (Ref. Output:Meter object).

1.14.13.2.5 Space or Zone ITE CPU Electricity Rate at Design Inlet Conditions [W]

1.14.13.2.6 ITE CPU Electricity Rate at Design Inlet Conditions [W]

1.14.13.2.7 Space or Zone ITE CPU Electricity Energy at Design Inlet Conditions [J]

1.14.13.2.8 ITE CPU Electricity Energy at Design Inlet Conditions [J]

The electric power (or energy) input to the ITE equipment CPU (total power input less cooling fan power) if the air inlet temperature were held at the design condition. May be used to calculate "IT efficiency", the ratio of (IT energy consumed in the facility) / (IT energy that would have been consumed in the facility if the ITE were held at the reference temperature).

1.14.13.2.9 Space or Zone ITE Fan Electricity Rate [W]

1.14.13.2.10 ITE Fan Electricity Rate [W]

1.14.13.2.11 Space or Zone ITE Fan Electricity Energy [J]

1.14.13.2.12 ITE Fan Electricity Energy [J]

The electric power (or energy) input to the ITE cooling fan. The ITE Fan Electricity Energy output is also added to a meter object with Resource Type = Electricity, End Use Key = InteriorEquipment, Group Key = Building (Ref. Output:Meter object).

1.14.13.2.13 Space or Zone ITE Fan Electricity Rate at Design Inlet Conditions [W]

1.14.13.2.14 ITE Fan Electricity Rate at Design Inlet Conditions [W]

1.14.13.2.15 Space or Zone ITE Fan Electricity Energy at Design Inlet Conditions [J]

1.14.13.2.16 ITE Fan Electricity Energy at Design Inlet Conditions [J]

The electric power (or energy) input to the ITE cooling fan if the air inlet temperature were held at the design condition. May be used to calculate "IT efficiency", the ratio of (IT energy consumed in the facility) / (IT energy that would have been consumed in the facility if the ITE were held at the reference temperature).

1.14.13.2.17 Space or Zone ITE UPS Electricity Rate [W]

1.14.13.2.18 ITE UPS Electricity Rate [W]

1.14.13.2.19 Space or Zone ITE UPS Electricity Energy [J]

1.14.13.2.20 ITE UPS Electricity Energy [J]

The net electric power (or energy) input to the ITE equipment UPS (total power input less power delivered to ITE). The ITE UPS Electricity Energy output is also added to a meter object with Resource Type = Electricity, End Use Key = InteriorEquipment, Group Key = Building (Ref. Output:Meter object).

1.14.13.2.21 ITE Electricity Meters

The ITE CPU Electricity Energy, ITE Fan Electricity Energy, and ITE UPS Electricity Energy are each included separately in the following Electricity meters, with their respective end-use subcategory:

```
Electricity:Facility
Electricity:Building
Electricity:Zone:<Zone Name>
Electricity:SpaceType:<Space Type Name>
InteriorEquipment:Electricity
InteriorEquipment:Electricity:Zone:<Zone Name>
InteriorEquipment:Electricity:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:Electricity
<End-Use Subcategory>:InteriorEquipment:Electricity:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:Electricity:SpaceType:<Space Type Name>
```

1.14.13.2.22 Space or Zone ITE UPS Heat Gain to Zone Rate [W]

1.14.13.2.23 ITE UPS Heat Gain to Zone Rate [W]

1.14.13.2.24 Space or Zone ITE UPS Heat Gain to Zone Energy [J]

1.14.13.2.25 ITE UPS Heat Gain to Zone Energy [J]

The heat gain rate (or energy) to the zone from the UPS.

1.14.13.2.26 Space or Zone ITE Total Heat Gain to Zone Rate [W]

1.14.13.2.27 ITE Total Heat Gain to Zone Rate [W]

1.14.13.2.28 Space or Zone ITE Total Heat Gain to Zone Energy [J]

1.14.13.2.29 ITE Total Heat Gain to Zone Energy [J]

The heat gain rate (or energy) to the zone from the UPS and from the CPU and fans if the ITE. Air Inlet Connection Type is AdjustedSupply or ZoneAirNode. If RoomAirModel is selected, then only the heat gain from the UPS is added directly to the zone air heat balance, the heat gain from the CPU and fans will be added to the ITE air Outlet Room Air Model Node

1.14.13.2.30 Space or Zone ITE Standard Density Air Volume Flow Rate [m3/s]

1.14.13.2.31 ITE Standard Density Air Volume Flow Rate [m3/s]

Reports the average air volume flow rate through the ITE over the reporting interval. Standard density in EnergyPlus corresponds to 20°C dry bulb, dry air, and nominally adjusted for elevation.

1.14.13.2.32 Space or Zone ITE Current Density Air Volume Flow Rate [m3/s]

1.14.13.2.33 ITE Current Density Air Volume Flow Rate [m3/s]

Reports the average air volume flow rate through the ITE over the reporting interval, calculated using the current density at the air inlet node.

1.14.13.2.34 Space or Zone ITE Air Mass Flow Rate [kg/s]

1.14.13.2.35 ITE Air Mass Flow Rate [kg/s]

Reports the average air mass flow rate through the ITE over the reporting interval, calculated using the current density at the air inlet node.

1.14.13.2.36 ITE Air Inlet Dry-Bulb Temperature [C]

The dry-bulb temperature of the air entering the ITE.

1.14.13.2.37 ITE Air Inlet Dewpoint Temperature [C]

The dewpoint temperature of the air entering the ITE.

1.14.13.2.38 ITE Air Inlet Relative Humidity [%]

The dewpoint temperature of the air entering the ITE.

1.14.13.2.39 ITE Air Outlet Dry-Bulb Temperature [C]

The dry-bulb temperature of the air leaving the ITE.

1.14.13.2.40 Space or Zone ITE Average Supply Heat Index []

1.14.13.2.41 ITE Supply Heat Index []

The supply heat index (SHI) for this equipment. SHI is a dimensionless measure of recirculation of hot air into the cold air intake of the ITE. SHI = $(T_{\text{in}} - T_{\text{supply}})/(T_{\text{out}} - T_{\text{supply}})$ where T_{in} is the dry-bulb temperature of the air entering the ITE, Tout is the dry-bulb temperature of the air leaving the ITE, and T_{supply} is the dry-bulb temperature at the Supply Air Node. If a Supply Air Node Name is not specified for this object, then this output will not be reported.

1.14.13.2.42 Space or Zone ITE Any Air Inlet Operating Range Exceeded Time [hr]

1.14.13.2.43 ITE Air Inlet Operating Range Exceeded Time [hr]

Hours when the dry-bulb and/or dewpoint temperature of the air entering the ITE is outside the range specified by the ITE Environmental Class.

1.14.13.2.44 Space or Zone ITE Any Air Inlet Dry-Bulb Temperature Above Operating Range Time [hr]

1.14.13.2.45 ITE Air Inlet Dry-Bulb Temperature Above Operating Range Time [hr]

Hours when the dry-bulb temperature of the air entering the ITE is above the range specified by the ITE Environmental Class.

1.14.13.2.46 ITE Air Inlet Dry-Bulb Temperature Difference Above Operating Range [deltaC]

The temperature difference (in Δ°C) between the air inlet dry-bulb temperature and the maximum allowable dry-bulb temperature specified by the ITE Environmental Class. Only positive values are reported. When the dry-bulb temperature of the air entering the ITE is below the maximum specified by the ITE Environmental Class, this output will be zero.

1.14.13.2.47 Space or Zone ITE Any Air Inlet Dry-Bulb Temperature Below Operating Range Time [hr]

1.14.13.2.48 ITE Air Inlet Dry-Bulb Temperature Below Operating Range Time [hr]

Hours when the dry-bulb temperature of the air entering the ITE is above the range specified by the ITE Environmental Class.

1.14.13.2.49 ITE Air Inlet Dry-Bulb Temperature Difference Below Operating Range [deltaC]

The temperature difference (in Δ°C) between the air inlet dry-bulb temperature and the minimum allowable dry-bulb temperature specified by the ITE Environmental Class. Only negative values are reported. When the dry-bulb temperature of the air entering the ITE is above the minimum specified by the ITE Environmental Class, this output will be zero.

1.14.13.2.50 Space or Zone ITE Any Air Inlet Dewpoint Temperature Above Operating Range Time [hr]

1.14.13.2.51 ITE Air Inlet Dewpoint Temperature Above Operating Range Time [hr]

Hours when the dewpoint temperature of the air entering the ITE is above the range specified by the ITE Environmental Class.

1.14.13.2.52 ITE Air Inlet Dewpoint Temperature Difference Above Operating Range [deltaC]

The temperature difference (in Δ°C) between the air inlet dewpoint temperature and the maximum allowable dewpoint temperature specified by the ITE Environmental Class. Only positive values are reported. When the dewpoint temperature of the air entering the ITE is below the maximum specified by the ITE Environmental Class, this output will be zero.

1.14.13.2.53 Space or Zone ITE Any Air Inlet Dewpoint Temperature Below Operating Range Time [hr]

1.14.13.2.54 ITE Air Inlet Dewpoint Temperature Below Operating Range Time [hr]

Hours when the dewpoint temperature of the air entering the ITE is above the range specified by the ITE Environmental Class.

1.14.13.2.55 ITE Air Inlet Dewpoint Temperature Difference Below Operating Range [deltaC]

The temperature difference (in Δ°C) between the air inlet dewpoint temperature and the minimum allowable dewpoint temperature specified by the ITE Environmental Class. Only negative values are reported. When the dewpoint temperature of the air entering the ITE is above the minimum specified by the ITE Environmental Class, this output will be zero.

1.14.13.2.56 Space or Zone ITE Any Air Inlet Relative Humidity Above Operating Range Time [hr]

1.14.13.2.57 ITE Air Inlet Relative Humidity Above Operating Range Time [hr]

Hours when the relative humidity of the air entering the ITE is above the range specified by the ITE Environmental Class.

1.14.13.2.58 ITE Air Inlet Relative Humidity Difference Above Operating Range [%]

The temperature difference (in Δ°C) between the air inlet relative humidity and the maximum allowable relative humidity specified by the ITE Environmental Class. Only positive values are reported. When the relative humidity of the air entering the ITE is below the maximum specified by the ITE Environmental Class, this output will be zero.

1.14.13.2.59 Space or Zone ITE Any Air Inlet Relative Humidity Below Operating Range Time [hr]

1.14.13.2.60 ITE Air Inlet Relative Humidity Below Operating Range Time [hr]

Hours when the relative humidity of the air entering the ITE is above the range specified by the ITE Environmental Class.

1.14.13.2.61 ITE Air Inlet Relative Humidity Difference Below Operating Range [%]

The difference (in Δ%) between the air inlet relative humidity and the minimum allowable relative humidity (RH) specified by the ITE Environmental Class. Only negative values are reported. When the relative humidity of the air entering the ITE is above the minimum specified by the ITE Environmental Class, this output will be zero.

1.14.13.2.62 Zone ITE Adjusted Return Air Temperature [C]

Return air temperature after adjustment by ITE objects.

1.14.14 ZoneBaseboard:OutdoorTemperatureControlled

This object specifies outside temperature-controlled (OTC) baseboard heating. The capacities (high and low) are specified in W at the temperatures specified. The schedule allows both capacities to change hourly on a proportional basis. This baseboard heater does not operate if the outdoor dry-bulb is above the high temperature limit. Between the high temperature and the low temperature, the capacity is interpolated (linear) between the high and the low capacity values. Below the low temperature, the capacity is set at the low capacity value. This allows the user to add baseboard heat to a perimeter zone starting at a prescribed temperature and then slowly increases this capacity to a max value.

Example:

Temperature High = 10 C	Capacity High = 100,000 W
Temperature Low = 0 C	Capacity Low = 500,000 W

Table 1.34: Outdoor Temperature Controlled Baseboard Heat Temperature vs. Capacity

Outdoor Dry-bulb(C)	Baseboard Output(W)
> 10	0
10	100,000
8	180,000
6	260,000
4	340,000
2	420,000
0	500,000
< 0	500,000

1.14.14.1 Inputs

1.14.14.1.1 Field: Name

The name of the ZoneBaseboard:OutdoorTemperatureControlled object.

1.14.14.1.2 Field: Zone or ZoneList or Space or SpaceList Name

This field applies this ZoneBaseboard:OutdoorTemperatureControlled object to one or more thermal zones or spaces in the building. The names of the actual baseboard heat objects may be concatenated as <Space Name> <ZoneBaseboard Object Name>. See Specifying Applicable Zone(s) or Space(s) for more details.

1.14.14.1.3 Field: Schedule Name

This field is the name of the schedule that modifies the capacities (high and low) for baseboard heat equipment (see next four fields). The schedule values can be any positive number. The actual energy input for the baseboard equipment in a zone as defined by this statement depends on the actual outdoor temperature and where that temperature is in the range of Low Temperature to High Temperature.

1.14.14.1.4 Field: Capacity at Low Temperature

This is the baseboard equipment capacity (Watts) at the low temperature limit. This is the maximum capacity of the baseboard equipment in full load operation. This field is autosizable.

1.14.14.1.5 Field: Low Temperature

If the outdoor dry-bulb temperature (degrees Celsius) is at or below the low temperature the baseboard heater operates at the low temperature capacity. This field is autosizable. The lowest design outdoor dry bulb temperature is chosen, if autosized.

1.14.14.1.6 Field: Capacity at High Temperature

This is the baseboard equipment capacity (Watts) at the high temperature limit. This field is autosizable. The capacity at low temperature is prorated against the reference low and high temperature fields, if autosized.

1.14.14.1.7 Field: High Temperature

If the outdoor dry-bulb temperature (degrees Celsius) is greater than the high temperature the baseboard heater will not operate. This field is autosizable. If autosized, this is equal to the design zone heating setpoint temperature described below, so that the capacity at high temperature is zero.

1.14.14.1.8 Field: Fraction Radiant

This field is a decimal number between 0.0 and 1.0 and is used to characterize the type of heat being given off by baseboard heat equipment in a zone. The number specified in this field will be multiplied by the total energy consumed by the baseboard heat equipment to give the amount of long wavelength radiation gain to the zone.

1.14.14.1.9 Field: End-Use Subcategory

Allows you to specify a user-defined end-use subcategory, e.g., "Perimeter Baseboards", etc. A new meter for reporting is created for each unique subcategory (ref: Output:Meter objects). Any text may be used here to categorize the end-uses in the ABUPS End Uses by Subcategory table and in the LEED Summary table EAp2-4/5 Performance Rating Method Compliance. If this field is omitted or blank, the baseboard equipment will be assigned to the "General" end-use subcategory.

1.14.14.1.10 Field: Design Zone Heating Setpoint

This is heating setpoint temperature in the zone where the unit serves. This is used to autosize high temperature and capacity at high temperature fields. The default value is 20°C.

An IDF example:

```
ZoneBaseboard:OutdoorTemperatureControlled,
  SPACE4-1 BBHeat 1,  !- Name
  SPACE4-1,           !- Zone Name
  EQUIP-1,            !- SCHEDULE Name
  1500,               !- Capacity at low temperature {W}
  0,                  !- Low Temperature {C}
  500,                !- Capacity at high temperature {W}
  10,                 !- High Temperature {C}
  0.5,                !- Fraction Radiant
  Baseboard Heat,     !- End-Use Subcategory
  20;                 !- Design Zone Heating Setpoint {C}
```

1.14.14.2 Outputs

ZoneBaseboard:OutdoorTemperatureControlled objects have output variables for individual objects and for zone totals. The following outputs are available:

- Zone,Average,Baseboard Electricity Rate [W]
- Zone,Sum,Baseboard Electricity Energy [J]
- Zone,Sum,Baseboard Radiant Heating Energy [J]
- Zone,Average,Baseboard Radiant Heating Rate [W]
- Zone,Sum,Baseboard Convective Heating Energy [J]
- Zone,Average,Baseboard Convective Heating Rate [W]
- Zone,Sum,Baseboard Total Heating Energy [J]
- Zone,Average,Baseboard Total Heating Rate [W]
- Zone,Average,Space Baseboard Electricity Rate [W]
- Zone,Sum,Space Baseboard Electricity Energy [J]
- Zone,Sum,Space Baseboard Radiant Heating Energy [J]
- Zone,Average,Space Baseboard Radiant Heating Rate [W]

- Zone,Sum,Space Baseboard Convective Heating Energy [J]
- Zone,Average,Space Baseboard Convective Heating Rate [W]
- Zone,Sum,Space Baseboard Total Heating Energy [J]
- Zone,Average,Space Baseboard Total Heating Rate [W]
- Zone,Average,Zone Baseboard Electricity Rate [W]
- Zone,Sum,Zone Baseboard Electricity Energy [J]
- Zone,Sum,Zone Baseboard Radiant Heating Energy [J]
- Zone,Average,Zone Baseboard Radiant Heating Rate [W]
- Zone,Sum,Zone Baseboard Convective Heating Energy [J]
- Zone,Average,Zone Baseboard Convective Heating Rate [W]
- Zone,Sum,Zone Baseboard Total Heating Energy [J]
- Zone,Average,Zone Baseboard Total Heating Rate [W]

1.14.14.2.1 Baseboard Electricity Rate [W]

This field is the electric power for the ZoneBaseboard:OutdoorTemperatureControlled object in Watts.

1.14.14.2.2 Baseboard Electricity Energy [J]

The outdoor temperature controlled baseboard heat option is assumed to be fueled by electricity. This field is the same as the Baseboard Total Heating Energy (above) in joules. This energy is included in the following meters:

```
Electricity:Facility
Electricity:Building
Electricity:Zone:<Zone Name>
Electricity:SpaceType:<Space Type Name>
InteriorEquipment:Electricity
InteriorEquipment:Electricity:Zone:<Zone Name>
InteriorEquipment:Electricity:SpaceType:<Space Type Name>
<End-Use Subcategory>:InteriorEquipment:Electricity
<End-Use Subcategory>:InteriorEquipment:Electricity:Zone:<Zone Name>
<End-Use Subcategory>:InteriorEquipment:Electricity:SpaceType:<Space Type Name>
```

1.14.14.2.3 Baseboard Radiant Heating Rate [W]

1.14.14.2.4 Baseboard Radiant Heating Energy [J]

These output variables are the amount of radiant heat gain for the ZoneBaseboard:OutdoorTemperatureControlled object in Watts (for rate) or Joules. This is determined by the current heat gain from the heater to the zone and the "Fraction Radiant" specified in the input. The radiant gains (long wavelength) are distributed to the surfaces using an area weighting scheme.

1.14.14.2.5 Baseboard Convective Heating Rate [W]

1.14.14.2.6 Baseboard Convective Heating Energy [J]

These output variables are the amount of convective heat gain for the ZoneBaseboard:OutdoorTemperatureControlled object in Watts (for rate) or Joules. This is determined by the current heat gain from the heater to the zone and the "Fraction Radiant" specified in input (1-FractionRadiant = FractionConvected). The convective heat gain is added to the zone air heat balance directly.

1.14.14.2.7 Baseboard Total Heating Rate [W]

1.14.14.2.8 Baseboard Total Heating Energy [J]

These output variables are the amount of heat gain for the ZoneBaseboard:OutdoorTemperatureControlled object in Watts (for rate) or Joules. This is determined by the sum of the radiant and convective heat gains from the baseboard heat.

1.14.14.2.9 Space or Zone Baseboard Electricity Rate [W]

This field is the electric power for all ZoneBaseboard:OutdoorTemperatureControlled objects within the space or zone in Watts.

1.14.14.2.10 Space or Zone Baseboard Electricity Energy [J]

The outdoor temperature controlled baseboard heat option is assumed to be fueled by electricity. This field is the same as the Baseboard Total Heating Energy (above) in joules.

1.14.14.2.11 Space or Zone Baseboard Radiant Heating Rate [W]

1.14.14.2.12 Space or Zone Baseboard Radiant Heating Energy [J]

These output variables are the amount of radiant heat gain for all ZoneBaseboard:OutdoorTemperatureControlled objects within the space or zone in Watts (for rate) or Joules. This is determined by the current heat gain from the heater to the space or zone and the "Fraction Radiant" specified in the input. The radiant gains (long wavelength) are distributed to the surfaces using an area weighting scheme.

1.14.14.2.13 Space or Zone Baseboard Convective Heating Rate [W]

1.14.14.2.14 Space or Zone Baseboard Convective Heating Energy [J]

These output variables are the amount of convective heat gain for all ZoneBaseboard:OutdoorTemperatureControlled objects within the space or zone in Watts (for rate) or Joules. This is determined by the current heat gain from the heater to the space or zone and the "Fraction Radiant" specified in input (1-FractionRadiant = FractionConvected). The convective heat gain is added to the space or zone air heat balance directly.

1.14.14.2.15 Space or Zone Baseboard Total Heating Rate [W]

1.14.14.2.16 Space or Zone Baseboard Total Heating Energy [J]

These output variables are the amount of heat gain for all ZoneBaseboard:OutdoorTemperatureControlled objects within the space or zone in Watts (for rate) or Joules. This is determined by the sum of the radiant and convective heat gains from the baseboard heat.

1.14.15 SwimmingPool:Indoor

The Indoor Swimming Pool object is used to describe the indoor swimming pools that are exposed to the internal environment. There are several rules that should be noted regarding the specification of an indoor pool in EnergyPlus. First, the pool is linked to a surface that must be a floor. The pool is assumed to cover the entire floor to which it is linked. If the pool only covers part of the floor in the actual building, then the user must break the floor up into multiple sections.

As pools attempt to achieve a particular water temperature and have a variety of heat losses, heating equipment is necessary to maintain the proper setpoint temperature. In EnergyPlus, the pool itself becomes part of the demand side of a plant loop with heating equipment on the supply side providing whatever heating is needed to maintain the desired temperature. This heating equipment as well as the loop connections must be entered separately and the input shown in this section only details what is needed to specify the pool itself.

There are a variety of rules that limit the application of indoor swimming pools in EnergyPlus. The following are a list of these rules:

- The pool must reference a valid surface in the input file. This surface must be a floor and cannot be other surface types like ceilings, walls, windows, etc.
- The pool cannot refer to a surface that is also a radiant system, ventilated slab, or another pool.
- The surface that the pool references must be modeled using conduction transfer functions (CTF).
- The pool cannot utilize movable insulation or have a heat source or sink associated with it (something used to model low temperature radiant systems).

The following information is useful for defining and modeling an indoor pool in EnergyPlus. For more information on the algorithm used for this model or details on some of the input parameters, please reference the Indoor Swimming Pool section of the EnergyPlus Engineering Reference document.

1.14.15.1 Inputs

1.14.15.1.1 Field: Name

This is a unique name associated with the indoor swimming pool.

1.14.15.1.2 Field: Surface Name

This is the name of the surface (floor) where the pool is located. Pools are not allowed on any surfaces other than a floor. For more rules on surfaces that can be used for pools, please see the information in this section on indoor pools above.

1.14.15.1.3 Field: Average Depth

This field is the average depth of the pool in meters. If the pool has variable depth, the average depth should be specified to achieve the proper volume of water in the pool.

1.14.15.1.4 Field: Activity Factor Schedule Name

This field references a schedule that contains values for pool activity. This parameter can be varied using the schedule named here, and it has an impact on the amount of evaporation that will take place from the pool to the surrounding zone air. For example values of the activity factor and what impact it will have on the evaporation of water from the pool, please refer to the Indoor Swimming Pool section of the EnergyPlus Engineering Reference document. If left blank, the activity factor will be assumed to be unity. Note that the activity factor schedule should not be set equal to an occupancy schedule since an activity factor of zero means that no evaporation will take place from the pool.

1.14.15.1.5 Field: Make-up Water Supply Schedule Name

The schedule named by this field establishes a cold water temperature [C] for the water that replaces the water which is lost from the pool due to evaporation. If blank, water temperatures are calculated by the Site:WaterMainsTemperature object. This field (even if blank) overrides the Cold Water Supply Temperature Schedule in all of the listed WaterUse:Equipment objects.

1.14.15.1.6 Field: Cover Schedule Name

This schedule defines when the pool water cover is available and affects the evaporation, convection, and radiation rate calculations. A schedule value of 0.0 means that the pool is not covered. A schedule value of 1.0 means the pool is 100% covered. The pool may be fully covered, fully open (uncovered), or partially covered (a value between 0.0 and 1.0). The user also has the option to control the evaporation, convection, short-wavelength radiation, and long-wavelength radiation factors when the pool is covered. These terms are discussed in the next four fields.

1.14.15.1.7 Field: Cover Evaporation Factor

This input field can optionally be used to modify the pool evaporation rate and is used in conjunction with the pool cover factor defined by the Pool Cover Schedule field (see above). The value for this parameter can normally range from 0.0 to 1.0, where 1 means that the pool cover completely eliminates evaporation from the pool surface, 0 means the pool cover has no effect on evaporation, and fractions in between 0 and 1 result in a fractional reduction in evaporation by the pool cover. So, if this parameter is 0.5 and the pool is 50% covered, the overall reduction in evaporation from a fully uncovered pool is 25% or 0.25.

1.14.15.1.8 Field: Cover Convection Factor

This input field can optionally be used to modify the pool convection rate and is used in conjunction with the pool cover factor defined by the Pool Cover Schedule field (see above). The value for this parameter can normally range from 0.0 to 1.0, where 1 means that the pool cover completely eliminates convection from the pool surface, 0 means the pool cover has no effect on convection, and fractions in between 0 and 1 result in a fractional reduction in convection by the pool cover. So, if this parameter is 0.5 and the pool is 50% covered, the overall reduction in convection from a fully uncovered pool is 25% or 0.25.

1.14.15.1.9 Field: Cover Short-Wavelength Radiation Factor

This input field can optionally be used to modify the pool short-wavelength radiation rate and is used in conjunction with the pool cover factor defined by the Pool Cover Schedule field (see above). The value for this parameter can normally range from 0.0 to 1.0, where 1 means that the pool cover completely eliminates short-wavelength radiation from the pool surface, 0 means the pool cover has no effect on short-wavelength radiation, and fractions in between 0 and 1 result in a fractional reduction in short-wavelength radiation by the pool cover. So, if this parameter is 0.5 and the pool is 50% covered, the overall reduction in short-wavelength radiation from a fully uncovered pool is 25% or 0.25. Note that with radiation terms that whatever portion of the short-wavelength radiation is blocked by the cover is transferred via convection to the surrounding zone air.

1.14.15.1.10 Field: Cover Long-Wavelength Radiation Factor

This input field can optionally be used to modify the pool long-wavelength radiation rate and is used in conjunction with the pool cover factor defined by the Pool Cover Schedule field (see above). The value for this parameter can normally range from 0.0 to 1.0, where 1 means that the pool cover completely eliminates long-wavelength radiation from the pool surface, 0 means the pool cover has no effect on long-wavelength radiation, and fractions in between 0 and 1 result in a fractional reduction in long-wavelength radiation by the pool cover. So, if this parameter is 0.5 and the pool is 50% covered, the overall reduction in long-wavelength radiation from a fully uncovered pool is 25% or 0.25. Note that with radiation terms that whatever portion of the long-wavelength radiation is blocked by the cover is transferred via convection to the surrounding zone air.

1.14.15.1.11 Field: Pool Water Inlet Node

This input is the name of the node on the demand side of a plant loop that leads into the pool. From the standpoint of an EnergyPlus input file, the pool is placed on a plant demand loop, and the pump and heater reside on the plant supply loop. The pool heater and pump must be defined by other existing EnergyPlus input.

1.14.15.1.12 Field: Pool Water Outlet Node

This input is the name of the node on the demand side of a plant loop that leads out of the pool. From the standpoint of an EnergyPlus input file, the pool is placed on a plant demand loop, and the pump and heater reside on the plant supply loop. The pool heater and pump must be defined by other existing EnergyPlus input.

1.14.15.1.13 Field: Pool Water Maximum Flow Rate

This input is the maximum water volumetric flow rate in m^3/s going between the pool and the water heating equipment. This along with the pool setpoint temperature and the heating plant equipment outlet temperature will establish the maximum heat addition to the pool. This flow rate to the pool will be varied in an attempt to reach the desired pool water setpoint temperature (see Setpoint Temperature Schedule below).

1.14.15.1.14 Field: Pool Miscellaneous Equipment Power

This input defines the power consumption rate of miscellaneous equipment such as the filtering and chlorination technology associated with the pool. The units for this input are in power consumption per flow rate of water through the pool from the heater or $W/(m^3/s)$. This field will be multiplied by the flow rate of water through the pool to determine the power consumption of this equipment. Any heat generated by this equipment is assumed to have no effect on the pool water itself.

1.14.15.1.15 Field: Setpoint Temperature Schedule

Pools attempt to maintain a particular water temperature. In EnergyPlus, this field defines the setpoint temperature for the desired pool water temperature. It is input as a schedule to allow the user to vary the pool setpoint temperature as desired. The equipment defined to provide heating for the pool will deliver the necessary hot water to the pool, up to the capacity of that equipment defined by other input by the user.

1.14.15.1.16 Field: Maximum Number of People

This field defines the maximum occupancy of people actually in the pool and thus will be used with the next two inputs to determine how much heat people contribute to the pool heat balance. People who are not in the pool should be modeled separately using the standard People description for zones.

1.14.15.1.17 Field: People Schedule

This field defines a schedule that establishes how many people are in the pool at any given time. The current value of this schedule is multiplied by the maximum number of people in the previous field determines how many people are currently in the pool.

1.14.15.1.18 Field: People Heat Gain Schedule

This field defines the amount of heat given off by an average person in the pool in Watts. This field is a schedule so that this heat gain can be allowed to vary as the type of activity in a pool can vary greatly and thus the amount of heat gain per person also varies. This parameter times the number of people in the pool determines how much heat is added to the pool. All heat given off by people is added to the heat balance of the pool water.

An example of an indoor swimming pool definition is:

```
SwimmingPool:Indoor,
  Test Pool,            !- Name
  F1-1,                 !- Surface Name
  1.5,                  !- Average Depth {m}
  PoolActivitySched,    !- Pool Activity Schedule
  MakeUpWaterSched,     !- MakeUp Water Temperature Schedule
  PoolCoverSched,       !- Pool Cover Schedule
  0.0,                  !- Cover Evaporation Factor
  0.2,                  !- Cover Convection Factor
  0.9,                  !- Cover Short-Wavelength Radiation Factor
  0.5,                  !- Cover Long-Wavelength Radiation Factor
  Pool Water Inlet Node,   !- Water Inlet Node (Plant/Heater)
  Pool Water Outlet Node,  !- Water Outlet Node (Plant/Heater)
  0.1,                  !- Maximum flow rate from water heating system {m3/s}
  0.6,                  !- Miscellaneous Equipment Power Factor {W/(m3/s)}
  PoolSetpointTempSched, !- Pool Water Setpoint Temperature Schedule
  15,                   !- Maximum Number of People in Pool
  PoolOccupancySched,   !- Pool People Schedule
  PoolOccHeatGainSched; !- Pool People Heat Gain Schedule
```

1.14.15.2 Outputs

- HVAC, Average, Indoor Pool Makeup Water Rate [m3/s]
- HVAC, Sum, Indoor Pool Makeup Water Volume [m3]
- HVAC, Average, Indoor Pool Makeup Water Temperature [C]
- HVAC, Average, Indoor Pool Water Temperature [C]
- HVAC, Average, Indoor Pool Inlet Water Temperature [C]
- HVAC, Average, Indoor Pool Inlet Water Mass Flow Rate [kg/s]
- HVAC, Average, Indoor Pool Miscellaneous Equipment Power [W]
- HVAC, Sum, Indoor Pool Miscellaneous Equipment Energy [J]
- HVAC, Average, Indoor Pool Water Heating Rate [W]
- HVAC, Sum, Indoor Pool Water Heating Energy [J]
- HVAC, Average, Indoor Pool Radiant to Convection by Cover [W]
- HVAC, Average, Indoor Pool People Heat Gain [W]
- HVAC, Average, Indoor Pool Current Activity Factor []
- HVAC, Average, Indoor Pool Current Cover Factor []
- HVAC, Average, Indoor Pool Evaporative Heat Loss Rate [W]
- HVAC, Sum, Indoor Pool Evaporative Heat Loss Energy [J]
- HVAC, Average, Indoor Pool Saturation Pressure at Pool Temperature [Pa]
- HVAC, Average, Indoor Pool Partial Pressure of Water Vapor in Air [Pa]
- HVAC, Average, Indoor Pool Current Cover Evaporation Factor []
- HVAC, Average, Indoor Pool Current Cover Convective Factor []
- HVAC, Average, Indoor Pool Current Cover SW Radiation Factor []
- HVAC, Average, Indoor Pool Current Cover LW Radiation Factor []

1.14.15.2.1 Indoor Pool Makeup Water Rate [m^3/s]

The water consumption rate for the makeup water of indoor swimming pool.

1.14.15.2.2 Indoor Pool Makeup Water Volume [m^3]

The water consumption for the makeup water of indoor swimming pool.

1.14.15.2.3 Indoor Pool Makeup Water Temperature [C]

The temperature of the makeup water of indoor swimming pool.

1.14.15.2.4 Indoor Pool Water Temperature [C]

The average calculated pool water temperature during the simulation at the time frequency requested.

1.14.15.2.5 Indoor Pool Inlet Water Temperature [C]

The temperature of the water being sent to the pool from the plant heating equipment.

1.14.15.2.6 Indoor Pool Inlet Water Mass Flow Rate [kg/s]

The mass flow rate of water being sent to the pool from the plant heating equipment. Typically this water is being passed through a heater and miscellaneous equipment.

1.14.15.2.7 Indoor Pool Miscellaneous Equipment Power [W]

The miscellaneous equipment power includes the power consumption of pool filter and chlorinator in Watts.

1.14.15.2.8 Indoor Pool Miscellaneous Equipment Energy [J]

The miscellaneous equipment power consumption includes the energy consumption of pool filter and chlorinator in Joules.

1.14.15.2.9 Indoor Pool Water Heating Rate [W]

This is the rate of heating provided by the plant loop to the pool in Watts.

1.14.15.2.10 Indoor Pool Water Heating Energy [J]

This is the amount of heating provided by the plant loop to the pool in Joules over the time step requested.

1.14.15.2.11 Indoor Pool Radiant to Convection by Cover [W]

The pool cover may block some or all of short- and long-wavelength radiation incident on the pool. To account for this and to not have the cover result in energy that is not accounted for by the model, the radiation that is blocked by the cover is converted to a convective gain (or loss) to/from the zone air. This output field reports this value.

1.14.15.2.12 Indoor Pool Current Activity Factor []

This is the current activity factor as defined by the user input schedule.

1.14.15.2.13 Indoor Pool Current Cover Factor []

This is the current cover factor as defined by the user input schedule.

1.14.15.2.14 Indoor Pool Evaporative Heat Loss Rate [W]

This is the rate of evaporative heat loss (latent) to the zone from the pool in Watts.

1.14.15.2.15 Indoor Pool Evaporative Heat Loss Energy [J]

This is the amount of evaporative heat loss (latent) to the zone from the pool in in Joules over the time step requested.

1.14.15.2.16 Indoor Pool Saturation Pressure at Pool Temperature [Pa]

This is the saturation pressure of water vapor in air at the pool water temperature.

1.14.15.2.17 Indoor Pool Partial Pressure of Water Vapor in Air [Pa]

This is the partial pressure of water vapor in air at the current zone air conditions for dry bulb temperature and humidity ratio.

1.14.15.2.18 Indoor Pool Current Cover Evaporation Factor []

This is the current value of the cover evaporation factor that is used as a modifier for the actual evaporation. A value of zero means no evaporation will take place while a value of unity means the maximum allowed evaporation will take place. This value is based on the current cover condition as well as the user input for the cover evaporation factor.

1.14.15.2.19 Indoor Pool Current Cover Convective Factor []

This is the current value of the cover convective factor that is used as a modifier for the actual convection. A value of zero means the cover will block all convection while a value of unity means that the cover will not affect convection from the water surface at all. This value is based on the current cover condition as well as the user input for the cover convective factor.

1.14.15.2.20 Indoor Pool Current Cover SW Radiation Factor []

This is the current value of the cover short wavelength radiation factor that is used as a modifier for the actual short wavelength radiation. A value of zero means the cover will block all short wavelength radiation while a value of unity means that the cover will not affect short wavelength radiation from the water surface at all. This value is based on the current cover condition as well as the user input for the cover short wavelength radiation factor.

1.14.15.2.21 Indoor Pool Current Cover LW Radiation Factor []

This is the current value of the cover long wavelength radiation factor that is used as a modifier for the actual long wavelength radiation. A value of zero means the cover will block all long wavelength radiation while a value of unity means that the cover will not affect long wavelength radiation from the water surface at all. This value is based on the current cover condition as well as the user input for the cover long wavelength radiation factor.

1.14.16 ZoneContaminantSourceAndSink:CarbonDioxide

The ZoneContaminantSourceAndSink:CarbonDioxide object allows users to input carbon dioxide sources or sinks in a zone. Note that carbon dioxide generation within a zone can also be specified using People and GasEquipment objects. Multiple ZoneContaminantSourceAndSink:CarbonDioxide objects can be specified for the same zone.

1.14.16.1 Inputs

1.14.16.1.1 Field: Name

The name of the ZoneContaminantSourceAndSink:CarbonDioxide object. The name for each ZoneContaminantSourceAndSink:CarbonDioxide object must be unique.

1.14.16.1.2 Field: Zone Name

This field is the name of the zone (ref: **Zone**) and links a particular ZoneContaminantSourceAndSink:CarbonDioxid object to a thermal zone in the building.

1.14.16.1.3 Field: Design Generation Rate

This field denotes the design carbon dioxide generation rate (m^3/s). The design value is modified by the schedule fraction (see Field: Schedule Name). The resulting volumetric generation rate is converted to mass generation rate using the current zone indoor air density at each time step. The rate can be either positive or negative. A positive value represents a source rate (CO_2 addition to the zone air) and a negative value represents a sink rate (CO_2 removal from the zone air).

When the mass design generation rate is available, a conversion is required to meet input requirement with volumetric flow rate. This can be accomplished by the mass flow rate divided by the density of carbon dioxide.

1.14.16.1.4 Field: Schedule Name

This field is the name of the schedule (ref: Schedules) that modifies the design carbon dioxide generation rate (see previous field). The schedule values can be any positive number between 0.0 and 1.0. For each simulation time step, the actual CO_2 generation rate in a zone is the product of the Design Generation Rate field (above) and the value specified by this schedule.

An IDF example is provided below:

```
ZoneContaminantSourceAndSink:CarbonDioxide,
  NORTH_ZONE CO2,       !- Name
  NORTH_ZONE,           !- Zone Name
  1.e-6,                !- Design Generation Rate {m3/s}
  CO2 Source Schedule;  !- Schedule Name
```

1.14.16.2 Outputs

```
HVAC,Average, Contaminant Source or Sink CO2 Gain Volume Flow Rate [m3/s]
HVAC,Average, Zone Contaminant Source or Sink CO2 Gain Volume Flow Rate [m3/s]
```

1.14.16.2.1 Contaminant Source or Sink CO2 Gain Volume Flow Rate [m3/s]

This output is the net carbon dioxide internal gain/loss in m^3/s for an individual ZoneContaminantSourceAndSink:C object.

1.14.16.2.2 Zone Contaminant Source or Sink CO2 Gain Volume Flow Rate [m3/s]

This output is the net carbon dioxide internal gain/loss in m^3/s for all ZoneContaminantSourceAndSink:CarbonDiox objects in a zone.

1.14.17 ZoneContaminantSourceAndSink:Generic:Constant

The ZoneContaminantSourceAndSink:Generic:Constant object specifies the generic contaminant generation rate and removal rate coefficient in a zone. The associated fraction schedules are required for allowing users to change the magnitude of sources and sinks. The object is equivalent to the combination of the constant coefficient model and the burst source model defined in the sources and sinks element types of CONTAM 3.0. The basic equation used to calculate generic contaminant source and sink for the constant model is given below:

$$S_f(t) = G_f(t) \cdot F_G - R_f(t) \cdot C_f(t) \cdot 10^{-6} \cdot F_R \tag{1.31}$$

where

S_f = Contaminant generic contaminant source strength [m^3/s]

G_f = Generic contaminant generation rate [m^3/s]

F_G = Fraction value from the source fraction schedule at a given time [dimensionless]
R_f = Generic contaminant effective removal rate [m^3/s]
F_R = Fraction value from the sink fraction schedule at a given time [dimensionless]
C_f = Generic contaminant concentration value at a given previous time step [ppm]

1.14.17.1 Inputs

1.14.17.1.1 Field: Name

This field represents a unique identifying name.

1.14.17.1.2 Field: Zone Name

This field signifies the Name of the zone with constant generic contaminant source and sink.

1.14.17.1.3 Field: Design Generation Rate

This field denotes the full generic contaminant design generation rate (m^3/s). The design generation rate is the maximum amount of generic contaminant expected at design conditions. The design value is modified by the schedule fraction (see Field:Generation Schedule Name).

When the mass generation rate is available, the rate must be converted to a volume flow rate. Use the mass flow rate divided by the vapor density of the generic contaminant.

1.14.17.1.4 Field: Generation Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the maximum design generation rate (G_f). This fraction between 0.0 and 1.0 is noted as F_G in the above equation.

1.14.17.1.5 Field: Design Removal Coefficient

This field denotes the full generic contaminant design removal coefficient (m^3/s). The design removal rate is the maximum amount of generic contaminant expected at design conditions times the generic contaminant concentration in the same zone. The design value is modified by the schedule fraction (see Field:Removal Schedule Name).

1.14.17.1.6 Field: Removal Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the maximum design generation rate (R_f). This fraction between 0.0 and 1.0 is noted as F_R in the above equation.

An IDF example is provided below:

```
ZoneContaminantSourceAndSink:Generic:Constant,
  NORTH ZONE GC,         !- Name
  NORTH ZONE,            !- Zone Name
  1.0E-6,                !- Design Generation Rate {m3/s}
  GC Source Schedule,    !- Generation Schedule Name
  1.0E-7,                !- Design Removal Coefficient {m3/s}
  GC Removal Schedule;   !- Removal Schedule Name
```

1.14.17.2 Outputs

When a ZoneContaminantSourceAndSink:Generic:Constant object is specified, the following output variables are available:

ZONE,Average, Generic Air Contaminant Constant Source Generation Volume Flow Rate [m3/s]

1.14.17.2.1 Generic Air Contaminant Constant Source Generation Volume Flow Rate [m3/s]

This output is the average generic contaminant generation rate from each ZoneContaminantSourceAndSink:Generic object. The generation rate is a sum of generation and removal rates. The zone air generic contaminant level at the previous zone time step is used in the removal rate calculation.

1.14.18 SurfaceContaminantSourceAndSink:Generic:PressureDriven

The SurfaceContaminantSourceAndSink:Generic:PressureDriven object specifies the generic contaminant generation rate coefficient, which is used to calculate the generation rate due to the pressure difference across the surface. The object is equivalent to the pressure driven model defined in the sources and sinks element types of CONTAM 3.0. This object assumes to work with the AirflowNetwork model. The surface has to be defined in the AirflowNetwork:Multizone:Surface. Although the model is designed to be applied to radon and soil gas entry, it is expanded to be applied to all contaminant transport, including generic contaminant. However, it should be used in caution. The basic equation used to calculate generic contaminant source for the pressure driven constant model is provided below:

$$S_f(t) = H_f(t) \cdot F_G \cdot (P_j - P_i)^n \tag{1.32}$$

where
S_f = Generic contaminant source strength [m^3/s]
H_f = Generic contaminant generation rate coefficient [m^3/s]
F_G = Fraction value from the source fraction schedule at a given time [dimensionless]
n = Flow power exponent
P_i = Zone pressure [Pa]
P_j = Pressure in an adjacent zone for a interior surface or outdoor for an exterior surface [Pa]

1.14.18.1 Inputs

1.14.18.1.1 Field: Name

The field signifies the unique identifying name.

1.14.18.1.2 Field: Surface Name

This field represents the name of the surface as a generic contaminant source using the pressure driven model.

1.14.18.1.3 Field: Design Generation Rate Coefficient

This field denotes the generic contaminant design generation coefficient (m^3/s). The design generation rate is the maximum amount of generic contaminant expected at design conditions times the pressure difference with a power exponent across a surface. The design value is modified by the schedule fraction (see Field:Generation Schedule Name).

1.14.18.1.4 Field: Generation Exponent

This field denotes the flow power exponent, n, in the contaminant source equation. The valid range is 0.0 to 1.0,

An IDF example is provided below:

```
SurfaceContaminantSourceAndSink:Generic:PressureDriven,
  WEST ZONE GC BLD,      !- Name
  Zn001:Wall001,         !- Surface Name
  1.0E-6,                !- Design Generation Rate Coefficient {m3/s}
```

```
GC Source Schedule,   !- Generation Schedule Name
0.5;                  !- Generation Exponent
```

1.14.18.2 Outputs

When a SurfaceContaminantSourceAndSink:Generic:PressureDriven object is specified, the following output variables are available:

- ZONE,Average,Generic Air Contaminant Pressure Driven Generation Volume Flow Rate [m3/s]

1.14.18.2.1 Generic Air Contaminant Pressure Driven Generation Volume Flow Rate [m3/s]

This output is the average generic contaminant generation rate from each SurfaceContaminantSource-AndSink:Generic:PressureDriven object.

1.14.19 ZoneContaminantSourceAndSink:Generic:CutoffModel

The ZoneContaminantSourceAndSink:Generic contaminant:CutoffModel object specifies the generic contaminant generation rate based on the cutoff concentration model. The basic equation used to calculate generic contaminant source for the pressure driven constant model is given below:

$$S_f(t) = \left\{ \begin{array}{ll} G_f(t) * F_G * \left(1 - \frac{C_f(t)}{C_{cutoff}}\right) & C_f < C_{cutoff} \\ 0 & C_f \geq C_{cutoff} \end{array} \right\} \tag{1.33}$$

where
S_f = Generic contaminant source strength [m^3/s]
G_f = Generic contaminant generation rate [m^3/s]
F_G = Fraction value from the source fraction schedule at a given time [dimensionless]
C_{cutoff} = Cutoff concentration at which emission ceases [ppm]
C_f = Generic contaminant concentration value at a given previous time step [ppm]

1.14.19.1 Inputs

1.14.19.1.1 Field: Name

The field signifies the unique identifying name.

1.14.19.1.2 Field: Zone Name

This field represents the name of the zone with generic contaminant source and sink using the cutoff model.

1.14.19.1.3 Field: Design Generation Rate Coefficient

This field denotes the full generic contaminant design generation rate (m^3/s). The design generation rate is the maximum amount of generic contaminant expected at design conditions. The design value is modified by the schedule fraction (see Field:Generation Schedule Name).

When the mass generation rate is available, the rate must be converted to a volume flow rate. Use the mass flow rate divided by the vapor density of the generic contaminant.

1.14.19.1.4 Field: Generation Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the maximum design generation rate (G_f). This fraction between 0.0 and 1.0 is noted as F_G in the above equation.

1.14.19.1.5 Field: Cutoff Generic Contaminant at which Emission Ceases

This field is the generic contaminant cutoff concentration level where the source ceases its emission.

An IDF example is provided below:

```
ZoneContaminantSourceAndSink:Generic:CutoffModel,
  NORTH ZONE GC CutoffModel,            !- Name
  NORTH ZONE,                           !- Zone Name
  1.0E-5,                               !- Design Generation Rate Coefficient {m3/s}
  GC Source Schedule,                   !- Schedule Name
  100000;                               !- Cutoff Generic Contaminant at which Emission Ceases {ppm}
```

1.14.19.2 Outputs

When a ZoneContaminantSourceAndSink:Generic:CutoffModel object is specified, the following output variables are available:

ZONE,Average, Generic Air Contaminant Cutoff Model Generation Volume Flow Rate [m3/s]

1.14.19.2.1 Generic Air Contaminant Cutoff Model Generation Volume Flow Rate [m3/s]

This output is the average generic contaminant generation rate from each SurfaceContaminantSource-AndSink:Generic:CutoffModel object.

1.14.20 ZoneContaminantSourceAndSink:Generic:DecaySource

The ZoneContaminantSourceAndSink:Generic:DecaySource object specifies the generic contaminant generation rate based on the decay source model. The basic equation used to calculate generic contaminant source for the decay source model is given below:

$$S_f(t) = G_f(t) F_G \exp(-t/t_c) \tag{1.34}$$

where

S_f = Generic contaminant source strength [m^3/s]

G_f = Initial generic contaminant generation rate [m^3/s]

F_G = Fraction value from the source fraction schedule at a given time [dimensionless]

t = Time since the start of emission [second]

t_c = Decay time constant [second]

1.14.20.1 Inputs

1.14.20.1.1 Field: Name

This field is the unique identifying name.

1.14.20.1.2 Field: Zone Name

This field represents the name of the zone with a generic contaminant decaying source.

1.14.20.1.3 Field: Initial Emission Rate

This field denotes the initial generic contaminant design emission rate (m^3/s). The generation is controlled by a schedule, as defined in the next field. Generic contaminant emission begins when the schedule changes from a zero to a non-zero value (between zero and one). The initial emission rate is equal to the schedule value times the initial generation rate. A single schedule may be used to initiate several emissions at different times.(see Field:Generation Schedule Name).

1.14.20.1.4 Field: Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the maximum design emission rate (G_f). This fraction between 0.0 and 1.0 is noted as F_G in the above equation. When the value is equal to 1, the generation rate is used and time is reset to zero. When the value is equal to zero, the schedule value is ignored in the equation.

1.14.20.1.5 Field: Decay Time Constant

This field is the time at which the generation rate reaches 0.37 of the original rate.

> Note: The variable t, time since the start of emission, will be reset to zero, when a new run period starts, or the generation schedule value is equal to zero.

An IDF example is provided below:

```
ZoneContaminantSourceAndSink:Generic:DecaySource,
  WEST ZONE GC DecaySource, !- Name
  WEST ZONE,              !- Zone Name
  1.0E-5,                 !- Initial Emission Rate {m3/s}
  GC Source Schedule,     !- Schedule Name
  100000;                 !- Delay Time Constant {s}
```

1.14.20.2 Outputs

When a ZoneContaminantSourceAndSink:Generic:DecaySource object is specified, the following output variables are available:

- Zone,Average,Generic Air Contaminant Decay Model Generation Volume Flow Rate [m3/s]
- Zone,Average,Generic Air Contaminant Decay Model Generation Emission Start Elapsed Time [s]

1.14.20.2.1 Generic Air Contaminant Decay Model Generation Volume Flow Rate [m3/s]

This output is the average generic contaminant decay rate from each SurfaceContaminantSourceAndSink:Generic:DecaySource object.

1.14.20.2.2 Generic Air Contaminant Decay Model Generation Emission Start Elapsed Time [s]

This output is the decay time since the start of emission. The start time is either at the beginning of each run period, including design day simulations, or the time when the generation schedule value is zero.

1.14.21 SurfaceContaminantSourceAndSink:Generic:BoundaryLayerDiffusion

The SurfaceContaminantSourceAndSink:Generic:BoundaryLayerDiffusion object specifies the generic contaminant generation rate from surface diffusion. The object is equivalent to the boundary layer diffusion model driven model defined in the sources and sinks element types of CONTAM 3.0.

The boundary layer diffusion controlled reversible sink/source model with a linear sorption isotherm follows the descriptions presented in [Axley 1991]. The boundary layer refers to the region above the surface of a material through which a concentration gradient exists between the near-surface concentration and the air-phase concentration. The rate at which a contaminant is transferred onto a surface (sink) is defined as:

$$S_f = h \cdot \rho \cdot A\left(C_f - \frac{C_s}{k}\right) * 10^{-6} \tag{1.35}$$

where
h = Average film mass transfer coefficient over the sink [m/s]
ρ = Film density of air [kg/m^3]
A = Surface area of the adsorbent [m^2]
C_f = Concentration in the air at the previous time step [ppm]
C_s = Concentration in the adsorbent [ppm]
k = Henry adsorption constant or partition coefficient [dimensionless]

1.14.21.1 Inputs

1.14.21.1.1 Field: Name

This field signifies a unique identifying name.

1.14.21.1.2 Field: Surface Name

This field denotes the name of the surface as a generic contaminant reversible source or sink using the boundary layer diffusion model.

1.14.21.1.3 Field: Mass Transfer Coefficient

This field specifies the average mass transfer coefficient of the contaminant generic contaminant within the boundary layer (or film) above the surface of the adsorbent.

1.14.21.1.4 Field: Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the mass transfer coefficient with the value between 0.0 and 1.0.

1.14.21.1.5 Field: Henry Adsorption Constant or Partition Coefficient

This field denotes the generic contaminant Henry partition coefficient in the units of dimensionless. The coefficient relates the concentration of the contaminant generic contaminant in the bulk-air to that at the surface of the adsorption material.

An IDF example is provided below:

```
SurfaceContaminantSourceAndSink:Generic:BoundaryLayerDiffusion,
  WEST ZONE GC BLD,      !- Name
  Zn001:Wall001,         !- Surface Name
  1.0E-2,                !- Mass Transfer Coefficient {m/s}
  GC Source Schedule,    !- Schedule Name
  2.0;                   !- Henry adsorption constant or partition coefficient
```

1.14.21.2 Outputs

When a SurfaceContaminantSourceAndSink:Generic:BoundaryLayerDiffusion object is specified, the following output variables are available:

- ZONE,Average, Generic Air Contaminant Boundary Layer Diffusion Generation Volume Flow Rate [m3/s]
- ZONE,Average, Generic Air Contaminant Boundary Layer Diffusion Inside Face Concentration [ppm]

1.14.21.2.1 Generic Air Contaminant Boundary Layer Diffusion Generation Volume Flow Rate [m3/s]

This output is the average generic contaminant generation rate from each SurfaceContaminantSourceAndSink:Generic:BoundaryLayerDiffusion object.

1.14.21.2.2 Generic Air Contaminant Boundary Layer Diffusion Inside Face Concentration [ppm]

This output is the average generic contaminant level at the interior surface.

1.14.22 SurfaceContaminantSourceAndSink:Generic:DepositionVelocitySink

The SurfaceContaminantSourceAndSink:Generic:DepositionVelocitySink object specifies the generic contaminant removal rate from a surface. The object is equivalent to the deposition velocity sink model defined in CONTAM 3.0 sources and sinks element types.

The deposition velocity model provides for the input of a sink's characteristic in the familiar term of deposition velocity. The removal stops when the sink concentration level is higher than the zone air concentration level. The deposition velocity model equation is:

$$S_f(t) = \nu_d A_s m \cdot C_f(t) \cdot 10^{-6} \cdot F_R \tag{1.36}$$

where

$S_f(t)$ = Removal rate at time t [m^3/s]

ν_d = Deposition velocity [m/s]

A_s = Deposition surface area [m^2]

m = Element multiplier [dimensionless]

$C_f(t)$ = Concentration of contaminant generic contaminant at the previous time step [ppm]

$\mathrm{F_R}$ = Schedule or control signal value at time t [-]

1.14.22.1 Inputs

1.14.22.1.1 Field: Name

This field denotes a unique identifying name.

1.14.22.1.2 Field: Surface Name

This field represents the name of the surface as a generic contaminant sink using the deposition velocity sink model.

1.14.22.1.3 Field: Deposition Velocity

This field specifies the deposition velocity to the surface of the adsorbent in the units of m/s.

1.14.22.1.4 Field: Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the maximum design removal rate (S_f). This fraction between 0.0 and 1.0 is noted as F_R in the above equation.

An IDF example is provided below:

```
SurfaceContaminantSourceAndSink:Generic:DepositionVelocitySink,
  EAST ZONE GC DVS,      !- Name
  Zn002:Wall001,         !- Surface Name
  1.0E-3,                !- Deposition Velocity {m/s}
  GC Source Schedule;    !- Schedule Name
```

1.14.22.2 Outputs

The following output variables are available when the SurfaceContaminantSourceAndSink:Generic:-DepositionVelocitySink object is specified.

- ZONE,Average, Generic Air Contaminant Deposition Velocity Removal Volume Flow Rate [m3/s]

1.14.22.2.1 Generic Air Contaminant Deposition Velocity Removal Volume Flow Rate [m3/s]

This output is the average generic contaminant generation rate from each SurfaceContaminantSource-AndSink:Generic:DepositionVelocitySink object.

1.14.23 ZoneContaminantSourceAndSink:Generic:DepositionRateSink

The ZoneContaminantSourceAndSink:Generic:DepositionRateSink object specifies the generic contaminant removal rate from a zone. The object is equivalent to the deposition rate sink model defined in CONTAM 3.0 sources and sinks element types.

The deposition rate model provides for the input of a sink's characteristic in the familiar term of deposition rate in a zone. The removal stops when the sink concentration level is higher than the zone air concentration level. The deposition rate model equation is:

$$S_f(t) = k_d V_z \cdot C_f(t) \cdot 10^{-6} \cdot F_R \cdot m \tag{1.37}$$

Where

$S_f(t)$ = Removal rate at time t [m^3/s]

k_d = Deposition rate [1/T]

V_z = Zone volume [m^3]

m = Element multiplier [dimensionless]

$C_f(t)$ = Concentration of generic contaminant at the previous time *step* [ppm]

F_R = Schedule or control signal value at time t [dimensionless]

1.14.23.1 Inputs

1.14.23.1.1 Field: Name

This field denotes a unique identifying name.

1.14.23.1.2 Field: Zone Name

This field represents the name of the zone as a generic contaminant sink using the deposition rate sink model.

1.14.23.1.3 Field: Deposition Rate

This field specifies the deposition rate to the zone in the units of 1/s.

1.14.23.1.4 Field: Schedule Name

This field is the name of the schedule (ref: Schedule) that modifies the maximum design removal rate (S_f). This fraction between 0.0 and 1.0 is noted as F_R in the above equation.

An IDF example is provided below:

```
ZoneContaminantSourceAndSink:Generic:DepositionRateSink,
  NORTH ZONE GC DRS,  !- Name
  NORTH ZONE,         !- Zone Name
  1.0E-5,             !- Deposition Rate {m3/s}
  GC Source Schedule; !- Schedule Name
```

1.14.23.2 Outputs

When the ZoneContaminantSourceAndSink:Generic:DepositionRateSink object is specified, the following output variables are available:

- ZONE,Average, Generic Air Contaminant Deposition Rate Removal Volume Flow Rate [m3/s]

1.14.23.2.1 Generic Air Contaminant Deposition Rate Removal Volume Flow Rate [m3/s]

This output is the average generic contaminant generation rate from each ZoneContaminantSourceAndSink:Generic:DepositionRateSink object.

1.15 Group – Daylighting

Daylighting can be invoked in EnergyPlus primarily by the use of the objects: **Daylighting:Controls** and **Daylighting:ReferencePoint**. Other objects related to daylighting are **Daylighting:DElight:ComplexFenestrati** **Output:IlluminanceMap**, and **OutputControl:IlluminanceMap:Style**. These are described in the following object descriptions.

> Note that two different methodologies are used in the Daylighting:Controls object (SplitFlux and DElight) and may be intermixed in a single IDF but may not be used in the same zone.

1.15.1 Daylighting:Controls

When this object is used, daylighting illuminance levels are calculated and then used to determine how much the electric lighting can be reduced. The daylight illuminance level in a zone depends on many factors, including sky condition; sun position; calculation point; location, size, and glass transmittance of windows; window shading devices; and reflectance of interior surfaces. Reduction of electric lighting depends on daylight illuminance level, illuminance set point, fraction of zone controlled and type of lighting control. Two different methods of computing the daylighting illuminance and subsequent lighting reduction called SplitFlux and DElight. After the object description below is are sections that describe the two methodologies in more detail.

1.15.1.1 Inputs

1.15.1.1.1 Field: Name

The name of the Daylighting:Controls object.

1.15.1.1.2 Field: Zone or Space Name

The name of the Zone or Space to which the following daylighting-related input applies. If a zone name is specified, the zone must lie completely within a single solar enclosure.